



DRAFT

OCPP 2.0.1 Edition 3
Errata 2025-11 DRAFT

Table of Contents

Disclaimer	1
Scope	2
Terminology and Conventions	2
0. Part 0	3
1. Part 1	4
1.1. Page 9 - (2025-01) - 3.4. Monitoring.	4
2. Part 2	5
2.1. Page 5 - (2025-01) - Updated limitations on BasicAuthPassword to increase security.	5
2.2. Page 14 - (2025-04) - Section 2.7. ISO 15118 support - Aligned ISO 15118 timeout table with updated version in OCPP	
2.1	6
2.3. Page 15 - (2025-01) - Improved text FR.04	6
2.4. Page 19 - (2025-01) - Removed requirement A00.FR.003 as the precondition never occurs	6
2.5. Page 19/41 - (2025-01) - Allow downgrading security profile from 3 to 2.	7
2.6. Page 22/27 - (2025-11) - Slight deviation in A00 requirement phrasing which is aligned	9
2.7. Page 26 - (2025-01) - 1.3.7. TLS with Client Side Certificates Profile - Requirements.	10
2.8. Page 28 - (2025-01) - Certificate Hierarchy	10
2.9. Page 31 - (2025-01) - A02/A03 - Updated error handling	10
2.10. Page 35 - (2025-11) - A02.FR.19 - Updated precondition	10
2.11. Page 38 - (2025-11) - A03.FR.19 - Updated precondition	11
2.12. Page 34 - (2025-01) - Disallow client certificates future validity date	11
2.13. Page 58/60 - (2025-01) - Missing requirement information about omitting the value for WriteOnly variables	15
2.14. Page 63 - (2025-11) - B09.FR.02/04/05 - Added optional reasonCode [1083]	15
2.15. Page 64 - (2025-09) - B09.FR.31/31 - Improved definition	16
2.16. Page 66 - (2025-11) - B10.FR.02 - Added optional <i>reasonCode</i>	16
2.17. Page 66 - (2025-04) - B10.FR.03/04/10 - Migrate to new NetworkConnectionProfile.	17
2.18. Page 67 - (2025-06) - B11 - Clarify meaning of OnIdle for Reset	17
2.19. Page 69 - (2025-01) - B11 - Reset without Ongoing Transaction - Requirements	18
2.20. Page 73 - (2025-01) - B12 - Reset with Ongoing Transaction - Requirements	18
2.21. Page 77 - (2025-01) - 1.6 Relationship between authorization and transaction.	19
2.22. Page 80 - (2025-01) - C01 - EV Driver Authorization using RFID - Requirements.	19
2.23. Page 80 - (2025-01) - C01 - EV Driver Authorization using RFID - Requirements.	20
2.24. Page 99 - (2025-01) - C09- Authorization by GroupId - Requirements	20
2.25. Page 101 - (2025-01) - Updated requirements related to clarify the relation between AuthCacheLifeTime and cacheExpiryDateTime	20
2.26. Page 109 - (2025-11) - C15 - Offline Authorization of unknown Id - Scenario description	21
2.26.1. C15 - Offline Authorization of unknown Id.	21
2.27. Page 113 - (2025-01) - C16 - Stop Transaction with a Master Pass - Requirements	21
2.28. Page 129 - (2025-01) - Updated sequence diagram E01 S5	22
2.29. Page 151 - (2025-02) - E06.FR.05 for DataSigned as TxStopPoint is invalid	23
2.30. Page 143 - (2025-11) - E04.FR.07/08 removed	23
2.31. Page 152 - (2025-01) - E07 - Improved scenario description names	24
2.32. Page 154 - (2025-01) - E07 - Transaction locally stopped by IdToken	24
2.33. Page 154 - (2025-11) - E07.FR.01 - Added note referring to C09	25
2.34. Page 155 - (2025-09) - E07.FR.07 - Improved precondition	26
2.35. Page 164 - (2025-11) - E10.FR.01/02 improved precondition [806]	26
2.35.1. E10 - When cable disconnected on EV-side: Suspend Transaction	26
2.36. Page 173 - (2025-11) - Updated Header - Section 2.2	27
2.37. Page 179 - (2025-11) - F01.FR.01/02 - No authorization for type = Central	27
2.38. Page 184 - (2025-11) - F02.FR.09/10 - No authorization for type = Central	28
2.39. Page 190 - (2025-11) - F05 - New requirement for connectors without lock [1013]	29
2.40. Page 198 - (2025-01) - G01 - Status Notification - Requirements	29
2.41. Page 197 - (2025-01) - G01 - Status Notification - State transition overview for connecting/disconnecting	29
2.42. Page 208 - (2025-01) - H. Reservation - Introduction.	30
2.43. Page 188 - (2025-11) - F03.FR.07 - Improved precondition and changed requirement definition	30
2.44. Page 188 - (2025-11) - F03.FR.08 - Improved precondition and changed requirement definition	31
2.45. Page 180 - (2025-11) - F01.FR.20/F02.FR.22 - EVSE can be reported on cable plug-in [786].	31

2.46. Page 213 - (2025-02) - H02 - Added missing requirements	31
2.47. Page 214/215 - (2025-01) - Improved use case scenario descriptions and added S3	33
2.48. Page 226 - (2025-01) - I06.FR.02 Improved requirement text	36
2.49. Page 231 - (2025-01) - Updated section Multiple Locations/Phases	36
2.50. Page 220 - (2025-11) - I01 - Scenario description update	37
2.50.1. I01 - Show EV Driver-specific Tariff Information	37
2.51. Page 221 - (2025-11) - I01 - Updated use case description	37
2.52. Page 243 - (2025-01) - Improved section on external Smart Charging Control Signals	37
2.53. Page 248 - (2025-01) - 3.7 Avoiding Phase Conflicts	39
2.54. Page 275 - (2025-06) - Updated remark of K11	39
2.55. Page 251 - (2025-06) - Updated note of K01.FR.05	39
2.56. Page 251 - (2025-06) - Add cross-references to K01.FR.06 and K01.FR.39	40
2.57. Page 254 - (2025-06) - K01.FR.50 requirement is a SHALL	40
2.58. Page 255 - (2025-11) - K01 added implicit requirement about <i>recurrencyKind</i> [777]	41
2.59. Page 257 - (2025-06) - K02 Updated remark of use case about merging profiles	42
2.60. Page 282 - (2025-01) - K15 - ISO 15118-2 Charging with load leveling - Requirements	42
2.61. Page 282 - (2025-09) - K15.FR.20 is not part of OCPP 2.0.1 [1061]	42
2.62. Page 286 - (2025-01) - K16 - Renegotiation initiated by CSMS - Requirements	43
2.63. Page 265 - (2025-11) - K06 - Updated use case description	43
2.64. Page 266 - (2025-11) - K06.FR.01 - Updated Requirement definition	43
2.65. Page 278 - (2025-11) - K13 - Updated description prerequisites	43
2.66. Page 284 - (2025-09) - K15 Added rule for composite schedules in case of multiple charging schedules [1002]	43
2.66.1. K15 - ISO 15118-2	44
2.67. Page 259/260 - (2025-01) - K03 - Updated use case description and sequence diagram	44
2.68. Page 274/275/276/277 - (2025-01) - K11/K12 - Updated use case descriptions and sequence diagrams	46
2.69. Page 278/279 - (2025-01) - K13 - Updated requirement preconditions	48
2.70. Page 279 - (2025-01) - K14 - Updated use case scenario description	49
2.71. Page 284 - (2025-09) - K16 use case description update	49
2.71.1. Page 285	50
2.72. Page 292 - (2025-01) - Use case L01 - Added clarification to step 3 about when to start downloading the firmware	50
2.73. Page 306 - (2025-01) - M. ISO 15118 Certificate Management	50
2.74. Page 308 - (2025-01) - Update introduction sequence diagram ISO 15118	51
2.75. Page 330 - (2025-01) - N03 Set Monitoring Base: Improved text of Remark	53
2.76. Page 331 - (2025-01) - N03.FR.04: text improvement	53
2.77. Page 350 - (2025-02) - O01 - Added missing requirements	53
2.78. Page 446 - (2025-01) - ActiveNetworkProfile is incorrectly marked as optional	54
2.79. Page 327 - (2025-09) - N01.FR.12 - Improved definition	54
2.80. Page 328 - (2025-09) - N02: changed empty to absent	54
2.81. Page 453 - (2025-01) - References to <i>monitorValue</i> changed to <i>value</i> [354]	55
2.82. Page 454 - (2025-01) - N04.FR.06 Improved limit definition of thresholds [353]	55
2.83. Page 456 - (2025-02) - New configuration variable to allow TLS wildcard certificates	55
2.84. Page 333 - (2025-11) - N04 Added requirement	56
2.85. Page 340 - (2025-11) - Added remark to N09	56
2.86. Page 341 - (2025-11) - N09.FR.06 Improvement requirement definition	56
2.87. Page 341 - (2025-11) - N09.FR.06 Improvement requirement definition	56
2.88. Page 458 - (2025-01) - Added optional variable to allow the Charging Station to report its supported idTokenTypes	57
2.89. Page 462 - (2025-01) - Added note to EnergyTransfer description as TxStartPoint	57
2.90. Page 467 - (2025-09) - Error in description of AlignedData interval variables [1043]	57
2.90.1. AlignedDataInterval	58
2.90.2. AlignedDataTxEndedInterval	58
2.91. Page 476 - (2025-01) - Added Connector component to AvailabilityState referenced variable	58
2.92. Page 452 - (2025-11) - TimeOffsetNextTransaction renamed [801]	59
2.92.1. TimeOffsetNextTransition	59
2.93. Page 397 - (2025-11) - issuerKeyHash in CertificateHashDataType [826]	59
2.94. CertificateHashDataType	59
2.95. Page 459 (2025-11) - AuthCache.Entries variable [942]	59
2.95.1. AuthCacheEntries	59
3. Part 3	61
4. Part 4	62

4.1. Page 4 - (2025-11) - Updates to RFC References	62
4.2. Page 6 - (2025-01) - 3.1.1. The connection URL	62
4.3. Page 7 - (2025-01) - 3.1.2. OCPP version	62
4.4. Page 8 - (2025-01) - 3.3. WebSocket Compression	62
4.5. Page 10 - (2025-01) - 4.1.3. The message type	62
4.6. Page 10 - (2025-01) - 4.1.3. The message type	63
4.7. Page 10 - (2024-09) - 4.1.4. Message ID	63
4.8. Page 13 - (2025-01) - 4.2.3. CALLERROR	63
4.9. Page 15 - (2025-01) - 5. Connection	63
4.10. TLS fragment length	63
4.11. Page 15 - (2025-01) - 5.3. WebSocket Ping in relation to OCPP Heartbeat	64
4.12. Page 15 - (2025-01) - 5 Connection - Added section about TLS fragment length	64
4.13. Page 16 - (2025-04) - 5.3 Reconnecting - reset backoff wait timer	64
4.14. Page 18 - (2025-02) - 6.3 Connection loss - Allow Local Controller to keep connection open	64
5. Part 5	66
5.1. General - (2025-02) - Renamed OCTT to Test System	66
5.2. Page 7-48 - (2025-09) - Add additional support for different types of Charging Stations	66
5.3. Page 7 - (2025-02) - Changed definition of C-01 Support for offline authorization of transactions	66
5.4. Page 9 - (2024-11) - Optional feature list for charging station - Change name R-3	67
5.5. Page 9 - (2025-02) - Optional feature list for charging station - Added AQ-10 to make TC_N_48_CS conditional	67
5.6. Page 9/11/27/48/52 - (2025-02) - Removed feature ISO-3, added additional question AQ-11	67
5.7. Page 9/10 - (2025-06) - Optional feature list for charging station & CSMS - Add optional feature for Security Profile 1	68
5.8. Page 13 - (2025-09) - Added TC_B_14_CS (as optional) to the Core profile	69
5.9. Page 13 - (2025-02) - Updated Id's of the additional questions of CSMSs in the appendix	69
5.10. Page 15 - (2025-04) - Addition of new testcase for CSMS to test WebSocket Subprotocol negotiation	69
5.11. Page 17/33 - (2025-02) - Test Cases Local Authorization List Management & Authorization Cache - Update conditions	69
5.12. Page 19 - (2024-09) - TC_E_04_CS Updated condition for test case to exclude it for MacAddress and ISO 15118 PnC	73
5.13. Page 22 - (2024-09) - TC_E_17_CS Updated condition for test case to correctly specify the applicable TxStopPoint combinations	73
5.14. Page 26 - (2025-11) - TC_F_01 / TC_F_02 / TC_F_03	74
5.15. Page 33 - (2025-06) - Fixed incorrect feature no reference	75
5.16. Page 39 - (2025-02) - TC_N_48_CS Made conditional	75
5.17. Page 40 - (2024-11) - TC_H_13_CS Updated invalid condition for test case	75
5.18. Page 42 - (2025-02) - Removed TC_O_15_CS from certification program	75
5.19. Test Cases Advanced User Interface	75
5.20. Page 42 - (2025-06) - Make optional feature R-1 available for Charging Stations	75
5.21. Page 44 - (2025-02) - Removed TC_A_13_CS and TC_A_13_CSMS from certification program	76
5.22. Page 48 - (2024-06) - Added additional questions to appendix	76
5.23. Page 48 - (2025-02) - Updated Id's of the additional questions for CSMSs in the appendix	77
5.24. Page 48 - (2025-04) - Duplicate AQ-11 id	77
5.25. Page 48 - (2025-09) - Added additional hardware features	77
6. Part 6	79
6.1. General	79
6.1.1. Page XX - (2024-11) - All testcases - Updated table structure of all testcases	79
6.1.2. Page XX - (2025-02) - Renamed OCTT to Test System	79
6.2. Charging Station	79
6.2.1. Page 4 - (2025-02) - TC_A_01_CS - Updated old identifierString reference in description	79
6.2.2. Page 7 - (2024-11) - TC_A_05_CS - Successfully reconnecting after every failed connection attempt	79
6.2.3. Page 7 - (2025-02) - TC_A_05_CS	80
6.2.4. Page 7 - (2025-04) - TC_A_05_CS - Updated before steps to take into account the AllowCSMSTLSWildcards variable	81
6.2.5. Page 20 - (2025-02) - TC_A_23_CS - CSMS returns a CertificateSigned message for each request	81
6.2.6. Page 22 - (2024-09) - TC_A_19_CS - Fixed references to ConfigurationSlot [020-4762]	82
6.2.7. Page 22 - (2025-06) - TC_A_19_CS - Added steps to validate the Charging Station does not downgrade back to security profile 1	83
6.2.8. Page 24 - (2025-02) - TC_A_20_CS - Testcase did not take into account that the used configuration slot could already be set	85
6.2.9. Page 24 - (2025-04) - TC_A_20_CS - SetNetworkConnectionProfile is allowed to be rejected	86
6.2.10. Page 25 - (2025-02) - TC_A_21_CS	86

6.2.11. Page 26 - (2025-02) - TC_A_22_CS	87
6.2.12. Page 51 - (2025-02) - TC_B_16_CS - Correctly validate result of reading WriteOnly component variables	87
6.2.13. Page 56 - (2024-09) - TC_B_20_CS - Added check on omitting evseld [4390]	88
6.2.14. Page 57 - (2024-09) - TC_B_21_CS - Added check on omitting evseld [4390]	88
6.2.15. Page 58 - (2024-09) - TC_B_22_CS - Added check on omitting evseld [4390]	89
6.2.16. Page 63 - (2025-02) - Changed reset to Immediate	89
6.2.17. Page 72-81 - (2025-04) - TC_B_45_CS & TC_B_46_CS & TC_B_47_CS & TC_B_49_CS & TC_B_50_CS & TC_A_19_CS - Clarified NetworkProfile configurationSlot usage	89
6.2.18. Page 72-81 - (2025-06) - TC_B_45_CS & TC_B_46_CS & TC_A_19_CS - ResetRequest will always be sent by the Test System to ensure the Charging Station switches NetworkProfile	93
6.2.19. Page 85 - (2025-04) - TC_B_53_CS - Check if all required values are provided.	94
6.2.20. Page 89 - (2025-02) - Add setting of NetworkProfileConnectionAttempts	94
6.2.21. Page 89/621 - (2025-04) - Addition of new testcase for CSMS to test WebSocket Subprotocol negotiation	95
6.2.22. Page 117/141 - (2025-02) - Added remote support for Authorization Cache & Local Authorization List testcases ..	95
6.2.23. Page 101 - (2025-04) - TC_C_14_CS - Fixing invalid component variable reference	96
6.2.24. Page 102 - (2025-04) - TC_C_15_CS - Fixing invalid component variable reference	97
6.2.25. Page 104 - (2025-04) - TC_C_16_CS - Fixing invalid component variable reference	97
6.2.26. Page 105 - (2025-04) - TC_C_17_CS - Fixing invalid component variable reference	97
6.2.27. Page 108 - (2025-04) - TC_C_18_CS - Fixing invalid component variable reference	97
6.2.28. Page 120 - (2025-04) - TC_C_34_CS - Making use of the Deauthorized reusable state.	97
6.2.29. Page 128 - (2025-04) - TC_C_41_CS - Fixing invalid component variable reference	98
6.2.30. Page 134 - (2025-04) - TC_C_44_CS - Fixing invalid component variable reference	98
6.2.31. Page 137 - (2025-04) - TC_C_46_CS - Fixing invalid component variable reference	98
6.2.32. Page 138 - (2024-09) - TC_C_47_CS - StoppedReason must be validated in Ended event [020-4467]	98
6.2.33. Page 145 - (2025-04) - TC_C_25_CS - Fixing invalid component variable reference	99
6.2.34. Page 146 - (2024-11) - TC_C_26_CS - Allow StatusNotification status = Occupied.	99
6.2.35. Page 147-153 - (2025-04) - TC_C_50_CS, TC_C_51_CS, TC_C_52_CS, TC_C_53_CS, TC_C_54_CS, TC_C_55_CS - Always re-install V2G certificates	100
6.2.36. Page 147 - (2025-02) - TC_C_50_CS	100
6.2.37. Page 148 - (2025-02) - TC_C_51_CS	101
6.2.38. Page 149 - (2025-02) - TC_C_52_CS	101
6.2.39. Page 150 - (2025-02) - TC_C_53_CS	102
6.2.40. Page 151 - (2024-09) - TC_C_54_CS - removed reusable state IdTokenCached [020-3510]	103
6.2.41. Page 151 - (2025-02) - TC_C_54_CS	103
6.2.42. Page 153 - (2024-09) - TC_C_55_CS - removed reusable state IdTokenCached [020-3510]	104
6.2.43. Page 153 - (2025-02) - TC_C_55_CS	104
6.2.44. Page 165/169 - (2025-02) - TC_E_01_CS/TC_E_09_CS/TC_E_10_CS/TC_E_12_CS/TC_E_13_CS - Extended the testcases until the Charging Station start charging	105
6.2.45. Page 174 - (2025-04) - TC_E_17_CS - Aligned configuration before steps with updated prerequisites	106
6.2.46. Page 174 - (2024-09) - TC_E_17_CS - Updated prerequisite for test case to correctly specify the applicable TxStopPoint combinations	107
6.2.47. Page 176 - (2024-11) - TC_E_39_CS - Missing StatusNotificationRequest/NotifyEventRequest.	107
6.2.48. Page 182 - (2025-04) - TC_E_52_CS - Testcase is not able to determine the authorization is refused in case TxStartPoint is not Authorized	108
6.2.49. Page 185 - (2024-09) - TC_E_35_CS - StoppedReason must be validated in Ended event [020-4467]	108
6.2.50. Page 188 - (2025-02) - TC_E_22_CS - Stop transaction options - EnergyTransfer stopped - will end transaction ..	109
6.2.51. Page 189 - (2025-06) - TC_E_14_CS - StoppedReason validation too strict for remote	109
6.2.52. Page 199 - (2025-04) - TC_E_27_CS - Remove manual action between step 4/5	110
6.2.53. Page 204 - (2025-02) - TC_E_31_CS - Add steps for when running the testcase in Remote mode	110
6.2.54. Page 214 - (2024-06) TC_E_43_CS Move reusable state TransactionEventsInQueueEnded to Before [768]	111
6.2.55. Page 217 - (2025-04) - TC_E_46_CS - Testcase updated to use the specialized Authorized15118 reusable state ..	112
6.2.56. Page 221 - (2024-06) TC_F_04_CS Made mandatory in part 5, but prerequisite in part 6 was not updated.	112
6.2.57. Page 221 - (2025-02) TC_F_04_CS Prerequisite only if supported.	113
6.2.58. Page 295 - (2024-09) - TC_J_XX_CS Meter Values	113
6.2.59. Page 236 - (2025-02) - TC_F_19_CS - The testcase ends while the firmware update is still ongoing	113
6.2.60. Page 272 - (2025-02) - TC_H_08_CS Reserve an unspecified EVSE - Accepted	114
6.2.61. Page 279/280 - (2025-06) - TC_H_15_CS & TC_H_16_CS can only be executed when the connector type of the Charging Station is part of the connectorEnumType.	115
6.2.62. Page 282 - (2025-02) - TC_H_17_CS - made more explicit on what to validate	115

6.2.63. Page 297 - (2025-02) - TC_J_02_CS Clock-aligned Meter Values - reporting multiple phases	117
6.2.64. Page 297 - (2024-09) - TC_J_02_CS Clock-aligned Meter Values	117
6.2.65. Page 306 - (2024-06) - TC_J_10_CS - Remove reference to non-existing requirements [4697]	118
6.2.66. Page 318 - (2025-04) TC_K_09_CS: Removed validFrom/To from test.	118
6.2.67. Page 343 - (2025-04) TC_K_23_CS: Removed validFrom/To from test.	119
6.2.68. Page 337 - (2025-04) TC_K_28_CS: Removed validFrom/To from test.	119
6.2.69. Page 345 - (2025-02) TC_K_XX_CS: Use realistic values for composite schedules	120
6.2.70. Page 345 - (2024-06) TC_K_35_CS Get Charging Profile - Evseld > 0 + chargingProfilePurpose [773].	121
6.2.71. Page 352 - (2025-04) - TC_K_39_CS - Validation of scheduleStart	121
6.2.72. Page 353 - (2025-04) - TC_K_40_CS: startSchedule improvements	122
6.2.73. Page 355 - (2025-04) - TC_K_41_CS: startSchedule improvement	124
6.2.74. Page 354/355 - (2025-02) - TC_K_40_CS & TC_K_41_CS - Updated composite schedule validation	125
6.2.75. Page 355 - (2025-04) - TC_K_41_CS: Added missing EnergyTransferStarted reusable state.	127
6.2.76. Page 359 - (2025-04) - TC_K_53_CS: Added missing validations.	128
6.2.77. Page 360 - (2025-02) - TC_K_54_CS: EVConnected must be before Authorization	129
6.2.78. Page 362 - (2025-02) - TC_K_56_CS: EVConnected must be before Authorization	130
6.2.79. Page 364 - (2025-02) - TC_K_57_CS: EVConnected must be before Authorization	130
6.2.80. Page 362 - (2024-09) - TC_K_56_CS Removed expecting triggerReason=ChargingRateChanged [776].	130
6.2.81. Page 366 - (2025-02) - TC_K_58_CS	131
6.2.82. Page 384 - TC_L_10_CS - Allow Download/InstallationFailed upon AcceptedCanceled	131
6.2.83. Page 387 - TC_L_06_CS - SecurityEventNotification and FirmwareStatusNotification can be sent in any order.	132
6.2.84. Page 422 - (2025-02) - TC_M_15_CS - V2GCertificateChain is not installed before being retrieved	133
6.2.85. Page 430 - (2025-06) - TC_M_24_CS - A GetCertificateStatusRequest is also sent for the subCAs	133
6.2.86. Page 436 - (2025-02) - TC_N_01_CS - Made used component variable configurable	134
6.2.87. Page 455 - (2025-02) - TC_N_12_CS - Updating test case for using configuration variables	134
6.2.88. Page 456 - (2025-02) - TC_N_13_CS - Updating test case for using more specific configuration variables.	136
6.2.89. Page 463 - (2025-02) TC_N_20_CS - Updating test case for using more specific configuration variables.	138
6.2.90. Page 468 - (2024-06) TC_N_23_CS Offline Notification - OfflineMonitoringEventQueuingSeverity set higher than severityLevel of the monitor [772].	141
6.2.91. Page 470 - (2024-09) - TC_N_24_CS - Referring to incorrect use case and requirements [O20-4793]	142
6.2.92. Page 492 - (2025-02) - TC_N_39_CS - Test case now searches suitable variables to do test with	143
6.2.93. Page 472 - (2024-12) - TC_N_26_CS - Made test case more explicit and more time before ending	145
6.2.94. Page 470 - (2024-12) - TC_N_24_CS - Test case now searches suitable variable to do test with	146
6.2.95. Page 470 - (2025-02) - TC_N_24_CS - Updating test case for using more specific configuration variables.	148
6.2.96. Page 482 - (2025-02) - TC_N_63_CS - Clear Customer Information - add manual action to stop session	149
6.2.97. Page 487 - (2025-04) - TC_N_36_CS - LogStatusNotification(AcceptedCanceled) allowed before GetLogResponse.	150
6.2.98. Page 482 - (2025-04) - TC_N_63_CS - Added missing reusable state EnergyTransferStarted at before steps	151
6.2.99. Page 493 - (2024-09) - TC_N_41_CS - Set Variable Monitoring - Return to FactoryDefault.	152
6.2.100. Page 482 - (2024-09) - TC_N_63_CS - Clear Customer Information - Clear and report - customerCertificate.	152
6.2.101. Page 482 - (2025-02) - TC_N_63_CS - Added missing configuration state and authorize explicit using Plug and Charge (PnC)	153
6.2.102. Page 493 - (2024-09) - TC_N_41_CS - Set Variable Monitoring - Return to FactoryDefault.	154
6.2.103. Page 493 - (2025-02) - TC_N_41_CS - Made less dependent on test case configuration variables, enables predefines monitors	154
6.2.104. Page 495 - (2024-11) - TC_N_43_CS - Remove incorrect tool validation StatusInfo	158
6.2.105. Page 495 - (2025-02) - TC_N_43_CS - Updating test case for using more specific configuration variables	158
6.2.106. Page 497 - (2025-02) - TC_N_45_CS - Updating test case for using more specific configuration variables	161
6.2.107. Page 501 - (2025-02) - TC_N_51_CS - Updating test case for using more specific configuration variables	163
6.2.108. Page 503 - (2025-02) - TC_N_52_CS - Updating test case for using more specific configuration variables	165
6.2.109. Page 503 - (2025-11) - TC_N_52_CS - Updating test case to check configured monitor works.	167
6.2.110. Page 504 - (2025-02) - TC_N_53_CS - Updating test case for using more specific configuration variables	169
6.2.111. Page 505 - (2025-02) - TC_N_56_CS - Made test case configurable using configuration variables	170
6.2.112. Page 506 - (2025-06) - TC_O_XX_CS - Updated configurations	171
6.2.113. Page 520 - (2025-02) - TC_O_15_CS - Test case removed	175
6.2.114. Page 530 - (2025-02) - TC_O_28_CS - Transaction id should be specified for DisplayMessage	175
6.2.115. Page 533 - (2025-02) - TC_O_32_CS - Made notes about display behaviour more explicit.	175
6.2.116. Page 544 - (2025-02) - TC_O_39_CS - Wait for StatusNotificationRequest or NotifyEventRequest.	176
6.2.117. Page 555 - (2024-11) - Remove StatusNotificationRequest from Authorized reusable state Main B steps.	179

6.2.118. Page 559 - (2025-11) - Reusable state EnergyTransferSuspended - Removed invalid note	179
6.2.119. Page 560 - (2025-04) - Reusable states StopAuthorized & Deauthorized	180
6.2.120. Deauthorized	182
6.2.121. Page 566 - (2025-02) - Reusable state RenegotiateChargingLimits	182
6.2.122. Page 573/151 - (2025-04) - Removed Main steps B from IdTokenCached reusable state and added IdTokenCached15118	183
6.2.123. Page 574 - (2025-04) - IdTokenLocalAuthList memory state - set Enable to true if implemented	185
6.2.124. Page 575 - (2024-09) - Reusable state RenewChargingStationCertificate expects a reconnection [784]	185
6.2.125. Page 575 - (2025-02) - Reusable state RenewChargingStationCertificate must not do a Reset [5281]	186
6.3. CSMS	188
6.3.1. Page 593 - (2025-04) - TC_A_11_CSMS - Added post scenario validation for clarification.	188
6.3.2. Page 593 - (2024-09) - TC_A_11_CSMS - Reconnect using new client certificate	188
6.3.3. Page 596 - (2024-09) - TC_A_14_CSMS - Update Charging Station Certificate by request of CSMS - Invalid certificate	188
6.3.4. Page 597 - (2025-04) - TC_A_19_CSMS - Added main steps and clarified tool validations	189
6.3.5. Page 597 - (2024-09) - TC_A_19_CSMS - Added additional information regarding the use of the client certificates .	190
6.3.6. Page 597 - (2024-09) - TC_A_19_CSMS - Removed validation of OcppCsmsUrl [O20-4355]	191
6.3.7. Page 637 - (2024-11) - TC_C_50_CSMS - Changed reference to configured valid idToken to a specific eMAID idToken.	191
6.3.8. Page 639 - (2024-09) - TC_C_52_CSMS - TC does not use <Configured contract_certificate>	192
6.3.9. Page 639 - (2025-02) - TC_C_52_CSMS - Certificate needs at least one subCA	192
6.3.10. Page 640 - (2025-04) - TC_D_01_CSMS - Missing tool validation that the idTokenInfo must be provided for all list entries.	193
6.3.11. Page 712 - (2024-09) - TC_I_01_CSMS - Show EV Driver running total cost.	193
6.3.12. Page 715 - (2025-02) - TC_I_02_CSMS - Added explicit information about CSMS tariff configuration and sending in needed metervalue	194
6.3.13. Page 726-760 - (2025-04) - TC_K_XX_CSMS - Improved tool validations to be sure valid Charging Profiles are used	195
6.3.14. Page 728 - (2024-09) - TC_K_03_CSMS - Not requiring validFrom/To fields in charging profile [O20-4592] and chargingProfileKind must be Absolute [O20-4591].	202
6.3.15. Page 733 - (2024-09) - TC_K_10_CSMS - Not requiring validFrom/To fields in charging profile [O20-4592]	202
6.3.16. Page 734 - (2024-09) - TC_K_15_CSMS - Not requiring validFrom/To fields in charging profile [O20-4592]	203
6.3.17. Page 752 - (2025-06) - TC_K_53_CSMS - Validate that the CSMS let's the Charging Station charging, according to the negotiated limits.	204
6.3.18. Page 753 - (2024-09) - TC_K_55_CSMS, TC_K_57_CSMS, TC_K_58_CSMS, TC_K_59_CSMS Removed triggerReason = ChargingRateChanged [776]	204
6.3.19. Page 760 - (2024-11) - TC_K_70_CSMS - Updated tool validation chargingProfiles and added preparation step. .	204
6.3.20. Page 805/806 - (2025-06) - TC_N_01_CSMS & TC_N_02_CSMS - omit filter fields that are not tested	205
6.3.21. Page 806 - (2025-06) - TC_N_02_CSMS - component and variable instance need to be omitted	206
6.3.22. Page 824 - (2025-04) - TC_N_62_CSMS - Check only single identifier is provided.	206
6.3.23. Page 830 - (2024-11) - TC_N_46_CSMS - Updated tool validation sendLocalListRequest	207
6.3.24. Page 830 - (2025-02) - TC_N_46_CSMS - Aligning configuration variable usage.	207
6.3.25. Page 854 - (2025-02) - TC_O_27_CSMS - Fixing validations to be more specific for test case	209
6.3.26. Page 855 - (2025-02) - TC_O_28_CSMS - Fixing validations to be more specific for test case	209

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Version History

Version	Date	Description
2025-11 DRAFT	2025-10-20	Includes errata for Part 1, 2, 4, 5 and 6 of OCPP 2.0.1 Edition 3.
2025-09	2025-09-30	Includes errata for Part 1, 2, 4, 5 and 6 of OCPP 2.0.1 Edition 3.
2025-06	2025-07-08	Includes errata for Part 2, 5 and 6 of OCPP 2.0.1 Edition 3.
2025-04	2025-04-30	Includes errata for Part 2, 5 and 6 of OCPP 2.0.1 Edition 3.
2025-02	2025-03-06	Includes errata for Part 2, 5 and 6 of OCPP 2.0.1 Edition 3.
2025-01	2025-01-23	Includes errata for Part 1-4 of OCPP 2.0.1 Edition 3
2024-11	2024-11-14	Includes errata for Part 5 and Part 6 of OCPP 2.0.1 Edition 3
2024-09	2024-09-25	Includes errata for Part 4, Part 5 and Part 6 of OCPP 2.0.1 Edition 3
2024-06	2024-06-27	Includes errata for Part 5 and Part 6.

Scope

This document contains errata on the OCPP 2.0.1 documentation. These errata have to be read as an addition to the release of OCPP 2.0.1 Edition 3.

The errata do not affect any schemas of OCPP messages. Certain errata do contain changes to requirements or even new requirements, but only in cases where a requirement contains an obvious error and would not or could not be implemented literally. New requirements are only added when they were already implicitly there. These changes have been discussed in or were proposed by the Technology Working Group of the Open Charge Alliance.

The appendices of the OCPP specification can be updated without requiring a new OCPP release. This mainly concerns the components and variables of the OCPP device model, which can be extended with new components or variables, as long as they are optional.

Terminology and Conventions

Bold: when needed to clarify differences, bold text might be used.

The errata entries are sorted by page number of the affected section of the specification document. When an errata entry affects multiple parts of the specification, then the various changes are grouped together with subsections referring to the pages affected by those changes.

This is version 2025-11 DRAFT of the errata. The errata of this version are marked with "(2025-11 DRAFT)" in the section title.

In some cases the issue number by which it was reported, is added in square brackets at the end of the section title, e.g. "[349]". For retrieval of the issue in the issue tracking system prefix the number with "OCPP20M", like "[OCPP20M-349]".

0. Part 0

Currently no new errata for OCPP 2.0.1 Edition 3 part 0.

DRAFT

1. Part 1

1.1. Page 9 - (2025-01) - 3.4. Monitoring

The first sub-bullet is further clarified that it refers to the absolute difference:

	Description
Old	When the dataType of the variable is integer or decimal, this value represents the difference to be reached to trigger the monitor.
New	When the dataType of the variable is integer or decimal, this value represents the absolute difference to be reached to trigger the monitor.

2. Part 2

2.1. Page 5 - (2025-01) - Updated limitations on BasicAuthPassword to increase security

A "passwordString" should not be limited to any set of symbols as this limits its security greatly. There is also no reason for limiting the password size.

Table 1. Primitive Datatypes

Datatype	Description
passwordString	This is a UTF-8 encoded case-sensitive string. that can only contain characters from the following character set: "a-z", "A-Z", "0-9" or any of the following limited set of symbols: * _ - : + ! @

	ID	Precondition	Requirement definition
Old text	A00.FR.205		The password SHALL be stored in the BasicAuthPassword configuration variable. It SHALL be a randomly chosen passwordString with a sufficiently high entropy, consisting of minimum 16 and maximum 40 characters (alpha-numeric characters and the special characters allowed by passwordString). The password SHALL be sent as a UTF-8 encoded string (NOT encoded into octet string or base64).
New text	A00.FR.205		The password SHALL be stored in the BasicAuthPassword configuration variable. It SHALL be a randomly chosen passwordString with a sufficiently high entropy, consisting of minimum 16 and a maximum as defined by the <i>maxLimit</i> of configuration variable BasicAuthPassword , which must be at least 40 characters and at most 64. The password SHALL be sent as a UTF-8 encoded string (NOT encoded into octet string or base64).
Old text	A00.FR.304	A00.FR.302	The password SHALL be stored in the BasicAuthPassword Configuration Variable. It SHALL be a randomly chosen passwordString with a sufficiently high entropy, consisting of minimum 16 and maximum 40 characters (alpha-numeric characters and the special characters allowed by passwordString). The password SHALL be sent as a UTF-8 encoded string (NOT encoded into octet string or base64). (Same as A00.FR.205)
New text	A00.FR.304	A00.FR.302	The password SHALL be stored in the BasicAuthPassword Configuration Variable. It SHALL be a randomly chosen passwordString with a sufficiently high entropy, consisting of minimum 16 and a maximum as defined by the <i>maxLimit</i> of configuration variable BasicAuthPassword , which must be at least 40 characters and at most 64. The password SHALL be sent as a UTF-8 encoded string (NOT encoded into octet string or base64). (Same as A00.FR.205)

BasicAuthPassword

The basic authentication password is used for HTTP Basic Authentication. The configuration value is write-only, so that it cannot be accidentally stored in plaintext by the CSMS when it reads out all configuration values.

Required	no		
Component	componentName	SecurityCtrlr	
Variable	variableName	BasicAuthPassword	
	variableAttributes	mutability	WriteOnly
	variableCharacteristics	dataType	string
		maxLimit	At least 40, at most 64.

Description	The basic authentication password is used for HTTP Basic Authentication. The password SHALL be a randomly chosen passwordString with a sufficiently high entropy, consisting of minimum 16 and a maximum as defined by the <i>maxLimit</i> of <i>BasicAuthPassword</i>, which must be at least 40 characters and at most 64. The password SHALL be sent as a UTF-8 encoded string (NOT encoded into octet string or base64). This configuration variable is write-only, so that it cannot be accidentally stored in plaintext by the CSMS when it reads out all configuration variables. This configuration variable is required unless only "security profile 3 - TLS with client side certificates" is implemented.
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2.2. Page 14 - (2025-04) - Section 2.7. ISO 15118 support - Aligned ISO 15118 timeout table with updated version in OCPP 2.1

For reference, the current timing constraints for **ISO 15118-2:2014** are:

Table 2. ISO 15118-2 Timing constraints

Timeout	Default
Sequence Timeouts	60 seconds
Sequence Performance Timeouts	40 seconds
PaymentDetailsReq/Res	4.5 seconds
AuthorizationReq/Res	1.5 seconds
CertificateUpdateReq/Res	4.5 seconds
CertificateInstallationReq/Res	4.5 seconds

2.3. Page 15 - (2025-01) - Improved text FR.04

The requirement did not take into account the pending status and should refer to the applicable use cases B02 and B03.

	ID	Precondition	Requirement definition	Note
Old text	FR.04	When the CSMS did not accept the BootNotificationRequest from the Charging Station AND The Charging Station sends a message other than BootNotificationRequest	The CSMS SHALL respond with a RPC Framework: CALLERROR: SecurityError.	
New text	FR.04	When the CSMS rejected the BootNotificationRequest from the Charging Station AND The Charging Station sends a message other than BootNotificationRequest	The CSMS SHALL respond with a RPC Framework: CALLERROR: SecurityError.	See use cases B0 2 and B03 for details.

2.4. Page 19 - (2025-01) - Removed requirement A00.FR.003 as the precondition never occurs

Requirement A00.FR.003 describes a precondition that never occurs in which the CSMS connects to the Charging Station. The requirement is therefore removed.

ID	Precondition	Requirement definition
A00.FR.003	If the CSMS tries to connect with a different profile than the Charging Station is using	The Charging Station SHALL terminate the connection.

2.5. Page 19/41 - (2025-01) - Allow downgrading security profile from 3 to 2

For migration purposes a Charging Station implementer has now the option for their Charging Stations to allow downgrading the security profile from 3 to 2, because not all CSMSs support security profile 3. ENCS was consulted to identify if this would pose any security risks. This was not the case as security profile 2 still uses TLS, which provides sufficient security.

Table 3. A05 - Upgrade Charging Station Security Profile

No.	Type	Description
1	Name	Upgrade Charging Station Security Profile
2	ID	A05
	Functional block	A. Security
3	Objective(s)	The CSO wants to increase change the security of the OCPP connection between CSMS and a Charging Station.
4	Description	Use case when migrating from OCPP 1.6 without security profiles to OCPP 2.0.1. Before migrating to a security profile, the prerequisites, like installed certificates or password, need to be configured.
	Actors	CSMS, Charging Station
	Scenario description	1. The CSMS sets a new value for the NetworkConfigurationPriority Configuration Variable via SetVariablesRequest, such that the NetworkConnectionProfile for the new (higher) security profile becomes first in the list and the existing connection profile becomes second in the list. 2. The Charging Station responds with a SetVariablesResponse with status <i>Accepted</i> 3. The CSMS sends a ResetRequest(OnIdle) 4. The Charging Station reboots and connects via the new primary NetworkConnectionProfile
5	Prerequisite(s)	The CSO ensures that a NetworkConnectionProfile has been set using (higher) an allowed security profile AND that the prerequisite(s) for going to a higher the new security profile are met before sending the command to change to a higher the new security profile.
6	Postcondition(s)	The Charging Station was successfully upgraded to a higher new security profile.
7	Error handling	n/a
8	Remark(s)	For security reasons it is by default not allowed to revert to a lower Security Profile using OCPP. Only when the variable AllowSecurityProfileDowngrade is implemented and is set to true, it is allowed to downgrade from profile 3 to profile 2. Even in that case, it is not allowed to revert from profile 2 or profile 3 to security profile 1 using OCPP.

Changed A00 requirement:

	ID	Precondition	Requirement definition
Old text	A00.FR.005		Lowering the security profile that is used, to a less secure profile, is for security reasons, not part of the OCPP specification, and MUST be done through another method, not via OCPP. OCPP messages SHALL NOT be used for this (e.g. SetVariablesRequest or DataTransferRequest).
New text	A00.FR.005		Lowering the security profile that is used, to a less secure profile, is for security reasons, not recommended. The Charging Station SHALL only allow to lower the security profile if the variable AllowSecurityProfileDowngrade is implemented and set to <i>true</i>. In that case, the Charging Station SHALL only allow to downgrade from profile 3 to profile 2. The Charging Station SHALL NOT allow to downgrade from profile 2 or profile 3 to profile 1 using the OCPP protocol.

Added A05 requirements:

	ID	Precondition	Requirement definition
New requirement	A05.FR.08	The variable <code>AllowSecurityProfileDowngrade</code> is implemented and set to true AND The currently active 'SecurityProfile' is 3 AND The Charging Station receives <code>SetVariablesRequest</code> for <code>NetworkConfigurationPriority</code> containing profile slots for <code>NetworkConnectionProfiles</code> with a 'securityProfile' value equal to 2.	The Charging Station SHALL respond with <code>SetVariablesResponse</code> (Accepted)
New requirement	A05.FR.09	The variable <code>AllowSecurityProfileDowngrade</code> is implemented and set to true AND The currently active 'SecurityProfile' is higher than 1 AND The Charging Station receives <code>SetVariablesRequest</code> for <code>NetworkConfigurationPriority</code> containing profile slots for <code>NetworkConnectionProfiles</code> with a 'securityProfile' value equal to 1.	The Charging Station SHALL respond with <code>SetVariablesResponse</code> (Rejected)
New requirement	A05.FR.10	The variable <code>AllowSecurityProfileDowngrade</code> is not implemented or implemented and set to false AND The Charging Station receives <code>SetVariablesRequest</code> for <code>NetworkConfigurationPriority</code> containing profile slots for <code>NetworkConnectionProfiles</code> with a 'securityProfile' value lower than the currently active security profile	The Charging Station SHALL respond with <code>SetVariablesResponse</code> (Rejected)

Changed/added B09 requirements:

	ID	Precondition	Requirement definition
Old text	B09.FR.04	On receipt of the <code>SetNetworkProfileRequest</code> AND the <code>NetworkConnectionProfile</code> contains a lower securityProfile than stored at the configuration variable <code>SecurityProfile</code>	The Charging Station SHALL respond by sending a <code>SetNetworkProfileResponse</code> message, with status <i>Rejected</i>
New text	B09.FR.04	The variable <code>AllowSecurityProfileDowngrade</code> is not implemented or implemented and set to false AND The Charging Station receives a <code>SetNetworkProfileRequest</code> AND the <code>NetworkConnectionProfile</code> contains a lower securityProfile than the currently active security profile	The Charging Station SHALL respond by sending a <code>SetNetworkProfileResponse</code> message, with status <i>Rejected</i>

	ID	Precondition	Requirement definition
New requirement	B09.FR.31	The variable AllowSecurityProfileDowngrade is implemented and set to true AND The currently active 'SecurityProfile' is 3 AND The Charging Station receives a SetNetworkProfileRequest AND the NetworkConnectionProfile contains a securityProfile with a value of 2.	The Charging Station SHALL respond with SetVariablesResponse (Accepted)
New requirement	B09.FR.32	The variable AllowSecurityProfileDowngrade is implemented and set to true AND The currently active 'SecurityProfile' is higher than 1 AND The Charging Station receives a SetNetworkProfileRequest AND the NetworkConnectionProfile contains a securityProfile with a value of 1.	The Charging Station SHALL respond with SetVariablesResponse (Rejected)

Added referenced variable:

AllowSecurityProfileDowngrade

Required	no		
Component	componentName	SecurityCtrlr	
Variable	variableName	AllowSecurityProfileDowngrade	
	variableAttributes	mutability	ReadWrite/ReadOnly
	variableCharacteristics	dataType	boolean
Description	If this variable is implemented and set to <i>true</i> , then the Charging Station allows downgrading the security profile from 3 to 2. For security reasons it is not allowed to revert from profile 2 or profile 3 to security profile 1 using OCPP.		

2.6. Page 22/27 - (2025-11) - Slight deviation in A00 requirement phrasing which is aligned

There are multiple requirements that describe the validation of the commonName of the server certificate. These should have been equal to each other, but there is a slight deviation in phrasing, which should be aligned. The intention was that the commonName matches the FQDN of the server, this has been confirmed by the TWG.

Changed A00 requirements:

	ID	Precondition	Requirement definition
Old text	A00.FR.309		The Charging Station SHALL verify that the commonName includes the CSMS's FQDN.
New text	A00.FR.309		The Charging Station SHALL verify that the commonName matches the CSMS's FQDN.
Old text	A00.FR.510		For the CSMS certificate, the subject field SHALL contain the FQDN of the endpoint of the server in the CN (commonName) RDN.

	ID	Precondition	Requirement definition
New text	A00.FR.510		For the CSMS certificate, the subject field SHALL match the FQDN of the endpoint of the server in the CN (commonName) RDN.

2.7. Page 26 - (2025-01) - 1.3.7. TLS with Client Side Certificates Profile - Requirements

A new requirement is added to support A00.FR.429:

New requirement

Table 4. Security Profile 3 - TLS with Client Side Certificates - Requirements

ID	Precondition	Requirement definition
A00.FR.430	If the Charging Station certificate has expired	The Charging Station SHOULD still attempt to establish a connection with the CSMS and leave the decision to accept the connection up to the CSMS.

2.8. Page 28 - (2025-01) - Certificate Hierarchy

A note has been added to the specification to warn OCPP implementers about the potential risks involved with the installation of CA bundles.

NOTE

It is not recommended to have preinstalled well-known root CA certificates on a Charging Station like in operating systems or browsers, like for example a CA bundle. Only root and intermediate certificates part of the Charging Station Operator hierarchy should be used for the OCPP connection, as described by section [Certificate Hierarchy](#). Trusting many additional well-known root CA certificates creates security risks.

2.9. Page 31 - (2025-01) - A02/A03 - Updated error handling

NOT RECOMMENDED in the error handling is confusing as there is already a requirement that explicitly states that the Charging Station SHALL retry after "CertSigningWaitMinimum".

7	Error handling	The CSMS accepts the CSR request from the Charging Station, before forwarding it to the CA. But when the CA cannot be reached, or rejects the CSR, the Charging Station will never be known. The CSMS may do some checks on the CSR, but cannot do all the checks that a CA does, and it does not prevent connection timeout to the CA. When something like this goes wrong, either the CA is offline or the CSR send by the Charging Station is not correct, according to the CA. In both cases this is something an operator at the CSO needs to be notified of. The operator then needs to investigate the issue. When resolved, the operator can re-run A02. It is NOT RECOMMENDED to let the Charging Station retry when the certificate is not send within X minutes or hours. When the CSR is incorrect, that will not be resolved automatically. It is possible that only a new firmware will fix this.
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2.10. Page 35 - (2025-11) - A02.FR.19 - Updated precondition

Precondition formulated differently with the corresponding variable name.

	ID	Precondition	Requirement definition	Note
Old	A02.FR.19	A02.FR.18 AND The maximum number of increments is reached	The Charging Station SHALL stop resending the SignCertificateRequest , until it is requested by the CSMS via a TriggerMessageRequest for SignChargingStationCertificate, SignV2GCertificate or SignCombinedCertificate.	

	ID	Precondition	Requirement definition	Note
New	A02.FR.19	A02.FR.18 AND + CertSigningRepeatTimes is reached	The Charging Station SHALL stop resending the SignCertificateRequest , until it is requested by the CSMS via a TriggerMessageRequest for SignChargingStationCertificate, SignV2GCertificate or SignCombinedCertificate. (Same as A02.FR.19)	

2.11. Page 38 - (2025-11) - A03.FR.19 - Updated precondition

Precondition formulated differently with the corresponding variable name.

	ID	Precondition	Requirement definition	Note
Old	A03.FR.19	A03.FR.18 AND The maximum number of increments is reached	The Charging Station SHALL stop resending the SignCertificateRequest , until it is requested by the CSMS via a TriggerMessageRequest for SignChargingStationCertificate, SignV2GCertificate or SignCombinedCertificate.	
New	A03.FR.19	A03.FR.18 AND + CertSigningRepeatTimes is reached	The Charging Station SHALL stop resending the SignCertificateRequest , until it is requested by the CSMS via a TriggerMessageRequest for SignChargingStationCertificate, SignV2GCertificate or SignCombinedCertificate. (Same as A02.FR.19)	

2.12. Page 34 - (2025-01) - Disallow client certificates future validity date

Currently the specification is unclear on whether accepting client certificates with a future validity date is intended behavior. If it would be allowed behavior, several problems were identified. For example, there would be no way for the CSMS to manage these (ghost) client certificates using OCPP. ENCS was consulted if there would be any security risks when not allowing accepting client certificates with a future validity date. ENCS identified no security risks, therefore it was decided to not allow accepting client certificates with a future validity date. In addition clarifications have been made to describe how to handle non-happy flow scenarios in which the Charging Station is unable to successfully connect using the new client certificate in combination with the NetworkConnectionProfile mechanism described at use case B10.

Changed/added A02 requirements:

	ID	Precondition	Requirement definition	Note
Old text	A02.FR.06		The Charging Station SHALL verify the validity of the signed certificate in the CertificateSignedRequest message, checking at least the period when the certificate is valid, the properties in Certificate Properties , and that it is part of the Charging Station Operator certificate hierarchy as described in Certificate Hierarchy .	

	ID	Precondition	Requirement definition	Note
New text	A02.FR.06		The Charging Station SHALL verify the validity of the signed certificate in the CertificateSignedRequest message, checking that the current date (at the time of the update) is within the certificate's validity period, the properties in Certificate Properties , and that it is part of the Charging Station Operator certificate hierarchy as described in Certificate Hierarchy .	When providing a newly signed client certificate with a start period that equals the current time, the CSMS should take into account that there might be a slight discrepancy in the time between the Charging Station and CSMS. This could cause the Charging Station to reject the new certificate, because in case a small time difference exists, the validity period might (just) be in the future for the device.
Old text	A02.FR.08		The Charging Station SHALL switch to the new certificate as soon as the current date and time is after the 'Not valid before' field in the certificate (e.g. by closing the websocket and TLS connection and reconnecting with the new certificate).	
New text	A02.FR.08	If the certificate is valid.	The Charging Station SHALL respond to the CertificateSignedRequest with status Accepted AND the Charging Station SHALL switch to the new certificate by reconnecting the websocket and TLS connection.	
Old text	A02.FR.09	If the Charging Station contains more than one valid certificate of the <i>ChargingStationCertificate</i> type.	The Charging Station SHALL use the newest certificate, as measured by the start of the validity period.	
New text	A02.FR.09		<Requirement removed>	
Old text	A02.FR.10	A02.FR.09 AND When the Charging Station has validated that the new certificate works	The Charging Station MAY discard the old certificate. It is RECOMMENDED to store old certificates for one month, as fallback.	
New text	A02.FR.10	(A02.FR.08 OR A02.FR.28) AND The Charging Station successfully connected to the CSMS using either one of the certificates.	The Charging Station SHALL discard the client certificate that is NOT in use.	This is to prevent having multiple client certificates installed at the Charging Station, which the CSMS is unable to manage.
New requirement	A02.FR.28	A02.FR.08 AND the charging station was not able to successfully connect to any of the configured entries of NetworkConfigurationPriority using the new certificate AND The Charging Station supports either one or both reconnection mechanisms described at requirements; B10.FR.07 and B10.FR.08.	The Charging Station SHALL for the reconnection mechanism described at B10.FR.07 fallback to the old client certificate AND for the reconnection mechanism described at B10.FR.08 alternate between using the old and new client certificate after all NetworkConfigurationPriority entries.	As described by requirement B10.FR.09, the Charging Station SHOULD NOT stop trying to reconnect to the CSMS. This is to prevent the Charging Station from becoming a stranded asset.

	ID	Precondition	Requirement definition	Note
New requirement	A02.FR.29	A02.FR.10 AND The Charging Station discarded the new client certificate.	The Charging Station SHOULD send a SecurityEventNotification DiscardedRenewedClientCertificate to the CSMS.	Otherwise the CSMS is not aware that the Charging Station discarded the new client certificate and the CSMS should again trigger a client certificate renewal.

Changed/added A03 requirements:

	ID	Precondition	Requirement definition	Note
Old text	A03.FR.06		The Charging Station SHALL verify the validity of the signed certificate in the CertificateSignedRequest message, checking at least the period when the certificate is valid, the properties in Certificate Properties , and that it is part of the Charging Station Operator certificate hierarchy as described in Certificate Hierarchy . (Same as A02.FR.06)	
New text	A03.FR.06		The Charging Station SHALL verify the validity of the signed certificate in the CertificateSignedRequest message, checking that the current date (at the time of the update) is within the certificate's validity period, the properties in Certificate Properties , and that it is part of the Charging Station Operator certificate hierarchy as described in Certificate Hierarchy . (Same as A02.FR.06)	When providing a newly signed client certificate with a start period that equals the current time, the CSMS should take into account that there might be a slight discrepancy in the time between the Charging Station and CSMS. This could cause the Charging Station to reject the new certificate, because in case a small time difference exists, the validity period might (just) be in the future for the device.
Old text	A03.FR.08		The Charging Station SHALL switch to the new certificate as soon as the current date and time is after the 'Not valid before' field in the certificate (e.g. by closing the websocket and TLS connection and reconnecting with the new certificate). (Same as A02.FR.08)	
New text	A03.FR.08	If the certificate is valid.	The Charging Station SHALL respond to the CertificateSignedRequest with status Accepted AND the Charging Station SHALL switch to the new certificate by reconnecting the websocket and TLS connection. (Same as A02.FR.08)	
Old text	A03.FR.09	If the Charging Station contains more than one valid certificate of the ChargingStationCertificate type.	The Charging Station SHALL use the newest certificate, as measured by the start of the validity period. (Same as A02.FR.09)	
New text	A03.FR.09		<Requirement removed>	
Old text	A03.FR.10	A03.FR.09 AND When the Charging Station has validated that the new certificate works	The Charging Station MAY discard the old certificate. It is RECOMMENDED to store old certificates for one month, as fallback. (Same as A02.FR.10)	

	ID	Precondition	Requirement definition	Note
New text	A03.FR.10	(A03.FR.08 OR A03.FR.24) AND The Charging Station successfully connected to the CSMS using either one of the certificates.	The Charging Station SHALL discard the client certificate that is NOT in use. (Same as A02.FR.10)	This is to prevent having multiple client certificates installed at the Charging Station, which the CSMS is unable to manage.
New requirement	A03.FR.24	A03.FR.08 AND the charging station was not able to successfully connect to any of the configured entries of NetworkConfigurationPriority using the new certificate AND The Charging Station supports either one or both reconnection mechanisms described at requirements; B10.FR.07 and B10.FR.08.	The Charging Station SHALL for the reconnection mechanism described at B10.FR.07 fallback to the old client certificate AND for the reconnection mechanism described at B10.FR.08 alternate between using the old and new client certificate after all NetworkConfigurationPriority entries. (Same as A02.FR.28)	As described by requirement B10.FR.09, the Charging Station SHOULD NOT stop trying to reconnect to the CSMS. This is to prevent the Charging Station from becoming a stranded asset.
New requirement	A03.FR.25	A03.FR.10 AND The Charging Station discarded the new client certificate.	The Charging Station SHOULD send a SecurityEventNotification DiscardedRenewedClientCertificate to the CSMS. (Same as A02.FR.29)	Otherwise the CSMS is not aware that the Charging Station discarded the new client certificate and the CSMS may need to trigger a new client certificate renewal.

Changed/added B10 requirements:

	ID	Precondition	Requirement definition	Note
Old text	B10.FR.07	B10.FR.03 AND All NetworkProfileConnectionAttempts for every entry of NetworkConfigurationPriority failed.	The Charging Station SHOULD fallback and start 'reconnecting' to the NetworkConnectionProfile for which the last successful connection was made.	'reconnecting' in this requirement, refers to the reconnection mechanism described at section 5.3. Reconnecting from "Part 4 - JSON over WebSockets implementation guide".
New text	B10.FR.07	B10.FR.09	The Charging Station SHOULD fallback and start 'reconnecting' to the NetworkConnectionProfile for which the last successful connection was made.	'reconnecting' in this requirement, refers to the reconnection mechanism described at section 5.3. Reconnecting from "Part 4 - JSON over WebSockets implementation guide".
New requirement	B10.FR.08	B10.FR.09	The Charging Station SHOULD restart connecting with all configured entries of the NetworkConfigurationPriority	
New requirement	B10.FR.09	B10.FR.03 AND All NetworkProfileConnectionAttempts for every entry of NetworkConfigurationPriority failed.	The Charging Station SHOULD NOT stop trying to reconnect to the CSMS. The Charging Station SHOULD implement either one or both reconnecting mechanisms described at requirements; B10.FR.07 and B10.FR.08.	This is to prevent the Charging Station from becoming a stranded asset.

2.13. Page 58/60 - (2025-01) - Missing requirement information about omitting the value for WriteOnly variables

The specification does describe already that the value for WriteOnly variables need to be omitted, however this information is not part of a requirement yet.

Changed B07 requirement:

	ID	Precondition	Requirement definition	Note
Old text	B07.FR.03	B07.FR.01	The Charging Station SHALL send the requested information via one or more NotifyReportRequest messages to the CSMS.	It is good practice to send the report data in as few messages as possible in order to limit data overhead.
New text	B07.FR.03	B07.FR.01	The Charging Station SHALL send the requested information, excluding the value of WriteOnly variables , via one or more NotifyReportRequest messages to the CSMS.	It is good practice to send the report data in as few messages as possible in order to limit data overhead.

Changed B08 requirement:

	ID	Precondition	Requirement definition
Old text	B08.FR.03	B08.FR.01	The Charging Station SHALL send the requested information via one or more NotifyReportRequest messages to the CSMS.
New text	B08.FR.03	B08.FR.01	The Charging Station SHALL send the requested information, excluding the value of WriteOnly variables , via one or more NotifyReportRequest messages to the CSMS.

2.14. Page 63 - (2025-11) - B09.FR.02/04/05 - Added optional reasonCode [1083]

A mention of adding an optional *reasonCode* when a *SetNetworkProfileRequest* is rejected, has been added. This is a minor modification with respect to the errata version (2025-09). The *reasonCode* to use is no longer part of the requirement, but a recommendation in the note.

Changed requirements

ID	Precondition	Requirement definition	Note
B09.FR.02	On receipt of the SetNetworkProfileRequest	The Charging Station SHALL validate the content. If the content is invalid, the Charging Station SHALL respond by sending a SetNetworkProfileResponse message, with status <i>Rejected</i> and an optional statusInfo.reasonCode .	Recommended <i>reasonCode</i> to use is "InvalidNetworkConf". The field <i>additionalInfo</i> is recommended to be used to convey which configuration parameter is invalid and why. Matches B09.FR.34 for NetworkConfiguration.
B09.FR.04	The variable AllowSecurityProfileDowngrade is not implemented or implemented and set to false AND the Charging Station receives a SetNetworkProfileRequest AND the NetworkConnectionProfile contains a lower securityProfile than the currently active security profile	The Charging Station SHALL respond by sending a SetNetworkProfileResponse message, with status <i>Rejected</i> and optional statusInfo.reasonCode .	Recommended <i>reasonCode</i> to use is "NoSecurityDowngrade". The field <i>additionalInfo</i> can be used to provide more details. Matches B09.FR.35 for NetworkConfiguration.

ID	Precondition	Requirement definition	Note
B09.FR.05	When the value of <i>configurationSlot</i> in <i>SetNetworkProfileRequest</i> does not match an entry in <i>valuesList</i> of <i>NetworkConfigurationPriority</i>	The Charging Station SHALL respond by sending a <i>SetNetworkProfileResponse</i> message with status <i>Rejected</i> with optional <i>statusInfo.reasonCode</i> .	Recommended <i>reasonCode</i> to use is "InvalidConfSlot". The field <i>additionalInfo</i> can be used to provide more details.

2.15. Page 64 - (2025-09) - B09.FR.31/31 - Improved definition

Requirement B09.FR.31 has been rephrased to reject downgrading from any security profile to profile 1, rather than allowing a downgrade from 3 to 2 and disallowing from 2 to 1. As a result B09.FR.32 has now been removed.

Changed requirements

	ID	Precondition	Requirement definition	Note
Old	B09.FR.31	The variable <i>AllowSecurityProfileDowngrade</i> is implemented and set to true AND The currently active 'SecurityProfile' is 3 AND The Charging Station receives a <i>SetNetworkProfileRequest</i> AND the <i>NetworkConnectionProfile</i> contains a securityProfile with a value of 2.	The Charging Station SHALL respond with <i>SetVariablesResponse</i> (Accepted)	
New	B09.FR.31	The variable <i>AllowSecurityProfileDowngrade</i> is implemented and set to true AND the currently active 'SecurityProfile' is higher than 1 AND the Charging Station receives a <i>SetNetworkProfileRequest</i> with a <i>NetworkConnectionProfile</i> with <i>securityProfile = 1</i>	The Charging Station SHALL respond with <i>SetNetworkProfileResponse</i> with status <i>Rejected</i> and optional <i>statusInfo.reasonCode = "NoSecurityDowngrade"</i>	
Delete	B09.FR.32	The variable <i>AllowSecurityProfileDowngrade</i> is implemented and set to true AND The currently active 'SecurityProfile' is higher than 1 AND The Charging Station receives a <i>SetNetworkProfileRequest</i> AND the <i>NetworkConnectionProfile</i> contains a securityProfile with a value of 1.	The Charging Station SHALL respond with <i>SetVariablesResponse</i> (Rejected)	

2.16. Page 66 - (2025-11) - B10.FR.02 - Added optional *reasonCode*

For consistency with B09 the optional *reasonCode* has been added to B09.FR.02.

Changed requirement

	ID	Precondition	Requirement definition	Note
Old	B10.FR.02	On receipt of a <i>SetVariablesRequest</i> , containing Configuration Variable <i>NetworkConfigurationPriority</i> AND any of the NetworkProfile slots in the message does not contain a valid configuration	The Charging Station SHALL send <i>SetVariablesResponse</i> with status <i>Rejected</i> .	The optional element <i>statusInfo</i> can be used to provide more information.

	ID	Precondition	Requirement definition	Note
New	B10.FR.02	On receipt of a SetVariablesRequest , containing Configuration Variable NetworkConfigurationPriority AND any of the NetworkProfile slots in the message does not contain a valid configuration	The Charging Station SHALL send SetVariablesResponse with status <i>Rejected</i> and an optional <i>statusInfo.reasonCode</i> .	Recommended <i>reasonCode</i> to use is "InvalidNetworkConf". The field <i>additionalInfo</i> is recommended to be used to convey which configuration slot and which configuration parameter is invalid and why.

2.17. Page 66 - (2025-04) - B10.FR.03/04/10 - Migrate to new NetworkConnectionProfile

Changed requirements

	ID	Precondition	Requirement definition	Note
Old	B10.FR.03	B10.FR.04 AND When connecting fails	The Charging Station SHALL make the number of attempts as configured in NetworkProfileConnectionAttempts per entry of NetworkConfigurationPriority .	
New	B10.FR.03	After a reboot OR When connection to CSMS is lost	The Charging Station SHALL make the number of attempts as configured in NetworkProfileConnectionAttempts per entry of NetworkConfigurationPriority .	
Old	B10.FR.04	B10.FR.01 OR B09.FR.01 AND After a reboot	The Charging Station SHALL begin connecting to the first entry of NetworkConfigurationPriority	Same as A05.FR.05
New	B10.FR.04	(B10.FR.01 OR B09.FR.01) AND After a reboot	The Charging Station SHALL begin connecting to the first entry of NetworkConfigurationPriority	Same as A05.FR.05

The following requirement is added to make explicit that a BootNotification must be sent, or else Charging Station might connect to a new CSMS without it, in which case CSMS would respond with a CALLERROR(SecurityEvent).

New requirement

ID	Precondition	Requirement definition	Note
B10.FR.10 (new)	B10.FR.03 AND Charging Station successfully connected after having switched to a different NetworkConnectionProfile	Charging Station SHALL send a BootNotificationRequest to CSMS to reestablish its registration status, even if it has not rebooted since last being accepted by any CSMS.	Charging Station does not need to check whether the CSMS it connected to, is actually one that it has not connected to before.

2.18. Page 67 - (2025-06) - B11 - Clarify meaning of OnIdle for Reset

The "idle state" is defined in Terminology as: "In both use cases and sequence diagrams, Idle status is referred as the state in which a Charging Station is not performing any use case related tasks. Condition during which the equipment can promptly provide a primary function but is not doing so." This is a broader concept, than having an active transaction. A remark is added to the use case to explain that.

The sentence about persistent states and ResetResponse did not belong in Remarks section.

No.	Type	Description
1	Name	Reset - Without Ongoing Transaction
...	...	

No.	Type	Description
8	Remark(s)	Persistent states: for example, EVSE set to <i>Unavailable</i> SHALL persist. + [line through]#The Charging Station responds with <i>ResetResponse</i>. OnIdle refers to the "idle state" of a charging station. This is when the Charging Station is not performing any use case related tasks that might interfere with a reset process. The most obvious case is being involved in an active transaction, but there are other conditions when the Charging Station is not idle, for example, when a firmware update process is ongoing, a log file is uploaded to CSMS, a reservation is pending or a cable is still locked in the Charging Station.

2.19. Page 69 - (2025-01) - B11 - Reset without Ongoing Transaction - Requirements

The definition of B11.FR.06 has been improved:

	ID	Precondition	Requirement definition
Old text	B11.FR.06	B11.FR.01 AND For example there is a firmware update ongoing that cannot be interrupted.	The Charging Station SHALL respond with a status <i>Rejected</i> .
New text	B11.FR.06	B11.FR.01 AND Charging Station is at this moment not able to perform a reset	The Charging Station SHALL respond with a status <i>Rejected</i> .

2.20. Page 73 - (2025-01) - B12 - Reset with Ongoing Transaction - Requirements

The requirement B12.FR.04, 08 and added B12.FR.10 have been improved:

At the (2025-04) version, the old text requirement definitions have been corrected, because they already contained the revision instead of the original text from the edition 3 specification. In addition, requirement B12.FR.10 was newly added to edition 3, not updated.

	ID	Precondition	Requirement definition
Old text (Corrected in 2025-04)	B12.FR.04	If no <i>evseld</i> is supplied AND If any transaction is in progress and an Immediate Reset is received.	The Charging Station SHALL attempt to terminate any transaction in progress and send a <i>TransactionEventRequest</i> (<i>eventType = Ended</i>) message before performing a reboot.
New text	B12.FR.04	If no <i>evseld</i> is supplied AND If any transaction is in progress and an Immediate Reset is received.	The Charging Station SHALL attempt to terminate any transaction in progress and send a <i>TransactionEventRequest</i> (<i>eventType = Ended</i>) message with <i>triggerReason = ResetCommand</i> and <i>transactionInfo.stoppedReason = ImmediateReset</i> for each terminated transaction before performing a reboot.
Old text (Corrected in 2025-04)	B12.FR.08	If an <i>evseld</i> is supplied AND If a transaction is in progress on the EVSE and an Immediate Reset is received.	The Charging Station SHALL attempt to terminate the transaction in progress on the EVSE and send a <i>TransactionEventRequest</i> (<i>eventType = Ended</i>) message before performing a reset.
New text	B12.FR.08	If an <i>evseld</i> is supplied AND If a transaction is in progress on the EVSE and an Immediate Reset is received.	The Charging Station SHALL attempt to terminate the transaction in progress on the EVSE and send a <i>TransactionEventRequest</i> (<i>eventType = Ended</i>) message with <i>triggerReason = ResetCommand</i> and <i>transactionInfo.stoppedReason = ImmediateReset</i> before resetting the EVSE.

	ID	Precondition	Requirement definition
Added (Corrected in 2025-04)	B12.FR.10	B12.FR.02 AND Charging Station is at this moment not able to perform an Immediate reset for a reason other than the fact that a transaction is in progress	The Charging Station SHALL return a ResetResponse Rejected

2.21. Page 77 - (2025-01) - 1.6 Relationship between authorization and transaction

A new section has been added after 1.5 Unknown Offline Authorization

This section is informative.

The purpose of authorization is twofold. It ensures in the first place, that energy is only offered to a known user (represented by the *idToken*), which is essential for billing. In the second place, it ensures that only the user who was authorized in the first place (or a member of the same group of users) is allowed to unplug the cable. This is an important safeguard against cable theft in situations where the charging station does not have a fixed cable and the user brings its own charging cable.

Authorization and the duration of the authorization period are not strictly tied to a transaction: it is possible to have transactions without explicit authorization, e.g. in the case of a charging station that can be started with a push button. In that case one could say that there is a permanent authorization for anyone to charge.

The start of the authorization period:

- can take place before a transaction is started (e.g. when a cable is not yet connected), or
- can cause a transaction to be started (e.g. when authorization is defined as the start of a transaction by setting TxStartPoint = Authorized), or
- can happen after a transaction has already started (e.g. when connection of the cable is defined as the start of a transaction by setting TxStartPoint = EVConnected).

(See chapter E.1.1 "Flexible transaction start/stop" for a description of transaction start and stop points.)

In any case, authorization (or authorization period) ends when the same *idToken* is presented again for authorization, or when the transaction ends. This means that ending of the authorization period:

- can happen during a transaction without ending the transaction (e.g. when *idToken* is presented again, but TxStopPoint = EVConnected), or
- can cause the transaction to end (e.g. when *idToken* is presented again and TxStopPoint = Authorized), or
- can be caused by the end of the transaction (e.g. when *idToken* is not presented for authorization, but the cable is disconnected and TxStopPoint = EVConnected), or
- can be caused by cable plug-out if no transaction was started.

A Charging Station defines when authorization starts (i.e. upon receiving the AuthorizeResponse, or when authorizing locally via authorization cache or local authorization list) and when authorization ends (i.e. when *idToken* is presented a second time, or when the transaction ends). Charging Station notifies CSMS about this, as follows:

- If authorization occurs before start of the transaction, Charging Station tells CSMS that authorization has taken place, by including the *idToken* in the first TransactionEventRequest of the transaction.
- If authorization occurs within a transaction or at the start of a transaction, Charging Station reports this by including the *idToken* in TransactionEventRequest together with a *triggerReason* = Authorized.
- The end of authorization is reported in a TransactionEventRequest with a *triggerReason* = StopAuthorized or Deauthorized, or by reporting the end of the transaction.

2.22. Page 80 - (2025-01) - C01 - EV Driver Authorization using RFID - Requirements

A requirement has been added to define when two *idTokens* are considered equal:

New requirement

ID	Precondition	Requirement definition	Note
C01.FR.25		Two <code>[idtokentype]</code> elements are considered to be equal when they have the same value for the fields <code>idToken.idToken</code> and <code>idToken.type</code>	<i>additionalInfo</i> is not taken into account when comparing. See C01.FR.02, C01.FR.03, C01.FR.05 for <i>idToken</i> requirements where <i>idTokens</i> are compared.

2.23. Page 80 - (2025-01) - C01 - EV Driver Authorization using RFID - Requirements

A requirement has been added to make explicit that authorization ends after EVConnectionTimeout:

New requirement

ID	Precondition	Requirement definition	Note
C01.FR.26	When an <i>idToken</i> has been authorized and the EV Driver does not plug in the charging cable before the timeout set by the Configuration Variable: EVConnectionTimeOut	Charging Station SHALL end the authorization of <i>idToken</i>	See also E03.FR.05 and F02.FR.07/08 for additional behavior in case a transaction had already been started.

2.24. Page 99 - (2025-01) - C09- Authorization by GroupId - Requirements

A GroupIdToken is controlled by CSMS. This should be reflected in the type.

New requirement

ID	Precondition	Requirement definition
C09.FR.13		The field <i>idToken.type</i> of a GroupIdToken SHOULD be Central

2.25. Page 101 - (2025-01) - Updated requirements related to clarify the relation between AuthCacheLifeTime and cacheExpiryDateTime

The OCPP specification already describes the relation between AuthCacheLifeTime and cacheExpiryDateTime at section 1.3 Authorization Cache, however the requirements are missing this information.

Changed C10 requirements:

	ID	Precondition	Requirement definition	Note
Old text	C10.FR.10	When the validity of an Authorization Cache entry expires.	The Authorization Cache entry SHALL be removed from the cache or changed to Expired.	
New text	C10.FR.10	NOT C10.FR.13 AND when more than <code>AuthCacheLifeTime</code> seconds have passed since <i>idTokenInfo</i> was last updated	The Authorization Cache entry SHALL be removed from the cache or changed to Expired.	A <i>cacheExpiryDateTime</i> in the past will prevent an <i>idToken</i> from being stored in the authorization cache, or remove it from authorization cache if it was already present. This is used e.g. for prepaid accounts that should not be kept in authorization cache.

	ID	Precondition	Requirement definition	Note
Old text	C10.FR.13	When IdTokenInfoType contains a value for <i>cacheExpiryDateTime</i>	The time a token is considered to be present in the cache is determined by <i>cacheExpiryDateTime</i> . This variable indicates the date and time after which a token expires in the Authorization Cache.	This expiry of the cache is not the same as the expiration date that is set for the IdToken (e.g. RFID card expiry date).
New text	C10.FR.13	When IdTokenInfoType contains a value for <i>cacheExpiryDateTime</i> and current time is greater than <i>idTokenInfo.cacheExpiryDateTime</i>	The Authorization Cache entry SHALL be removed from the cache or changed to Expired .	This expiry of the cache is not the same as the expiration date that is set for the IdToken (e.g. RFID card expiry date).

2.26. Page 109 - (2025-11) - C15 - Offline Authorization of unknown Id - Scenario description

In the Scenario Description, points 2 and 3 have been reversed, placing the Local list above the cache in terms of precedence.

2.26.1. C15 - Offline Authorization of unknown Id

No.	Type	Description
1	Name	Offline Authorization of unknown Id
2	ID	C15
	Parent use case	C12 - Start Transaction - Cached Id
3	Objective(s)	To allow automatic authorization of any "unknown" identifiers that cannot be explicitly authorized by Authorization Cache entries.
4	Description	This use case describes the scenario of presented "unknown" identifiers, other than are present in an Authorization Cache or Local Cache entry using <i>OfflineTxForUnknownIdEnabled</i> .
	Actors	Charging Station, EV Driver
	Scenario description	1. The EV Driver wants to start charging the EV and presents the IdToken. 2. The Charging Station checks the Local Authorization List, the IdToken is not present in the Local Authorization List. 3. The Charging Station checks the Authorization Cache, the IdToken is not present in the Authorization Cache. 4. The Charging Station accepts the unknown IdToken if <i>OfflineTxForUnknownIdEnabled</i> is set True 5. The Charging Station rejects the unknown IdToken if <i>OfflineTxForUnknownIdEnabled</i> is set False

2.27. Page 113 - (2025-01) - C16 - Stop Transaction with a Master Pass - Requirements

Requirements about actual stopping of transactions have been added/updated:

Updated requirement

Table 5. C16 - Stop Transaction with a Master Pass - Requirements

	ID	Precondition	Requirement definition
Old text	C16.FR.01	User presents an IdToken that has a groupid equal to <i>MasterPassGroupId</i> AND The Charging Station has a UI with input capabilities.	The Charging Station SHALL "show" the Master Pass UI to let user select which transaction to stop.

	ID	Precondition	Requirement definition
New text	C16.FR.01	User presents an IdToken that has a groupId equal to MasterPassGroupId AND The Charging Station has a UI with input capabilities .	The Charging Station SHALL "show" the Master Pass UI to let user select which transaction to stop.
Old text	C16.FR.02	User presents an IdToken that has a groupId equal to MasterPassGroupId AND the Charging Station does NOT have a UI.	The Charging Station SHALL stop all ongoing transactions as described in use case E07.
New text	C16.FR.02	User presents an IdToken that has a groupId equal to MasterPassGroupId AND the Charging Station does NOT have a UI.	The Charging Station SHALL stop all ongoing transactions as described in use case E07.

New requirements

ID	Precondition	Requirement definition
C16.FR.07	C16.FR.01 OR C16.FR.02	Charging Station SHALL stop the transaction as described in use case E07.
C16.FR.07	C16.FR.07	Charging Station SHALL set <i>transactionInfo.stoppedReason</i> = MasterPass in TransactionEventRequest with <i>eventType</i> = Ended.

2.28. Page 129 - (2025-01) - Updated sequence diagram E01 S5

After all added clarifications on the PowerPathClosed TxStartPoint the corresponding sequence diagram was not updated to reflect this.

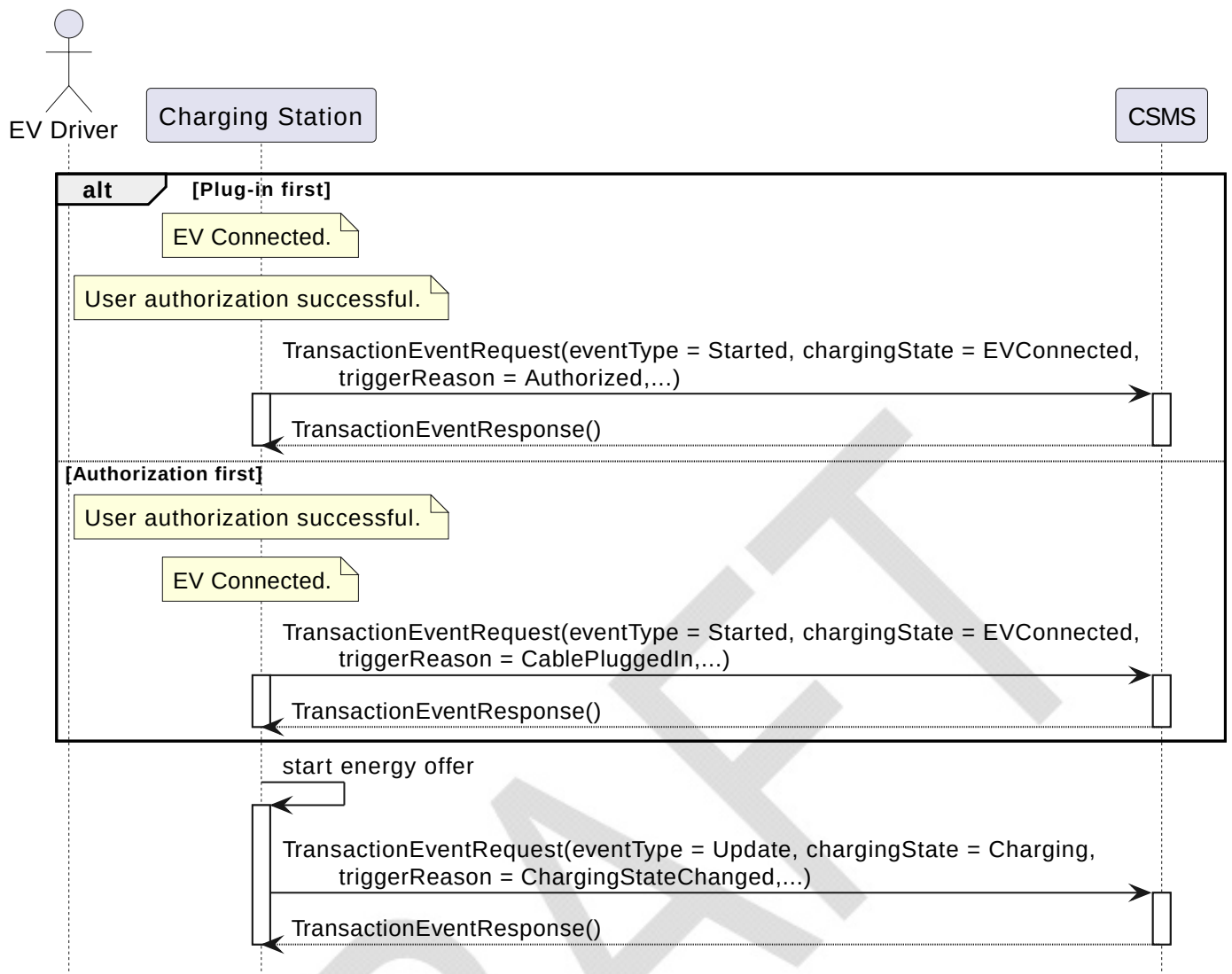


Figure 43. Sequence Diagram: Start Transaction options - PowerPathClosed

2.29. Page 151 - (2025-02) - E06.FR.05 for DataSigned as TxStopPoint is invalid

DataSigned cannot be used as a TxStopPoint. This requirement is therefore invalid and confusing when present.

Deleted requirement

	ID	Precondition	Requirement definition
Delete	E06.FR.05	TxStopPoint contains: DataSigned AND Charging Station can no longer retrieve signed meter values.	The Charging Station SHALL stop the transaction and send a TransactionEventRequest (eventType = Ended) to the CSMS.

2.30. Page 143 - (2025-11) - E04.FR.07/08 removed

Requirements E04.FR.07/08 are already covered by E12.FR.04/05, which is the better place for this. They are therefore removed from E04 to avoid future inconsistencies.

Deleted requirements

ID	Precondition	Requirement definition	Note
E04.FR.07	E04.FR.06 AND <i>Offline</i> AND The Charging Station is running low on memory	The Charging Station MAY drop TransactionEventRequest(eventType = Updated) messages.	Same as J02.FR.12
E04.FR.08	E04.FR.07	When dropping TransactionEventRequest(eventType = Updated) messages, the Charging Station SHALL drop intermediate messages first (2nd message, 4th message, 6th message etc.), not start dropping messages from the start or stop adding messages to the queue.	Same as J02.FR.13

2.31. Page 152 - (2025-01) - E07 - Improved scenario description names

The use case did not clearly indicate when the scenario and the alternative scenario are applicable.

No.	Type	Description
1	Name	Transaction locally stopped by IdToken
2	ID	E07
3	Objective(s)	The EV Driver wants to stop an ongoing transaction, by locally presenting his IdToken.
4	Description	This use case covers how the EV Driver can stop a transaction when he wants to leave the charging station.
	Actors	Charging Station, CSMS, EV Driver
	Scenario description Reporting StopAuthorized with end of transaction	<i>TxStopPoint = Authorized (or PowerPathClosed)</i> 1. The EV Driver presents IdToken a second time to end charging. 2. The Charging Station stops the energy transfer and if the cable is not permanently attached, the Charging Station unlocks the cable. 3. The Charging Station sends a TransactionEventRequest (eventType = Ended) with <i>triggerReason = StopAuthorized</i> and <i>stoppedReason = Local</i>. 4. The CSMS responds with a TransactionEventResponse.
	Alternative scenario(s) Reporting StopAuthorized in Update event first, then end transaction	<i>TxStopPoint = Authorized (or PowerPathClosed)</i> 1. The EV Driver presents IdToken a second time to end charging. 2. The Charging Station sends a TransactionEventRequest (eventType = Updated) with <i>triggerReason = StopAuthorized</i> 3. The CSMS responds with a TransactionEventResponse. 4. The Charging Station stops the energy transfer and if the cable is not permanently attached, the Charging Station unlocks the cable. 5. The Charging Station sends a TransactionEventRequest (eventType = Ended) with <i>triggerReason = ChargingStateChanged</i>, <i>transactionInfo.chargingState = EVConnected</i> 6. The CSMS responds with a TransactionEventResponse.
...		

2.32. Page 154 - (2025-01) - E07 - Transaction locally stopped by IdToken

Sequence diagram Fig. 55 is updated to fix an error.

The fourth TransactionEventRequest needs to have: *triggerReason = ChargingStateChanged*, *chargingState = EVConnected*

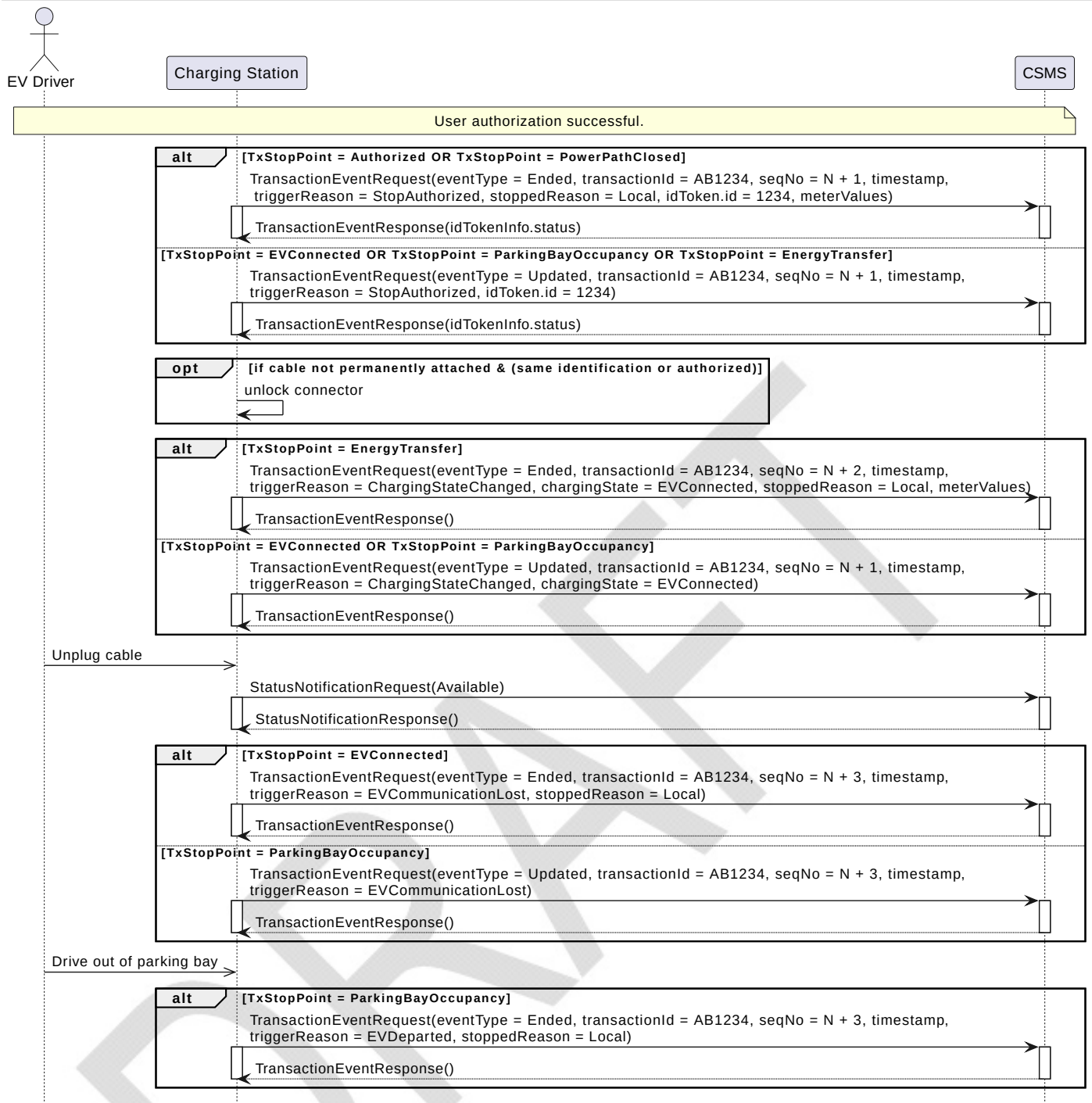


Figure 55. Sequence Diagram: Transaction locally stopped by IdToken with TransactionEventRequest reported strictly by TxStopPoint configuration

2.33. Page 154 - (2025-11) - E07.FR.01 - Added note referring to C09

The process of stopping with an idToken with the same groupId is explained in more detail in use case C09. A reference to this is added to the note.

Changed requirement

	ID	Precondition	Requirement definition	Note
Old	E07.FR.01	When an idToken is presented during a transaction that has been authorized AND (a) the presented idToken is the same as the idToken that started the authorization OR (b) when the presented idToken is in the Local Authorization List or Authorization Cache AND is valid AND has the same GroupIdToken as the IdToken that started the authorization.	The Charging Station SHALL end the authorization of the transaction, without first sending an AuthorizeRequest	The idToken that started the authorization can always be used to end the authorization. Ending authorization will end delivery of energy. Depending on the TxStopPoint ending of the authorization may also end the transaction. (See C01.FR.03)
New	E07.FR.01	When an idToken is presented during a transaction that has been authorized AND (a) the presented idToken is the same as the idToken that started the authorization OR (b) when the presented idToken is in the Local Authorization List or Authorization Cache AND is valid AND has the same GroupIdToken as the IdToken that started the authorization.	The Charging Station SHALL end the authorization of the transaction, without first sending an AuthorizeRequest	The idToken that started the authorization can always be used to end the authorization. Ending authorization will end delivery of energy. Depending on the TxStopPoint ending of the authorization may also end the transaction. Read use case C09 (C09.FR.05/07) for detailed description of stopping based on groupId.

2.34. Page 155 - (2025-09) - E07.FR.07 - Improved precondition

The precondition of E07.FR.07 was written as text, but it is more precise to refer another requirement.

	ID	Precondition	Requirement definition	Note
Old	E07.FR.07	As part of the normal transaction termination.	The Charging Station SHALL unlock the cable (if not permanently attached).	
New	E07.FR.07	E07.FR.02	The Charging Station SHALL unlock the cable (if not permanently attached).	

2.35. Page 164 - (2025-11) - E10.FR.01/02 improved precondition [806]

Requirements for E10 all have the prerequisite that configuration variable StopTxOnEVSideDisconnect = false, but this was not made explicit. The combination of StopTxOnEVSideDisconnect = false (transaction remains authorized) and UnlockOnEVSideDisconnect = true (charging station side is unlocked when EV side is unlocked) is not specified. This is made explicit in remark of use case.

2.35.1. E10 - When cable disconnected on EV-side: Suspend Transaction

No.	Type	Description
1	Name	When cable disconnected on EV-side: Suspend Transaction
2	ID	E10
...	...	...
8	Remark(s)	... The combination of StopTxOnEVSideDisconnect = false (transaction remains authorized) together with UnlockOnEVSideDisconnect = true (charging station side is unlocked when EV side is unlocked) is not specified and behavior is undefined.

Updated requirements

Old	ID	Precondition	Requirement definition	Note
Old	E10.FR.01	Cable not permanently attached	The Connector SHALL remain locked at the Charging Station until the EV Driver presents the IdToken.	
New	E10.FR.01	If StopTxOnEVSideDisconnect = false AND Cable not permanently attached	The Connector SHALL remain locked at the Charging Station until the EV Driver presents the IdToken.	
Old	E10.FR.02	Cable permanently attached AND Cable not plugged in within timeout	The Charging Station SHALL deauthorize the transaction.	
New	E10.FR.02	If StopTxOnEVSideDisconnect = false AND Cable permanently attached AND Cable not plugged in within timeout	The Charging Station SHALL deauthorize the transaction.	

2.36. Page 173 - (2025-11) - Updated Header - Section 2.2

Header is explicitly for use case: E15 - End of charging process. Here, the reference to ISO 15118 is already made, mentioning it in the section header is not necessary.

2.2. Interrupting and Stopping ~~ISO 15118~~ Charging === Page 173 - (2025-11) - Updated Header - Section 2.2

Header is explicitly for use case: E15 - End of charging process. Here, the reference to ISO 15118 is already made, mentioning it in the section header is not necessary.

2.2. Interrupting and Stopping ~~ISO 15118~~ Charging

2.37. Page 179 - (2025-11) - F01.FR.01/02 - No authorization for type = Central

According to C03.FR.01 no AuthorizeRequest shall be sent for idToken.type = Central. This was also the intention of F01.FR.01, because it states "... as if in response to a local action ...". That can never apply to a token type Central, but that is not immediately clear. The preconditions have therefore been made explicit for this. The same also applies to type = NoAuthorization.

Old	ID	Precondition	Requirement definition	Note
Old	F01.FR.01	If the value of AuthorizeRemoteStart = true AND Charging Station receives a RequestStartTransactionRequest	The Charging Station SHALL behave as if in response to a local action at the Charging Station to allow energy transfer after successful authorization of the IdToken given in RequestStartTransactionRequest message.	Charging Station will first respond to the request and then try to authorize the IdToken, using the Local Authorization List, Authorization Cache and/or an AuthorizeRequest . Energy transfer is only allowed after authorization was obtained.
New	F01.FR.01	Charging Station receives a RequestStartTransactionRequest AND (AuthorizeRemoteStart = true AND idToken.type is NOT Central or NoAuthorization)	The Charging Station SHALL behave as if in response to a local action at the Charging Station to allow energy transfer after successful authorization of the IdToken given in RequestStartTransactionRequest message.	Charging Station will first respond to the request and then try to authorize the IdToken, using the Local Authorization List, Authorization Cache and/or an AuthorizeRequest . Energy transfer is only allowed after authorization was obtained.

	ID	Precondition	Requirement definition	Note
Old	F01.FR.02	If the value of AuthorizeRemoteStart = false AND Charging Station receives a RequestStartTransactionRequest	The Charging Station SHALL allow energy transfer for the IdToken given in RequestStartTransactionRequest message without checking authorization.	Charging Station will first respond to the request, and send a TransactionEventRequest with the idToken to the CSMS, and the CSMS will check the authorization status of the IdToken when processing this TransactionEventRequest .
New	F01.FR.02	Charging Station receives a RequestStartTransactionRequest AND (AuthorizeRemoteStart = false OR idToken.type is Central or NoAuthorization)	The Charging Station SHALL allow energy transfer for the IdToken given in RequestStartTransactionRequest message without checking authorization.	Charging Station will first respond to the request, and send a TransactionEventRequest with the idToken to the CSMS, and the CSMS will check the authorization status of the IdToken when processing this TransactionEventRequest .

2.38. Page 184 - (2025-11) - F02.FR.09/10 - No authorization for type = Central

According to C03.FR.01 no AuthorizeRequest shall be sent for idToken.type = Central. This was also the intention of F02.FR.09, because it states "... as if in response to a local action ...". That can never apply to a token type Central, but that is not immediately clear. The preconditions have therefore been made explicit for this. The same also applies to type = NoAuthorization.

	ID	Precondition	Requirement definition	Note
Old	F02.FR.09	If the value of AuthorizeRemoteStart = true AND Charging Station receives a RequestStartTransactionRequest	The Charging Station SHALL behave as if in response to a local action at the Charging Station to start a transaction after successful authorization of the IdToken given in RequestStartTransactionRequest message.	Charging Station will first respond to the request and then try to authorize the IdToken, using the Local Authorization List, Authorization Cache and/or an AuthorizeRequest . A transaction is only started after authorization was obtained. Same as F01.FR.01
New	F02.FR.09	Charging Station receives a RequestStartTransactionRequest AND (AuthorizeRemoteStart = true AND idToken.type is NOT Central or NoAuthorization)	The Charging Station SHALL behave as if in response to a local action at the Charging Station to allow energy transfer after successful authorization of the IdToken given in RequestStartTransactionRequest message.	Charging Station will first respond to the request and then try to authorize the IdToken, using the Local Authorization List, Authorization Cache and/or an AuthorizeRequest . Energy transfer is only allowed after authorization was obtained. Same as F01.FR.01.

	ID	Precondition	Requirement definition	Note
Old	F02.FR.10	If the value of <code>AuthorizeRemoteStart</code> = false AND Charging Station receives a <code>RequestStartTransactionRequest</code>	The Charging Station SHALL start a transaction for the IdToken given in <code>RequestStartTransactionRequest</code> message without checking authorization.	Note that after the transaction has been started, the Charging Station will send a <code>TransactionEventRequest</code> with the idToken to the CSMS, and the CSMS will check the authorization status of the IdToken when processing this <code>TransactionEventRequest</code> . Same as F01.FR.02
New	F02.FR.10	Charging Station receives a <code>RequestStartTransactionRequest</code> AND (<code>AuthorizeRemoteStart</code> = false OR <code>idToken.type</code> is Central or <code>NoAuthorization</code>)	The Charging Station SHALL allow energy transfer for the IdToken given in <code>RequestStartTransactionRequest</code> message without checking authorization.	Charging Station will first respond to the request, and send a <code>TransactionEventRequest</code> with the idToken to the CSMS, and the CSMS will check the authorization status of the IdToken when processing this <code>TransactionEventRequest</code> . Same as F01.FR.02.

2.39. Page 190 - (2025-11) - F05 - New requirement for connectors without lock [1013]

New requirement

ID	Precondition	Requirement definition
F05.FR.07	F05.FR.01 AND The connector has no lock or a manual lock (e.g. sType1)	The Charging Station SHOULD respond with RPC Framework CALLERROR: NotSupported.

2.40. Page 198 - (2025-01) - G01 - Status Notification - Requirements

A requirement is added to make explicit that a plug-in on reserved connector does not automatically change status to Occupied.

ID	Precondition	Requirement definition
G01.FR.09	The connector is Reserved when an EV is connecting AND (No IdToken is presented OR EV driver presents an IdToken not matching the reservation)	Connector status SHALL not change.

2.41. Page 197 - (2025-01) - G01 - Status Notification - State transition overview for connecting/disconnecting

The following text from the table has to be removed:

Initial	Cable plugin	Cable unplug
Available	→ Occupied	-

Initial	Cable plugin	Cable unplug
Occupied	-	→ Available (→ Unavailable, if scheduled to become Unavailable)
Reserved	(→ Occupied , only if authorized for reserved IdToken)	-
Unavailable	-	-
Faulted	-	-

2.42. Page 208 - (2025-01) - H. Reservation - Introduction

The Introduction text has been updated:"

Old text	This Functional Block describes the reservation functionality of OCPP. The reservation functionality enables an EV Driver to make a reservation of a Charging Station/EVSE, ensuring an available Connector at a Charging Station when he arrives. With Charging Stations not being abundantly available, and EVs having limited range, EV Drivers plan their trips from Charging Station to Charging Station. They need to know for sure they can use a Charging Station they plan to go to. They don't like it when another EV Driver has started using the Charging Station in the time they were traveling to the Charging Station. For the EV Driver it is useful to be able to reserve a specific Type of Connector, or, when the EV Driver has no preference, an unspecified EVSE at a Charging Station. So he knows for sure he can charge at the Charging Station when he arrives.
New text	This Functional Block describes the reservation functionality of OCPP. The reservation functionality enables an EV Driver to reserve an EVSE at a Charging Station until a certain time in order to ensure that this EVSE cannot be occupied by another user. OCPP allows to reserve a specific EVSE at a Charging Station or a specific connector type. The EV Driver can also reserve an unspecified EVSE, in which case the Charging Station will make sure that at least one EVSE remains available for the EV Driver. Only available EVSEs can be reserved, since a Charging Station cannot know in advance when an occupied EVSE will become available again. This makes it impossible to guarantee a reservation for an EVSE that is currently occupied. NOTE: A CSMS would still be able to support the reservation functionality for occupied EVSEs by delaying the sending of the reservation message to the Charging Station until the EVSE becomes available, but there is no guarantee that it is available in time.

2.43. Page 188 - (2025-11) - F03.FR.07 - Improved precondition and changed requirement definition

Part of the requirement definition has been moved to the precondition.

	ID	Precondition	Requirement definition	Note
Old	F03.FR.07	When the Charging Station receives a RequestStopTransactionRequest	And the TransactionId can be matched to an active transaction; the Charging Station SHALL respond with a RequestStopTransactionResponse with status set to <i>Accepted</i> .	
New	F03.FR.07	When the Charging Station receives a RequestStopTransactionRequest AND the TransactionId can be matched to an active transaction.	And the TransactionId can be matched to an active transaction; The Charging Station SHALL respond with a RequestStopTransactionResponse with status set to <i>Accepted</i> .	

2.44. Page 188 - (2025-11) - F03.FR.08 - Improved precondition and changed requirement definition

Part of the requirement definition has been moved to the precondition.

	ID	Precondition	Requirement definition	Note
Old	F03.FR.08	When the Charging Station receives a RequestStopTransactionRequest	And the TransactionId can be matched to an active transaction; the Charging Station SHALL respond with a RequestStopTransactionResponse with status set to <i>Rejected</i> .	
New	F03.FR.08	When the Charging Station receives a RequestStopTransactionRequest AND the TransactionId can be matched to an active transaction.	And the TransactionId can be matched to an active transaction; The Charging Station SHALL respond with a RequestStopTransactionResponse with status set to <i>Rejected</i> .	

2.45. Page 180 - (2025-11) - F01.FR.20/F02.FR.22 - EVSE can be reported on cable plug-in [786]

Updated requirement

	ID	Precondition	Requirement definition	Note
Old	F01.FR.20	If the RequestStartTransactionRequest does not contain an <i>evseId</i> AND the Charging Station is capable of selecting an EVSE	The Charging Station SHALL select an EVSE to be used as a value for <i>evseId</i> for the operation	See also F01.FR.07 if Charging Station does not support starting at an arbitrary EVSE.
New	F01.FR.20	If the RequestStartTransactionRequest does not contain an <i>evseId</i> AND the Charging Station is capable of selecting an EVSE or accepts a plug-in at an arbitrary EVSE	The Charging Station SHALL report in later TransactionEventRequests the selected EVSE or the EVSE where cable is plugged in.	See also F01.FR.07 if Charging Station does not support starting at an arbitrary EVSE.

Updated requirement

	ID	Precondition	Requirement definition	Note
Old	F02.FR.22	If the RequestStartTransactionRequest does not contain an <i>evseId</i> AND the Charging Station is capable of selecting an EVSE	The Charging Station SHALL select an EVSE to be used as a value for <i>evseId</i> for the operation	See also F02.FR.15 if Charging Station does not support starting at an arbitrary EVSE. Same as F01.FR.20
New	F02.FR.22	If the RequestStartTransactionRequest does not contain an <i>evseId</i> AND the Charging Station is capable of selecting an EVSE or accepts a plug-in at an arbitrary EVSE	The Charging Station SHALL report in later TransactionEventRequests the selected EVSE or the EVSE where cable is plugged in.	See also F02.FR.15 if Charging Station does not support starting at an arbitrary EVSE. Same as F01.FR.20

2.46. Page 213 - (2025-02) - H02 - Added missing requirements

Added missing requirements explicitly specifying behaviour of Charging Station when a reservation is cancelled.

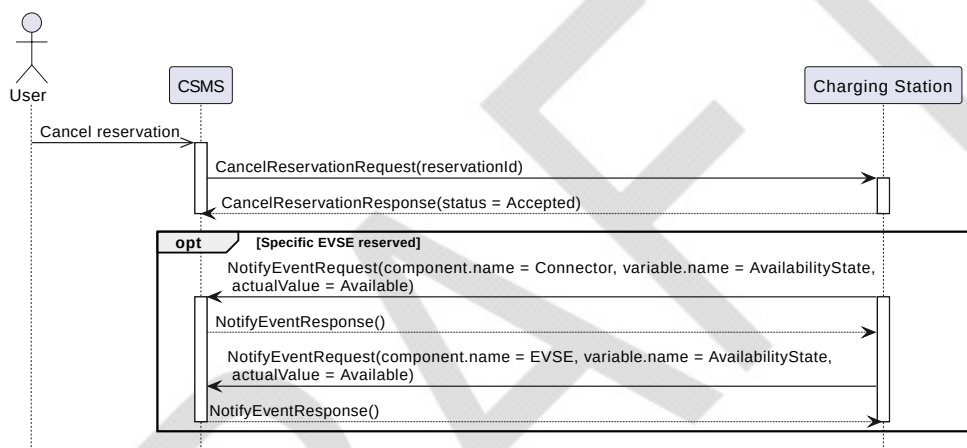
Removed details from scenario description:

No.	Type	Description
[...]		

No.	Type	Description
	Scenario description	1. EV Driver asks the CSMS to cancel a reservation. 2. To cancel a reservation, the CSMS sends CancelReservationRequest to the Charging Station. 3. If the Charging Station has a reservation matching the reservationId in the request PDU, it returns the status <i>Accepted</i>. 4. If a specific EVSE was reserved for this reservation, the Charging Station sends StatusNotificationRequest with the status <i>Available</i> or a NotifyEventRequest with <i>AvailabilityState</i> set to <i>Available</i> for all the Connectors of that EVSE. 4. If needed, the Charging Station sends StatusNotificationRequest with the status <i>Available</i> or a NotifyEventRequest with <i>AvailabilityState</i> set to <i>Available</i> for all the Connectors of EVSEs that became available. 5. The CSMS responds with StatusNotificationResponse or NotifyEventResponse to the Charging Station. 6. The reservation is canceled.
[...]		

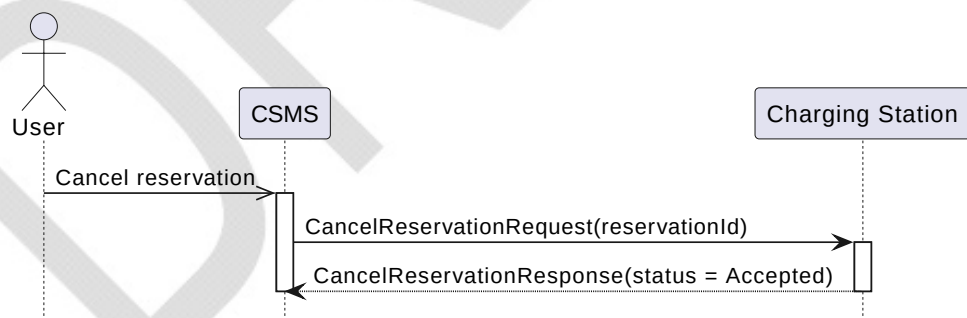
Removed details from sequence diagram

Old:



Sequence Diagram: Cancel Reservation

New:



Sequence Diagram: Cancel Reservation

New requirements

ID	Precondition	Requirement definition
H02.FR.03	H02.FR.02 AND If a specific EVSE was reserved for this reservation	The Charging Station SHALL allow charging again on this EVSE.
H02.FR.04	H02.FR.03	The Charging Station SHALL send a StatusNotificationRequest with status <i>Available</i> or a NotifyEventRequest with <i>AvailabilityState</i> set to <i>Available</i> to the CSMS for each connector, notifying the CSMS that all the connectors of this EVSE are available again for any EV Driver.

ID	Precondition	Requirement definition
H02.FR.05	H02.FR.02 AND If no specific EVSE was reserved for this reservation	The Charging Station SHALL allow charging on all EVSE which were not reserved explicitly.
H02.FR.06	H01.FR.05 AND before cancelling the reservation the amount of EVSEs reserved was equal to the amount of reservations	The Charging Station SHALL send for all connectors of all EVSEs which were not reserved explicitly: - a NotifyEventRequest with <i>component</i> = "Connector", <i>variable</i> = "AvailabilityState", <i>trigger</i> = "Delta", <i>actualValue</i> = "Available", OR - a StatusNotificationRequest with <i>connectorStatus</i> = Available.

2.47. Page 214/215 - (2025-01) - Improved use case scenario descriptions and added S3

The scenario descriptions have been updated to better explain the actual claiming of a reservation and have been clearly divided based on the configured TxStartPoint(s). In addition, A S3 has been added: 'Use an EVSE when Charging Station has a reservation for idToken, but connector status is Available.' This happens when reservation is for an unspecified EVSE and multiple EVSEs are available.

No.	Type	Description
1	Name	Use a reserved EVSE
2	ID	H03
3	Objective(s)	Use a reserved EVSE
4	Description	This use cases covers how a reserved EVSE can be used based on IdToken and GroupIdToken information.
	Actors	Charging Station, CSMS, EV Driver
S1	Scenario objective	Use an EVSE with connector status Reserved, that is reserved for this IdToken
	Scenario description	<i>TxStartPoint</i> = "Authorized"; <i>IdToken</i> presented first - The EV Driver presents an IdTokenType at the Charging Station that is the same as the reservation's IdTokenType. - Charging Station matches IdTokenType with the reservation. - Connector status becomes Available, since reservation has now been consumed. - Charging Station optionally authorizes the IdTokenType via an AuthorizeRequest. - If authorization accepted, or authorization step was skipped: - Charging Station starts a transaction as in E03 - Start Transaction - IdToken First. - Connector status will become Occupied when cable is connected.
	Scenario description #2	<i>TxStartPoint</i> = "EVConnected"; <i>Cable plugged in</i> first - The EV Driver connects the cable. - Charging Station starts a transaction, but EVSE connector status remains Reserved. - The EV Driver presents an IdTokenType at the Charging Station that is the same as the reservation's IdTokenType - Charging Station matches IdTokenType with the reservation - Connector status becomes Occupied, since reservation has now been consumed - Charging Station optionally authorizes the IdTokenType via an AuthorizeRequest - If authorization accepted, or authorization step was skipped: - Charging Station starts a transaction as in E02 - Start Transaction - Cable Plugin First

No.	Type	Description
5	Prerequisite(s)	EVSE has been reserved for IdToken and connector status is Reserved.
6	Postcondition(s)	n/a

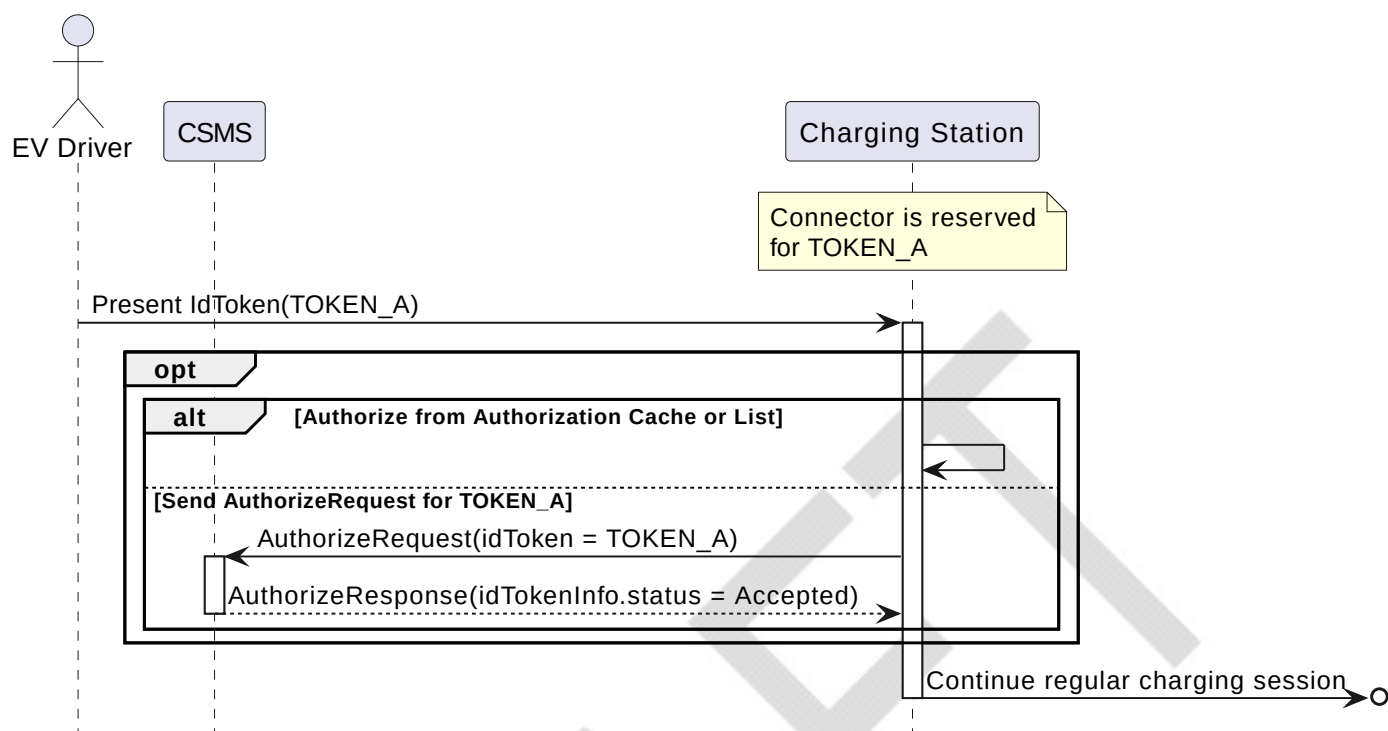


Figure 82. Sequence Diagram: Use a reserved EVSE with IdToken

S2	Scenario objective	Use an EVSE with connector status Reserved, that is reserved for this GroupIdToken
	Scenario description	<i>TxStartPoint = "Authorized"; IdToken presented first</i> 1. The EV Driver presents an IdTokenType at the Charging Station that is not the same as the reservation's IdTokenType, but the reservation contains a groupIdToken. 2. Charging Station authorizes the IdTokenType via an AuthorizeRequest, Local Authorization List or Authorization Cache, and checks if the groupIdToken of the IdTokenType matches with the reservation. 3. If groupIdTokens match: a. Connector status becomes Available, since reservation has now been consumed. b. Charging Station starts a transaction as in E03 - Start Transaction - IdToken First c. Connector status will become Occupied when cable is connected

	Scenario description #2	<i>TxStartPoint = "EVConnected"; Cable plugged in first</i> 1. The EV Driver connects the cable. 2. Charging Station starts a transaction, but connector status remains Reserved. 3. The EV Driver presents an IdTokenType at the Charging Station that is not the same as the reservation's IdTokenType, but the reservation contains a <i>groupIdToken</i>. 4. Charging Station authorizes the IdTokenType via an AuthorizeRequest, Local Authorization List or Authorization Cache, and checks if the <i>groupIdToken</i> of the IdTokenType matches with the reservation. 5. If <i>groupIdTokens</i> match: a. Connector status becomes Occupied, since reservation has now been consumed. b. Charging Station starts a transaction as in E02 - Start Transaction - Cable Plugin First
5	Prerequisite(s)	EVSE has been reserved for GroupIdToken. EVSE <i>connectorStatus</i> = Reserved.
6	Postcondition(s)	n/a

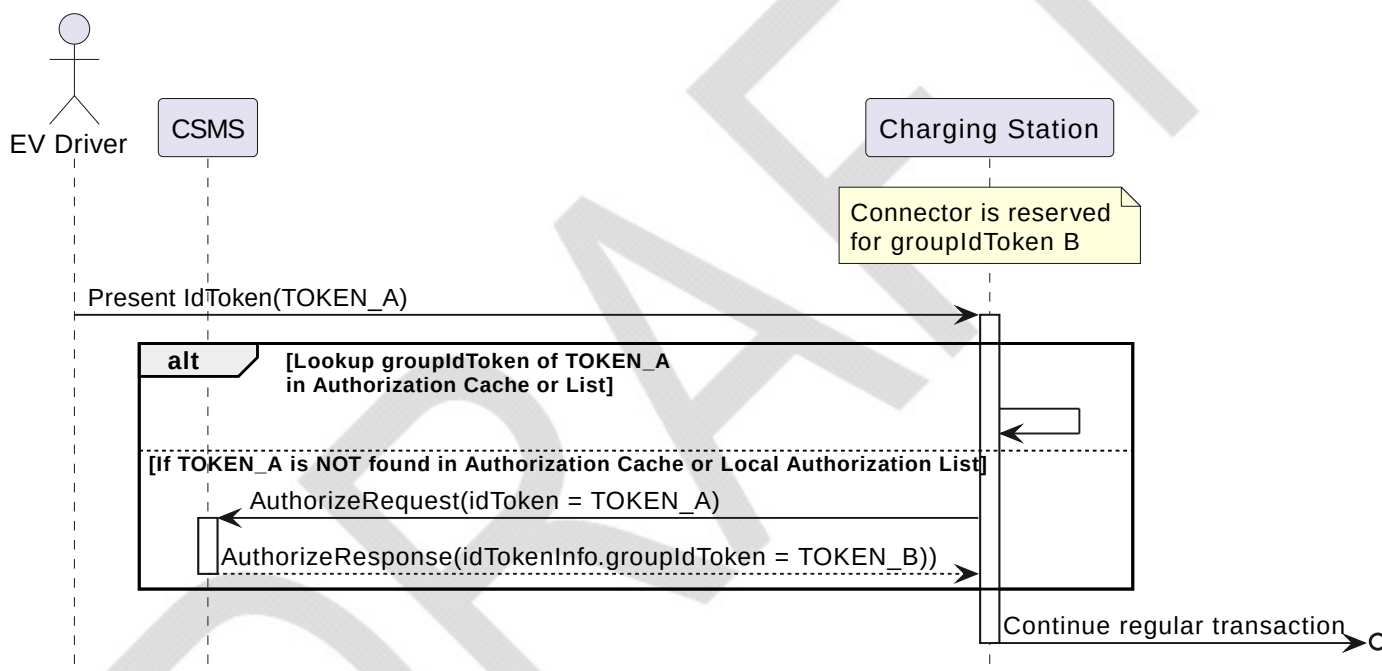


Figure 83. Sequence Diagram: Use a reserved EVSE with GroupId

S3	Scenario objective	Use an EVSE when Charging Station has a reservation for idToken, but connector status is Available. This happens when reservation is for an unspecified EVSE and multiple EVSEs are available.
	Scenario description	<i>TxStartPoint = "Authorized"; IdToken presented first</i> + Identical to scenario S1 above.

	Scenario description #2	<i>TxStartPoint = "EVConnected"; Cable plugged in first</i> 1. The EV Driver connects the cable 2. Charging Station reports connector status as Occupied 3. Charging Station starts a transaction 4. The EV Driver presents an IdTokenType at the Charging Station that is the same as the reservation's IdTokenType 5. Charging Station matches IdTokenType with the reservation 6. Charging Station optionally authorizes the IdTokenType via an AuthorizeRequest 7. If authorization accepted, or authorization step was skipped: a. Charging Station starts a transaction as in E02 - Start Transaction - Cable Plugin First
5	Prerequisite(s)	Unspecified EVSE has been reserved for idToken. EVSE connector status is Available.
6	Postcondition(s)	n/a

7	Error handling	n/a
8	Remark(s)	It is RECOMMENDED to validate the Identifier with an AuthorizeRequest after reception of ReserveNowRequest and before the start of the transaction. If an <i>idToken</i> is presented that does not match the reservation (and <i>groupIdTokens</i> do not match either), then this <i>idToken</i> is not authorized to charge. If TxStartPoint = Authorized or PowerPathClosed then a transaction would not be started in this case. If TxStartPoint = EVConnected or ParkingBayOccupancy then a transaction would be started by cable plug-in or occupancy of parking bay, but charging would not start. Assuming a TxStopPoint of EVConnected the transaction would be ended at cable plug-out.

Added note:

	ID	Precondition	Requirement definition	Note
Old text	H03.FR.09	When an <i>idToken</i> or <i>groupIdToken</i> is presented that matches a reservation	Charging Station SHALL consider the reservation to be used (consumed)	
New text	H03.FR.09	When an <i>idToken</i> or <i>groupIdToken</i> is presented that matches a reservation	Charging Station SHALL consider the reservation to be used (consumed)	The <i>(group)IdToken</i> can be presented locally at a card reader, but can also be part of a RequestStartTransaction

2.48. Page 226 - (2025-01) - I06.FR.02 Improved requirement text

Old	I06.FR.02	I06.FR.01 AND When there is updated tariff information available.	The CSMS SHALL respond with a TransactionEventResponse message to the Charging Station, containing the updated tariff information in the <i>PersonalMessage</i> field.
New	I06.FR.02	I06.FR.01 AND When there is updated tariff information available.	The CSMS SHALL respond with a TransactionEventResponse message to the Charging Station, containing the updated tariff information in the updatedPersonalMessage field.

2.49. Page 231 - (2025-01) - Updated section Multiple Locations/Phases

The section now specifies the 'relevant' configuration variables that apply.

Old	When a Charging Station can measure the same measurand on multiple locations or phases, all possible locations and/or phases SHALL be reported when configured in one of the relevant Configuration Variables.
New	When a Charging Station has measurands configured in SampledDataTxStarted/Updated/EndedMeasurands and/or AlignedDataMeasurands / AlignedDataTxEndedMeasurands , that can be measured on multiple locations or phases, then all possible locations and/or phases SHALL be reported.

2.50. Page 220 - (2025-11) - I01 - Scenario description update

Clarification that the personalMessage in the idTokenInfo field is used, as it may be confused with AuthorizeRequest; however, there is no PersonalMessage field in the Authorize response.

2.50.1. I01 - Show EV Driver-specific Tariff Information

No.	Type	Description
1	Name	Show EV Driver-specific Tariff Information
2	ID	I01
...	...	...
	Scenario description	1. The EV Driver wants to charge an EV, he presents his IdTokenType. 2. The Charging Station sends AuthorizeRequest to the CSMS to request authorization. 3. Upon receipt of AuthorizeRequest, the CSMS responds with AuthorizeResponse. This response message indicates whether or not the IdTokenType is accepted by the CSMS, and reports the EV Driver-specific tariff in the IdTokenInfo.personalMessage field.
...	...	...

2.51. Page 221 - (2025-11) - I01 - Updated use case description

Clarification that the personalMessage in the idTokenInfo field is used, as it may be confused with AuthorizeRequest; however, there is no PersonalMessage field in the Authorize response.

ID.	Precondition	Requirements
I01.FR.01		The CSMS MAY send EV Driver-specific tariff information in the IdTokenInfo.personalMessage field of an AuthorizeResponse message.

2.52. Page 243 - (2025-01) - Improved section on external Smart Charging Control Signals

To be inline with the EMS scenario additions to OCPP 2.1, this section also has been improved for OCPP 2.0.1.

Old text:

The OCPP protocol is originally developed for communication between a CSMS and one or more Charging Stations. As described in the above, this means that a Charging Station Operator (CSO) CSMS controls a Charging Station and, based on the charging limits of both the EV and the Charging Station, the CSO determines how fast the EV is charged. However, in some situations / applications of OCPP enabled Charging Stations, these are not the only 2 factors that determine the charging speed. Other inputs that determine charging speed could be DSO signals (e.g. via IEC 61850 [IEC61850-7-420], IEC 60870 [IEC60870-5-104], DNP3 [DNP3] or OpenADR [OPENADR]) or signals from a Building / Home Energy Management System. Although these signals are out of scope for OCPP, it seems clear from an OCPP perspective that the CSMS is to be informed of changes in charging by external signals. However, this also leads to a number of questions, such as how to deal with conflicting signals. The figure below presents an example setup with an Energy Management System, where the external signals are visualized both in a setup with direct communication to the Charging Station as well as a multiple Charging Station setup using a Local Controller:

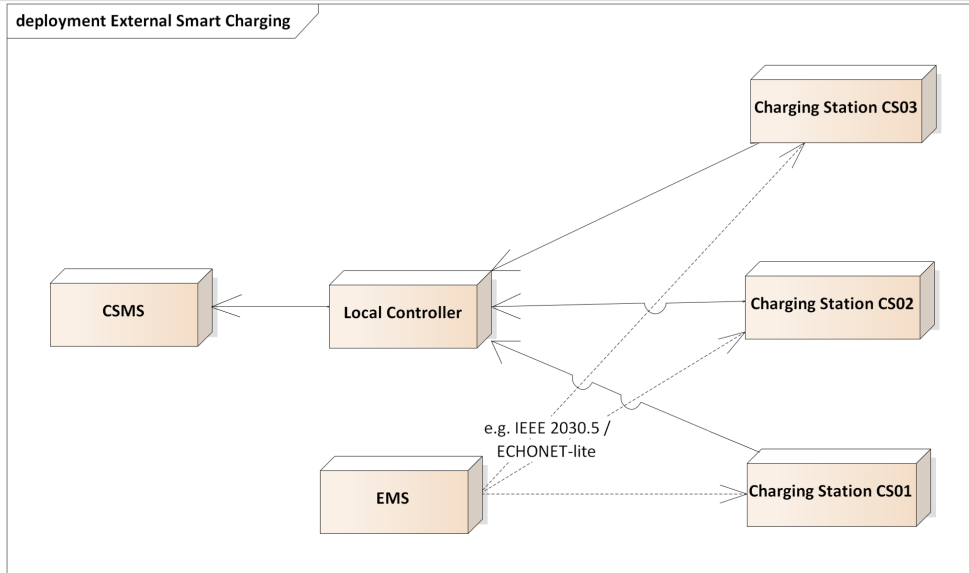


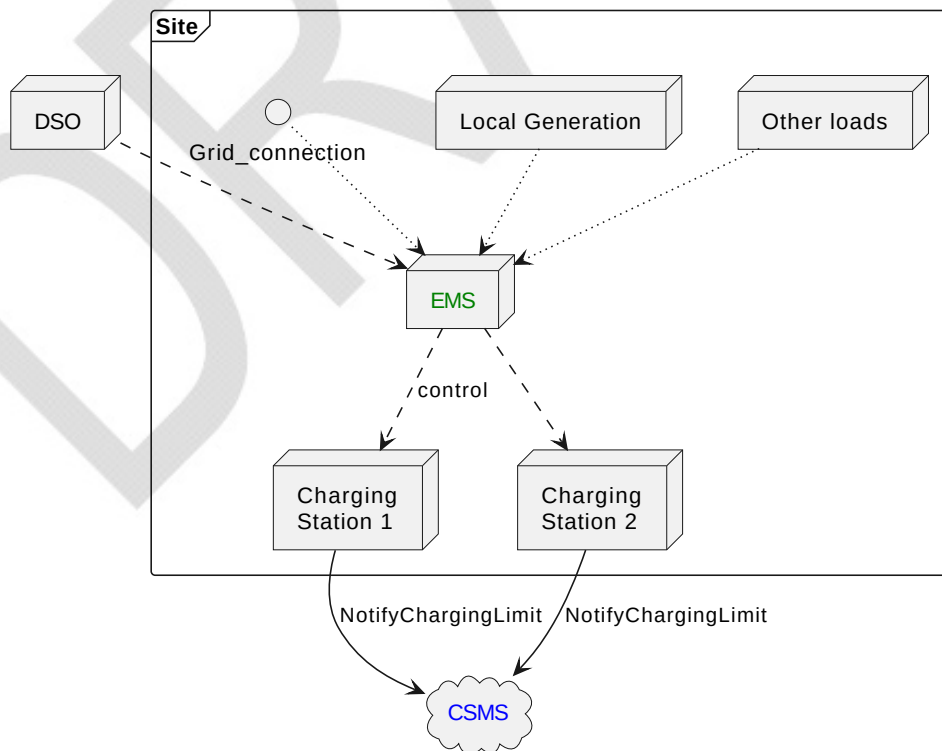
Figure 97. External Smart Charging

New text:

The OCPP protocol is developed for communication between a CSMS and one or more Charging Stations. As described in the above, this means that a CSMS of a Charging Station Operator (CSO) controls a Charging Station and, based on the charging limits of both the EV and the Charging Station, the CSO controls how fast the EV is charged. In some situations there are other factors that might control charging power: A DSO can send signals to change charging power (e.g. via IEC 61850 [IEC61850-7-420], IEC 60870 [IEC60870-5-104], DNP3 [DNP3] or OpenADR [OPENADR]), or a Home Energy Management System or a smart meter may be in place to limit charging power.

An external actor can connect to a Charging Station with any protocol that is supported by the Charging Station for this purpose, like Modbus, EEBUS, and even OCPP. This control signal can be a single limit value or a schedule. In both cases Charging Station will represent the limit internally as a charging profile of purpose `ChargingStationExternalConstraints`.

A CSMS may need to be informed of changes in charging rate as a result of external signals. OCPP provides a `NotifyChargingLimitRequest` message to report such changes.



EMS control directly to Charging Stations

2.53. Page 248 - (2025-01) - 3.7 Avoiding Phase Conflicts

The following paragraph about Avoiding Phase Conflicts has been added:

In the situation where a `ChargingStationMaxProfile` or a `ChargingStationExternalConstraints` define a value for `numberPhases` or `phaseToUse`, then a possible conflict might arise if such values are also specified in a `TxDefaultProfile` or `TxProfile`. The following rules apply in that case:

numberPhases

The lowest value for a schedule period of all applicable profiles is used for the composite schedule period. If `ChargingStationMaxProfile` has `numberPhases` = 3 and `TxProfile` has `numberPhases` = 1, then the value 1 is used. The same applies to the reverse situation.

phaseToUse

When there is a conflicting value of `phaseToUse` between the schedule periods of applicable profiles, then there is no way to create a composite schedule period. For example, a CSMS should not submit a charging profile of purpose `ChargingStationMaxProfile` for `phaseToUse` = 1 and then a `TxProfile` for `phaseToUse` = 3, because the charging station will not know which value has preference. Therefore, a `SetChargingProfileRequest` that causes such a conflict will have to be rejected.

When a relative `TxProfile` is being used and different phases occur in various schedule periods, then it may become difficult to detect if and where such a phase conflict occurs. A charging station should only accept a `SetChargingProfileRequest` when it can be certain, that there is a no risk of a phase conflict. This means, that when the charging station is not able to verify that no phase conflict occurs in any schedule period (which can happen when the `TxProfile` is received for a transaction, but charging has not yet started, so that start time of the first schedule period is not known), that it cannot accept a charging profile if any of the schedule periods contains a value for `phaseToUse` that differs from the value used in the `ChargingStationMaxProfile` or `ChargingStationExternalConstraints`.

NOTE | A value of `phaseToUse` may only be used when `numberOfPhases` = 1.

2.54. Page 275 - (2025-06) - Updated remark of K11

Added sentence to Remarks a new charging profile for an update of external limit can use the same charging profile id.

No.	Type	Description
...	...	...
8	Remarks	[...] If the external limit is represented by an Absolute or Relative <code>ChargingStationExternalConstraints</code> charging profile, then every update of the external limit requires (K11.FR.06) that the existing <code>ChargingStationExternalConstraints</code> charging profile is replaced by a new one. This one can use the same <code>chargingProfile.id</code> , however.

2.55. Page 251 - (2025-06) - Updated note of K01.FR.05

Note suggested that `ChargingStationExternalConstraints` cannot be replaced at all. Updated note to clarify that a `ChargingStationExternalConstraints` cannot be replaced by CSMS.

	ID	Precondition	Requirement definition	Note
Old	K01.FR.05	When a SetChargingProfileRequest with an already known ChargingProfile.id is received AND the existing ChargingProfile does NOT have chargingProfilePurpose = ChargingStationExternalConstraints	The Charging Station SHALL replace the existing ChargingProfile with the one specified.	ChargingStationExternal Constraints profile cannot be replaced.
New	K01.FR.05	When a SetChargingProfileRequest with an already known ChargingProfile.id is received AND the existing ChargingProfile does NOT have chargingProfilePurpose = ChargingStationExternalConstraints	The Charging Station SHALL replace the existing ChargingProfile with the one specified.	ChargingStationExternal Constraints profile cannot be replaced by CSMS .

2.56. Page 251 - (2025-06) - Add cross-references to K01.FR.06 and K01.FR.39

Requirement K01.FR.06 and K01.FR.39 are similar, but located far apart in the table. It is convenient to add a cross-reference between both.

	ID	Precondition	Requirement definition	Note
Old	K01.FR.06	When chargingProfilePurpose is NOT TxProfile	The CSMS SHALL NOT send a ChargingProfile with a stackLevel - chargingProfilePurpose - evseld combination that already exists in another ChargingProfile (with different id) on the Charging Station and has an overlapping validity period.	This is to ensure that no two charging profiles with same stack level and purpose can be valid at the same time.
New	K01.FR.06	When chargingProfilePurpose is NOT TxProfile	The CSMS SHALL NOT send a ChargingProfile with a stackLevel - chargingProfilePurpose - evseld combination that already exists in another ChargingProfile (with different id) on the Charging Station and has an overlapping validity period.	This is to ensure that no two charging profiles with same stack level and purpose can be valid at the same time. (See also K01.FR.39)
Old	K01.FR.39	When chargingProfilePurpose is TxProfile	The CSMS SHALL NOT send a ChargingProfile with a stackLevel - transactionId combination that already exists in another ChargingProfile (with different id) with purpose TxProfile.	This is to ensure that no two charging profiles with same stack level and purpose can be valid at the same time.
New	K01.FR.39	When chargingProfilePurpose is TxProfile	The CSMS SHALL NOT send a ChargingProfile with a stackLevel - transactionId combination that already exists in another ChargingProfile (with different id) with purpose TxProfile.	This is to ensure that no two charging profiles with same stack level and purpose can be valid at the same time. (See also K01.FR.06)

2.57. Page 254 - (2025-06) - K01.FR.50 requirement is a SHALL

Physics determines how to convert power to current. This cannot be a "should" requirement, but is a SHALL.

	ID	Precondition	Requirement definition	Note
Old	K01.FR.49	When a SetChargingProfileRequest without a value for <i>numberPhases</i> is received AND the EVSE is of type AC	The Charging Station SHALL assume <i>numberPhases</i> = 3 as a default value.	
New	K01.FR.49	When a SetChargingProfileRequest without a value for <i>numberPhases</i> is received AND the EVSE is of type AC	The Charging Station SHALL assume <i>numberPhases</i> = 3 as a default value.	Regions with a single phase network should always provide <i>numberPhases</i> = 1, otherwise 3 phases will be assumed.
Old	K01.FR.50	When a SetChargingProfileRequest with a chargingRateUnit = W is received AND The ChargingSchedule is used for AC charging	The Charging Station SHOULD calculate the phase current limit via: $\text{Current per phase} = \text{Power} / (\text{Line Voltage} * \text{Number of Phases})$.	The "Line Voltage" used in the calculation is not the measured voltage, but the set voltage for the area (for example, 230 or 110 V). The "Number of Phases" is the <i>numberPhases</i> from the ChargingSchedulePeriod . It is usually more convenient to use chargingRateUnit = A for AC charging.
New	K01.FR.50	When a SetChargingProfileRequest with a chargingRateUnit = W is received AND The charging profile is used for AC charging	The Charging Station SHALL calculate the phase current limit via: $\text{Current per phase} = \text{limit} / (\text{Line Voltage} * \text{numberPhases})$, in which <i>limit</i> and <i>numberPhases</i> are the values from the ChargingSchedulePeriod .	The "Line Voltage" used in the calculation is not the measured voltage, but the set voltage for the area (for example, 230 or 110 V). . The <i>limit</i> and <i>numberPhases</i> are the values from the ChargingSchedulePeriod . When <i>numberPhases</i> is not specified, a value of 3 is assumed (see K01.FR.49). It is usually more convenient to use chargingRateUnit = A for AC charging, since in that case the limit does not change depending on number of phases in use.

2.58. Page 255 - (2025-11) - K01 added implicit requirement about *recurrencyKind* [777]

The implicit requirement that it only makes sense to supply a *recurrencyKind* when the *chargingProfileKind* = *Recurring* has now been made explicit.

New requirement

ID	Precondition	Requirement definition	Note
K01.FR.60	Only when <i>chargingProfileKind</i> of a ChargingProfile is <i>Recurring</i>	A value for <i>recurrencyKind</i> SHALL be supplied in the ChargingProfile .	

2.59. Page 257 - (2025-06) - K02 Updated remark of use case about merging profiles

The description of merging profiles in the remark was not complete. It has been updated to refer to the appropriate requirement.

No.	Type	Description
...	...	...
8	Remark(s)	[...] The final schedule constraints that apply to a transaction are determined by merging the profiles with purposes <i>ChargingStationMaxProfile</i> with the profile <i>TxProfile</i> or <i>TxDefaultProfile</i> in case no profile of purpose <i>TxProfile</i> is provided. Zero or more of the following <i>ChargingProfile</i> purposes MAY have been previously received from the CSMS: <i>ChargingStationMaxProfile</i> or <i>TxDefaultProfile</i>. as described in requirement SC.01 in Chapter 4. Smart Charging Signals to a Charging Station from Multiple Actors . [...]

2.60. Page 282 - (2025-01) - K15 - ISO 15118-2 Charging with load leveling - Requirements

Note: This erratum has been superseded by erratum: [Page 282 - \(2025-09\) - K15.FR.20 is not part of OCPP 2.0.1 \[1061\]](#)

Added recommendation for timestamp when offline:

New / Updated requirements

	ID	Precondition	Requirements	Note
Changed	K15.FR.01	When the Charging Station receives charging needs from the EV	The Charging Station SHALL send a <i>NotifyEVChargingNeedsRequest</i> to the CSMS.	See also K15.FR.20
New	K15.FR.20	K15.FR.01 AND Charging Station is offline	Charging Station SHOULD add <i>timestamp</i> to the <i>NotifyEVChargingNeedsRequest</i> with the time when charging needs were received from EV	This will tell CSMS how old this data is, if it was not immediately sent because of an offline period.
New	K15.FR.21	K15.FR.10	Charging Station SHOULD set <i>selectedScheduleTupleId</i> to the Id of the <i>chargingSchedule</i> that EV selected from the provided <i>ChargingProfileType(s)</i> .	

2.61. Page 282 - (2025-09) - K15.FR.20 is not part of OCPP 2.0.1 [1061]

Note: This erratum supersedes erratum: [Page 282 - \(2025-01\) - K15 - ISO 15118-2 Charging with load leveling - Requirements](#)

The above-mentioned requirement about K15.FR.20 as a new requirement in OCPP 2.0.1 is wrong. K15.FR.20 does not apply to OCPP 2.0.1, because the *timestamp* field was only added in OCPP 2.1.

Delete requirement

	ID	Precondition	Requirements	Note
Delete	K15.FR.20	K15.FR.01 AND Charging Station is offline	Charging Station SHOULD add <i>timestamp</i> to the <i>NotifyEVChargingNeedsRequest</i> with the time when charging needs were received from EV	This will tell CSMS how old this data is, if it was not immediately sent because of an offline period.

2.62. Page 286 - (2025-01) - K16 - Renegotiation initiated by CSMS - Requirements

The following requirement was added:

New requirement

ID	Precondition	Requirements	NOTE
K16.FR.14	K16.FR.05	Charging Station SHOULD set <i>selectedScheduleTupleId</i> to the Id of the <i>chargingSchedule</i> that EV selected from the provided <i>ChargingProfileType(s)</i> .	

2.63. Page 265 - (2025-11) - K06 - Updated use case description

Updated the description to clarify that it is also possible that the TxProfile expires.

No.	Type	Description
1	Name	Offline Behavior Smart Charging During Transaction
2	ID	K06
3	Objective(s)	To enable the Charging Station to continue to use the current <i>ChargingProfile</i> for the duration of the transaction while it is <i>Offline</i> .
4	Description	If a Charging Station goes <i>Offline</i> after having received a transaction-specific <i>ChargingProfile</i> with purpose <i>TxProfile</i> , then it continues to use this profile for the duration of the transaction or until the <i>TxProfile</i> reaches its expiration, depending on which occurs first.

2.64. Page 266 - (2025-11) - K06.FR.01 - Updated Requirement definition

Updated the requirement definition to clarify that it is also possible that the TxProfile expires.

ID	Precondition	Requirement definition
K06.FR.01	If the Charging Station goes <i>Offline</i> after having received a transaction-specific <i>ChargingProfile</i> with purpose <i>TxProfile</i> .	The Charging Station SHALL continue to use this profile for the duration of the transaction or until the <i>TxProfile</i> reaches its expiration, depending on which occurs first.

2.65. Page 278 - (2025-11) - K13 - Updated description prerequisites

Updated prerequisites for this specification explicitly allow for an OCPP connection to an EMS.

No.	Type	Description
1	Name	Reset / Release External Charging Limit
2	ID	K13
3	Objective(s)	To release a charging limit that was previously imposed.
4	...	...
5	Prerequisites	Previously, a charging limit was sent to the Charging Station under consideration. An external system that can set/clear a charging limit/schedule on the Charging Station. via another connection than OCPP.

2.66. Page 284 - (2025-09) - K15 Added rule for composite schedules in case of multiple charging schedules [1002]

In the theoretical situation that 2 TxProfiles are submitted with different stack levels and multiple charging schedules (which can only be the case for an ISO 15118 session) and, because of different durations of these schedules, parts of each of these

schedules will be valid at one point or another, then how is the composite schedule calculated? It is not 3 * 3 composite schedules (all possible combinations), but only 3 composite schedules, because schedule #1 is always combined with schedule #1, #2 with #2 and #3 with #3. Other *chargingProfilePurposes*, like *ChargingStationMaxProfile* need also to be taken into account when calculating the composite schedule.

A new requirement is added to define this behavior.

2.66.1. K15 - ISO 15118-2

New requirement

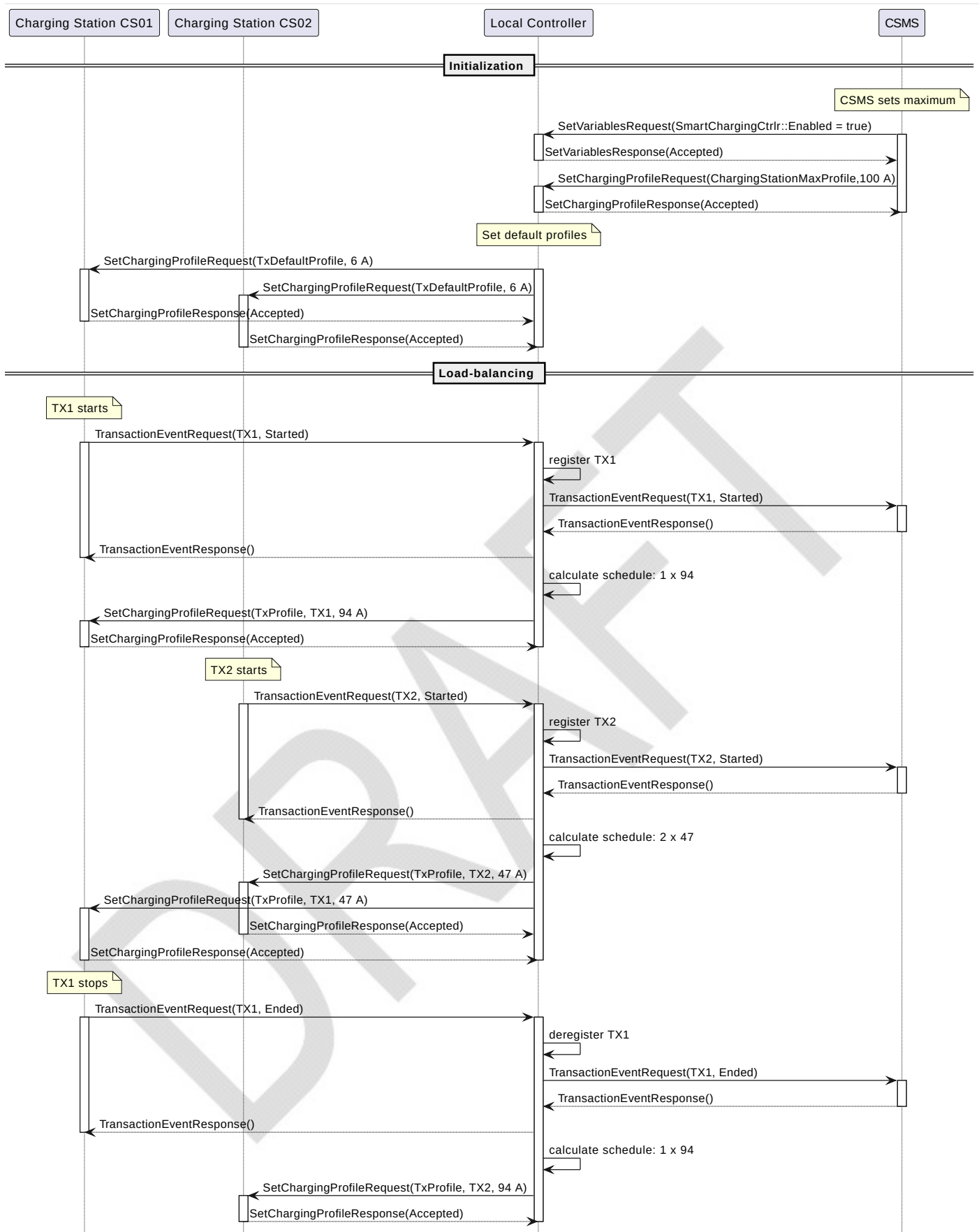
ID	Precondition	Requirements	Note
K15.FR.22	When calculating <i>CompositeSchedule(s)</i> to create a <i>SAScheduleList</i> for ISO 15118-2 to send to EV AND multiple <i>ChargingProfileTypes</i> of <i>chargingProfilePurpose</i> = <i>TxProfile</i> with different <i>stackLevels</i> are valid AND some or all these <i>ChargingProfileTypes</i> have more than one <i>chargingSchedule</i>	Charging Station SHALL create up to three <i>CompositeSchedules</i> as defined in K08.FR.04, by combining the first <i>chargingSchedule</i> with the first <i>chargingSchedule</i> of other stack levels, the second with second (if existing), the third with the third (if existing), based on their order in the <i>ChargingProfileTypes</i> .	This is about a corner case when multiple <i>TxProfiles</i> with different stack levels and multiple charging schedules have been sent to the Charging Station.

2.67. Page 259/260 - (2025-01) - K03 - Updated use case description and sequence diagram

The use case description and sequence diagram have been updated to provide more information on how local load-balancing can be performed. (Requirements for K03 have not changed).

No.	Type	Description
1	Name	Local Smart Charging
2	ID	K03
3	Objective(s)	To illustrate the process of local load-balancing by a Local Cluster.
4	Description	This use case is an example of how local load-balancing can be performed. It does not imply that other approaches would not be correct. The process has been simplified for clarity and should not be regarded as prescriptive. A Local Controller is configured with a value for maximum current for the total cluster by CSMS via a charging profile of type <i>ChargingStationMaxProfile</i> to the Local Controller, or an EMS may have set a <i>ChargingStationExternalConstraints</i> charging profile. The Local Controller divides the maximum current among the active transactions. Whenever a transaction starts or finishes, the Local Controller will update the charging profiles of the remaining transactions to divide the maximum current equally. For simplicity's sake, this use case does not differentiate on departure time or state of charge of vehicles, nor does it take the actual energy consumption of vehicles into account.
	Actors	Charging Station (CS01, CS02), Local Controller (LC), CSMS

No.	Type	Description
	Scenario description	Assume no transactions are active in the local cluster and the maximum current for the local cluster has been configured to be 100 A. The charging stations all have a <code>TxDefaultProfile</code> that allows a current of only 6 A, so that vehicles cannot immediately start charging at full power before the LC had the chance to set a charging profile. 1. A transaction starts on charging station CS01. It sends a <code>TransactionEventRequest(Started)</code> to LC. 2. LC is configured to do local load-balancing (i.e. its <code>SmartChargingCtrlr.Enabled = true</code>), so it registers the transaction id TX1 of the transaction that has been started on CS01, before forwarding the message on the websocket for CS01 towards CSMS. 3. LC sends a <code>SetChargingProfileRequest</code> to CS01 with <code>chargingProfilePurpose = TxProfile</code>, <code>chargingProfileKind = Relative</code>, <code>transactionId = TX1</code> and a <code>chargingSchedule</code> with a <code>chargingRateUnit = A</code> that contains one <code>chargingSchedulePeriod</code> with a <code>limit</code> of 94 A, so that the entire quota is available to this transaction minus the <code>TxDefaultProfile</code> amount for new transactions. 4. Another transaction starts on charging station CS02. It sends a <code>TransactionEventRequest(Started)</code> to LC. 5. LC registers the new transaction id TX2 and forwards the message on the websocket for CS02 to CSMS. 6. LC divides the available quota by allowing each transaction a maximum of 47 A. 7. LC sends a <code>SetChargingProfile</code> message to CS01 that updates the existing <code>TxProfile</code> and sets the <code>limit</code> to 47 A. 8. LC sends new <code>SetChargingProfile</code> to CS02 with <code>chargingProfilePurpose = TxProfile</code>, <code>chargingProfileKind = Relative</code>, <code>transactionId = TX2</code> and a <code>chargingSchedule</code> with a <code>chargingRateUnit = A</code> that contains one <code>chargingSchedulePeriod</code> with a <code>limit</code> of 47 A. 9. The transaction of CS01 finishes. It sends a <code>TransactionEventRequest(Ended)</code> to LC. 10. LC registers that transaction TX1 on CS01 has finished and forwards the message on the websocket for CS01 to CSMS. 11. LC now allows the maximum to TX2. It sends a <code>SetChargingProfile</code> message to CS02 that updates the existing <code>TxProfile</code> and sets the <code>limit</code> to 94 A. (Note, that the <code>TxProfile</code> for TX1 on CS01 has automatically ceased to exist upon termination of the transaction.)
5	Prerequisites	The LC has been configured with a fixed maximum current level. The <code>SmartChargingCtrlr</code> component of Local Controller has been Enabled, which will trigger the Local Controller to read and interpret <code>TransactionEventRequest</code> messages from connected Charging Stations.
6	Post conditions	
7	Error Handling	
8	Remarks	As described in Part 1, a Local Controller replicates all websockets from Charging Stations in the cluster towards CSMS, and forwards messages from Charging Station to CSMS on the appropriate websocket (and vice versa). This allows the Local Controller to read messages, such as a <code>TransactionEventRequest</code> message, from the Charging Station. The Local Controller for local smart charging can be implemented in different ways, for example: as a separate physical component or as part of a "master" Charging Station controlling a number of other Charging Stations. The Local Controller MAY or MAY NOT have any EVSEs of its own. The limits on Charging Stations in a Local Smart Charging group can either be pre-configured in the Local Controller in one way or another, or they can be set by the CSMS. The Local Controller contains the logic to distribute this capacity among the connected EVSEs by adjusting their limits as needed.



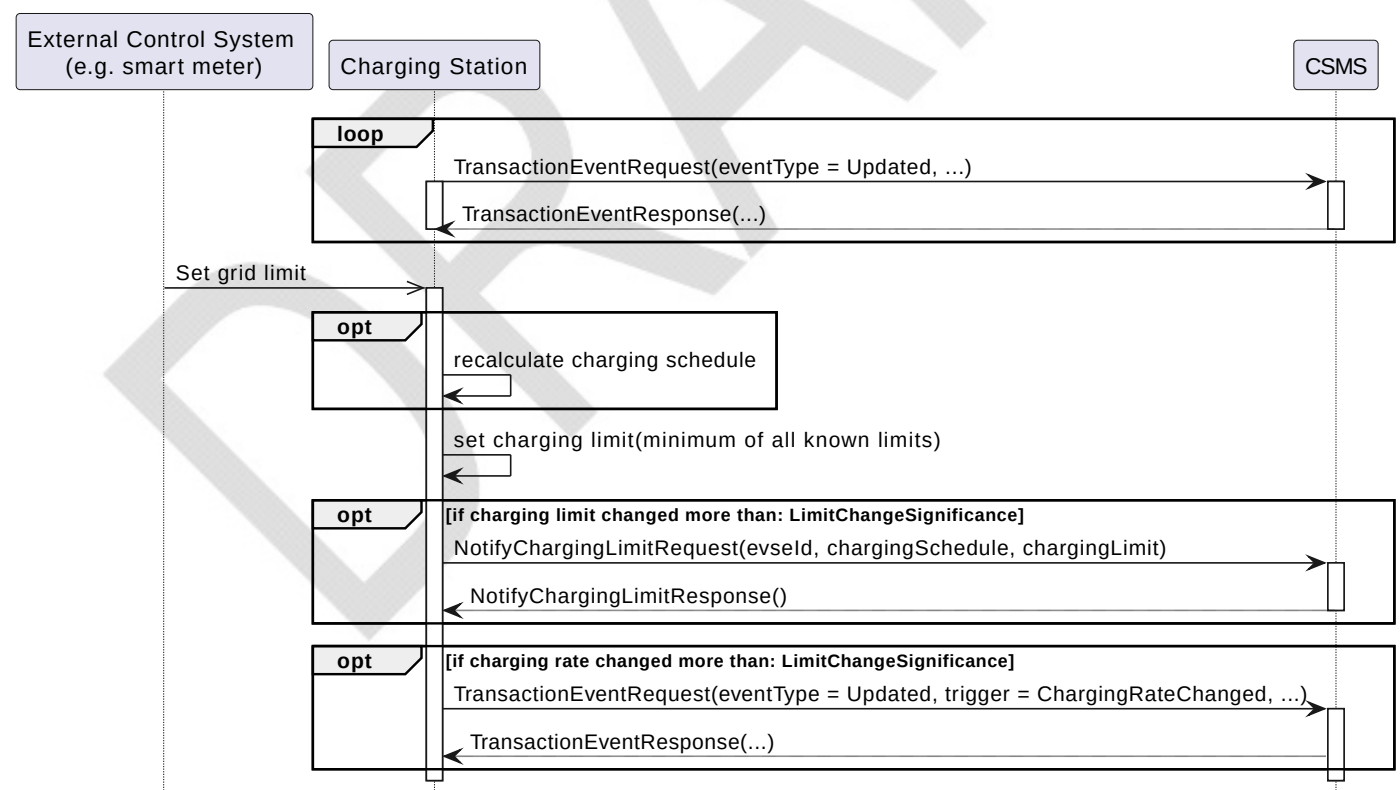
Local Controller performing local load-balancing

2.68. Page 274/275/276/277 - (2025-01) - K11/K12 - Updated use case descriptions and sequence diagrams

The use case descriptions and sequence diagrams have been updated to describe the more likely scenario of a smart meter or EMS as external actor.

K11 - Set / Update External Charging Limit With Ongoing Transaction

No.	Type	Description
1	Name	Set / Update External Charging Limit With Ongoing Transaction
2	ID	K11
3	Objective(s)	To inform the CSMS of a charging schedule or charging limit imposed by an External Control System on the Charging Station with ongoing transaction(s).
4	Description	An External Control System sends a charging limit/schedule to a Charging Station. This limit is sent to the CSMS. The External Control System can be a DSO, but also a smart meter or a home energy management system. The interface between External Control System and Charging Station is not specified. It can be any protocol that is supported by Charging Station for this purpose, even OCPP.
	Actors	External Control System, Charging Station, CSMS
	Scenario description	1. External control system sends charging limit/schedule to Charging Station. 2. Optional: Charging Station calculates new charging schedule. 3. Charging Station adjusts the charging speed of the ongoing transaction(s). 4. If the charging limit changed by more than: LimitChangeSignificance, the Charging Station sends a NotifyChargingLimitRequest message to CSMS with optionally the set charging limit/schedule. 5. The CSMS responds with NotifyChargingLimitResponse to the Charging Station. 6. If the charging rate changes by more than: LimitChangeSignificance, the Charging Station sends a TransactionEventRequest message to inform the CSMS. 7. The CSMS responds with TransactionEventResponse to the Charging Station.
5	Prerequisites	Charging Station is not in error state. The external system can set/clear a charging limit/schedule on the Charging Station via a direct connection to the Charging Station.
...		

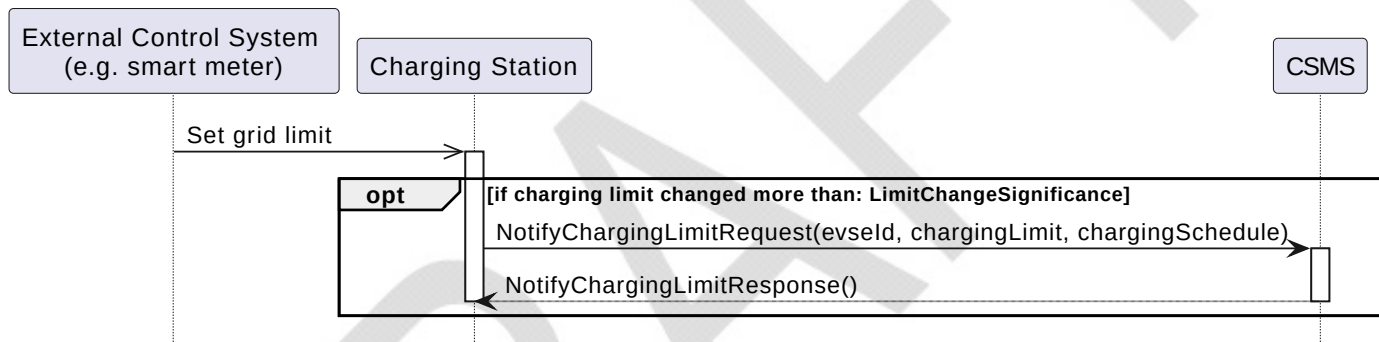


Sequence diagram of the use case "Setting / Updating External Charging Limit with Ongoing Transaction"

K12 - Set / Update External Charging Limit Without Ongoing Transaction

No.	Type	Description
1	Name	Set / Update External Charging Limit Without Ongoing Transaction

No.	Type	Description
2	ID	K12
3	Objective(s)	To inform the CSMS of a charging schedule or charging limit imposed by an external system on the Charging Station for new transactions or on the grid connection.
4	Description	To inform the CSMS of a charging schedule or charging limit imposed by an external system on the Charging Station for new transactions or on the grid connection. The External Control System can be a DSO, but also a smart meter or a home energy management system. The interface between External Control System and Charging Station is not specified. It can be any protocol that is supported by Charging Station for this purpose, even OCPP.
	Actors	External Control System, Charging Station, CSMS
	Scenario description	1. External Control System sends a charging limit to Charging Station (not during a transaction). 2. Optional: Charging Station calculates new charging schedule. 3. Charging Station adjusts the charging speed. 4. If the charging limit changed by more than: LimitChangeSignificance, the Charging Station sends a NotifyChargingLimitRequest message to CSMS with optionally the set charging limit/schedule. 5. The CSMS responds with a NotifyChargingLimitResponse to the Charging Station.
5	Prerequisites	Charging Station is not in error state. The external system can set/clear a charging limit/schedule on the Charging Station via a direct connection to the Charging Station.
...		



Sequence diagram of the use case "Set / Update External Charging Limit Without Ongoing Transaction"

2.69. Page 278/279 - (2025-01) - K13 - Updated requirement preconditions

Moved precondition "transaction is ongoing" from K13.FR.01 to K13.FR.03.

Changed K13 requirements:

	ID	Precondition	Requirements
Old text	K13.FR.01	A transaction is ongoing AND External charging limit is released/removed	The Charging Station SHALL NOT limit charging anymore based on the previously received limit.
New text	K13.FR.01	External charging limit is released/removed	The Charging Station SHALL NOT limit charging anymore based on the previously received limit.
Old text	K13.FR.03	K13.FR.01 AND Charging rate changed by more than: LimitChangeSignificance	The Charging Station SHALL send a TransactionEventRequest message to the CSMS with trigger = ChargingRateChanged .

	ID	Precondition	Requirements
New text	K13.FR.03	K13.FR.01 AND A transaction is ongoing AND Charging rate changed by more than: LimitChangeSignificance	The Charging Station SHALL send a TransactionEventRequest message to the CSMS with trigger = ChargingRateChanged.

2.70. Page 279 - (2025-01) - K14 - Updated use case scenario description

Improved scenario step description 5 and 7.

No.	Type	Description
1	Name	Handle external charging limit with a local controller
2	ID	K14
3	Objective(s)	To adjust the charging limits according to the External Control System requirements.
4	Description	An external control system sends a charging limit to the Local Controller. The Local Controller notifies the CSMS, calculates the new charging schedules and sends a SetChargingProfileRequest messages to all Charging Stations for which the charging profile has changed.
	Actors	External control system, Local Controller, Charging Station, CSMS
	Scenario description	1. External control system sends a charging limit/schedule to Local Controller. 2. Local Controller sends a NotifyChargingLimitRequest message to the CSMS. 3. Local Controller calculates new Charging Profiles for all connected Charging Stations. 4. Local Controller sends a SetChargingProfileRequest message to all Charging Stations for which the charging profile has changed. 5. External control releases a charging limit/schedule to Local Controller. 6. Local Controller sends a ClearedChargingLimitRequest message to the CSMS. 7. Local Controller clears Charging Profiles for all connected Charging Stations. 8. Local Controller sends a ClearChargingProfileRequest messages to all affected Charging Stations.
		...

2.71. Page 284 - (2025-09) - K16 use case description update

The use case description refers to SetChargingProfile in step 7, but that is too restricting. It is the composite schedule that is provided to EV.

No.	Type	Description
...	...	...
	Scenario description	1 CSMS sends a SetChargingProfileRequest to the Charging Station. 2 Charging Station responds with a SetChargingProfileResponse to the CSMS. 3 When EV sends the next CurrentDemandReq (for DC) or ChargingStatusReq (for AC), the Charging Station will respond with evseNotification = ReNegotiation. 4 EV sends a PowerDeliveryReq with chargeProgress = ReNegotiate to confirm this. 5 Charging Station responds with a PowerDeliveryRes. 6 EV sends a ChargeParameterDiscoveryReq. 7 Charging Station responds with a ChargeParameterDiscoveryRes with an SAScheduleList that contains the composite schedule(s) for the EVSE ChargingSchedule data from the SetChargingProfileRequest. 8 EV sends a PowerDeliveryReq with chargeProgress = Start (with an optional charging profile) to confirm this. 9 Charging Station responds with PowerDeliveryRes and, if charging was suspended at start of the renegotiation, will resume power delivery. 10 If EV provided a charging profile in the previous step, then Charging Station will send a NotifyEVChargingScheduleRequest to the CSMS.

No.	Type	Description
...	...	...

2.71.1. Page 285

	ID	Precondition	Requirements	NOTE
Old	K16.FR.02	K16.FR.01	Charging Station SHALL initiate schedule renegotiation with EV.	In ISO 15118 this is done by replying with EVSENotification=ReNegotiation to a CurrentDemandReq (for DC) or ChargingStatusReq (for AC) message.
New	K16.FR.02	When the composite schedule for the EVSE changes	Charging Station SHALL initiate schedule renegotiation with EV.	This can be caused by a Set/ClearChargingProfileRequest or a change in ChargingStationExternalConstraints/Local Generation charging profiles. In ISO 15118 this is done by replying with EVSENotification=ReNegotiation to a CurrentDemandReq (for DC) or ChargingStatusReq (for AC) message.
Old	K16.FR.03	K16.FR.02	Charging Station SHALL provide the ChargingSchedule data to the EV.	In ISO 15118 this is done in the ChargeParameterDiscoverRes message.
New	K16.FR.03	K16.FR.02	Charging Station SHALL provide the composite schedule(s) ChargingSchedule data to the EV.	In ISO 15118 this is done in the ChargeParameterDiscoverRes message.

2.72. Page 292 - (2025-01) - Use case L01 - Added clarification to step 3 about when to start downloading the firmware

No.	Type	Description
1	Name	Secure Firmware Update
2	ID	L01
3	Objective(s)	Download and install a Secure firmware update.
4	Description	Illustrate how a Charging Station processes a Secure firmware update.
	Actors	CSMS, Charging Station
	Scenario description	1. The CSMS sends an UpdateFirmwareRequest message that contains the location of the firmware, the time after which it should be retrieved, and information on how many times the Charging Station should retry downloading the firmware. 2. The Charging Station verifies the validity of the certificate against the Manufacturer root certificate. 3. If the certificate is valid AND the retrieveDateTime has passed, the Charging Station starts downloading the firmware, and sends a FirmwareStatusNotificationRequest with status Downloading. If the certificate is not valid or could not be verified, the Charging Station aborts the firmware update process and sends a UpdateFirmwareResponse with status InvalidCertificate and a SecurityEventNotificationRequest with the security event InvalidFirmwareSigningCertificate (See part 2 appendices for the full list of security events). ...

2.73. Page 306 - (2025-01) - M. ISO 15118 Certificate Management

The functional block ISO 15118 Certificate Management is renamed to Certificate Management since this also contains certificate management of non-ISO 15118 related certificates.

2.74. Page 308 - (2025-01) - Update introduction sequence diagram ISO 15118

Not all ISO 15118 message sequences were complete, so the sequence diagram has been updated to show a more complete version of the message flow.

DRAFT

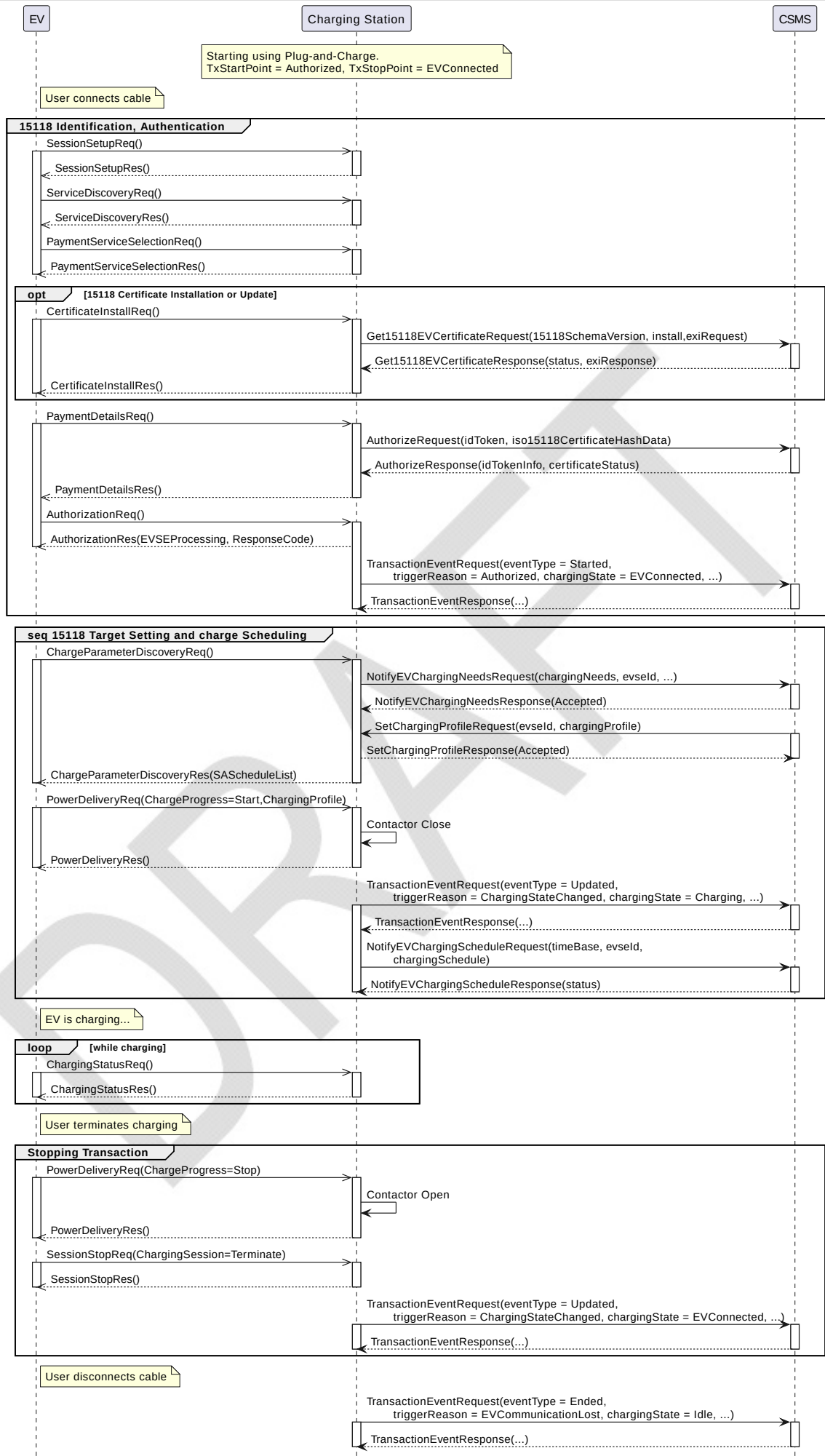


Figure 122. Sequence with Authorization and Scheduling with ISO 15118-2

NOTE

The time-out on the ChargeParameterDiscoveryReq is 2 seconds, but this can be prolonged up to 60 seconds to wait for charging profile (SAScheduleList) to be provided by the CSMS. See ISO 15118-2 [ISO15118-2].

2.75. Page 330 - (2025-01) - N03 Set Monitoring Base: Improved text of Remark

Old	8	Remark(s)	Upon receipt of a SetMonitoringBaseRequest for HardWiredOnly or FactoryDefault the Charging Station will discard of any previously configured custom monitors and will activate the monitoring settings that are related to given MonitoringBase. For a MonitoringBase = All the Charging Station will activate all pre-configured monitors and leave previously configured custom monitors intact. This includes the custom monitors that were created when changing an existing pre-configured monitor. When the set of pre-configured monitors for All and FactoryDefault is the same, then the difference between the two is, that with FactoryDefault all custom monitors are deleted before the factory default pre-configured monitors are restored.
New	8	Remark(s)	Upon receipt of a SetMonitoringBaseRequest for: • <i>monitoringBase</i> = HardWiredOnly: the Charging Station will deactivate all pre-configured monitors and remove any previously configured custom monitors. Only the HardWiredMonitor monitors remain. • <i>monitoringBase</i> = FactoryDefault: the Charging Station will (re)activate all PreconfiguredMonitor monitors and remove all custom monitors. • <i>monitoringBase</i> = All: the Charging Station will activate all pre-configured monitors and leave previously configured CustomMonitor monitors intact. This includes the custom monitors that were created when changing an existing pre-configured monitor.

2.76. Page 331 - (2025-01) - N03.FR.04: text improvement

Improved requirement text for N03.FR.04.

Old	N03.FR.04	N03.FR.01 AND When the Charging Station received a setMonitoringBaseRequest with <i>monitoringBase</i> FactoryDefault	Then the Charging Station SHALL delete all custom monitors (including overruled pre-configured monitors) and activate the default monitoring settings as recommended by the manufacturer.
New	N03.FR.04	N03.FR.01 AND When the Charging Station received a setMonitoringBaseRequest with <i>monitoringBase</i> FactoryDefault	Then the Charging Station SHALL delete all custom monitors (including overruled pre-configured monitors) and activate the pre-configured monitors of the Charging Station.

Page 434 - MonitoringBaseEnumType description update

The description has been updated to be better aligned with the use case.

Value	Description
All	Activate all pre-configured monitors while leaving custom monitors intact, including those that overrule a pre-configured monitor.
FactoryDefault	(Re)activate the default monitoring settings has recommended by the manufacturer. This is a subset of all pre-configured monitors. of Charging Station and remove all custom monitors.
HardWiredOnly	Removes all custom monitors and disables all pre-configured monitors.

2.77. Page 350 - (2025-02) - 001 - Added missing requirements

Added missing requirements explicitly specifying behaviour of Charging Station it contains one or more displays.

New requirements

ID	Precondition	Requirement definition
<i>Multiple Display support</i>		
O01.FR.20	When Charging Station has multiple displays AND Charging Station receives a [setDisplaymessagerequest] without a <i>display</i> element in its MessageInfoType	Charging Station SHOULD use the message for the main display(s)
O01.FR.21	When receiving a GetBaseReportRequest AND Charging Station has one or more displays	Charging Station SHOULD include in the report a Display component for every display it contains.
O01.FR.22	When Charging Station receives a [setDisplaymessagerequest] with Display element referencing an unknown Display in its MessageInfoType	Charging Station SHOULD respond with a [setDisplaymessageresponse] with <i>status</i> = Rejected.
O01.FR.23	When Charging Station receives a [setDisplaymessagerequest] with Display element referencing a known Display in its MessageInfoType	Charging Station SHOULD use the message only for the specified display.

2.78. Page 446 - (2025-01) - ActiveNetworkProfile is incorrectly marked as optional

The referenced variable OCPPCommCtrlr.ActiveNetworkProfile is incorrectly marked as optional. As the description states this variable needs to be implemented when the Charging Station supports the NetworkConnectionProfile use cases B09/B10. These use cases are an integral part of the Core of OCPP. In addition, all other NetworkConnectionProfile related referenced variables are already required; OCPPCommCtrlr.NetworkConfigurationPriority and OCPPCommCtrlr.NetworkProfileConnectionAttempts. Therefore, OCPPCommCtrlr.ActiveNetworkProfile should also be marked as required.

ActiveNetworkProfile

Required	yes		
Component	componentName	OCPPCommCtrlr	
Variable	variableName	ActiveNetworkProfile	
	variableAttributes	mutability	ReadOnly
	variableCharacteristics	dataType	integer
Description	This variable indicates the NetworkConnectionProfile configuration slot the Charging Station currently uses for its connection with the CSMS.		

2.79. Page 327 - (2025-09) - N01.FR.12 - Improved definition

Updated requirement definition to clarify the AcceptedCancelled status.

	ID	Precondition	Requirement definition
Old	N01.FR.12	When a Charging Station is assembling or uploading the log file AND the Charging Station receives a new GetLogRequest	The Charging Station SHOULD cancel the ongoing log file upload AND respond with status AcceptedCancelled.
New	N01.FR.12	When a Charging Station is assembling or uploading the log file AND the Charging Station receives a new GetLogRequest	The Charging Station SHOULD cancel the ongoing log file upload AND respond GetLogResponse with status AcceptedCancelled.

2.80. Page 328 - (2025-09) - N02: changed empty to absent.

A number of requirements previously stated "empty" when they should have indicated "absent." For example, the phrases referring to monitoringCriteria and componentVariables being "empty" are incorrect. These arrays cannot be empty; they must be absent

instead. This correction has been applied to all occurrences throughout section N02.

2.81. Page 453 - (2025-01) - References to *monitorValue* changed to *value* [354]

The specification in N Diagnostics consistently refers to a variable *monitorValue*, but this variable is called *value* in JSON schemas.

All occurrences of *monitorValue* in N Diagnostics have been changed to *value*.

2.82. Page 454 - (2025-01) - N04.FR.06 Improved limit definition of thresholds [353]

The requirement has been defined more exact, because limits for upper and lower differ slightly.

	ID	Precondition	Requirement definition	Note
Old	N04.FR.06	When the Charging Station receives a SetVariableMonitoringRequest with monitor type UpperThreshold or LowerThreshold AND the <i>monitorValue</i> is lower or higher than the range of the given Variable	The Charging Station SHALL set the attributeStatus field in the corresponding SetMonitoringResult to: Rejected .	More information can be provided in the optional <i>statusInfo</i> element.
New	N04.FR.06	When the Charging Station receives a SetVariableMonitoringRequest with (monitor type = UpperThreshold AND <i>monitorValue</i> < <i>minLimit</i> OR <i>monitorValue</i> > <i>maxLimit</i>) OR (monitor type = LowerThreshold AND <i>monitorValue</i> < <i>minLimit</i> OR <i>monitorValue</i> > <i>maxLimit</i>)	The Charging Station SHALL set the attributeStatus field in the corresponding SetMonitoringResult to: Rejected .	<i>minLimit</i> and <i>maxLimit</i> refer to the [cmn_variablecharacteristicstyp] for the [cmn_variabletype] . Be aware that setting a UpperThreshold to the <i>maxLimit</i> or setting a LowerThreshold to the <i>minLimit</i> will result in a monitor that will never trigger. More information on the reason of rejection can be provided in the optional <i>statusInfo</i> element.

2.83. Page 456 - (2025-02) - New configuration variable to allow TLS wildcard certificates

New configuration key

AllowCSMSTLSWildcards

Required	no		
Component	componentName	SecurityCtrlr	
Variable	variableName	AllowCSMSTLSWildcards	
	variableAttributes	mutability	ReadWrite
	variableCharacteristics	dataType	boolean

Description	This variable allows a Charging Station to support non-compliant OCPP behavior and connect to a CSMS that uses a wildcard TLS server certificate for the OCPP connection. If this variable is present it SHALL be ReadWrite. If this variable is not implemented or has value false, the OCPP-compliant behavior is that a Charging Station rejects a connection from a CSMS that presents a wildcard certificate. It is highly RECOMMENDED to not allow wildcard certificates.
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2.84. Page 333 - (2025-11) - N04 Added requirement

This requirement allows the Charging Station to reject the value zero when the Charging Station receives a [SetVariableMonitoringRequest](#) with type [Delta](#).

New requirement

ID	Precondition	Requirements	Note
N04.FR.20	When the Charging Station receives a SetVariableMonitoringRequest with type Delta and the value is zero.	The Charging Station SHOULD set the attributeStatus field in the corresponding SetMonitoringResult to: Rejected.	N04.FR.14 describes rejecting the negative value. This requirement allows the Charging Station to also reject the value zero.

2.85. Page 340 - (2025-11) - Added remark to N09

The fact that *customerCertificate* can be supplied to CustomerInformationRequest does not imply that there is a need to stop plug-and-charge contract certificates on a charging station. This is clarified in the remark of the use case.

No.	Type	Description
1	Name	Get Customer Information
...	...	...
8	Remark(s)	The fact that <i>customerCertificate</i> can be supplied to [customerinformationrequest] does not imply that there is a requirement to store plug-and-charge contract certificates on a charging station.

2.86. Page 341 - (2025-11) - N09.FR.06 Improvement requirement definition

Minor improvement of requirement definition to clarify that [NotifyCustomerInformationRequest](#) data field is set to an empty string.

	ID	Precondition	Requirement definition	Note
Old	N09.FR.06	N09.FR.02 AND the Charging Station has no information stored about the customer referred to by the customer identifier.	The Charging Station SHALL send one NotifyCustomerInformationRequest message to the CSMS indicating that no data was found.	
New	N09.FR.06	N09.FR.02 AND the Charging Station has no information stored about the customer referred to by the customer identifier.	The Charging Station SHALL send one NotifyCustomerInformationRequest message to the CSMS with the data field set to an empty string.	

2.87. Page 341 - (2025-11) - N09.FR.06 Improvement requirement definition

Minor improvement of requirement definition to clarify that [NotifyCustomerInformationRequest](#) data field is set to an empty string.

	ID	Precondition	Requirement definition	Note
Old	N09.FR.06	N09.FR.02 AND the Charging Station has no information stored about the customer referred to by the customer identifier.	The Charging Station SHALL send one NotifyCustomerInformationRequest message to the CSMS indicating that no data was found.	
New	N09.FR.06	N09.FR.02 AND the Charging Station has no information stored about the customer referred to by the customer identifier.	The Charging Station SHALL send one NotifyCustomerInformationRequest message to the CSMS with the data field set to an empty string.	

2.88. Page 458 - (2025-01) - Added optional variable to allow the Charging Station to report its supported idTokenTypes

Currently there is no method for the Charging Station to report which idTokenTypes it supports. This would be very useful for a CSMS to know and will improve the automated onboarding of Charging Stations.

SupportedIdTokenTypes

Required	no		
Component	componentName	AuthCtrlr	
Variable	variableName	SupportedIdTokenTypes	
	variableAttributes	mutability	ReadOnly
	variableCharacteristics	dataType	MemberList
		valuesList	List of IdTokenEnumType.
Description	The subset of the list of supported IdTokenTypes as defined by IdTokenEnumtype, that is supported by the Charging Station.		

2.89. Page 462 - (2025-01) - Added note to EnergyTransfer description as TxStartPoint

A note has been added to the description of the EnergyTransfer TxStartPoint to warn for potential skews of the values of the energy meter readings associated with start of the transaction.

Value	Description
EnergyTransfer	Energy is being transferred between EV and EVSE. Note: Since energy needs to start flowing first to cause the transaction to be started, there is a small time gap (order of milliseconds) between the start of energy transfer and start of transaction. Depending on the implementation this may potentially skew the value of the energy meter reading associated with start of the transaction. Use PowerPathClosed as TxStartPoint to avoid this situation.

2.90. Page 467 - (2025-09) - Error in description of AlignedData interval variables [1043]

The Interval and TxEndedInterval variables of AlignedDataCtrlr mention an incorrect time and duration format (ISO8601) that is not supported by OCPP.

2.90.1. AlignedDataInterval

...	...
Description	Size (in seconds) of the clock-aligned data interval, intended to be transmitted in the MeterValuesRequest or TransactionEventRequest message. This is the size (in seconds) of the set of evenly spaced aggregation intervals per day, starting at 00:00:00 (midnight). For example, a value of 900 (15 minutes) indicates that every day should be broken into 96 15-minute intervals. When clock-aligned data is being transmitted, the interval in question is identified by the start time and (optional) duration interval value, represented according to the ISO8601 standard. A value of "0" (numeric zero), by convention, is to be interpreted to mean that no clock-aligned data should be transmitted.

2.90.2. AlignedDataTxEndedInterval

...	...
Description	Size (in seconds) of the clock-aligned data interval, intended to be transmitted in the TransactionEventRequest (eventType = Ended) message. This is the size (in seconds) of the set of evenly spaced aggregation intervals per day, starting at 00:00:00 (midnight). For example, a value of 900 (15 minutes) indicates that every day should be broken into 96 15-minute intervals. When clock-aligned data is being collected, the interval in question is identified by the start time and (optional) duration interval value, represented according to the ISO8601 standard. All intervals are transmitted (if so enabled) at the end of the transaction in 1 TransactionEventRequest (eventType = Ended) message. This is not a recommended practice, since the size of the message can become very large.

2.91. Page 476 - (2025-01) - Added Connector component to AvailabilityState referenced variable

It was already possible to report the AvailabilityState of the Connector component, however the definition was missing at this table.

AvailabilityState

Required	yes		
Components	componentName	ChargingStation	
		EVSE	
		Connector	
	evse	* (for EVSE and Connector)	
Variable	variableName	AvailabilityState	
	variableAttributes	mutability	ReadOnly
		data type	optionList
	variableCharacteristics	valuesList	Available, Occupied, Reserved, Unavailable, Faulted
Description	This variable reports current availability state for the ChargingStation, EVSE and Connector. When this variable reports the Connector AvailabilityState, it replicates the connectorStatus values as would be reported by the StatusNotification messages. An EVSE or Connector component is addressed on its own tier. So, EVSE #1 is addressed as component EVSE on tier <code>evse.id = 1</code>, and EVSE #1, Connector #1 is addressed as component Connector on tier <code>evse.id = 1</code>, <code>evse.connectorId = 1</code>.		

2.92. Page 452 - (2025-11) - TimeOffsetNextTransition renamed [801]

Configuration variable `TimeOffsetNextTransition` is defined as instance "NextTransition" of variable `TimeOffset` of `ClockCtrlr`: `ClockCtrlr.TimeOffset[NextTransition]`. This is not possible, however, because a variable `ClockCtrlr.TimeOffset` without an instance already exists. A variable either has instances or not, but not both. This (optional and rarely used) variable has therefore been renamed into `ClockCtrlr.TimeOffsetNextTransition`.

2.92.1. TimeOffsetNextTransition

Required	no		
Component	componentName	ClockCtrlr	
Variable	variableName	TimeOffsetNextTransition	
	variableInstance	NextTransition	
	variableAttributes	mutability	ReadWrite
	variableCharacteristics	dataType	string
Description	Next local time offset in the format: "+01:00", "-02:00" etc. New offset that will be set on the next time offset transition as configured via 'NextTimeOffsetTransitionDateTime' . This can be used to manually configure the offset for the start or end of the daylight saving time period.		

2.93. Page 397 - (2025-11) - issuerKeyHash in CertificateHashDataType [826]

The field type of `issuerKeyHash` was incorrectly presented as `string[0..128]`, but it must be a (case-insensitive) `identifierString[0..128]`, like the other hashes.
This change does not affect the JSON schemas, because both are represented as strings in JSON.

2.94. CertificateHashDataType

Class

	Field Name	Field Type	Card.	Description
Old	issuerKeyHash	string[0..128]	1..1	Required. The hash of the DER encoded public key: the value (excluding tag and length) of the subject public key field in the issuer's certificate.
New	issuerKeyHash	identifierString[0..128]	1..1	Required. The hash of the DER encoded public key: the value (excluding tag and length) of the subject public key field in the issuer's certificate.

2.95. Page 459 (2025-11) - AuthCache.Entries variable [942]

A variable to represent the number of idTokens in authorization cache has been added.

2.95.1. AuthCacheEntries

Required	no	
Component	componentName	AuthCacheCtrlr

Variable	variableName	Entries	
	variableAttributes	mutability	ReadOnly
	variableCharacteristics	dataType	integer
Description	Amount of IdTokens currently in the Authorization Cache.		

DRAFT

3. Part 3

Currently no new errata for OCPP 2.0.1 part 3.

DRAFT

4. Part 4

4.1. Page 4 - (2025-11) - Updates to RFC References

RFC 2617 has been obsoleted, and the updated RFC 7617 includes necessary improvements and clarifications regarding the HTTP Authentication framework, ensuring compliance with current standards.

Obsolete reference

Reference	Description
[RFC2617]	"HTTP Authentication: Basic and Digest Access Authentication". http://tools.ietf.org/html/rfc2617

New reference

Reference	Description
[RFC7617]	"The 'Basic' HTTP Authentication Scheme". https://tools.ietf.org/html/rfc7617

4.2. Page 6 - (2025-01) - 3.1.1. The connection URL

The following clarifying text was added:

Old text	[...] percent-encoded [...]
New text	[...] percent-encoded / URL encoded [...]

4.3. Page 7 - (2025-01) - 3.1.2. OCPP version

The following text has been rewritten for clarity:

Old text	The OCPP version should not be part of the OCPP-J endpoint URL string if you want to select the OCPP version to use via the websocket protocol negotiation mechanism, as explained in Server Response .
New text	If the OCPP version is part of the OCPP-J endpoint URL it SHALL not determine the OCPP version to use, because the OCPP version is selected via the websocket protocol negotiation mechanism, as explained in section 3.3 [server-response] .

4.4. Page 8 - (2025-01) - 3.3. WebSocket Compression

The following text was duplicate. The duplication has been removed and the text has been clarified:

Old text	OCPP Requires the CSMS (and Local Controller) to support RFC 7692, WebSocket compression is seen as a relative simple way to reduce mobile data usage. For a Charging Station this is not a hard requirement, as this might be more complex to implement on an embedded platform, but as this is seen as efficient solution to reduce mobile data usage, it is RECOMMENDED to be implemented on a Charging Station that uses a mobile data connection. OCPP Requires the CSMS (and Local Controller) to support RFC 7692, WebSocket compression is seen as a relative simple way to reduce mobile data usage. For a Charging Station this is not a hard requirement, as this might be more complex to implement on an embedded platform, but as this is seen as efficient solution to reduce mobile data usage, it is RECOMMENDED to be implemented on a Charging Station that uses a mobile data connection.
New text	The CSMS (and Local Controller) SHALL support RFC 7692 , WebSocket compression, which is a relative simple way to reduce mobile data usage. For a Charging Station this is not a hard requirement, as this might be more complex to implement on an embedded platform. It is RECOMMENDED to be implemented on a Charging Station, because it is an efficient solution to reduce mobile data usage.

4.5. Page 10 - (2025-01) - 4.1.3. The message type

OLD: |CALL |2 | Request message |CALLRESULT |3 | Response message

New:

4.6. Page 10 - (2025-01) - 4.1.3. The message type

Refer to 'system', instead of 'server'.

Old text	When a server receives a message with a Message Type Number not in this list, it SHALL ignore the message payload. Each message type may have additional required fields.
New text	When a system receives a message with a Message Type Number not in this list, it SHALL ignore the message payload. Each message type may have additional required fields.

4.7. Page 10 - (2024-09) - 4.1.4. Message ID

Change the following text in paragraph 4.1.4:

Old text	The message ID serves to identify a request. A message ID for any CALL message MUST be different from all message IDs previously used by the same sender for any other CALL messages on any WebSocket connection using the same unique Charging Station identifier. This also applies to retries of messages.
New text	The message ID serves to identify a request. A message ID for any CALL message MUST be different from all message IDs previously used by the same sender for any other CALL messages on any WebSocket connection using the same unique Charging Station identifier. The message ID for a retried message (e.g. when no response was received within timeout) MAY be identical to the message ID of the original message.

4.8. Page 13 - (2025-01) - 4.2.3. CALLERROR

In the example, the ErrorDescription text has been updated:

Old text	"SetDisplayMessageRequest not implemented"
New text	"SetDisplayMessageRequest not supported "

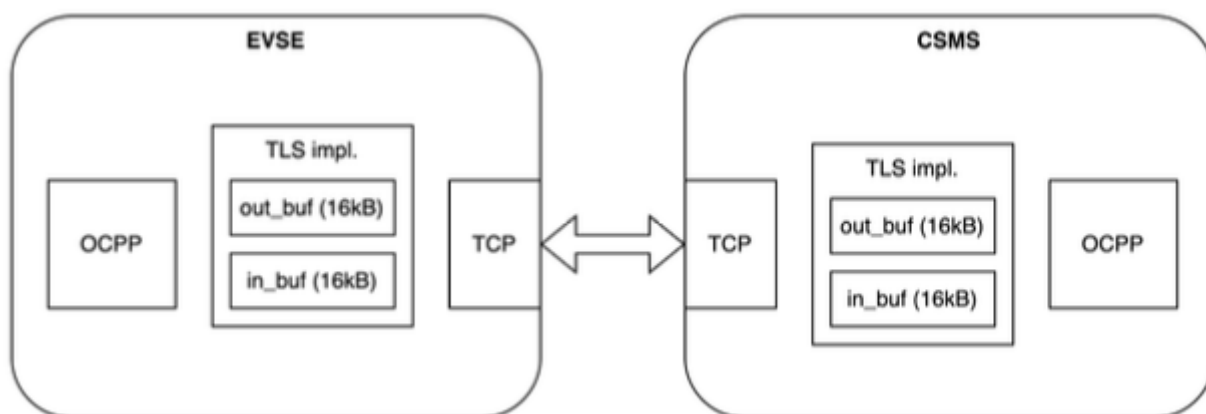
4.9. Page 15 - (2025-01) - 5. Connection

The following clarifying text has been added related to the TLS fragment length:

4.10. TLS fragment length

TLS involves sending "Records" between peers. Records can be of type "Handshake", "Alert", "ChangeCipherSpec", "Heartbeat" or "Application". OCPP messages are sent in Application records. The payload contains a "fragment" of the application data. The record layer fragments information blocks into TLSPlaintext records carrying data in chunks of 2^{14} bytes (16kB) or less.

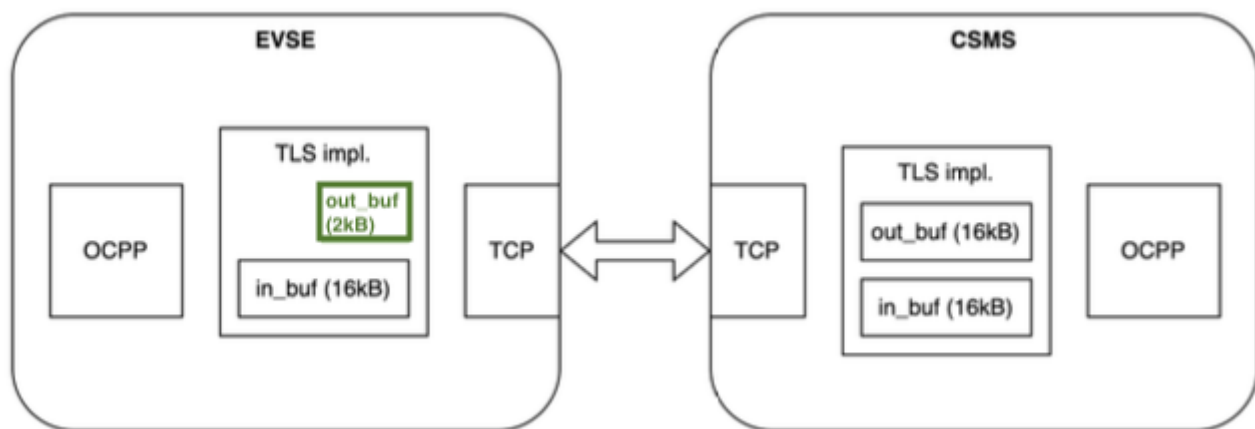
TLS peers need to maintain an input and an output buffer to store an entire fragment of 16 kB. For a low resource device it is a large cost to allocate 32 kB for the TLS connection.



Peers allocating standard 16 kB TLS buffers

A TLS extension is defined in TLS Extensions RFC6066 Section 4, that allows the client to ask for a different maximum fragment length than the default 16kB. A client can ask for a maximum fragment length of 0.5 kB, 1 kB, 2 kB or 4 kB. This TLS extension is, however, not widely supported and native managed cloud TLS termination services typically don't support this.

A resource-constrained Charging Station SHOULD try to negotiate a smaller TLS maximum fragment size, and if that is not accepted by the peer, then Charging Station MAY unilaterally decide to allocate less memory to its TLS output buffer. A TLS maximum fragment length of 2 kB is suggested based on data collection during certification tests, which shows that 99% of the messages fit in a 2 kB buffer.



Charging Station allocating a 2 kB TLS output buffer

4.11. Page 15 - (2025-01) - 5.3. WebSocket Ping in relation to OCPP Heartbeat

The following clarifying text was added:

New text	A Heartbeat message checks connectivity end-to-end, whereas a Websocket ping/pong only checks from point-to-point. This makes a difference in an extended network topology with a Local Controller between Charging Station and CSMS.
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4.12. Page 15 - (2025-01) - 5 Connection - Added section about TLS fragment length

As a result of the discussions at the OCPP lite taskgroup, a section has been added providing guidance on implementing TLS fragment negotiation.

4.13. Page 16 - (2025-04) - 5.3 Reconnecting - reset backoff wait timer

The RetryBackOffWaitMinimum timer is to be used the first time it tries to connect. A sentence has been added to below paragraph to make it explicit that it needs to be reset after successful connection.

The first reconnection attempts SHALL be after a back-off time of: <code>RetryBackOffWaitMinimum</code> seconds, plus a random value with a maximum of <code>RetryBackOffRandomRange</code> seconds. After every failed reconnection attempt the Charging Station SHALL double the previous back-off time, with a maximum of <code>RetryBackOffRepeatTimes</code>, adding a new random value with a maximum of <code>RetryBackOffRandomRange</code> seconds to every reconnection attempt. After <code>RetryBackOffRepeatTimes</code> reconnection attempts, the Charging Station SHALL keep reconnecting with the last back-off time, not increasing it any further. After a successful connection the backoff wait timer SHALL be reset to <code>RetryBackOffWaitMinimum</code> seconds.

4.14. Page 18 - (2025-02) - 6.3 Connection loss - Allow Local Controller to keep connection open

The sentence in this section was too strict about requiring a Local Controller to always close the connection with its charging stations when the connection with CSMS is lost. The sentence has been updated in order to allow for Local Controller

implementations that are able to manage the local charging stations locally (for a limited time) when the connection with CSMS is down.

Old text	Whenever one or more WebSocket connections between CSMS and the Local Controller are lost, the Local Controller SHALL close all corresponding WebSockets to the Charging Stations that are connected to it.
New text	Whenever one or more WebSocket connections between CSMS and the Local Controller are lost, the Local Controller SHALL close all corresponding WebSockets to the Charging Stations that are connected to it, unless the Local Controller is capable of responding to Charging Station requests, and forwards transaction-related requests to the CSMS once the connection is restored.

DRAFT

5. Part 5

5.1. General - (2025-02) - Renamed OCTT to Test System

Updated (2025-04)
~~In the entire document, the term~~ A few references to "OCTT" have been replaced by "Test System" (mostly "OCTT Id" has been replaced by "TC id").

5.2. Page 7-48 - (2025-09) - Add additional support for different types of Charging Stations

Added support for additional different types of Charging Stations:

- Wireless inductive Charging Stations
 - For this a product subtype Wireless Charging Station has been added.
- Charging Stations with different connectorTypes (like sType1) without (automatic) mechanized locking mechanisms
 - For this an additional question has been added: AQ-18: Does your Charging Station have at least one connector with an (automatic) mechanized locking mechanism on Charging Station side? (this is always true for connectorTypes; sType2 and sType3)
- Charging Stations with a rfid reader to start a transaction and a stop button to stop the transaction.
 - For this new stop local authorization methods C-70, C-71, C-72 and C-75 have been added, which makes it possible to have different authorization methods for the start and for the stop.

As a result of above additions/updates, many testcase conditions have been updated to incorporate them accordingly.

5.3. Page 7 - (2025-02) - Changed definition of C-01 Support for offline authorization of transactions

The current definition is does not take into account the supported local authorization methods in relation to the features to locally store idTokens.

Old	C-01	Support for offline authorization of transactions	Optional. Supporting this feature depends on whether at least one of the following is supported; - Certification Profile: Local Authorization List Management - C-02: Support for allowing offline authorization for unknown ids - C-49: Authorization Cache (AuthCacheEnabled)
-----	------	---	---

New	C-01	Support for offline authorization of transactions	Conditional. Supporting this feature depends on whether at least one of the following feature combinations is supported; - Certification Profile: Local Authorization List Management AND at least one of the following local authorization options; C-30 or C-31 or C-32 or C-34. - C-02: Support for allowing offline authorization for unknown ids AND at least one of the following local authorization options; C-30 or C-31 or C-32 or C-33 or C-34. - C-49: Authorization Cache AND at least one of the following local authorization options; C-30 or C-31 or C-32 or C-34. - C-35: Local Authorization - NoAuthorization (Because there is no authorization, no local authorization mechanism is needed.)
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5.4. Page 9 - (2024-11) - Optional feature list for charging station - Change name R-3

The specified name of feature R-3 is not correct. It should reflect the ability to disable reservations.

Old	R-3	Reservation support (ReservationEnabled)	Configuration Variable for H01
New	R-3	Support for disabling Reservations (ReservationEnabled)	Configuration Variable for H01

5.5. Page 9 - (2025-02) - Optional feature list for charging station - Added AQ-10 to make TC_N_48_CS conditional

Added AQ-10 to make TC_N_48_CS conditional.

Id	Additional questions for lab testing
AQ-10	Does your Charging Station support setting a Delta monitor on the WriteOnly component.variable SecurityCtrlr.BasicAuthPassword?

5.6. Page 9/11/27/48/52 - (2025-02) - Removed feature ISO-3, added additional question AQ-11

Page 9 - Optional feature list for charging station

	ISO 15118 support		
Removed	ISO-3	Combined charging station Certificate (for both OCPP and ISO 15118)	Optional

Page 11 - Optional feature list for CSMS

	ISO 15118 support		
Removed	ISO-3	Combined charging station certificate (for both OCPP and ISO 15118)	Optional

Page 27 - Test Cases Core

		Trigger message					
Old	TC_F_27	NotImplemented	C	M	For CS: can only be done when SignCombinedCertificate is notimplemented	NOT ISO-3	
New	TC_F_27	NotImplemented	C	M	For CS: can only be done when SignCombinedCertificate is notimplemented	NOT AQ-11	

Page 48 - Questions for Charging Stations

	Id	Additional questions for lab testing
Added	AQ-11	Does your Charging Station support a combined charging station Certificate (for both OCPP and ISO 15118)

Page 52 - Appendix C: Features vs. OCPP use cases

	Id	Feature	Related use cases
	ISO 15118 support		
Removed	ISO-3	Combined charging station certificate (for both OCPP and ISO 15118)	A02/A03

5.7. Page 9/10 - (2025-06) - Optional feature list for charging station & CSMS - Add optional feature for Security Profile 1

Security profile 1 has become optional for Core certification.

Added	C-61	Security Profile 1 - Unsecured Transport with Basic Authentication			Optional		
TC_A_20	No valid CSMSRootCertificate installed	C		If the last CSMSRootCertificate can be removed and Security Profile 1 is supported.	AQ-1 and C-61	Can the last CSMSRootCertificate be removed? Security Profile 1 - Unsecured Transport with Basic Authentication	

5.8. Page 13 - (2025-09) - Added TC_B_14_CS (as optional) to the Core profile

TC_B_14_CS was already a mandatory part of the Advanced Device Management profile, however it should also be possible to support the Summary inventory as part of the Core profile. Therefore, it will be added as an optional part of the Core profile.

Added to Core	TC_B_14	SummaryInventory	C			C-56	
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5.9. Page 13 - (2025-02) - Updated Id's of the additional questions of CSMSs in the appendix

The following Id's of additional questions for CSMSs have been updated:

Old	TC_B_30	Pending/Rejected - SecurityError	M	C	For CSMS: if CSMS can be configured to first respond to a BootNotificationRequest with status Pending or Rejected	C-44 or NOT AQ-6	BootNotification Pending or Does the CSMS reject unknown Charging Stations during websocket connection setup?
New	TC_B_30	Pending/Rejected - SecurityError	M	C	For CSMS: if CSMS can be configured to first respond to a BootNotificationRequest with status Pending or Rejected	C-44 or NOT AQ-16	BootNotification Pending or Does the CSMS reject unknown Charging Stations during websocket connection setup?
Old	TC_B_31	Pending/Rejected - TriggerMessage		C	For CSMS: if CSMS can be configured to first respond to a BootNotificationRequest with status Pending or Rejected	C-44 or NOT AQ-6	BootNotification Pending or Does the CSMS reject unknown Charging Stations during websocket connection setup?
New	TC_B_31	Pending/Rejected - TriggerMessage		C	For CSMS: if CSMS can be configured to first respond to a BootNotificationRequest with status Pending or Rejected	C-44 or NOT AQ-16	BootNotification Pending or Does the CSMS reject unknown Charging Stations during websocket connection setup?

5.10. Page 15 - (2025-04) - Addition of new testcase for CSMS to test WebSocket Subprotocol negotiation

New	TC_B_58	WebSocket Subprotocol negotiation		M			
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5.11. Page 17/33 - (2025-02) - Test Cases Local Authorization List Management & Authorization Cache - Update conditions

Testcases marked as mandatory have conditions that are not applicable and should be removed. The CWG decided that online local authorization list and authorization cache testcase can also be tested with remote authorization. In addition, local auth method C-34 MacAddress has been added to the offline conditions as possible method to use.

Authorization Cache

	Store Authorization Data in the Authorization Cache					
TC_C_32	Persistent over reboot	C		If the Charging Station has an authorization cache, then it must support this use case	C-49	Authorization Cache
TC_C_33	Update on AuthorizeResponse	C		If the Charging Station has an authorization cache, then it must support this use case	C-49	Authorization Cache
TC_C_34	Update on TransactionResponse	C		If the Charging Station has an authorization cache, then it must support this use case	C-49	Authorization Cache
TC_C_36	AuthCacheCtrlr.LocalPreAuthorize = false	C		If the Charging Station has an authorization cache and AuthCacheEnabled is implemented	C-49	Authorization Cache
TC_C_46	AuthCacheLifeTime	C		If the Charging Station has an authorization cache and supports to set a lifetime for its entries.	C-49 and C-53	Authorization Cache & AuthCacheLifeTime
	Clear Authorization Data in Authorization Cache					
TC_C_37	Accepted	C	M	If the Charging Station has an authorization cache, then it must support this use case	C-49	Authorization Cache
TC_C_38	Rejected	C	M	If the Charging Station has an authorization cache and AuthCacheEnabled is implemented	C-49	Authorization Cache
	Authorization by GroupId					
TC_C_41	Success with Authorization Cache	C		For CS: - The Charging Station supports at least one of the following local start authorization options: C-30, C-31, C-32, C-34 - If the Charging Station has an authorization cache.	C-49 and (C-30 or C-31 or C-32 or C-34)	Authorization Cache
TC_C_44	Invalid status with Authorization Cache	C		For CS: - The Charging Station supports at least one of the following local start authorization options: C-30, C-31, C-32, C-34 - If the Charging Station has an authorization cache.	C-49 and (C-30 or C-31 or C-32 or C-34)	Authorization Cache
	Authorization through authorization cache					
TC_C_08	Accepted	C	M	If the Charging Station has an authorization cache	C-49	Authorization Cache
TC_C_09	Invalid & Not Accepted	C		If the Charging Station has an authorization cache	C-49	Authorization Cache
TC_C_12	Invalid & Accepted	C		If the Charging Station has an authorization cache	C-49	Authorization Cache
TC_C_10	Blocked	C		If the Charging Station has an authorization cache	C-49	Authorization Cache
TC_C_11	Expired	C		If the Charging Station has an authorization cache	C-49	Authorization Cache
TC_C_13	Accepted but cable not connected yet.	C		If the Charging Station has an authorization cache	C-49	Authorization Cache

	Store Authorization Data in the Authorization Cache					
TC_C_15	StopTxOnInvalidId = false, MaxEnergyOnInvalidId > 0	C		If the Charging Station has an authorization cache AND the Charging Station supports at least one of the following local start authorization options: C-30, C-31, C-32, C-34 If MaxEnergyOnInvalidId is implemented.	C-49 and C-03 and (C-30 or C-31 or C-32 or C-34)	Authorization Cache & MaxEnergyOnInvalidId
TC_C_16	StopTxOnInvalidId = true	C		If the Charging Station has an authorization cache AND the Charging Station supports at least one of the following local start authorization options: C-30, C-31, C-32, C-34	C-49 and (C-30 or C-31 or C-32 or C-34)	Authorization Cache
TC_C_17	StopTxOnInvalidId = false	C		If the Charging Station has an authorization cache AND the Charging Station supports at least one of the following local start authorization options: C-30, C-31, C-32, C-34	C-49 and (C-30 or C-31 or C-32 or C-34)	Authorization Cache
TC_C_18	StopTxOnInvalidId = true, MaxEnergyOnInvalidId > 0	C		If the Charging Station has an authorization cache AND the Charging Station supports at least one of the following local start authorization options: C-30, C-31, C-32, C-34 If MaxEnergyOnInvalidId is implemented.	C-49 and C-03 and (C-30 or C-31 or C-32 or C-34)	Authorization Cache & MaxEnergyOnInvalidId
TC_C_20	Invalid		M			
TC_C_57	AuthCacheDisablePostAuthorize	C		If the Charging Station supports the option for disabling remote authorization for cached invalid idTokens AND has an authorization cache	C-59 and C-49	
	Local start transaction - Authorization first					
						
TC_E_52	DisableRemoteAuthorization	C		If the Charging Station supports the option for disabling remote authorization and The Charging Station supports at least one of the following local start authorization options C-30, C-31, C-32, C_34 and Either Authorization Cache or Local Authorization List is supported.	C-58 and (C-30 or C-31 or C-32 or C-34) and (C-49 or Local Authorization List Management)	Local Authorization - using RFID ISO14443 / RFID ISO15693 / KeyCode / MacAddress & Authorization Cache & Local Authorization List.
						
	Offline Behaviour					

	Store Authorization Data in the Authorization Cache					
TC_E_45	Stop transaction during offline period - Same GroupId	C		For CS: the Charging Station supports at least one of the following local start authorization options: C-30, C-31, C-32, C-34 and Local Authorization List or Authorization Cache	(C-30 or C-31 or C-32 or C-34) AND (Local Authorization List Management or C-49)	Local Authorization - using RFID ISO14443 / RFID ISO15693 / KeyCode / MacAddress and Local Authorization List or Authorization Cache

Local Authorization List Management

					Related features	
TC Id	OCPP Compliance Testing Tool scenario	Conf. Test for Charging Station	Conf. test for CSMS	Condition / remark	Feature no.	Feature
	Offline authorization through local authorization list					
TC_C_21	Accepted	C		The Charging Station must support at least one of the following local start authorization options: C-30, C-31, C-32, C-34	C-30 or C-31 or C-32 or C-34	
TC_C_22	Invalid	C		The Charging Station must support at least one of the following local start authorization options: C-30, C-31, C-32, C-34	C-30 or C-31 or C-32 or C-34	
TC_C_23	Blocked	C		The Charging Station must support at least one of the following local start authorization options: C-30, C-31, C-32, C-34	C-30 or C-31 or C-32 or C-34	
TC_C_24	Expired	C		The Charging Station must support at least one of the following local start authorization options: C-30, C-31, C-32, C-34	C-30 or C-31 or C-32 or C-34	
TC_C_25	Local Authorization List > Authorization Cache	C		The Charging Station must support at least one of the following local start authorization options: C-30, C-31, C-32, C-34	C-30 or C-31 or C-32 or C-34	
	Online authorization through local authorization list					
TC_C_27	Accepted	M		<Removed>	<Removed>	
TC_C_28	Invalid & Not Accepted	M		<Removed>	<Removed>	

					Related features	
TC_C_31	Invalid & Accepted	M		<Removed>	<Removed>	
TC_C_29	Blocked	M		<Removed>	<Removed>	
TC_C_30	Expired	M		<Removed>	<Removed>	
TC_C_58	LocalAuthListDisablePostAuthorize	C		The Charging Station supports the option for disabling remote authorization for invalid idTokens stored at the Local Authorization List.	LA-3	
	Authorization by GroupId					
TC_C_40	Success with Local Authorization List	C	M	For CS: - The Charging Station supports at least one of the following local start authorization options: C-30, C-31, C-32, C-34	C-30 or C-31 or C-32 or C-34	
TC_C_43	Invalid status with Local Authorization List	C	M	For CS: - The Charging Station supports at least one of the following local start authorization options: C-30, C-31, C-32, C-34	C-30 or C-31 or C-32 or C-34	

5.12. Page 19 - (2024-09) - TC_E_04_CS Updated condition for test case to exclude it for MacAddress and ISO 15118 PnC

This test case cannot be performed with the local authorization option MacAddress or ISO 15118 PnC.

		Local start transaction - Authorization first				
Old	TC_E_04	Success	C	M	Applicable if one or more of the local start authorization options is implemented.	C-30 - C-35 or ISO 15118 support Authorization options for local start
New	TC_E_04	Success	C	M	Applicable if one or more of the local start authorization options is implemented.	(C-30 or C-31 or C-32 or C-33 or C-35) Authorization options for local start

5.13. Page 22 - (2024-09) - TC_E_17_CS Updated condition for test case to correctly specify the applicable TxStopPoint combinations

This testcase allows for a limited set of TxStopPoint combinations, otherwise it is not applicable.

		Local start transaction - Authorization first					
Old	TC_E_17	Deauthorized - EV side disconnect	C	M	- TxStopPoint can either be ReadOnly with a subset of the values or have a valueList of supported values, that contains a subset. This testcase is applicable if the value Authorized or PowerPathClosed is a supported value. - StopTxOnEVSideDisconnect needs to ReadWrite or ReadOnly with value true	(C-10.2 or C-10.3) and C-06.2 and AQ-9	Supported Transaction Stop points
New	TC_E_17	Deauthorized - EV side disconnect	C	M	This testcase is applicable if the value Authorized is a supported value for TxStopPoint AND EVConnected, PowerPathClosed and EnergyTransfer must not be set as TxStopPoint AND StopTxOnEVSideDisconnect true must be a supported value.	C-10.2 and C-06.2 and AQ-9 and NOT (NOT C-52 AND (10.1 OR C-10.3 OR 10.4))	Supported Transaction Stop points

5.14. Page 26 - (2025-11) - TC_F_01 / TC_F_02 / TC_F_03

Unclear meaning of -(or)

The intended meaning is a logical OR between all subsequent optional features, but this unclear with the current description, therefore it will be clarified by fully writing out the formula as intended.

		Local start transaction - Authorization first					
Old	TC_F_01	Cable plugin first	C	M	If the Charging Station does not have a cable lock.	NOT AQ-2 and (C-36 -(or) C-39)	Authorization options for remote start
New	TC_F_01	Cable plugin first	C	M	If the Charging Station does not have a cable lock.	NOT AQ-2 and (C-36 or C-37 or C-38 or C-39)	Authorization options for remote start
Old	TC_F_02	Remote start first - AuthorizeRemoteStart is true	C	M	If AuthorizeRemoteStart can be set to true	C-48.1 and (C-36 -(or) C-39)	Authorization options for remote start
New	TC_F_02	Remote start first - AuthorizeRemoteStart is true	C	M	If AuthorizeRemoteStart can be set to true	C-48.1 and (C-36 or C-37 or C-38 or C-39)	Authorization options for remote start
Old	TC_F_03	Remote start first - AuthorizeRemoteStart is false	C	M	If AuthorizeRemoteStart can be set to false	C-48.2 and (C-36 -(or) C-39)	Authorization options for remote start
New	TC_F_03	Remote start first - AuthorizeRemoteStart is false	C	M	If AuthorizeRemoteStart can be set to false	C-48.2 and (C-36 or C-37 or C-38 or C-39)	Authorization options for remote start

5.15. Page 33 - (2025-06) - Fixed incorrect feature no reference

TC_A_21 referred to AQ-1, instead of AQ-3. This has been fixed.

TC_A_21	No valid ChargingStationCertificate installed	C		If the last ChargingStationCertificate can be removed (Via other means than OCPP).	AQ-3	
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5.16. Page 39 - (2025-02) - TC_N_48_CS Made conditional

This test case is only applicable if the Charging Station supports Delta monitoring on the *SecurityCtrlr.BasicAuthPassword* component variable.

Old	TC_N_48	Variable monitoring on write only	M				
New	TC_N_48	Variable monitoring on write only	C		CS: if the CS supports Delta monitoring on the <i>SecurityCtrlr.BasicAuthPassword</i>	AQ-10	

5.17. Page 40 - (2024-11) - TC_H_13_CS Updated invalid condition for test case

The condition should have been reversed.

Old	TC_H_13	Rejected	C		Depending on configuration variable ReservationNonSpecificEVSE	R-2	Support reservations of unspecified EVSE
New	TC_H_13	Rejected	C		Depending on the Charging Station not supporting the configuration variable ReservationNonSpecificEVSE	NOT R-2	Support reservations of unspecified EVSE

5.18. Page 42 - (2025-02) - Removed TC_O_15_CS from certification program

5.19. Test Cases Advanced User Interface

							Related features
Removed	TC_O_15	Language preference of the EV Driver	M				

5.20. Page 42 - (2025-06) - Make optional feature R-1 available for Charging Stations

The ConnectorEnumType list does not contains all connectorTypes. At OCPP 2.1 this enum has been changed to a string and can be extended, however this is not possible for OCPP 2.0.1. Therefore the reservation of connectorType testcases will not work for connectorTypes that are not part of the enum. It is not a major issue, because the CSMS can always reserve a specific EVSE based on the connectorTypes specified at the device model.

						Related features	
Old	TC_H_15	Success	C	C		CSMS: R-1 CS: R-2	For CSMS: Support reservations of connectorType For CS: Support for reservation of unspecified EVSE
New	TC_H_15	Success	C	C		R-1	Support reservations of connectorType
Old	TC_H_16	Amount of available connectors of a type equals the amount of reservations	C			R-2	Support for reservation of unspecified EVSE
New	TC_H_16	Amount of available connectors of a type equals the amount of reservations	C			R-1	Support reservations of connectorType

Added for Charging Station:

Added	R-1	Support reservations of connectorType	Conditional. Supporting this feature depends on whether at least one connectorType is supported that is part of the ConnectorEnumType list from part 2 specification.
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5.21. Page 44 - (2025-02) - Removed TC_A_13_CS and TC_A_13_CSMS from certification program

Test Cases ISO 15118 Support

						Related features	
Removed	TC_A_13	Success - Combined Certificate	C	C	If Combined Charging Station Certificate is supported.	ISO-3	Combined Charging Station Certificate

5.22. Page 48 - (2024-06) - Added additional questions to appendix

Note: This erratum is extended by erratum: [Page 48 - \(2025-02\) - Updated Id's of the additional questions for CSMSs in the appendix](#)

The following additional questions are added for CSMSs:

Id	Additional questions for lab testing
AQ-3	Does your CSMS support Absolute values for the following Charging Profiles:
AQ-3.1	TxDefaultProfile
AQ-3.2	ChargingStationMaxProfile

Id	Additional questions for lab testing
AQ-4	Does your CSMS support Recurring values for the following Charging Profiles:
AQ-4.1	<i>TxDefaultProfile</i>
AQ-4.2	<i>ChargingStationMaxProfile</i>

5.23. Page 48 - (2025-02) - Updated Id's of the additional questions for CSMSs in the appendix

Note: This erratum extends erratum: [Page 48 - \(2024-06\) - Added additional questions to appendix](#)

Note: This erratum is extended by erratum: [Page 48 - \(2025-04\) - Duplicate AQ-11 id](#)

The following Id's of additional questions for CSMSs have been updated:

Old Id	New Id
AQ-1	AQ-11
AQ-2	AQ-12
AQ-3.1	AQ-13.1
AQ-3.2	AQ-13.2
AQ-4	AQ-14
AQ-4.1	AQ-14.1
AQ-4.2	AQ-14.2
AQ-6	AQ-16

5.24. Page 48 - (2025-04) - Duplicate AQ-11 id

Note: This erratum extends erratum: [Page 48 - \(2025-02\) - Updated Id's of the additional questions for CSMSs in the appendix](#)

At the 2025-02 errata sheet release, both for Charging Station and CSMS an additional question was added with id AQ-11, but these should be kept unique. The following Id's of additional questions for CSMSs have been updated:

Old Id	New Id
AQ-11	AQ-17

5.25. Page 48 - (2025-09) - Added additional hardware features

Id	Hardware Feature
HFS-11	DC power level (kW)
HFS-12	Number of displays

DRAFT

6. Part 6

6.1. General

6.1.1. Page XX - (2024-11) - All testcases - Updated table structure of all testcases

The table structure of all testcases have been updated. This has been done for multiple reasons:

- It improves readability by providing more space for the main steps.
- It decreases the chance of testcase tables being broken, resulting in missing steps at the bottom of a testcase.
- It makes it easier for the Technical Editors to update testcases.

6.1.2. Page XX - (2025-02) - Renamed OCTT to Test System

In the entire document, the term "OCTT" has been replaced by "Test System" .

6.2. Charging Station

6.2.1. Page 4 - (2025-02) - TC_A_01_CS - Updated old identifierString reference in description

Tool validations
* Step 1: [...] BasicAuthPassword may only contain alpha-numeric characters and the special characters allowed by identifierString. BasicAuthPassword may only contain alpha-numeric characters and the special characters allowed by passwordString.
Post scenario validations:
N/a

6.2.2. Page 7 - (2024-11) - TC_A_05_CS - Successfully reconnecting after every failed connection attempt

It is needed to reconnect after every (intended) failed connection, otherwise the retryBackoffTime may double itself several time, resulting in a very large number that may exceed the configured timeouts.

Table 6. Test Case Id: TC_A_05_CS

Test case name	TLS - server-side certificate - Invalid certificate
Test case Id	TC_A_05_CS
Use case Id(s)	A00
Requirement(s)	A00.FR.309,A00.FR.310,A00.FR.311,A00.FR.412,A00.FR.413,A00.FR.414
System under test	Charging Station
Description	The CSMS uses a server-side certificate to identify itself to the Charging Station, when using security profile 2 or 3.
Purpose	To verify whether the Charging Station is able to terminate the connection when the received server certificate is invalid.
Prerequisite(s)	- The charging station supports security profile 2 and/or 3 - The active NetworkConnectionProfile uses either security profile 2 OR 3. <Removed>

Test case name	TLS - server-side certificate - Invalid certificate	
Before (Preparations)	Configuration State: OCCPPCommCtrlr.NetworkConfigurationPriority only contains <Value from ActiveNetworkProfile>	
	Memory State: N/a	
	Reusable State(s): N/a	
Main (Test scenario)	Charging Station	CSMS
	1. The OCTT aborts the connection with the Charging Station.	
	2. The Charging Station initiates a TLS handshake and sends a Client Hello to the OCTT.	3. The OCTT responds with a Server Hello With a <Configured valid server certificate>
	Note(s) : - The OCTT will use this as an indication of the time it takes the Charging Station to reconnect.	
	4. The OCTT aborts the connection with the Charging Station.	
	5. The Charging Station initiates a TLS handshake and sends a Client Hello to the OCTT.	6. The OCTT responds with a Server Hello With a <Configured invalid server certificate>
	7. The Charging Station deems the server certificate invalid and terminates the connection.	
	Note : The OCTT will wait two times the measured reconnection time from step 3, before switching the server certificate back to the valid server certificate. The reason for this is that the OCTT is not always able to detect a failed connection attempt.	
	8. The Charging Station initiates a TLS handshake and sends a Client Hello to the OCTT.	9. The OCTT responds with a Server Hello With a <Configured valid server certificate>
	Note(s) : - The OCTT will accept the connection to prevent doubling of the RetryBackOffWaitMinimum.	
	10 The Charging Station sends a SecurityEventNotificationRequest	11 The OCTT responds with a SecurityEventNotificationResponse
Note(s) : - Steps 4 to 11 are repeated per configured invalid server certificate. - In case default certificates are being used, the OCTT will use three different invalid server certificates; "Not signed by installed Root certificate", "Expired", "CommonName that does not equal the FQDN of the server". - In case custom certificates are being used, the OCTT will loop through all certificates configured at the 'CSMS Keystore Invalid'.		
Tool validations	* Step 11: Message: SecurityEventNotificationRequest - type must be <i>InvalidCsmsCertificate</i>	
	Post scenario validations: N/a	

6.2.3. Page 7 - (2025-02) - TC_A_05_CS

Before (Preparations)

Configuration State: OCPPCommCtrlr.NetworkProfileConnectionAttempts is 3 OCPPCommCtrlr.NetworkConfigurationPriority only contains <Value from ActiveNetworkProfile>
Memory State: N/a
Reusable State(s): N/a

Main (Test scenario)	
Charging Station	CSMS
[...]	
5. The Charging Station initiates a TLS handshake and sends a Client Hello to the Test System.	6. The Test System responds with a Server Hello With a <Configured invalid server certificate> With a <Generated invalid server certificate>
[...]	
10 The Charging Station sends a SecurityEventNotificationRequest	11 The Test System responds with a SecurityEventNotificationResponse
Note(s): Steps 4 to 11 are repeated per configured invalid server certificate. In case default certificates are being used, the Test System will use three different invalid server certificates; "Not signed by installed Root certificate", "Expired", "CommonName that does not equal the FQDN of the server". In case custom certificates are being used, the Test System will loop through all certificates configured at the 'CSMS Keystore Invalid'. The Test System will loop through steps 4 to 11 for a set of generated invalid certificates; "Expired", "Future validity date", "Not signed by installed CSMS Root certificate", "CommonName that does not equal the FQDN of the server", "CommonName containing a wildcard hostname matching the FQDN".	

6.2.4. Page 7 - (2025-04) - TC_A_05_CS - Updated before steps to take into account the AllowCSMSTLSWildcards variable

Test case name	TLS - server-side certificate - Invalid certificate
Test case Id	TC_A_05_CS
...	

Before (Preparations)
Configuration State: OCPPCommCtrlr.NetworkProfileConnectionAttempts is 3 OCPPCommCtrlr.NetworkConfigurationPriority only contains <Value from ActiveNetworkProfile> SecurityCtrlr.AllowCSMSTLSWildcards is false (If implemented)
Memory State: N/a
Reusable State(s): N/a

6.2.5. Page 20 - (2025-02) - TC_A_23_CS - CSMS returns a CertificateSigned message for each request

TC_A_23_CS: Update Charging Station Certificate by request of CSMS - CertificateSignedRequest Timeout

Test case name	Update Charging Station Certificate by request of CSMS - CertificateSignedRequest Timeout
Test case Id	TC_A_23_CS
...	...
Purpose	To verify if the Charging Station is able to send a new signCertificateRequest when it did not receive a certificateSignedRequest after the configured timeout. CSMS will after a delay send a CertificateSignedRequest for each SignCertificateRequest that it has accepted.
...	...

Before (Preparations)

Main (Test scenario)	
Charging Station	CSMS
...	...
12. The Charging Station responds with a CertificateSignedResponse	11. The Test System sends a CertificateSignedRequest With certificateChain <Certificate generated from the received CSR from step 3 and signed by the provided CSMS Root certificate> certificateType ChargingStationCertificate
14. The Charging Station responds with a CertificateSignedResponse	13. The Test System sends a CertificateSignedRequest With certificateChain <Certificate generated from the received CSR from step 6 and signed by the provided CSMS Root certificate> certificateType ChargingStationCertificate
16. The Charging Station responds with a CertificateSignedResponse	15. The Test System sends a CertificateSignedRequest With certificateChain <Certificate generated from the received CSR from step 9 and signed by the provided CSMS Root certificate> certificateType ChargingStationCertificate

Tool validations
...
* Step 12, 14, 16: Message: CertificateSignedResponse - status must be Accepted or Rejected
Post scenario validations: Note: It does not matter whether Charging Station accepts first or last or all certificates. At least one CertificateSignedResponse must have status Accepted

6.2.6. Page 22 - (2024-09) - TC_A_19_CS - Fixed references to ConfigurationSlot [020-4762]

Test case name	Upgrade Charging Station Security Profile - Accepted
Test case Id	TC_A_19_CS

Test case name	Upgrade Charging Station Security Profile - Accepted	
Main (Test scenario)	Charging Station	CSMS
	2. The Charging Station responds with a SetNetworkProfileResponse	1. The OCTT sends a SetNetworkProfileRequest with configurationSlot is <Configured configurationSlot> or <Configured configurationSlot2> depending on which one is already in use - connectionData.messageTimeout <Configured messageTimeout> - connectionData.ocppCsmsUrl <Configured ocppCsmsUrl> - connectionData.ocppInterface <Configured ocppInterface> - connectionData.ocppVersion OCPP20 - connectionData.securityProfile <Configured securityProfile + 1>
	4. The Charging Station responds with a SetVariablesResponse	3. The OCTT sends a SetVariablesRequest with variable.name is "NetworkConfigurationPriority" component.name is "OCPPCommCtrlr" attributeValue is "<Configured configurationSlot2>,<Configured configurationSlot>"
	...	...
Tool validations	* Step 2: Message SetNetworkProfileResponse - status Accepted * Step 4: Message SetVariablesResponse - setVariableResult[0].attributeStatus Accepted OR RebootRequired * Step 6: Message ResetResponse - status Accepted * Step 11: Message GetVariablesResponse - getVariableResult[0].attributeValue <Configured securityProfile + 1> * Step 13: Message GetVariablesResponse - getVariableResult[0].attributeValue Does not contain the configurationSlot with the previous (lower) security profile Post scenario validations: - N/a	

6.2.7. Page 22 - (2025-06) - TC_A_19_CS - Added steps to validate the Charging Station does not downgrade back to security profile 1.

Added steps to validate the Charging Station does not downgrade back to security profile 1 after having upgraded to security profile 2.

Test case name	Upgrade Charging Station Security Profile - Accepted
Test case Id	TC_A_19_CS
...	

Main (Test scenario)	
Charging Station	CSMS

Main (Test scenario)	
2. The Charging Station responds with a SetNetworkProfileResponse	1. The Test System sends a SetNetworkProfileRequest with configurationSlot is <Configured configurationSlot> or <Configured configurationSlot2> depending on which one is already in use - connectionData.messageTimeout <Configured messageTimeout> - connectionData.ocppCsmsUrl <Configured ocppCsmsUrl> - connectionData.ocppInterface <Configured ocppInterface> - connectionData.ocppVersion OCPP20 - connectionData.securityProfile <Configured securityProfile + 1> <u>Note(s):</u> - The Test System checks the ActiveNetworkProfile variable to see which slot is currently active. - The Test System prevents overwriting the NetworkProfile at the active slot, as this is not recommended.
4. The Charging Station responds with a SetVariablesResponse	3. The Test System sends a SetVariablesRequest with variable.name is "NetworkConfigurationPriority" component.name is "OCPPCommCtrlr" attributeValue is <configurationSlot set at Step 1, the other configured configurationSlot>
6. The Charging Station responds with a ResetResponse	5. The Test System sends a ResetRequest with type OnIdle
	7. The Test System restarts the WebSocket server using <Configured securityProfile + 1>
8. The Charging Station reconnects to the Test System using <Configured securityProfile + 1>	9. The Test System accepts the connection attempt.
10. Execute Reusable State <i>Booted</i>	
12. The Charging Station responds with GetVariablesResponse	11. Test System sends GetVariablesRequest with: - variable.name = "SecurityProfile" - component.name = "SecurityCtrlr"
14. The Charging Station responds with GetVariablesResponse	13. Test System sends GetVariablesRequest with: - variable.name = "NetworkConfigurationPriority" - component.name = "OCPPCommCtrlr"
The following steps are only executed when this testcase is upgrading from Security Profile 1 to Security Profile 2.	
16. The Charging Station does NOT reconnect to the Test System using Security Profile 1.	15. The Test System closes the connection and restarts the WebSocket server using Security profile 1 and waits the <Configured long operation timeout>.
18. The Charging Station reconnects to the Test System using Security Profile 2.	17. The Test System restarts the WebSocket server using Security Profile 2.

Tool validations
* Step 2: Message SetNetworkProfileResponse - status <i>Accepted</i> * Step 4: Message SetVariablesResponse - setVariableResult[0].attributeStatus <i>Accepted OR RebootRequired</i> * Step 6: Message ResetResponse - status <i>Accepted</i> * Step 12: Message GetVariablesResponse - getVariableResult[0].attributeValue <i><Configured securityProfile + 1></i> * Step 14: Message GetVariablesResponse - getVariableResult[0].attributeValue Does not contain the configurationSlot with the previous (lower) security profile
Post scenario validations: - N/a

6.2.8. Page 24 - (2025-02) - TC_A_20_CS - Testcase did not take into account that the used configuration slot could already be set

The testcase did not take into account that the NetworkConnectionProfile slot updated during the main steps, might already be set at the OCPPCommCtrlr.NetworkConfigurationPriority. This would cause setting the networkConnectionProfile to be rejected already.

Test case name	Upgrade Charging Station Security Profile - No valid CSMSRootCertificate installed
Test case Id	TC_A_20_CS
...	...
Prerequisite(s)	- The Test System connectionData configuration for SUT Charging Station only allows for ip addresses the Test System is able to bind. - The Charging Station support s at least 2 security profiles, one of which is security profile 1. - The Charging Station does not have a valid CSMSRootCertificate installed. - The first Test System connectionData configuration slot must be configured for security profile 1. - The second Test System connectionData configuration slot must be configured for security profile 2 or 3. - The Charging Station is connected using security profile 1. - When starting this testcase the Test System will start another websocket server for the second connectionData slot.

Before (Preparations)
Configuration State: OCPPCommCtrlr.NetworkConfigurationPriority is <i><ActiveNetworkProfile slot></i> (All others are removed)
Memory State: N/a
Reusable State(s): N/a

Main (Test scenario)	
Charging Station	CSMS

Main (Test scenario)	
2. The Charging Station responds with a SetNetworkProfileResponse	1. The Test System sends a SetNetworkProfileRequest with - configurationSlot is <Configured configurationSlot2> or <Configured configurationSlot> (the one currently not used for the active connection) - connectionData.messageTimeout <Configured messageTimeout2> - connectionData.ocppCsmsUrl <ocppCsmsUrl that is not currently active> - connectionData.ocppInterface <Configured ocppInterface2> - connectionData.ocppVersion OCPP20 - connectionData.securityProfile <Configured securityProfile2>
4. The Charging Station responds with a SetVariablesResponse	3. The Test System sends a SetVariablesRequest with variable.name is "NetworkConfigurationPriority" component.name is "OCPPCommCtrlr" attributeValue is <configurationSlot set at step 1>, <previous configurationSlot>

6.2.9. Page 24 - (2025-04) - TC_A_20_CS - SetNetworkConnectionProfile is allowed to be rejected

Test case name	Upgrade Charging Station Security Profile - No valid CSMSRootCertificate installed
Test case Id	TC_A_20_CS
...	...

Tool validations
* Step 2: Message SetNetworkProfileResponse - status Accepted or Rejected * Step 4: Message SetVariablesResponse - setVariableResult[0].attributeStatus Rejected
Post scenario validations: - N/a

6.2.10. Page 25 - (2025-02) - TC_A_21_CS

TC_A_21_CS: Upgrade Charging Station Security Profile - No valid ChargingStationCertificate installed

Test case name	Upgrade Charging Station Security Profile - No valid ChargingStationCertificate installed
...	...
Description	The CSMS is able to change the connectionData at the Charging Station. By doing this it is able to upgrade the connection to a security profile 3.
Purpose	To verify if the Charging Station is able to reject upgrading to a security profile 3 when it does not have a valid ChargingStationCertificate installed.
Prerequisite(s)	- The Test System connectionData configuration for SUT Charging Station only allows for ip addresses the Test System is able to bind. - The Charging Station support at least 2 security profiles. - The Charging Station does not have a valid ChargingStationCertificate installed. - The Charging Station has a valid CSMSRootCertificate installed. - The second Test System connectionData configuration slot must be configured for security profile 3. - When starting this testcase the Test System will start another webSocket server for the second connectionData slot.

Main (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a SetNetworkProfileResponse	1. The Test System sends a SetNetworkProfileRequest with - configurationSlot is <Configured configurationSlot> or <Configured configurationSlot> depending on which one is already in use - connectionData.messageTimeout <Configured messageTimeout2> - connectionData.ocppCsmsUrl <ocppCsmsUrl that is not currently active> - connectionData.ocppInterface <Configured ocppInterface2> - connectionData.ocppVersion OCPP20 - connectionData.securityProfile <Configured securityProfile2> 3

6.2.11. Page 26 - (2025-02) - TC_A_22_CS

TC_A_22_CS: Upgrade Charging Station Security Profile - Downgrade security profile - Rejected

Test case name	Upgrade Charging Station Security Profile - Downgrade security profile - Rejected
[...]	
Description	The CSMS is able to change the connectionData at the Charging Station. It tries to downgrade the connection to a lower security profile 1.
Purpose	To verify if the Charging Station is able to reject downgrading to a lower security profile than the currently active security profile. 1.
Prerequisite(s)	- The Test System connectionData configuration for SUT Charging Station only allows for ip addresses the Test System is able to bind. - The Charging Station supports security profile 2 and/or 3. - The second Test System connectionData configuration slot must be configured for a security profile lower than the first Test System connectionData configuration slot. - The Charging Station has a connection using security profile 2 or 3. - When starting this testcase the Test System will start another webSocket server for the second connectionData slot.

Main (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a SetNetworkProfileResponse	1. The Test System sends a SetNetworkProfileRequest with: - configurationSlot is <Configured configurationSlot> or <Configured configurationSlot> depending on which one is already in use - configurationSlot is <Configured configurationSlot> or <Configured configurationSlot2> depending on which one is already in use - connectionData.messageTimeout <Configured messageTimeout2> - connectionData.ocppCsmsUrl <ocppCsmsUrl that is not currently active> - connectionData.ocppInterface <Configured ocppInterface2> - connectionData.ocppVersion OCPP20 - connectionData.securityProfile <Configured securityProfile2> connectionData.securityProfile 1

6.2.12. Page 51 - (2025-02) - TC_B_16_CS - Correctly validate result of reading WriteOnly component variables

Test case name	Get Custom Report - with component criteria
Test case Id	TC_B_16_CS
Use case Id(s)	B08

Test case name	Get Custom Report - with component criteria
Requirement(s)	B08.FR.01, B08.FR.03, B08.FR.04, B08.FR.07, B08.FR.09, B089.FR.10, B08.FR.12, B08.FR.13, B08.FR.14 , B06.FR.09
[...]	

Tool validations
[...]
* Step 6: Message: GetVariablesResponse with: For component variables where NotifyReportRequest.reportData.variableAttribute.mutability from step 3 is not <i>WriteOnly</i> - attributeStatus = <i>Accepted</i> - attributeValue = <i>true</i> For component variables where NotifyReportRequest.reportData.variableAttribute.mutability from step 3 is <i>WriteOnly</i> - attributeStatus = <i>Rejected</i> - attributeValue = <i><omitted></i>
[...]

6.2.13. Page 56 - (2024-09) - TC_B_20_CS - Added check on omitting evseld [4390]

Test case name	Reset Charging Station - Without ongoing transaction - OnIdle	
Test case Id	TC_B_20_CSMS	
...		
Main (Test scenario)	Charging Station	CSMS
	...	
Tool validations	* Step 1: Message ResetRequest - evseld must be omitted	
	* Step 4: Message BootNotificationResponse - status <i>Accepted</i>	
	Post scenario validations: - N/a	

6.2.14. Page 57 - (2024-09) - TC_B_21_CS - Added check on omitting evseld [4390]

Test case name	Reset Charging Station - With Ongoing Transaction - OnIdle	
Test case Id	TC_B_21_CSMS	
...		
Main (Test scenario)	Charging Station	CSMS
	...	
Tool validations	* Step 1: Message ResetRequest - type <i>OnIdle</i> - evseld must be omitted	
	* Step 8: Message BootNotificationResponse - status <i>Accepted</i> Post scenario validations: - N/a	

6.2.15. Page 58 - (2024-09) - TC_B_22_CS - Added check on omitting evseld [4390]

Test case name	Reset Charging Station - With Ongoing Transaction - Immediate	
Test case Id	TC_B_22_CSMS	
...		
Main (Test scenario)	Charging Station	CSMS
	...	
Tool validations	* Step 1: Message ResetRequest - type <i>Immediate</i> - evseld is omitted	
	* Step 6: Message BootNotificationResponse - status <i>Accepted</i>	
	Post scenario validations: - N/a	

6.2.16. Page 63 - (2025-02) - Changed reset to Immediate

The reset in step #1 has been changed to "Immediate", because having a reservation pending is not an "idle" situation. As a result a Charging Station would respond with status "Scheduled" instead of resetting.

Test case name	Reset Charging Station - Reserved persists reset
Test case Id	TC_B_24_CS
...	

Main (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a ResetResponse	1. The Test System sends a ResetRequest with type <i>Immediate</i>
...	

6.2.17. Page 72-81 - (2025-04) - TC_B_45_CS & TC_B_46_CS & TC_B_47_CS & TC_B_49_CS & TC_B_50_CS & TC_A_19_CS - Clarified NetworkProfile configurationSlot usage

The Test system has two endpoints it is able to switch between. Depending on which one is active at the start of the testcase, it makes the SUT switch to the other endpoint.

Test case name	Migrate to new ConnectionProfile - Success - Same CSMS Root
Test case Id	TC_B_45_CS
...	

Main (Test scenario)	
Charging Station	CSMS

Main (Test scenario)	
2. The Charging Station responds with a SetNetworkProfileResponse	1. The Test System sends a SetNetworkProfileRequest with configurationSlot is <Configured configurationSlot> or <Configured configurationSlot2> depending on which one is already in use - connectionData.messageTimeout <Configured messageTimeout> or <Configured messageTimeout2> - connectionData.ocppCsmsUrl <ocppCsmsUrl that is not currently active> - connectionData.ocppInterface <Configured ocppInterface> or <Configured ocppInterface2> - connectionData.ocppVersion OCPP20 - connectionData.securityProfile <Configured securityProfile> or <Configured securityProfile2> Note(s) : - The Test System checks the ActiveNetworkProfile variable to see which slot is currently active. - The Test System prevents overwriting the NetworkProfile at the active slot, as this is not recommended.
4. The Charging Station responds with a SetVariablesResponse	3. The Test System sends a SetVariablesRequest with variable.name is "NetworkConfigurationPriority" component.name is "OCPPCommCtrlr" attributeValue is <configurationSlot set at Step 1, the other configured configurationSlot>
...	
7. Execute Reusable State <i>Booted</i>	
Note(s): - The Charging Station connects to the slot configured at step 1.	

Test case name	Migrate to new ConnectionProfile - Fallback mechanism - Same CSMS Root
Test case Id	TC_B_46_CS
...	

Main (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a SetNetworkProfileResponse	1. The Test System sends a SetNetworkProfileRequest with configurationSlot is <Configured configurationSlot> or <Configured configurationSlot2> depending on which one is already in use - connectionData.messageTimeout <Configured messageTimeout> or <Configured messageTimeout2> - connectionData.ocppCsmsUrl <ocppCsmsUrl that is not currently active> - connectionData.ocppInterface <Configured ocppInterface> or <Configured ocppInterface2> - connectionData.ocppVersion OCPP20 - connectionData.securityProfile <Configured securityProfile> or <Configured securityProfile2> Note(s) : - The Test System checks the ActiveNetworkProfile variable to see which slot is currently active. - The Test System prevents overwriting the NetworkProfile at the active slot, as this is not recommended.

Main (Test scenario)	
4. The Charging Station responds with a SetVariablesResponse	3. The Test System sends a SetVariablesRequest with variable.name is "NetworkConfigurationPriority" component.name is "OCPPCommCtrlr" attributeValue is <configurationSlot set at Step 1, the other configured configurationSlot>
...	
9. Execute Reusable State <i>Booted</i>	
<u>Note(s):</u> - The Charging Station connects to the second slot configured at the NetworkConfigurationPriority at step 3.	

Test case name	Migrate to new ConnectionProfile - Fallback after NetworkProfileConnectionAttempts per NetworkConfigurationPriority failed - New CSMS Root - New CSMS
Test case Id	TC_B_47_CS
...	

Main (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a SetNetworkProfileResponse	1. The Test System sends a SetNetworkProfileRequest with configurationSlot is <Configured configurationSlot> or <Configured configurationSlot2> depending on which one is already in use - connectionData.messageTimeout <Configured messageTimeout> or <Configured messageTimeout2> - connectionData.ocppCsmsUrl <ocppCsmsUrl that is not currently active> - connectionData.ocppInterface <Configured ocppInterface> or <Configured ocppInterface2> - connectionData.ocppVersion OCPP20 - connectionData.securityProfile <Configured securityProfile> or <Configured securityProfile2> <u>Note(s):</u> - The Test System checks the ActiveNetworkProfile variable to see which slot is currently active. - The Test System prevents overwriting the NetworkProfile at the active slot, as this is not recommended.
4. The Charging Station responds with a SetVariablesResponse	3. The Test System sends a SetVariablesRequest with variable.name is "NetworkConfigurationPriority" component.name is "OCPPCommCtrlr" attributeValue is <configurationSlot set at Step 1, the other configured configurationSlot>
...	

Test case name	Migrate to new ConnectionProfile - Fallback after NetworkProfileConnectionAttempts per NetworkConfigurationPriority failed - Same CSMS Root
Test case Id	TC_B_49_CS
...	

Main (Test scenario)	
Charging Station	CSMS

Main (Test scenario)	
2. The Charging Station responds with a SetNetworkProfileResponse	1. The Test System sends a SetNetworkProfileRequest with configurationSlot is <Configured configurationSlot> or <Configured configurationSlot2> depending on which one is already in use - connectionData.messageTimeout <Configured messageTimeout> or <Configured messageTimeout2> - connectionData.ocppCsmsUrl <ocppCsmsUrl that is not currently active> - connectionData.ocppInterface <Configured ocppInterface> or <Configured ocppInterface2> - connectionData.ocppVersion OCPP20 - connectionData.securityProfile <Configured securityProfile> or <Configured securityProfile2> Note(s) : - The Test System checks the ActiveNetworkProfile variable to see which slot is currently active. - The Test System prevents overwriting the NetworkProfile at the active slot, as this is not recommended.
4. The Charging Station responds with a SetVariablesResponse	3. The Test System sends a SetVariablesRequest with variable.name is "NetworkConfigurationPriority" component.name is "OCPPCommCtrlr" attributeValue is <configurationSlot set at Step 1, the other configured configurationSlot>
...	

Test case name	Migrate to new ConnectionProfile - Success - New CSMS Root - New CSMS
Test case Id	TC_B_50_CS
...	

Main (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a SetNetworkProfileResponse	1. The Test System sends a SetNetworkProfileRequest with configurationSlot is <Configured configurationSlot> or <Configured configurationSlot2> depending on which one is already in use - connectionData.messageTimeout <Configured messageTimeout> or <Configured messageTimeout2> - connectionData.ocppCsmsUrl <ocppCsmsUrl that is not currently active> - connectionData.ocppInterface <Configured ocppInterface> or <Configured ocppInterface2> - connectionData.ocppVersion OCPP20 - connectionData.securityProfile <Configured securityProfile> or <Configured securityProfile2> Note(s) : - The Test System checks the ActiveNetworkProfile variable to see which slot is currently active. - The Test System prevents overwriting the NetworkProfile at the active slot, as this is not recommended.

Main (Test scenario)	
4. The Charging Station responds with a SetVariablesResponse	3. The Test System sends a SetVariablesRequest with variable.name is "NetworkConfigurationPriority" component.name is "OCPPCommCtrlr" attributeValue is <configurationSlot set at Step 1, the other configured configurationSlot>
...	

Test case name	Upgrade Charging Station Security Profile - Accepted
Test case Id	TC_A_19_CS
...	

Main (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a SetNetworkProfileResponse	1. The Test System sends a SetNetworkProfileRequest with configurationSlot is <Configured configurationSlot> or <Configured configurationSlot2> depending on which one is already in use - connectionData.messageTimeout <Configured messageTimeout> - connectionData.ocppCsmsUrl <Configured ocppCsmsUrl> - connectionData.ocppInterface <Configured ocppInterface> - connectionData.ocppVersion OCPP20 - connectionData.securityProfile <Configured securityProfile + 1> Note(s) : - The Test System checks the ActiveNetworkProfile variable to see which slot is currently active. - The Test System prevents overwriting the NetworkProfile at the active slot, as this is not recommended.
4. The Charging Station responds with a SetVariablesResponse	3. The Test System sends a SetVariablesRequest with variable.name is "NetworkConfigurationPriority" component.name is "OCPPCommCtrlr" attributeValue is <configurationSlot set at Step 1, the other configured configurationSlot>
...	

6.2.18. Page 72-81 - (2025-06) - TC_B_45_CS & TC_B_46_CS & TC_A_19_CS - ResetRequest will always be sent by the Test System to ensure the Charging Station switches NetworkProfile

ResetRequest will always be sent by the Test System to ensure the Charging Station switches NetworkProfile.

Test case Id	TC_B_45_CS / C_B_46_CS / TC_A_19_CS
...	

Main (Test scenario)	
Charging Station	CSMS
...	

Main (Test scenario)	
6. The Charging Station responds with a ResetResponse	5. The Test System sends a ResetRequest with type OnIdle Note(s): This step will only be executed when the status RebootRequired is returned at step 4, or if the charging does not automatically reboot.
...	

6.2.19. Page 85 - (2025-04) - TC_B_53_CS - Check if all required values are provided

The testcase currently does not explicitly describe that the value fields for standardized component variables should not be omitted (with EVSE.Power as an exception. For this only the maxLimit is required). These are optional at the JSON schema, because they are allowed to omit in case of WriteOnly variables.

Test case name	Get Base Report - Test mandatory DM variables via FullInventory	
Test case Id	TC_B_53_CS	
...		
Main (Test scenario)	Charging Station	CSMS
	...	...
Tool validations	...	
	Post scenario validations: The Test System checks that all implemented standardized components / variables are implemented correctly according to the OCPP specification: - The components / variables that are required according to the OCPP specification are implemented. - For each component/variable, where variableCharacteristics.dataType is set to OptionList, SequenceList or MemberList, the variableCharacteristics.valuesList is not omitted or empty. - For each component/variable, where variableCharacteristics.dataType is OptionList, SequenceList or MemberList, the variableAttribute.value is allowed based on the values in the provided variableCharacteristics.valuesList. - For variables with mutability set to <i>WriteOnly</i> the variableAttribute.value is omitted in the NotifyReportRequest. - For variables with mutability NOT set to <i>WriteOnly</i> the variableAttribute.value is NOT omitted in the NotifyReportRequest . There is one exception to this rule and that is for EVSE . Power . This variable only has a maxLimit .	

6.2.20. Page 89 - (2025-02) - Add setting of NetworkProfileConnectionAttempts

TC_B_57_CS: Network Reconnection - After connection loss

Test case name	Network Reconnection - After connection loss
Test case Id	TC_B_57_CS
...	...

Before (Preparations)
Configuration State: OCPPCommCtrlr.NetworkProfileConnectionAttempts is 3 OCPPCommCtrlr.RetryBackOffRepeatTimes is 2 OCPPCommCtrlr.RetryBackOffRandomRange is 0 OCPPCommCtrlr.RetryBackOffWaitMinimum is <Configured RetryBackOffWaitMinimum>

Before (Preparations)
Memory State: N/a
Reusable State(s): N/a

6.2.21. Page 89/621 - (2025-04) - Addition of new testcase for CSMS to test WebSocket Subprotocol negotiation

TC_B_58_CSMS: WebSocket Subprotocol negotiation

Test case name	WebSocket Subprotocol validation
Test case Id	TC_B_58_CSMS
Use case Id(s)	Part 4 - JSON over WebSockets implementation guide
Requirement(s)	Section 3.1.2. OCPP version
System under test	CSMS
Description	OCPP-J imposes extra constraints on the WebSocket subprotocol, detailed in the following section 3.1.2.
Purpose	To verify whether the CSMS is able to select a supported OCPP version, when also a different unsupported version is supported by the Charging Station and relays this selection via the Sec-WebSocket-Protocol header.
Prerequisite(s)	N/a

Before (Preparations)
Configuration State: N/a
Memory State: N/a
Reusable State(s): N/a

Main (Test scenario)	
Charging Station	CSMS
1. The Test System disconnects the WebSocket connection and reconnects by sending a HTTP upgrade request with the header; Sec-WebSocket-Protocol: ocpp0.1	2. The CSMS rejects the connection attempt and does NOT upgrade the connection to a WebSocket connection.
3. The Test System disconnects the WebSocket connection and reconnects by sending a HTTP upgrade request with the header; Sec-WebSocket-Protocol: ocpp0.1, ocpp<Selected OCPP version>	4. The CSMS accepts the connection attempt and upgrades the connection to a WebSocket connection.

Tool validations
* Step 4: The authorization header of the HTTP upgrade response must contain the header Sec-Websocket-Protocol, and it must comply to the following: - The header is formatted as follows; Sec-WebSocket-Protocol: ocpp<Selected OCPP version>
Post scenario validations: N/a

6.2.22. Page 117/141 - (2025-02) - Added remote support for Authorization Cache & Local Authorization List testcases

The CWG decided that online local authorization list and authorization cache testcase can also be tested with remote authorization.

...	...
Test case Id	TC_C_08_CS, TC_C_09_CS, TC_C_10_CS, TC_C_11_CS, TC_C_12_CS, TC_C_13_CS, TC_C_33_CS, TC_C_34_CS, TC_C_36_CS, TC_C_37_CS, TC_C_38_CS
...	...
Prerequisite(s)	AuthCacheCtrlr.Available is implemented with value true The Charging Station has an authorization cache AND the Charging Station supports at least one of the following local start authorization options: C-30, C-31, C-32, C-34 OR supports at least one of the following remote start authorization options: C-36, C-37

...	...
Test case Id	TC_C_57_CS
...	...
Prerequisite(s)	AuthCacheCtrlr.Available is implemented with value true AuthCacheCtrlr.DisablePostAuthorize is implemented AuthCacheCtrlr.DisablePostAuthorize is implemented AND The Charging Station has an authorization cache AND the Charging Station supports at least one of the following local start authorization options: C-30, C-31, C-32, C-34 OR supports at least one of the following remote start authorization options: C-36, C-37

...	...
Test case Id	TC_C_27_CS, TC_C_28_CS, TC_C_29_CS, TC_C_30_CS, TC_C_31_CS
...	...
Prerequisite(s)	- LocalAuthListCtrlr.LocalAuthListAvailable is implemented with value true AND - The Charging Station must support an authorization method other than NoAuthorization or Central

...	...
Test case Id	TC_C_58_CS
...	...
Prerequisite(s)	- LocalAuthListCtrlr.Available is implemented with value true - LocalAuthListCtrlr.DisablePostAuthorize is implemented AND - The Charging Station must support an authorization method other than NoAuthorization or Central

...	...
Test case Id	TC_C_32_CS
...	...
Prerequisite(s)	The Charging Station supports the Authorization Cache feature Authorization cache is stored in the non-volatile memory. The Charging Station has an authorization cache AND Authorization cache is stored in the non-volatile memory AND the Charging Station supports at least one of the following local start authorization options: C-30, C-31, C-32, C-34 OR supports at least one of the following remote start authorization options: C-36, C-37

Main (Test scenario)	
Main steps have been updated to use the Reusable State Authorized based on the configured scenario Local or Remote.	
...	

6.2.23. Page 101 - (2025-04) - TC_C_14_CS - Fixing invalid component variable reference

Test case name	Authorization through authorization cache - GroupID equal to MasterPassGroupId.
...	...
Prerequisite(s)	AuthCacheCtrlr.AuthCacheAvailable AuthCacheCtrlr.Available is implemented with value true The Charging station supports MasterPass feature. The Charging Station supports authorization methods other than NoAuthorization

6.2.24. Page 102 - (2025-04) - TC_C_15_CS - Fixing invalid component variable reference

Test case name	Authorization through authorization cache - StopTxOnInvalidId = false, MaxEnergyOnInvalidId > 0
...	...
Prerequisite(s)	- AuthCacheCtrlr.AuthCacheAvailable AuthCacheCtrlr.Available is implemented with value true - The Charging Station has MaxEnergyOnInvalidId implemented - At least one of the following must be supported; Local auth list, auth cache, StartTxUnknownIds. - The Charging Station supports authorization methods other than NoAuthorization

6.2.25. Page 104 - (2025-04) - TC_C_16_CS - Fixing invalid component variable reference

Test case name	Authorization through authorization cache - StopTxOnInvalidId = true
...	...
Prerequisite(s)	- AuthCacheCtrlr.AuthCacheAvailable AuthCacheCtrlr.Available is implemented with value true - At least one of the following must be supported; Local auth list, auth cache, StartTxUnknownIds. - The Charging Station supports authorization methods other than NoAuthorization

6.2.26. Page 105 - (2025-04) - TC_C_17_CS - Fixing invalid component variable reference

Test case name	Authorization through authorization cache - StopTxOnInvalidId = false
...	...
Prerequisite(s)	- AuthCacheCtrlr.AuthCacheAvailable AuthCacheCtrlr.Available is implemented with value true - At least one of the following must be supported; Local auth list, auth cache, StartTxUnknownIds. - The Charging Station supports authorization methods other than NoAuthorization

6.2.27. Page 108 - (2025-04) - TC_C_18_CS - Fixing invalid component variable reference

Test case name	Authorization through authorization cache - StopTxOnInvalidId = true, MaxEnergyOnInvalidId > 0
...	...
Prerequisite(s)	- AuthCacheCtrlr.AuthCacheAvailable AuthCacheCtrlr.Available is implemented with value true - The Charging Station has MaxEnergyOnInvalidId implemented. - At least one of the following must be supported; Local auth list, auth cache, StartTxUnknownIds. - The Charging Station supports authorization methods other than NoAuthorization

6.2.28. Page 120 - (2025-04) - TC_C_34_CS - Making use of the Deauthorized reusable state

Test case name	Store Authorization Data in the Authorization Cache - Update on TransactionResponse
Test case Id	TC_C_34_CS
...	...

Before (Preparations)
Configuration State: AuthCacheEnabled is true (If implemented) LocalPreAuthorize is true LocalAuthListEnabled is true StopTxOnInvalidId is true MaxEnergyOnInvalidId is 0

Before (Preparations)
Memory State: <i>IdTokenCached</i> for <Configured valid IdToken fields>
Reusable State(s): N/a

Main (Test scenario)	
Charging Station	CSMS
1. Execute Reusable State <i>Authorized</i> (Cached idToken, idTokenInfo.status invalid)	
2. Execute Reusable State <i>Deauthorized</i>	
3. Execute Reusable State <i>EVDIsconnected</i>	
4. Execute Reusable State <i>ParkingBayUnoccupied</i>	
5. Execute Reusable State <i>ParkingBayOccupied</i>	
6. Execute Reusable State <i>Authorized</i> (idTokenInfo.status invalid)	

6.2.29. Page 128 - (2025-04) - TC_C_41_CS - Fixing invalid component variable reference

Test case name	Authorization by GroupId - Success with Authorization Cache
...	...
Prerequisite(s)	- AuthCacheCtrlr.AuthCacheAvailable AuthCacheCtrlr.Available is implemented with value true - The Charging Station supports authorization methods other than NoAuthorization

6.2.30. Page 134 - (2025-04) - TC_C_44_CS - Fixing invalid component variable reference

Test case name	Authorization by GroupId - Invalid status with Authorization Cache
...	...
Prerequisite(s)	- AuthCacheCtrlr.AuthCacheAvailable AuthCacheCtrlr.Available is implemented with value true - The Charging Station supports authorization methods other than NoAuthorization

6.2.31. Page 137 - (2025-04) - TC_C_46_CS - Fixing invalid component variable reference

Test case name	Store Authorization Data in the Authorization Cache - AuthCacheLifeTime
...	...
Prerequisite(s)	- AuthCacheCtrlr.AuthCacheAvailable AuthCacheCtrlr.Available is implemented with value true - Configuration variable AuthCacheLifeTime is implemented

6.2.32. Page 138 - (2024-09) - TC_C_47_CS - StoppedReason must be validated in Ended event [020-4467]

Test case name	Stop Transaction with a Master Pass - With UI - All transactions	
Test case Id	TC_C_47_CS	
...		
Main (Test scenario)	Charging Station	CSMS
	...	...

Test case name	Stop Transaction with a Master Pass - With UI - All transactions
Tool validations	* Step 1: Message AuthorizeRequest - idToken.idToken <Configured masterpass_idtoken_idtoken> - idToken.type <Configured masterpass_idtoken_type> * Step 3: Message TransactionEventRequest - transactionInfo.stoppedReason MasterPass (in last TransactionEventRequest) - idToken omit or - idToken.idToken <Configured masterpass_idtoken_idtoken> and - idToken.type <Configured masterpass_idtoken_type> (once per stopped transaction) - eventType Ended (in last TransactionEventRequest) Post scenario validations: - N/a

6.2.33. Page 145 - (2025-04) - TC_C_25_CS - Fixing invalid component variable reference

Test case name	Offline authorization through local authorization list - Local Authorization List > Authorization Cache
...	...
Prerequisite(s)	- LocalAuthListCtrlr.LocalAuthListAvailable is implemented with value true - AuthCacheCtrlr.AuthCacheAvailable AuthCacheCtrlr.Available is implemented with value true - OfflineTxForUnknownIdEnabled is implemented. - The Charging Station supports authorization methods other than NoAuthorization

6.2.34. Page 146 - (2024-11) - TC_C_26_CS - Allow StatusNotification status = Occupied

The connector status change should be reported after the connection is restored. The Charging Station should be allowed to report StatusNotificationRequest status = Occupied (or NotifyEventRequest).

Test case name	Set Variable Monitoring - Periodic event	
Test case Id	TC_C_26_CS	
Use case Id(s)	C15 & C13	
Requirement(s)	C15.FR.02,C15.FR.06,C15.FR.08,C13.FR.04	
...		
Main (Test scenario)	Charging Station	CSMS
	...	...
	1. The Charging Station notifies the CSMS about the current state of all connectors.	2. The OCTT responds accordingly.
	...	...
	3. Execute Reusable State <i>StopAuthorized</i>	
	4. Execute Reusable State <i>EVConnectedPostSession</i>	
	5. Execute Reusable State <i>EVDisconnected</i>	

Test case name	Set Variable Monitoring - Periodic event
Tool validations	* Step 1: Message: StatusNotificationRequest - connectorStatus must be <i>Occupied</i> Message: NotifyEventRequest - eventData[0].trigger must be <i>Delta</i> - eventData[0].actualValue must be <i>Occupied</i> - eventData[0].component.name must be <i>Connector</i> - eventData[0].variable.name must be <i>AvailabilityState</i> ... Post scenario validations: N/A

6.2.35. Page 147-153 - (2025-04) - TC_C_50_CS, TC_C_51_CS, TC_C_52_CS, TC_C_53_CS, TC_C_54_CS, TC_C_55_CS - Always re-install V2G certificates

Note: This erratum revises erratum: [Page 147 - \(2025-02\) - TC_C_50_CS](#) and the related errata below

It is not possible to detect whether all expected certificates are installed, when reusing the SubCAs. Therefore the Test System always re-installs the V2G certificates.

Before (Preparations)
Memory State: <i>CertificateInstalled</i> for certificateType V2GRootCertificate <i>CertificateInstalled</i> for certificateType MORootCertificate <i>RenewV2GChargingStationCertificate</i> (If none are present, when checking with <code>GetInstalledCertificateIds.certificateType = V2GCertificateChain</code>)

6.2.36. Page 147 - (2025-02) - TC_C_50_CS

Note: This erratum and the related errata below are revised by erratum: [Page 147-153 - \(2025-04\) - TC_C_50_CS, TC_C_51_CS, TC_C_52_CS, TC_C_53_CS, TC_C_54_CS, TC_C_55_CS - Always re-install V2G certificates](#)

Before (Preparations)
Configuration State: TxCtrlr.TxStartPoint contains one or more of <i>PowerPathClosed, Authorized, EVConnected, ParkingBayOccupancy</i> AuthCtrlr.AuthEnabled is <i>true</i> (If implemented AND ReadWrite) AuthCtrlr.DisableRemoteAuthorization is <i>false</i> (If implemented) For the ISO15118Ctrlr of the EVSE used in the PnC transaction: ISO15118Ctrlr.CentralContractValidationAllowed is <i>false</i> ISO15118Ctrlr.ContractCertificateInstallationEnabled is <i>true</i> ISO15118Ctrlr.V2GCertificateInstallationEnabled is <i>true</i> ISO15118Ctrlr.PnCEnabled is <i>true</i> ISO15118Ctrlr.Secoid is configured <i>secoid</i> ISO15118Ctrlr.CountryName is <i>NL</i> ISO15118Ctrlr.OrganizationName is configured <i>vendorId</i> . Memory State: <i>CertificateInstalled</i> for certificateType V2GRootCertificate <i>CertificateInstalled</i> for certificateType MORootCertificate <i>RenewV2GChargingStationCertificate</i> (If none are present, when checking with <code>GetInstalledCertificateIds.certificateType = V2GCertificateChain</code>) Reusable State(s): State is <i>EVConnectedPreSession</i>

Main (Test scenario)	
Charging Station	CSMS
[...]	
3. The Charging Station sends a TransactionEventRequest <u>Note(s):</u> This step needs to be executed when TxStartPoint contains Authorized OR the transaction already started. So in the case TxStartPoint contains ParkingBayOccupancy or (EVConnected, in the case this testcase was initiated from state EVConnectedPreSession.)	4. The Test System responds with a TransactionEventResponse With idTokenInfo.status Accepted
5. Execute Reusable State EnergyTransferStarted	

6.2.37. Page 148 - (2025-02) - TC_C_51_CS

Before (Preparations)
Configuration State: TxCtrlr.TxStartPoint contains one or more of <i>PowerPathClosed, Authorized, EVConnected, ParkingBayOccupancy</i> AuthCtrlr.AuthEnabled is <i>true</i> (If implemented AND ReadWrite) AuthCtrlr.DisableRemoteAuthorization is <i>false</i> (If implemented) For the ISO15118Ctrlr of the EVSE used in the PnC transaction: ISO15118Ctrlr.CentralContractValidationAllowed is <i>false</i> ISO15118Ctrlr.PnCEnabled is <i>true</i>
Memory State: <i>CertificateInstalled</i> for certificateType <i>V2GRootCertificate</i> <i>CertificateInstalled</i> for certificateType <i>MORootCertificate</i> <i>RenewV2GChargingStationCertificate</i> (If none are present, when checking with <i>GetInstalledCertificateIds.certificateType = V2GCertificateChain</i>)
Reusable State(s): State is <i>EVConnectedPreSession</i>

6.2.38. Page 149 - (2025-02) - TC_C_52_CS

TC_C_52_CS: Authorization using Contract Certificates 15118 - Online - Central contract certificate validation - Accepted

Test case name	Authorization using Contract Certificates 15118 - Online - Central contract certificate validation - Accepted
Test case Id	TC_C_52_CS
Use case Id(s)	C07
Requirement(s)	C07.FR.01,C07.FR.02,C07.FR.06
System under test	Charging Station
Description	The Charging Station is able to authorize with contract certificates when it supports ISO 15118.
Purpose	To verify if the Charging Station is able to authorize, while not being able to locally validate the contract certificate and then send it to the CSMS.
Prerequisite(s)	- The V2G /MO Root certificate that is needed to validate the EV Contract certificate must NOT be installed at the Charging Station. - The Charging Station supports central contract validation.

Before (Preparations)	
Configuration State: TxCtrlr.TxStartPoint contains one or more of <i>PowerPathClosed, Authorized, EVConnected, ParkingBayOccupancy</i> AuthCtrlr.AuthEnabled is <i>true</i> (If implemented AND ReadWrite) AuthCtrlr.DisableRemoteAuthorization is <i>false</i> (If implemented) For the ISO15118Ctrlr of the EVSE used for the PnC transaction: ISO15118Ctrlr.CentralContractValidationAllowed is <i>true</i> ISO15118Ctrlr.ContractCertificateInstallationEnabled is <i>true</i> ISO15118Ctrlr.V2GCertificateInstallationEnabled is <i>true</i> ISO15118Ctrlr.PnCEnabled is <i>true</i> ISO15118Ctrlr.Secoid is configured <i>secoid</i> ISO15118Ctrlr.CountryName is <i>NL</i> ISO15118Ctrlr.OrganizationName is configured <i>vendorId</i> 	
Memory State: N/a CertificateInstalled for certificateType <i>V2GRootCertificate</i> RenewV2GChargingStationCertificate (If none are present, when checking with <i>GetInstalledCertificateIds.certificateType = V2GCertificateChain</i>)	
Reusable State(s): State is <i>EVConnectedPreSession</i>	
Main (Test scenario)	
Charging Station	CSMS
[...]	
3. The Charging Station sends a TransactionEventRequest <u>Note(s):</u> This step needs to be executed when TxStartPoint contains Authorized OR the transaction already started. So in the case TxStartPoint contains ParkingBayOccupancy or (EVConnected, in the case this testcase was initiated from state EVConnectedPreSession.)	4. The Test System responds with a TransactionEventResponse With idTokenInfo.status <i>Accepted</i>
5. Execute Reusable State <i>EnergyTransferStarted</i>	

6.2.39. Page 150 - (2025-02) - TC_C_53_CS

TC_C_53_CS: Authorization using Contract Certificates 15118 - Online - Central contract validation fails

Test case name	Authorization using Contract Certificates 15118 - Online - Central contract validation fails
Test case Id	TC_C_53_CS
Use case Id(s)	C07
Requirement(s)	N/a
System under test	Charging Station
Description	The Charging Station is able to authorize with contract certificates when it supports ISO 15118.
Purpose	To verify if the Charging Station is able to handle an invalid contract certificate.
Prerequisite(s)	- The V2G /MO Root certificate that is needed to validate the EV Contract certificate must NOT be installed at the Charging Station. - The Charging Station supports central contract validation.

Before (Preparations)
Configuration State: TxCtrlr.TxStartPoint contains one or more of <i>PowerPathClosed, Authorized, EVConnected, ParkingBayOccupancy</i> AuthCtrlr.AuthEnabled is <i>true</i> (If implemented AND ReadWrite) AuthCtrlr.DisableRemoteAuthorization is <i>false</i> (If implemented) For the ISO15118Ctrlr of the EVSE involved in the PnC transaction: ISO15118Ctrlr.CentralContractValidationAllowed is <i>true</i> ISO15118Ctrlr.PnCEnabled is <i>true</i>
Memory State: N/a <i>CertificateInstalled</i> for certificateType <i>V2GRootCertificate</i> <i>RenewV2GChargingStationCertificate</i> (If none are present, when checking with <i>GetInstalledCertificateIds.certificateType = V2GCertificateChain</i>)
Reusable State(s): State is <i>EVConnectedPreSession</i>

6.2.40. Page 151 - (2024-09) - TC_C_54_CS - removed reusable state IdTokenCached [O20-3510]

Test case name	Authorization using Contract Certificates 15118 - Offline - ContractValidationOffline is true	
Test case Id	TC_C_54_CS	
...		
Before (Preparations)	Configuration State:	
	...	
	Memory State:	
	<i>CertificateInstalled</i> for certificateType <i>V2GRootCertificate</i> <i>CertificateInstalled</i> for certificateType <i>MORootCertificate</i> <i>IdTokenCached</i> for <Configured valid IdToken fields> (If implemented) <i>IdTokenLocalAuthList</i> for <Configured valid IdToken fields> (If implemented)	
Main (Test scenario)	Reusable State(s):	
	N/a	
	Charging Station	CSMS
...	...	...

6.2.41. Page 151 - (2025-02) - TC_C_54_CS

Test case name	Authorization using Contract Certificates 15118 - Offline - ContractValidationOffline is true	
[...]		
Description	The Charging Station is able to authorize with contract certificates when it supports ISO 15118- for an EMAID that exists in authorization cache or local authorization list, while offline.	
[...]		

Before (Preparations)
[...]
Memory State: <i>CertificateInstalled</i> for certificateType <i>V2GRootCertificate</i> <i>CertificateInstalled</i> for certificateType <i>MORootCertificate</i> <i>RenewV2GChargingStationCertificate</i> (If none are present, when checking with <i>GetInstalledCertificateIds.certificateType = V2GCertificateChain</i>) <i>IdTokenCached</i> for <Configured valid IdToken fields> (If implemented) <i>IdTokenLocalAuthList</i> for <Configured valid IdToken fields> (If implemented)
Reusable State(s): N/a

Main (Test scenario)	
Charging Station	CSMS
[...]	
5. The Charging Station sends a TransactionEventRequest <u>Note(s):</u> - This step needs to be executed when TxStartPoint contains EVConnected OR the transaction already started. So in the case TxStartPoint contains ParkingBayOccupancy (in the case TxStartPoint contains ParkingBayOccupancy_)	6. The Test System responds with a TransactionEventResponse
7. The Charging Station sends a TransactionEventRequest <u>Note(s):</u> - This step needs to be executed when TxStartPoint contains Authorized OR the transaction already started. So in the case TxStartPoint contains ParkingBayOccupancy or EVConnected.	8. The Test System responds with a TransactionEventResponse With idTokenInfo.status Accepted
9. Execute Reusable State <i>EnergyTransferStarted</i>	

6.2.42. Page 153 - (2024-09) - TC_C_55_CS - removed reusable state IdTokenCached [020-3510]

Test case name	Authorization using Contract Certificates 15118 - Offline - ContractValidationOffline is false	
Test case Id	TC_C_55_CS	
...		
Before (Preparations)	Configuration State: ...	
	Memory State: CertificateInstalled for certificateType V2GRootCertificate CertificateInstalled for certificateType MORootCertificate IdTokenCached for <Configured valid IdToken fields> (If implemented) IdTokenLocalAuthList for <Configured valid IdToken fields> (If implemented)	
	Reusable State(s): N/a	
	Main (Test scenario)	Charging Station
	...	...
...		

6.2.43. Page 153 - (2025-02) - TC_C_55_CS

Test case name	Authorization using Contract Certificates 15118 - Offline - ContractValidationOffline is false
[...]	
Description	The Charging Station is able to authorize with contract certificates when it supports ISO 15118. The Charging Station will not authorize with contract certificates when offline.
[...]	

Before (Preparations)
Configuration State: TxCtrlr.TxStartPoint contains one or more of <i>PowerPathClosed, Authorized, EVConnected, ParkingBayOccupancy</i> AuthCtrlr.AuthEnabled is <i>true</i> (If implemented AND ReadWrite) AuthCacheCtrlr.Enabled is <i>true</i> OR LocalAuthListCtrlr.Enabled is <i>true</i> OfflineTxForUnknownIdEnabled is <i>true</i> (If implemented) OfflineTxForUnknownIdEnabled is <i>false</i> (If implemented) OfflineThreshold is <Configured <i>RetryBackOffWaitMinimum_duration</i> > + 60.0 RetryBackOffWaitMinimum is <Configured <i>RetryBackOffWaitMinimum_duration</i> > RetryBackOffRandomRange is 0 <u>Note:</u> <Configured <i>RetryBackOffWaitMinimum_duration</i> should be long enough to execute manual tasks> For the ISO15118Ctrlr of the EVSE involved in the PnC transaction: ISO15118Ctrlr.ContractValidationOffline is <i>false</i> ISO15118Ctrlr.PnCEnabled is <i>true</i>
Memory State: <i>CertificateInstalled</i> for certificateType <i>V2GRootCertificate</i> <i>CertificateInstalled</i> for certificateType <i>MORootCertificate</i> <i>RenewV2GChargingStationCertificate</i> (If none are present, when checking with <i>GetInstalledCertificateIds.certificateType = V2GCertificateChain</i>) <i>IdTokenCached</i> for <Configured valid <i>IdToken</i> fields> (If implemented) <i>IdTokenLocalAuthList</i> for <Configured valid <i>IdToken</i> fields> (If implemented)
Reusable State(s): N/a

Main (Test scenario)	
Charging Station	CSMS
[...]	
5. The Charging Station sends a TransactionEventRequest <u>Note(s):</u> - This step needs to be executed when TxStartPoint contains <i>EVConnected</i> OR the transaction already started - So in the case TxStartPoint contains <i>ParkingBayOccupancy</i> (in the case TxStartPoint contains <i>ParkingBayOccupancy</i>)	6. The Test System responds with a TransactionEventResponse
[...]	

6.2.44. Page 165/169 - (2025-02) -

TC_E_01_CS/TC_E_09_CS/TC_E_10_CS/TC_E_12_CS/TC_E_13_CS - Extended the testcases until the Charging Station start charging

This is needed to test the TransactionEventRequest with eventType Updated messages, while specific TxStartPoints are configured.

Test case name	Start transaction options - PowerPathClosed
Test case Id	TC_E_01_CS
...	...

Main (Test scenario)	
Charging Station	CSMS
1. Execute Reusable State <i>EVConnectedPreSession</i>	
2. Execute Reusable State <i>EnergyTransferStarted</i>	

Test case name	Start transaction options - EVConnected
Test case Id	TC_E_09_CS
...	...

Main (Test scenario)	
Charging Station	CSMS
1. Execute Reusable State <i>EVConnectedPreSession</i>	
2. Execute Reusable State <i>Authorized</i>	
3. Execute Reusable State <i>EnergyTransferStarted</i>	

Test case name	Start transaction options - Authorized - Local
Test case Id	TC_E_10_CS
...	...

Main (Test scenario)	
Charging Station	CSMS
1. Execute Reusable State <i>Authorized</i> (Local)	
2. Execute Reusable State <i>EVConnectedPreSession</i>	
3. Execute Reusable State <i>EnergyTransferStarted</i>	

Test case name	Start transaction options - ParkingBayOccupied
Test case Id	TC_E_12_CS
...	...

Main (Test scenario)	
Charging Station	CSMS
1. Execute Reusable State <i>ParkingBayOccupied</i>	
2. Execute Reusable State <i>Authorized</i>	
3. Execute Reusable State <i>EVConnectedPreSession</i>	
4. Execute Reusable State <i>EnergyTransferStarted</i>	

Test case name	Start transaction options - Authorized - Remote
Test case Id	TC_E_13_CS
...	...

Main (Test scenario)	
Charging Station	CSMS
1. Execute Reusable State <i>Authorized</i> (Remote)	
2. Execute Reusable State <i>EVConnectedPreSession</i>	
3. Execute Reusable State <i>EnergyTransferStarted</i>	

6.2.45. Page 174 - (2025-04) - TC_E_17_CS - Aligned configuration before steps with updated prerequisites

Note: This erratum extends erratum: [Page 174 - \(2024-09\) - TC_E_17_CS - Updated prerequisite for test case to correctly specify the applicable TxStopPoint combinations](#)

The prerequisite was updated, but the before configuration steps were not aligned with this change.

Test case name	Stop transaction options - Deauthorized - EV side disconnect
Test case Id	TC_E_17_CS
...	...

Before (Preparations)
Configuration State: If the mutability of TxStopPoint is <i>ReadWrite</i> then TxStopPoint contains PowerPathClosed AND/OR Authorized StopTxOnEVSideDisconnect is <i>true</i> UnlockOnEVSideDisconnect is <i>false</i> AuthCtrlr.AuthEnabled is <i>true</i> (If implemented AND <i>ReadWrite</i>) AuthCtrlr.DisableRemoteAuthorization is <i>false</i> (If implemented)

6.2.46. Page 174 - (2024-09) - TC_E_17_CS - Updated prerequisite for test case to correctly specify the applicable TxStopPoint combinations

Note: This erratum is extended by erratum: [Page 174 - \(2025-04\) - TC_E_17_CS - Aligned configuration before steps with updated prerequisites](#)

This testcase allows for a limited set of TxStopPoint combinations, otherwise it is not applicable.

Test case name	Stop transaction options - Deauthorized - EV side disconnect
Test case Id	TC_E_17_CS
...	
Old: Prerequisite(s)	- The Charging Station does NOT have the following configuration; The mutability of TxStopPoint is <i>ReadOnly</i> AND the value <i>Authorized</i> OR <i>PowerPathClosed</i> is NOT set OR (<i>EnergyTransfer</i> OR <i>DataSigned</i> OR <i>EVConnected</i> is set). - If the mutability of TxStopPoint is <i>_ReadWrite</i> , then the value <i>Authorized</i> OR <i>PowerPathClosed</i> must be supported.
New: Prerequisite(s)	This testcase is applicable if the value <i>Authorized</i> is a supported value for TxStopPoint AND <i>EVConnected</i> , <i>PowerPathClosed</i> and <i>EnergyTransfer</i> must not be set as TxStopPoint AND StopTxOnEVSideDisconnect <i>true</i> must be a supported value.

6.2.47. Page 176 - (2024-11) - TC_E_39_CS - Missing StatusNotificationRequest/NotifyEventRequest

Test case name	Stop transaction options - Deauthorized - timeout	
Test case Id	TC_E_39_CS	
...		
Main (Test scenario)	Charging Station	CSMS
	...	...
	<u>Manual Action:</u> Connect the EV and EVSE on EV side.	
	<u>Manual Action:</u> Connect the EV and EVSE on EVSE side.	
	3. The Charging Station notifies the CSMS about the status change of the connector.	4. The OCTT responds accordingly.
	...	...

Test case name	Stop transaction options - Deauthorized - timeout
Tool validations	* Step 1: Message: TransactionEventRequest - triggerReason must be <i>EVConnectTimeout</i> - eventType must be <i>Updated</i> if TxStartPoint is <i>ParkingBayOccupancy</i>, else <i>Ended</i> - transactionInfo.stoppedReason must be <i>Timeout</i> * Step 3: Message: StatusNotificationRequest - connectorStatus must be <i>Occupied</i> Message: NotifyEventRequest - eventData[0].trigger must be <i>Delta</i> - eventData[0].actualValue must be <i>Occupied</i> - eventData[0].component.name must be <i>Connector</i> - eventData[0].variable.name must be <i>AvailabilityState</i> * Step 5: Message: TransactionEventRequest - triggerReason can only be <i>CablePluggedIn</i> - transactionInfo.chargingState should not be <i>Charging</i> - eventType must be <i>Updated</i> if TxStartPoint is <i>ParkingBayOccupancy</i>, else <i>Ended</i> Post scenario validations: N/a

6.2.48. Page 182 - (2025-04) - TC_E_52_CS - Testcase is not able to determine the authorization is refused in case TxStartPoint is not Authorized

The testcase is not able to verify the authorization is refused when the TxStartPoint is not Authorized. Therefore the testcase is extended to also plugin the cable to determine whether the authorization was actually refused.

Test case name	Local start transaction - Authorization first - DisableRemoteAuthorization
Test case Id	TC_E_52_CS
...	

Main (Test scenario)	
Charging Station	CSMS
<u>Manual Action:</u> Present an idToken which is not configured in the Local Authorization List nor present in Authorization Cache.	
1. The Charging Station does NOT send a AuthorizeRequest	
2. Execute Reusable State <i>EVConnectedPreSession</i>	
3. The Charging Station does NOT start charging.	

6.2.49. Page 185 - (2024-09) - TC_E_35_CS - StoppedReason must be validated in Ended event [020-4467]

Test case name	Stop transaction options - PowerPathClosed - Remote stop	
Test case Id	TC_E_35_CS	
...		
Main (Test scenario)	Charging Station	CSMS
	...	...

Test case name	Stop transaction options - PowerPathClosed - Remote stop
Tool validations	* Step 2: Message: RequestStopTransactionResponse - status must be <i>Accepted</i> * Step 3: Message: TransactionEventRequest - triggerReason must be <i>RemoteStop</i> (for one of the TransactionEventRequests) - transactionInfo.stoppedReason must be <i>Remote</i> (for the last TransactionEventRequest) - eventType must be <i>Ended</i> (for the last TransactionEventRequest) Post scenario validations: N/a

6.2.50. Page 188 - (2025-02) - TC_E_22_CS - Stop transaction options - EnergyTransfer stopped - will end transaction

When TxStopPoint is EnergyTransfer this will end transaction when EV suspends energy.

TC_E_22_CS: Stop transaction options - EnergyTransfer stopped - SuspendedEV

Test case name	Stop transaction options - EnergyTransfer stopped - SuspendedEV
Test case Id	TC_E_22_CS
..	..

Tool validations
* Step 1: Message: TransactionEventRequest - triggerReason must be <i>ChargingStateChanged</i> - transactionInfo.chargingState must be <i>EVConnected</i> -OR - transactionInfo.chargingState must be <i>SuspendedEV</i> - AND - transactionInfo.stoppedReason must be <i>StoppedByEV</i> - eventType must be <i>Ended</i> (if chargingState is <i>EVConnected</i>) -OR - eventType must be <i>Updated</i> (if chargingState is <i>SuspendedEV</i>) Post scenario validations: N/a

6.2.51. Page 189 - (2025-06) - TC_E_14_CS - StoppedReason validation too strict for remote

Test case name	Stop transaction options - EVDisconnected - Charging Station side
Test case Id	TC_E_14_CS
...	

Tool validations
* Step 1: Message: StatusNotificationRequest - connectorStatus must be <i>Available</i> Message: NotifyEventRequest - eventData[0].trigger must be <i>Delta</i> - eventData[0].actualValue must be <i>Available</i> - eventData[0].component.name must be <i>Connector</i> - eventData[0].variable.name must be <i>AvailabilityState</i> * Step 3: Message: TransactionEventRequest - triggerReason must be <i>EVCommunicationLost</i> - transactionInfo.chargingState must be <i>Idle</i> - If the Test System is configured to stop transactions using a RequestStopTransactionRequest message then transactionInfo.stoppedReason must be <i>Remote</i> or <i>EVDisconnected</i>. Else transactionInfo.stoppedReason must be <i>Local</i>, <i>EVDisconnected</i> or be omitted. - eventType must be <i>Ended</i>
Post scenario validations: N/a

6.2.52. Page 199 - (2025-04) - TC_E_27_CS - Remove manual action between step 4/5

There seems to have been copy/paste errors between testcase TC_E_26_CS and TC_E_27_CS. The cable should not have been reconnected between these steps as we are trying to trigger the timeout.

Test case name	Disconnect cable on EV-side - Suspend transaction - Fixed cable connection timeout
Test case Id	TC_E_27_CS
...	

Main (Test scenario)	
Charging Station	CSMS
<u>Manual Action:</u> Disconnect the EV and EVSE on EV side (EVSE loses connection with EV).	
1. The Charging Station sends a TransactionEventRequest	2. The Test System responds with a TransactionEventResponse
3. The Charging Station notifies the CSMS about the current state of the connector.	4. The Test System responds accordingly.
<u>Note(s):</u> - This step needs to be executed when the Charging Station has a permanently attached cable on the Charging Station side.	
<u>Manual Action:</u> Reconnect the EV and EVSE on EV side. <u>Note(s):</u> - If the Charging Station has a permanently attached cable on the Charging Station side, then this step needs to be executed before the configured EVConnectionTimeout expires.	
5. The Charging Station sends a TransactionEventRequest	6. The Test System responds with a TransactionEventResponse
<u>Note(s):</u> Optionally the Charging Station can send a StatusNotificationRequest or NotifyEventRequest with status Available	

6.2.53. Page 204 - (2025-02) - TC_E_31_CS - Add steps for when running the testcase in Remote mode

As the Charging Station will go offline, the transaction must be stopped from EV side in case of remote authorization.

TC_E_31_CS: Check Transaction status - Transaction with id ended - with message in queue

Test case name	Check Transaction status - Transaction with id ended - with message in queue
Test case Id	TC_E_31_CS
...	...

Before (Preparations)

Configuration State:

...

UnlockOnEVSideDisconnect is *true* (If ReadWrite)

...

Main (Test scenario)

Charging Station	CSMS
	The OCTT closes the WebSocket connection AND does not accept a reconnect.

Manual Action: Present the same idToken as used to start the transaction.

Notes(s): Only if configured scenario is local

Manual Action: Stop the energy transfer via the EV.

Notes(s): Only if configured scenario is remote

Manual Action: Disconnect the EV and EVSE.

Manual Action: Drive EV out of parking bay. (Only needed if TxStopPoint is ParkingBayOccupancy)

...

Tool validations

...

* Step 3:

Message: **TransactionEventRequest**

The tool validations from the reusable states need to be used to verify whether all required TransactionEventRequests have been received.

From *StopAuthorized* through *ParkingBayUnoccupied* (in case of scenario *Local*).

And from *EnergyTransferSuspended* through *ParkingBayUnoccupied* (in case of scenario *Remote*).

Post scenario validations:

N/a

6.2.54. Page 214 - (2024-06) TC_E_43_CS Move reusable state TransactionEventsInQueueEnded to Before [768]

State TransactionEventsInQueueEnded is moved to Before stage.

Test Case Id: TC_E_43_CS

Test case name	Offline Behaviour - Transaction during offline period
Test case Id	TC_E_43_CS
Use case Id(s)	E12
Requirement(s)	E12.FR.01,E12.FR.02,E12.FR.06
System under test	Charging Station
Description	The Charging Station queues TransactionEvent messages to inform the CSMS that a transaction occurred while the Charging Station was Offline.
Purpose	To verify if the Charging Station is able to queue TransactionEvent messages while it was offline.
Prerequisite(s)	The Charging Station supports authorization methods other than NoAuthorization

Test case name	Offline Behaviour - Transaction during offline period	
Before (Preparations)	Configuration State: N/a	
	Memory State: N/a	
	Reusable State(s): State is TransactionEventsInQueueEnded	
Main (Test scenario)	Charging Station	CSMS
	1. Execute Reusable State TransactionEventsInQueueEnded	
	1. The Charging Stations sends a TransactionEventRequest Note(s): - The Charging Station will empty its Transaction message queue. This will contain one or more TransactionEventRequest messages	2. The OCTT responds with a TransactionEventResponse
Tool validations	* Step 1: All messages: TransactionEventRequest - offline must be <i>true</i> One of the messages: TransactionEventRequest - eventType <i>Started</i> One of the messages: TransactionEventRequest - eventType <i>Ended</i>	
	Post scenario validations: N/a	

NOTE | If the Charging Station supports ISO15118, this testcase needs to be executed using EIM.

6.2.55. Page 217 - (2025-04) - TC_E_46_CS - Testcase updated to use the specialized Authorized15118 reusable state

Testcase updated to use the specialized Authorized15118 reusable state for starting the transaction, instead of the standard Authorized reusable state.

Test case name	End of charging process 15118
Test case Id	TC_E_46_CS
...	

Before (Preparations)
Configuration State: N/a
Memory State: N/a
Reusable State(s): State is EVConnectedPreSession State is Authorized15118 State is EnergyTransferStarted

6.2.56. Page 221 - (2024-06) TC_F_04_CS Made mandatory in part 5, but prerequisite in part 6 was not updated

Removed Prerequisite(s):

Old	The Charging Station supports TxCtrlr.TxStartPoint ParkingBayOccupancy OR Authorized.
-----	---

New	N/a
-----	-----

And added to Preparations:

Before (Preparations)
Configuration State:
...
- TxCtrlr.TxStartPoint is <i>ParkingBayOccupancy</i> OR <i>Authorized</i> (If supported)

6.2.57. Page 221 - (2025-02) TC_F_04_CS Prerequisite only if supported

Update Preparation(s):

Updated Preparations to only set TxStartPoint if supported:

Before (Preparations)
Configuration State:
...
- TxCtrlr.TxStartPoint is <i>ParkingBayOccupancy</i> OR <i>Authorized</i> (If supported)

6.2.58. Page 295 - (2024-09) - TC_J_XX_CS Meter Values

Meter values cannot have *location* = "EV", unless it is for measurand "SoC".

For all test cases in J add the following Post scenario validation:

...	
Tool validations	...
	Post scenario validations:
	Message: MeterValuesRequest/TransactionEventRequest
	- ...
	- None of the provided sampledValue s shall have location = EV , except when measurand = SoC .

6.2.59. Page 236 - (2025-02) - TC_F_19_CS - The testcase ends while the firmware update is still ongoing

Testcase TC_F_19_CS starts a firmware update and ends the testcase while it is still ongoing. This may cause problems for the next testcase, without proper cleanup. Therefore an invalid signature is given to prevent a full firmware update as this can take quite some time. These messages are then handled by the cleanup, but these described steps are not part of the scope of the testcase and therefore the SUT is unable to fail during these steps.

Test case name	Trigger message - FirmwareStatusNotification - Downloading
Test case Id	TC_F_19_CS
...	...

Main (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a UpdateFirmwareResponse	1. The Test System sends a UpdateFirmwareRequest firmware.location is <Configured <i>firmware_location</i> > firmware.retrieveDateTime is <Current <i>dateTime</i> - 2 hours> firmware.installDateTime is omitted firmware.signingCertificate is <Configured <i>signingCertificate</i> > firmware.signature is <Configured <i>invalid firmware signature</i> >
...	
Note : Step 9 through 14 are cleanup to prevent an ongoing firmware update after the testcase is already ended. The behavior part of these steps is part of TC_L_06_CS and therefore not part of the scope for this testcase.	

Main (Test scenario)	
9. The Charging Station sends a FirmwareStatusNotificationRequest . With status <i>Downloaded</i>	10. The Test System responds with a FirmwareStatusNotificationResponse .
11. The Charging Station sends a FirmwareStatusNotificationRequest . With status <i>InvalidSignature</i>	12. The Test System responds with a FirmwareStatusNotificationResponse .
13. The Charging Station sends a SecurityEventNotificationRequest . With type <i>InvalidFirmwareSignature</i>	14. The Test System responds with a SecurityEventNotificationResponse .

6.2.60. Page 272 - (2025-02) - TC_H_08_CS Reserve an unspecified EVSE - Accepted

Wrong token was used for RequestStartTransaction.

TC_H_08_CS: Reserve an unspecified EVSE - Accepted

Test case name	Reserve an unspecified EVSE - Accepted
Test case Id	TC_H_08_CS
...	...

Before (Preparations)
...

Main (Test scenario)	
Charging Station	CSMS
...	
5. Execute Reusable State <i>Authorized</i>	
<u>Note(s):</u> - <i><<Configured valid_idtoken_idtoken></i> is used for the authorization.	
...	

Tool validations
* Step 2: Message: ReserveNowResponse - status must be <i>Accepted</i> * Step 3: (Optional) Message: StatusNotificationRequest - connectorStatus must be <i>Reserved</i> - evseld must be <i><Configured evseld></i> - connectorId must be <i><Configured connectorId></i> Message: NotifyEventRequest - trigger must be <i>Delta</i> - actualValue must be <i>"Reserved"</i> - component.name must be <i>"Connector"</i> - evse.id must be <i><Configured evseld></i> - eves.connectorId must be <i><Configured connectorId></i> - variable.name must be <i>"AvailabilityState"</i> (Optional) Message: NotifyEventRequest - eventData[0].trigger must be <i>Delta</i> - eventData[0].actualValue must be <i>Available</i> - eventData[0].component.name must be <i>EVSE</i> - eventData[0].variable.name must be <i>AvailabilityState</i>
Post scenario validations: N/a

6.2.61. Page 279/280 - (2025-06) - TC_H_15_CS & TC_H_16_CS can only be executed when the connector type of the Charging Station is part of the connectorEnumType.

The ConnectorEnumType list does not contains all connectorTypes. At OCPP 2.1 this enum has been changed to a string and can be extended, however this is not possible for OCPP 2.0.1. Therefore the reservation of connectorType testcases will not work for connectorTypes that are not part of the enum. It is not a major issue, because the CSMS can always reserve a specific EVSE based on the connectorTypes specified at the device model.

Test case name	Reserve a connector with a specific type - Success
Test case Id	TC_H_15_CS
Prerequisite(s)	- The configuration variable ReservationCtrlr.ReservationAvailable is implemented with value <i>true</i> - The Charging Station supports the reservation of a specific connector type, that is part of the ConnectorEnumType.

Test case name	Reserve a connector with a specific type - Amount of available connectors of a type equals the amount of reservations
Test case Id	TC_H_16_CS
Prerequisite(s)	- The configuration variable ReservationCtrlr.ReservationAvailable is implemented with value <i>true</i> - The Charging Station supports the reservation of a specific connector type, that is part of the ConnectorEnumType.

6.2.62. Page 282 - (2025-02) - TC_H_17_CS - made more explicit on what to validate

Tool validations

* Step 2:

Message: **CancelReservationResponse**

- **status** must be *Accepted*

* Step 3:

For each connector on the <Configured evseld> one of the following messages must be sent:

Message: **StatusNotificationRequest**

- **connectorStatus** must be *Available*
- **evseld** must be <Configured evseld>
- **connectorId** must be <Configured connectorId>

Message: **NotifyEventRequest**

- **trigger** must be *Delta*
- **actualValue** must be *"Available"*
- **component.name** must be *"Connector"*
- **evse.id** must be <Configured evseld>
- **eves.connectorId** must be <Configured connectorId>
- **variable.name** must be *"AvailabilityState"*

Post scenario validations:

N/a

6.2.63. Page 297 - (2025-02) - TC_J_02_CS Clock-aligned Meter Values - reporting multiple phases

An erratum was added on adding support for validating reporting measurands on multiple phases: [TC_J_02_CS Clock-aligned Meter Values \(2024-09\)](#). However the validation is too strict, so this erratum supersedes it. Reporting measurands per phase must be done, when the energy meter supports it and it is applicable for the specified measurand.

Test case name	Clock-aligned Meter Values - Transaction ongoing	
Test case Id	TC_J_02_CS	
Use case Id(s)	J01	
...		
Before (Preparations)	Configuration State: AlignedDataInterval is <Configured clock-aligned Meter Values interval> AlignedDataSendDuringIdle is false (If implemented) RegisterValuesWithoutPhases is false (If implemented)	
	Memory State: N/a	
	Reusable State(s): State is <i>EnergyTransferStarted</i>	
Main (Test scenario)	Charging Station	CSMS
	...	
Tool validations	Note: The following steps do not need to be sent in a specific order. * Step 1: Message: MeterValuesRequest - meterValue[0].sampledValue[0].context must be <i>Sample.Clock</i> - meterValue[0].sampledValue must contain <An element per configured measurand at the <i>AlignedDataMeasurands</i>> Notes : - The measurand field may be omitted when the measurand is "Energy.Active.Import.Register" - It is possible that measurands are reported on multiple locations or phases, based on the capabilities of the energy meter . * Step 3: Message: TransactionEventRequest - triggerReason must be <i>MeterValueClock</i> - metervalue[0].sampledValue[0].context must be <i>Sample.Clock</i> - metervalue[0].sampledValue must contain <An element per configured measurand at the <i>AlignedDataMeasurands</i>> Notes : - The measurand field may be omitted when the measurand is "Energy.Active.Import.Register" - It is possible that measurands are reported on multiple locations or phases, based on the capabilities of the energy meter . Post scenario validations: ...	

6.2.64. Page 297 - (2024-09) - TC_J_02_CS Clock-aligned Meter Values

Meter values must be reported for all phases.

Test case name	Clock-aligned Meter Values - Transaction ongoing
Test case Id	TC_J_02_CS
Use case Id(s)	J01
...	

Test case name	Clock-aligned Meter Values - Transaction ongoing	
Before (Preparations)	Configuration State: AlignedDataInterval is <Configured clock-aligned Meter Values interval> AlignedDataSendDuringIdle is false (If implemented) RegisterValuesWithoutPhases is false (If implemented)	
	Memory State: N/a	
	Reusable State(s): State is <i>EnergyTransferStarted</i>	
Main (Test scenario)	Charging Station	CSMS
	...	
Tool validations	Note: The following steps do not need to be sent in a specific order. * Step 1: Message: MeterValuesRequest - meterValue[0].sampledValue[0].context must be <i>Sample.Clock</i> - meterValue[0].sampledValue must contain <An element per configured measurand at the AlignedDataMeasurands for the number of phases reported in SupplyPhases . The measurand field may be omitted when the measurand is "Energy.Active.Import.Register"> * Step 3: Message: TransactionEventRequest - triggerReason must be <i>MeterValueClock</i> - meterValue[0].sampledValue[0].context must be <i>Sample.Clock</i> - meterValue[0].sampledValue must contain <An element per configured measurand at the AlignedDataMeasurands for the number of phases reported in SupplyPhases . The measurand field may be omitted when the measurand is "Energy.Active.Import.Register">	
	Post scenario validations: ...	

6.2.65. Page 306 - (2024-06) - TC_J_10_CS - Remove reference to non-existing requirements [4697]

Test case name	Sampled Meter Values - EventType Ended
Test case Id	TC_J_10_CS
Use case Id(s)	J02 & (E06,E07,E08,E09,E10,E12)
Requirement(s)	J02.FR.01,J02.FR.02,J02.FR.03,J02.FR.04,J02.FR.10, E06.FR.11, E06.FR.17 , E07.FR.08, E07.FR.13 ,E08.FR.09,E09.FR.05,E10.FR.04,E12.FR.07
...	

6.2.66. Page 318 - (2025-04) TC_K_09_CS: Removed validFrom/To from test

Test case name	Clear Charging Profile - Clearing a TxDefaultProfile - With ongoing transaction
Test case Id	TC_K_09_CS
...	

Before (Preparations)
Configuration State: SmartChargingCtrlr.LimitChangeSignificance is 1.0

Before (Preparations)
Memory State: SetChargingProfile with ChargingProfile 1: chargingProfilePurpose is TxDefaultProfile chargingProfileKind should be Absolute stackLevel should be 0 evseld <Configured evseld> validFrom <current dateTime - <Configured max time deviation> seconds> validTo <current dateTime + <Configured max time deviation> + 401 seconds> startSchedule <current dateTime> numberPhases <Configured numberPhases> ChargingSchedule: duration 400 + <Configured max time deviation> ...

6.2.67. Page 343 - (2025-04) TC_K_23_CS: Removed validFrom/To from test

Test case name	Set Charging Profile - StartSchedule
Test case Id	TC_K_23_CS
...	...

Main (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a SetChargingProfileResponse	1. The Test System sends a SetChargingProfileRequest with chargingProfile.chargingProfilePurpose is TxDefaultProfile chargingProfile.chargingProfileKind is Absolute evseld <configured evseld> chargingProfile.validFrom <current dateTime - <Configured max time deviation> + 50 seconds> chargingProfile.validTo <current dateTime + <Configured max time deviation> + 400 seconds> chargingProfile.chargingSchedule[0].startSchedule <current dateTime> + 60 seconds> chargingProfile.chargingSchedule[0].chargingSchedulePeriod[] . numberPhases <Configured numberPhases> chargingProfile.chargingSchedule[0].chargingSchedulePeriod[] . startPeriod 0 chargingProfile.chargingSchedule[0].chargingSchedulePeriod[0] . limit 6 * <limit multiplier> Note: Check [csKSmartChargingChargingProfileLimits] for <limit multiplier>
...	...

6.2.68. Page 337 - (2025-04) TC_K_28_CS: Removed validFrom/To from test

Test case name	Set Charging Profile - TxDefaultProfile with transaction ongoing
Test case Id	TC_K_28_CS
...	...

Main (Test scenario)	
Charging Station	CSMS

Main (Test scenario)	
2. The Charging Station responds with a SetChargingProfileResponse	1. The Test System sends a SetChargingProfileRequest with chargingProfile.chargingProfilePurpose is <i>TxDefaultProfile</i> chargingProfile.chargingProfileKind is <i>Absolute</i> chargingProfile.chargingSchedule[0].duration is $\langle 400 + \langle \text{Configured max time deviation} \rangle \text{ seconds} \rangle$ evseld $\langle \text{Configured evseld} \rangle$ chargingProfile.validFrom $\langle \text{current dateTime} - \langle \text{Configured max time deviation} \rangle \text{ seconds} \rangle$ chargingProfile.validTo $\langle \text{current dateTime} + \langle \text{Configured max time deviation} \rangle + 401 \text{ seconds} \rangle$ chargingProfile.chargingSchedule[0].startSchedule $\langle \text{current dateTime} - \langle \text{Configured max time deviation} \rangle \text{ seconds} \rangle$ chargingProfile.chargingSchedule[0].chargingSchedulePeriod[0].numberPhases $\langle \text{Configured numberPhases} \rangle$ chargingProfile.chargingSchedule[0].chargingSchedulePeriod[0].startPeriod 0 chargingProfile.chargingSchedule[0].chargingSchedulePeriod[0].limit $6 * \langle \text{limit multiplier} \rangle$...

6.2.69. Page 345 - (2025-02) TC_K_XX_CS: Use realistic values for composite schedules

In order to use realistic limits for charging profiles, the limit of a charging profile will be determined by $\langle \text{Configured chargingRateUnit} \rangle$ and $\langle \text{Configured numberPhases} \rangle$.

Added new section to section K Smart Charging

Determine Charging Profile Limit Multiplier

Not all chargers support setting limits in A or W. This can be configured with the configuration variable $\langle \text{Configured chargingRateUnit} \rangle$. To calculate the limit to be used, the following rules must be followed:

- If $\langle \text{Configured chargingRateUnit} \rangle$ is A, then $\langle \text{limit multiplier} \rangle$ is 1
- If $\langle \text{Configured chargingRateUnit} \rangle$ is W and $\langle \text{Configured numberPhases} \rangle$ is 1, then $\langle \text{limit multiplier} \rangle$ is 230
- If $\langle \text{Configured chargingRateUnit} \rangle$ is W and $\langle \text{Configured numberPhases} \rangle$ is 2, then $\langle \text{limit multiplier} \rangle$ is 460
- If $\langle \text{Configured chargingRateUnit} \rangle$ is W and $\langle \text{Configured numberPhases} \rangle$ is 3, then $\langle \text{limit multiplier} \rangle$ is 690

Example 1

Given a test case is configured with:

```
<Configured chargingRateUnit> W
<Configured numberPhases> 2
```

When the test case specifies:

```
chargingProfile.chargingSchedule.chargingSchedulePeriod.limit 6 * <limit multiplier>
```

Then it should set

```
chargingProfile.chargingSchedule.chargingSchedulePeriod.limit 2760
```

Example 2

Given a test case is configured with:

<Configured chargingRateUnit> A
<Configured numberPhases> 3

When the test case specifies:

chargingProfile.chargingSchedule.chargingSchedulePeriod.limit 6 * <limit multiplier>

Then it should set

chargingProfile.chargingSchedule.chargingSchedulePeriod.limit 6

All test cases using limits are updated

Test case name	[...]
Test case Id	TC_K_01_CS / TC_K_03_CS / TC_K_04_CS / TC_K_09_CS / TC_K_10_CS / TC_K_13_CS / TC_K_21_CS / TC_K_22_CS / TC_K_23_CS / TC_K_28_CS / TC_K_60_CS / TC_K_37_CS / TC_K_38_CS / TC_K_40_CS / TC_K_41_CS / TC_K_56_CS / TC_K_58_CS
[...]	

Main (Test scenario)	
Charging Station	CSMS
[...]	[...] Replace all *.limit assignments with: *.limit ? * <limit multiplier> Note: Check [csKSmartChargingChargingProfileLimits] for <limit multiplier>

Tool validations
* Step 2: Replace all *.limit validations with: *.limit <?> * <limit multiplier> Note: Check [csKSmartChargingChargingProfileLimits] for <limit multiplier>
Post scenario validations:
[...]

6.2.70. Page 345 - (2024-06) TC_K_35_CS Get Charging Profile - Evseld > 0 + chargingProfilePurpose [773]

Change initial charging state from "N/A" to:

	Charging State: State is <i>EnergyTransferStarted</i>
--	--

6.2.71. Page 352 - (2025-04) - TC_K_39_CS - Validation of scheduleStart

Test case name	Get Composite Schedule - No ChargingProfile installed on Charging Station
Test case Id	TC_K_39_CS
...	...

Tool validations
* Step 2: (Message: GetCompositeScheduleResponse) status <i>Accepted</i> evseld 0 scheduleStart <The time the GetCompositeScheduleRequest was transmitted +/- <Configured max time deviation>> duration is 300 chargingRateUnit <Configured chargingRateUnit> startPeriod 0
Post scenario validations: ...

6.2.72. Page 353 - (2025-04) - TC_K_40_CS: startSchedule improvements

Now using same startSchedule time for both profiles.

Test case name	Get Composite Schedule - Stacking ChargingProfiles
Test case Id	TC_K_40_CS
Use case Id(s)	K08
Requirement(s)	K08.FR.02,K08.FR.06
System under test	Charging Station
Description	The CSMS requests a composite schedule which is a combination of local limits and the prevailing Charging Profiles of the different chargingProfilePurposes and stack levels. 2 ChargingProfiles with same startSchedule and different stackLevels are submitted.

Before (Preparations)
Configuration State: N/a

Before (Preparations)	
Memory State: set <startScheduleTime> = <current dateTime> - <Configured max time deviation> SetChargingProfile with ChargingProfile 1: chargingProfilePurpose is TxDefaultProfile chargingProfileKind should be Absolute stackLevel should be 0 evseld <Configured evseld> validFrom <current dateTime> - <Configured max time deviation> seconds> validTo <current dateTime> + <Configured max time deviation> + 401 seconds> startSchedule <startScheduleTime> numberPhases <Configured numberPhases> ChargingSchedule: duration 400 + <Configured max time deviation> chargingRateUnit <Configured chargingRateUnit> startPeriod 0, limit 6 * <limit multiplier> startPeriod 100, limit 8 * <limit multiplier> startPeriod 200, limit 10 * <limit multiplier> Note: Check [csKSmartChargingChargingProfileLimits] for <limit multiplier> ChargingProfile 2: chargingProfilePurpose is TxDefaultProfile chargingProfileKind should be Absolute stackLevel should be 1 evseld <Configured evseld> validFrom <current dateTime> - <Configured max time deviation> seconds> validTo <current dateTime> + <Configured max time deviation> + 401 seconds> startSchedule <startScheduleTime> numberPhases <Configured numberPhases> ChargingSchedule: duration 150 + <Configured max time deviation> chargingRateUnit <Configured chargingRateUnit> startPeriod 0, limit 7 * <limit multiplier> Note: Check [csKSmartChargingChargingProfileLimits] for <limit multiplier> startPeriod 100, limit 9 * <limit multiplier> Note: Check [csKSmartChargingChargingProfileLimits] for <limit multiplier>	
Reusable State(s): N/a	

Main (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a GetCompositeScheduleResponse	1. The Test System sends a GetCompositeScheduleRequest with evseld <Configured evseld> duration is 350 chargingRateUnit <Configured chargingRateUnit>

Tool validations * Step 2: (Message: GetCompositeScheduleResponse) status Accepted evseld <Configured evseld> ChargingSchedule: duration 350 chargingRateUnit <Configured chargingRateUnit> scheduleStart <The time the GetCompositeScheduleRequest was transmitted> plus/minus <Configured max time deviation> Note: The period of time between the scheduleStart from the SetChargingProfileRequest with ChargingProfile 2 and the scheduleStart from the GetCompositeScheduleResponse is called x . Note: The period of time between the scheduleStart from the SetChargingProfileRequest with ChargingProfile 1 and the scheduleStart from the GetCompositeScheduleResponse is called y. startPeriod 0, limit 7 * <limit multiplier> (stackLevel 1) startPeriod (100 - x), limit 9 * <limit multiplier> (stackLevel 1) startPeriod (150 - x + <Configured max time deviation>), limit 8 * <limit multiplier> (stackLevel 0) startPeriod (200 - x), limit 10 * <limit multiplier> (stackLevel 0) Note: Check [csKSmartChargingChargingProfileLimits] for <limit multiplier>
Post scenario validations: N/a

6.2.73. Page 355 - (2025-04) - TC_K_41_CS: startSchedule improvement

Test case name	Get Composite Schedule - Combining chargingProfilePurposes
Test case Id	TC_K_41_CS
...	...

Before (Preparations) Configuration State: N/a Memory State: set <startScheduleTime> = <current dateTime> - <Configured max time deviation> seconds Note: Set MaxProfile for the next 24 hours: SetChargingProfile with ChargingProfile 1: chargingProfilePurpose is ChargingStationMaxProfile chargingProfileKind should be Absolute stackLevel should be 0 evseld 0 startSchedule <startScheduleTime> numberPhases <Configured numberPhases> ChargingSchedule: duration 86400 chargingRateUnit <Configured chargingRateUnit> startPeriod 0, limit 10 * <limit multiplier> Note: Check [csKSmartChargingChargingProfileLimits] for <limit multiplier>
--

Before (Preparations)	
Note: Set a default profile for 300 seconds ChargingProfile 2: chargingProfilePurpose is TxDefaultProfile chargingProfileKind should be Absolute stackLevel should be 0 evseld <Configured evseld> validFrom <current dateTime - <Configured max time deviation> seconds> validTo <current dateTime + <Configured max time deviation> + 401 seconds> startSchedule <startScheduleTime> numberPhases <Configured numberPhases> ChargingSchedule: duration 300 chargingRateUnit <Configured chargingRateUnit> startPeriod 0,60,120,180,260, limit 6,10,8,15,8 * <limit multiplier> Note: Check [csKSmartChargingChargingProfileLimits] for <limit multiplier>	Note: set a TxProfile for 260 seconds: ChargingProfile 3: chargingProfilePurpose is TxProfile chargingProfileKind should be Absolute stackLevel should be 0 evseld <Configured evseld> validFrom <current dateTime - <Configured max time deviation> seconds> validTo <current dateTime + <Configured max time deviation> + 401 seconds> startSchedule <startScheduleTime> numberPhases <Configured numberPhases> ChargingSchedule: duration 260 chargingRateUnit <Configured chargingRateUnit> startPeriod 0,50,140,200,240, limit 8,11,16,6,12 * <limit multiplier> Note: Check [csKSmartChargingChargingProfileLimits] for <limit multiplier>
Reusable State(s): N/a	

Main (Test scenario)	
Charging Station	CSMS
...	...

Tool validations
* Step 2: (Message: GetCompositeScheduleResponse) status Accepted evseld <Configured evseld> ChargingSchedule: duration 400 chargingRateUnit <Configured chargingRateUnit> scheduleStart <The time the GetCompositeScheduleRequest was transmitted +/- <Configured max time deviation>> Note: The period of time between the scheduleStart from the SetChargingProfileRequest with ChargingProfile 3 and the scheduleStart from the GetCompositeScheduleResponse is called x. Note: The period of time between the scheduleStart from the SetChargingProfileRequest with ChargingProfile 2 and the scheduleStart from the GetCompositeScheduleResponse is called y. startPeriod 0, limit 8 * <limit multiplier> (TxProfile) startPeriod (50 - x), limit 10 * <limit multiplier> (ChargingStationMaxProfile) startPeriod (200 - x), limit 6 * <limit multiplier> (TxProfile) startPeriod (240 - x), limit 10 * <limit multiplier> (ChargingStationMaxProfile) startPeriod (260 - x), limit 8 * <limit multiplier> (TxDefaultProfile) startPeriod (300 - x), limit 10 * <limit multiplier> (ChargingStationMaxProfile) Note: Check [csKSmartChargingChargingProfileLimits] for <limit multiplier>
Post scenario validations: N/a

6.2.74. Page 354/355 - (2025-02) - TC_K_40_CS & TC_K_41_CS - Updated composite schedule validation

The startPeriod validation did not take the (to the duration added) <configured max time deviation> into account.

Test case name	Get Composite Schedule - Stacking ChargingProfiles
Test case Id	TC_K_40_CS
...	...

Tool validations * Step 2: (Message: GetCompositeScheduleResponse) status Accepted evseld <Configured evseld> ChargingSchedule: duration 350 chargingRateUnit <Configured chargingRateUnit> scheduleStart <The time the GetCompositeScheduleRequest was transmitted \+/- <Configured max time deviation>> Note: If <Configured chargingRateUnit> is W, then the limit field will be multiplied by 1000. Note: The period of time between sending the second SetChargingProfileRequest and the scheduleStart from the GetCompositeScheduleResponse is called x. Note: The period of time between the scheduleStart from the SetChargingProfileRequest with ChargingProfile 3 and the scheduleStart from the GetCompositeScheduleResponse is called x. Note: The period of time between the scheduleStart from the SetChargingProfileRequest with ChargingProfile 2 and the scheduleStart from the GetCompositeScheduleResponse is called y. startPeriod 0, limit 7 startPeriod (100 - x), limit 9 startPeriod (150 - x) startPeriod (200 - x), limit 10 startPeriod 0, limit 8 startPeriod (50 - x), limit 10 startPeriod (200 - x), limit 6 startPeriod (240 - x), limit 10 startPeriod (260 - x + <Configured max time deviation>), limit 8 (TxDefaultProfile) startPeriod (300 - y + <Configured max time deviation>), limit 10 (ChargingStationMaxProfile) Post scenario validations: N/a
--

Test case name	Get Composite Schedule - Combining chargingProfilePurposes
Test case Id	TC_K_41_CS
...	...

Before (Preparations) Configuration State: N/a Memory State: SetChargingProfile with ChargingProfile 1: chargingProfilePurpose is ChargingStationMaxProfile chargingProfileKind should be Absolute stackLevel should be 0 evseld 0 startSchedule <current dateTime - <Configured max time deviation> seconds> numberPhases <Configured numberPhases> ChargingSchedule: duration <86400 <Configured max time deviation> seconds> chargingRateUnit <Configured chargingRateUnit> Note: If <Configured chargingRateUnit> is W, then the limit field will be multiplied by 1000. startPeriod 0, limit 10
--

Before (Preparations)	
ChargingProfile 2: chargingProfilePurpose is TxDefaultProfile chargingProfileKind should be Absolute stackLevel should be 0 evseld <Configured evseld> validFrom <current dateTime - <Configured max time deviation> seconds> validTo <current dateTime + <Configured max time deviation> + 401 seconds> startSchedule <current dateTime - <Configured max time deviation> seconds> numberPhases <Configured numberPhases> ChargingSchedule: duration <300 <Configured max time deviation> seconds> chargingRateUnit <Configured chargingRateUnit> Note: If <Configured chargingRateUnit> is W, then the limit field will be multiplied by 1000. startPeriod 0,60,120,180,260, limit 6,10,8,15,8	ChargingProfile 3: chargingProfilePurpose is TxProfile chargingProfileKind should be Absolute stackLevel should be 0 evseld <Configured evseld> validFrom <current dateTime - <Configured max time deviation> seconds> validTo <current dateTime + <Configured max time deviation> + 401 seconds> startSchedule <current dateTime - <Configured max time deviation> seconds> numberPhases <Configured numberPhases> ChargingSchedule: duration <260 <Configured max time deviation> seconds> chargingRateUnit <Configured chargingRateUnit> Note: If <Configured chargingRateUnit> is W, then the limit field will be multiplied by 1000. startPeriod 0,50,140,200,240, limit 8,11,16,6,12
Reusable State(s): N/a	

Tool validations
* Step 2: (Message: GetCompositeScheduleResponse) status Accepted evseld <Configured evseld> ChargingSchedule: duration 400 chargingRateUnit <Configured chargingRateUnit> Note: If <Configured chargingRateUnit> is W, then the limit field will be multiplied by 1000. Note: The period of time between sending the second SetChargingProfileRequest and the scheduleStart from the GetCompositeScheduleResponse is called x : startPeriod 0, limit 8 startPeriod (50 - x), limit 10 startPeriod (200 - x), limit 6 startPeriod (240 - x), limit 10 startPeriod (260 - x + <Configured max time deviation>), limit 8 (TxDefaultProfile) startPeriod (300 - x + <Configured max time deviation>), limit 10 (ChargingStationMaxProfile)
Post scenario validations: N/a

6.2.75. Page 355 - (2025-04) - TC_K_41_CS: Added missing EnergyTransferStarted reusable state

A transaction needs to be started to be able to set a TxProfile. The tool already started a transaction for this reason, but the testcase description lacks this information.

Test case name	Get Composite Schedule - Combining chargingProfilePurposes
Test case Id	TC_K_41_CS
...	

Before (Preparations)
Configuration State: N/a

Before (Preparations)
Memory State: ...
Reusable State(s): State is EnergyTransferStarted

6.2.76. Page 359 - (2025-04) - TC_K_53_CS: Added missing validations

Before (Preparations)
Configuration State: N/a
Memory State: N/a
Reusable State(s): State is Authorized (local) State is EVCConnectedPreSession

Main (Test scenario)	
Charging Station	CSMS
1. Execute Reusable State Authorized 15118	
2. The Charging Station sends a NotifyEVChargingNeedsRequest	3. The Test System responds with a NotifyEVChargingNeedsResponse with status Accepted
5. The Charging Station responds with a SetChargingProfileResponse	4. The Test System sends a SetChargingProfileRequest with chargingProfile.chargingProfilePurpose TxProfile chargingProfile.transactionId <transactionId> chargingProfile.chargingSchedule[0].id <Id generated by Test System> chargingProfile.chargingSchedule[0].chargingRateUnit <Configured chargingRateUnit> chargingProfile.chargingSchedule[0].chargingSchedulePeriod.startPeriod 0 chargingProfile.chargingSchedule[0].chargingSchedulePeriod.limit 6 * <limit multiplier> Note: Check [csKSmartChargingChargingProfileLimits] for <limit multiplier>
6. The Charging Station sends a NotifyEVChargingScheduleRequest	7. The Test System responds with a NotifyEVChargingScheduleResponse with status Accepted
4. Execute Reusable State EnergyTransferStarted	
8. The Charging Station sends a TransactionEventRequest .	9. The Test System responds with a TransactionEventResponse .

Tool validations
* Step 1: Message: TransactionEventRequest of <i>Authorized15118</i> step 3 - eventType must be <i>Updated</i> or <i>Started</i> - triggerReason must be <i>Authorized</i> * Step 2: Message: NotifyEVChargingNeedsRequest IF chargingNeeds.acChargingParameters is <i><omitted></i> THEN chargingNeeds.dcChargingParameters must be <i><not omitted></i> END IF IF chargingNeeds.dcChargingParameters is <i><omitted></i> THEN chargingNeeds.acChargingParameters must be <i><not omitted></i> END IF * Step 5: Message: SetChargingProfileResponse - status must be <i>Accepted</i> * Step 6: Message: NotifyEVChargingScheduleRequest - chargingSchedule must be within bounds of chargingSchedule of step 4 * Step 8: Message: TransactionEventRequest - triggerReason must be <i>ChargingStateChanged</i> - transactionInfo.chargingState must be <i>Charging</i>
Post scenario validations: N/a

6.2.77. Page 360 - (2025-02) - TC_K_54_CS: EVConnected must be before Authorization

TC_K_54_CS: Charging with load leveling based on High Level Communication - No SASchedule (rejected)

Test case name	Charging with load leveling based on High Level Communication - No SASchedule (rejected)
Test case Id	TC_K_54_CS
...	...

Before (Preparations)
Configuration State: N/a
Memory State: N/a
Reusable State(s): State is <i>EVConnectedPreSession</i> State is <i>Authorized</i> (local)

Main (Test scenario)	
Charging Station	CSMS
1. Execute Reusable State <i>Authorized15118</i>	
2. The Charging Station sends a NotifyEVChargingNeedsRequest.	3. The Test System responds with a NotifyEVChargingNeedsResponse. With status <i>Rejected</i>
[...]	

6.2.78. Page 362 - (2025-02) - TC_K_56_CS: EVConnected must be before Authorization

TC_K_56_CS: Charging with load leveling based on High Level Communication - Offline

Test case name	Charging with load leveling based on High Level Communication - Offline
Test case Id	TC_K_56_CS
...	...

Before (Preparations)
Configuration State: RetryBackOffWaitMinimum is <Configured RetryBackOffWaitMinimum>
Memory State: ...
Reusable State(s): State is EVConnectedPreSession State is Authorized (local)

6.2.79. Page 364 - (2025-02) - TC_K_57_CS: EVConnected must be before Authorization

TC_K_57_CS: Renegotiating a Charging Schedule - Initiated by EV

Test case name	Renegotiating a Charging Schedule - Initiated by EV
Test case Id	TC_K_57_CS
...	...

Before (Preparations)
Configuration State: N/a
Memory State: N/a
Reusable State(s): State is EVConnectedPreSession State is Authorized (local) State is RenegotiateChargingLimits

6.2.80. Page 362 - (2024-09) - TC_K_56_CS Removed expecting triggerReason=ChargingRateChanged [776]

A trigger reason ChargingStateChange must only be sent, when an external actor (not CSMS) changes the charging rate. Therefore, removed the check that triggerReason=ChargingStateChange is sent. Also added a check that the EV charging schedule fits within the given charging profile.

Test case name	Charging with load leveling based on High Level Communication - Offline	
Test case Id	TC_K_56_CS	
...	...	
Main (Test scenario)	Charging Station	CSMS 6. The OCTT responds with a TransactionEventResponse.
	...	

Test case name	Charging with load leveling based on High Level Communication - Offline
Tool validations	* Step 3: (Message: NotifyEVChargingScheduleRequest) evseld <Configured evseld> chargingSchedule.chargingSchedule[0].chargingRateUnit <Configured chargingRateUnit> chargingSchedule.chargingSchedule[0].chargingSchedulePeriod[0].startPeriod 0 If <Configured chargingRateUnit> is W: chargingSchedule.chargingSchedule[0].chargingSchedulePeriod[0].limit <= 8000 Else: chargingSchedule.chargingSchedule[0].chargingSchedulePeriod[0].limit <= 8 * Step 5: Message: TransactionEventRequest triggerReason must be ChargingStateChanged transactionInfo.chargingState must be Charging offline true Post scenario validations: N/a

6.2.81. Page 366 - (2025-02) - TC_K_58_CS

Before (Preparations)
Configuration State: N/a
Memory State: N/a
Reusable State(s): State is Authorized (local) State is EVConnectedPreSession State is RenegotiateChargingLimits State is EVConnectedPreSession State is Authorized15118 State is EnergyTransferStarted

6.2.82. Page 384 – TC_L_10_CS - Allow Download/InstallationFailed upon AcceptedCanceled

When a new firmware update is issued, the ongoing firmware update is canceled, but Charging Station may still send a FirmwareStatusNotification(DownloadFailed/InstallationFailed)

Test case name	Secure Firmware Update - AcceptedCanceled
Test case Id	TC_L_10_CS
...	...

Before (Preparations)
...

Main (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a UpdateFirmwareResponse With status Accepted	1. The Test System sends a UpdateFirmwareRequest with requestId = <#1> firmware.installDateTime <Current DateTime - 2 hours> firmware.location <Configured firmware_location> firmware.retrieveDateTime <Current DateTime - 2 hours> firmware.signingCertificate <Configured signingCertificate> firmware.signature <Configured signature>

Main (Test scenario)	
3. The Charging Station sends a FirmwareStatusNotificationRequest With requestId <#1> and status <i>Downloading</i>	4. The Test System responds with a FirmwareStatusNotificationResponse
6. The Charging Station responds with a UpdateFirmwareResponse With requestId <#1> and status <i>AcceptedCanceled</i>	5. The Test System sends a UpdateFirmwareRequest with requestId = <#2> firmware.installDateTime <Current DateTime - 2 hours> firmware.location <Configured firmware_location> firmware.retrieveDateTime <Current DateTime - 2 hours> firmware.signingCertificate <Configured signingCertificate> firmware.signature <Configured signature>
7. The Charging Station sends a FirmwareStatusNotificationRequest With requestId <#2> and status <i>Downloading</i>	8. The Test System responds with a FirmwareStatusNotificationResponse
...	...

Tool validations
* Step 2: Message UpdateFirmwareResponse - status <i>Accepted</i> * Step 3: Message FirmwareStatusNotificationRequest - status <i>Downloading</i> - requestId = <#1> * Step 6: Message UpdateFirmwareResponse - status <i>AcceptedCanceled</i> A FirmwareStatusNotificationRequest <i>DownloadFailed</i> or <i>InstallationFailed</i> may be sent for requestId <#1> before or after step 6. (The requestId at the FirmwareStatusNotificationRequest messages must refer to the id <#2> from the second UpdateFirmwareRequest from this point on) * Step 7: Message FirmwareStatusNotificationRequest - status <i>Downloading</i> ...
Post scenario validations: N/a

6.2.83. Page 387 - TC_L_06_CS - SecurityEventNotification and FirmwareStatusNotification can be sent in any order

The SecurityEventNotification with type *InvalidFirmwareSignature* and the FirmwareStatusNotification with status *InvalidSignature* can be sent in any order. These events are triggered simultaneously, so which message is transmitted first may vary.

Test case name	Secure Firmware Update - InvalidSignature
Test case Id	TC_L_06_CS
...	

Main (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a UpdateFirmwareResponse	1. The Test System sends a UpdateFirmwareRequest with firmware.installDateTime <Current DateTime - 2 hours> firmware.location <Configured firmware_location> firmware.retrieveDateTime <Current DateTime - 2 hours> firmware.signingCertificate <Configured signingCertificate> firmware.signature <Configured invalid firmware signature>

Main (Test scenario)	
3. The Charging Station sends a FirmwareStatusNotificationRequest . With status <i>Downloading</i>	4. The Test System responds with a FirmwareStatusNotificationResponse .
5. The Charging Station sends a FirmwareStatusNotificationRequest . With status <i>Downloaded</i>	6. The Test System responds with a FirmwareStatusNotificationResponse .
Note: Step 7 through 10 can be sent in a different order.	
7. The Charging Station sends a FirmwareStatusNotificationRequest . With status <i>InvalidSignature</i>	8. The Test System responds with a FirmwareStatusNotificationResponse .
9. The Charging Station sends a SecurityEventNotificationRequest . With type <i>InvalidFirmwareSignature</i>	10. The Test System responds with a SecurityEventNotificationResponse .

6.2.84. Page 422 - (2025-02) - TC_M_15_CS - V2GCertificateChain is not installed before being retrieved

Test case name	Retrieve certificates from Charging Station - V2GCertificateChain
Test case Id	TC_M_15_CS
...	

Before (Preparations)
Configuration State: N/a
Memory State: RenewV2GChargingStationCertificate
Reusable State(s): N/a

6.2.85. Page 430 - (2025-06) - TC_M_24_CS - A GetCertificateStatusRequest is also sent for the subCAs

According to requirement M06.FR.07, the Charging Station sends a GetCertificateStatusRequest for the V2G Charging Station certificate (leaf) and the subCAs.

Test case name	Get Charging Station Certificate status - Success
Test case Id	TC_M_24_CS
...	

Main (Test scenario)	
Charging Station	CSMS
1. The Charging Station sends a GetCertificateStatusRequest	2. The Test System responds with a GetCertificateStatusResponse with status <i>Accepted</i> ocspResult <OCSPResponse class as defined in IETF RFC 6960. DER encoded (as defined in IETF RFC 6960), and then base64 encoded.>
Note: Step 1/2 are repeated for the V2G Charging Station (leaf), the subCA1 and subCA2 certificates.	

6.2.86. Page 436 - (2025-02) - TC_N_01_CS - Made used component variable configurable

Before (Preparations)
Configuration State: The following monitors (on arbitrary variables) must be present as 'hard-wired' or 'preconfigured' or must have been configured by CSMS: -LowerThreshold -UpperThreshold -Delta -Periodic -PeriodicClockAligned The following monitors must have been configured by CSMS for Component Variable <Configured threshold monitor component variable> : - LowerThreshold using value <Configured threshold monitor component variable LowerThreshold trigger value> - UpperThreshold using value <Configured threshold monitor component variable UpperThreshold trigger value> - Periodic using value <Configured Clock Aligned MeterValues Interval> The following monitors must have been configured by CSMS for Component Variable <Configured numeric delta component variable> : - Delta using value <Configured numeric delta component variable Delta numeric trigger value> - PeriodicClockAligned using value <Configured Clock Aligned MeterValues Interval>
Memory State: N/a
Reusable State(s): N/a

6.2.87. Page 455 - (2025-02) - TC_N_12_CS - Updating test case for using configuration variables

Test case name	Set Variable Monitoring - Value out of range - Delta monitor
[...]	
Prerequisite(s)	Charging Station has implemented device model monitoring and MonitoringCtrlr::Enabled = true. This test case assumes the following component exists and can be monitored: -Component "EVSE", evse "1", variable "AvailabilityState", monitor type <i>Delta</i> Note: Variable <i>AvailabilityState</i> is mandatory for an EVSE and it is likely (but not guaranteed), that it can be monitored. This test case assumes a numeric component variable exists which can be monitored.
Main (Test scenario)	
Charging Station	CSMS

Main (Test scenario)	
2. Charging Station responds with: SetVariableMonitoringResponse	<i>Install monitors</i> 1. Test System sends SetVariableMonitoringRequest with: - setMonitoringData[0].value = -1 - setMonitoringData[0].type = <i>Delta</i> - setMonitoringData[0].severity = <i><Configured severity></i> - setMonitoringData[0].component.name = "EVSE" - setMonitoringData[0].component.evse.id = <i><Configured evseId></i> - setMonitoringData[0].variable.name = "AvailabilityState" - setMonitoringData[0].component = <i><Configured numeric delta component></i> - setMonitoringData[0].variable = <i><Configured numeric delta component variable></i>
Tool validations	
* Step 2: Message: SetVariableMonitoringResponse with (in arbitrary order): <pre> setMonitoringResult = { - id is absent - status = <i>Rejected</i> - type = <i>Delta</i> - severity = <i><Configured severity></i> - component.name = "EVSE" - component.evse.id = <i><Configured evseId></i> - variable.name = "AvailabilityState" - component = <i><Configured numeric delta component variable></i> - variable = <i><Configured numeric delta component variable></i> - statusInfo is absent or statusInfo.reasonCode = "ValueOutOfRange" or statusInfo.reasonCode = "ValuePositiveOnly" } </pre>	
Post scenario validations: N/A	

6.2.88. Page 456 - (2025-02) - TC_N_13_CS - Updating test case for using more specific configuration variables

TC_N_13_CS: Set Variable Monitoring - Value out of range - Threshold monitor

Test case name	Set Variable Monitoring - Value out of range - Threshold monitor
Test case Id	TC_N_13_CS
Use case Id(s)	N04
Requirement(s)	N04.FR.13 N04.FR.06
System under test	Charging Station
Description	CSMS tries to set a threshold monitor with a value that is out of range.
Purpose	To test that Charging Station checks that value is within range of variable.
Prerequisite(s)	Charging Station has implemented device model monitoring and MonitoringCtrlr::Enabled = true. This test case assumes the <Configured threshold monitor component variable> component variable exists and can be monitored and has variableCharacteristics.maxLimit < <Configured threshold monitor value> This test case assumes the <Configured threshold monitor component variable with maxLimit> component variable exists and can be monitored and has variableCharacteristics.maxLimit < <Configured threshold monitor component variable with maxLimit exceeding maxLimit value> + Note: Variable _Power(maxLimit) is mandatory for an EVSE, but the actual value not, but it is likely (but not guaranteed), that it can be monitored._

Before (Preparations)
Configuration State: N/a
Memory State: N/a
Reusable State(s): N/a

Main (Test scenario)	
Charging Station	CSMS
2. Charging Station responds with: SetVariableMonitoringResponse	Install monitors 1. Test System sends SetVariableMonitoringRequest with: setMonitoringData[0].value = <Configured threshold monitor value> - setMonitoringData[0].value = <Configured threshold monitor component variable with maxLimit exceeding maxLimit value> - setMonitoringData[0].type = UpperThreshold - setMonitoringData[0].severity = <Configured severity> setMonitoringData[0].component.name = <Configured threshold monitor component variable> setMonitoringData[0].component.evse.id = <Configured evseId> setMonitoringData[0].variable.name = <Configured threshold monitor component variable> - setMonitoringData[0].component = <Configured threshold monitor component variable with maxLimit> - setMonitoringData[0].variable = <Configured threshold monitor component variable with maxLimit>

Tool validations
* Step 2: Message: SetVariableMonitoringResponse with (in arbitrary order): setMonitoringResult = { - id is absent - status = <i>Rejected</i> - type = <i>UpperThreshold</i> - severity = <i><Configured severity></i> - component.name = <i><Configured threshold monitor component variable></i> - component.evse.id = <i><Configured evseId></i> - variable.name = <i><Configured threshold monitor component variable></i> - component = <i><Configured threshold monitor component variable with maxLimit></i> - variable = <i><Configured threshold monitor component variable with maxLimit></i> - statusInfo is absent or statusInfo.reasonCode = <i>"ValueOutOfRange"</i> }
Post scenario validations: N/A

6.2.89. Page 463 - (2025-02) TC_N_20_CS - Updating test case for using more specific configuration variables

TC_N_20_CS: Alert Event - Threshold value exceeded

Test case name	Alert Event - Threshold value exceeded
Test case Id	TC_N_20_CS
Use case Id(s)	N07
Requirement(s)	N07.FR.06, N07.FR.07, N07.FR.16, N07.FR.17
System under test	Charging Station
Description	A monitored variable exceeds a threshold monitor and causes a NotifyEventRequest message to be sent.
Purpose	To test that Charging Station supports threshold monitors
Prerequisite(s)	Charging Station has implemented device model monitoring and MonitoringCtrlr::Enabled = true.

Before (Preparations)
Configuration State: N/a
Memory State: This test requires the Monitoring Base to be set to <i>All</i> . - SetMonitoringBaseRequest with monitoringBase = <i>All</i> . Furthermore this test requires the existence of a <i>LowerThreshold</i> and <i>UpperThreshold</i> monitor on a (numerical) variable. Since it is not mandated which variables are required to be monitored, this test used the variable "Power" of component "EVSE". - setMonitoringData[0].value = <Configured threshold monitor value> - setMonitoringData[0].value = <Configured threshold monitor component variable UpperThreshold trigger value> - setMonitoringData[0].type = <i>UpperThreshold</i> - setMonitoringData[0].severity = <Configured severity> - setMonitoringData[0].component.name = <Configured threshold monitor component variable> - setMonitoringData[0].component.evse.id = <Configured evseId> - setMonitoringData[0].variable.name = <Configured threshold monitor component variable> - setMonitoringData[0].severity = 5 - setMonitoringData[0].component = <Configured threshold monitor component variable> - setMonitoringData[0].variable = <Configured threshold monitor component variable> Set MonitoringLevel to 8 Notes: - If componentVariable is set to "Power" or "Current", the value is set to the configured maxLimit 100.0 - Take a threshold that can easily be exceeded.
Reusable State(s): N/a

Main (Test scenario)	
Charging Station	CSMS
1. Execute Reusable State <i>EnergyTransferStarted</i> or manually trigger the monitor. <u>Notes:</u> If componentVariable is set to "Power" or "Current" <i>EnergyTransferStarted</i> will trigger the monitor. If another componentvariable is chosen a manual action is needed to trigger the monitor.	
2. Charging Station sends a NotifyEventRequest with: - Power exceeding upper threshold	3. Test System responds with a NotifyEventResponse

Main (Test scenario)	
5. Charging Station responds with a SetVariableMonitoringResponse with: - status Accepted	4. Test System sends a SetVariableMonitoringRequest with: - type LowerThreshold - component.name <Configured threshold monitor component variable> - component.evse.id <configured evseId> - variable.name <Configured threshold monitor component variable> - value <Configured threshold monitor2 value> - component <Configured threshold monitor component variable> - variable <Configured threshold monitor component variable> - value <Configured threshold monitor component variable LowerThreshold trigger value> <u>Notes:</u> - If componentVariable is set to "Power" or "Current", the value is set to the configured maxLimit 10.0 - Take a threshold that won't be exceeded.
6. Execute Reusable State <i>StopAuthorized</i> or manually trigger the second monitor. <u>Notes:</u> If componentVariable is set to "Power" or "Current" EnergyTransferStarted will trigger the monitor. If another componentvariable is chosen a manual action is needed to trigger the monitor.	
7. Charging Station sends: NotifyEventRequest for 2 events: - Returning below upper threshold (<i>cleared</i>) - Dropping below lower threshold	8. Test System responds: NotifyEventResponse
<u>Notes:</u> Steps 2, 3, 7, and 8 may be repeated if the data is sent using two requests instead of one. Depending on the configuration the Charging Station may also send other notifications during step 4 and 9.	

Tool validations
* Step 2: Message: NotifyEventRequest with: - generatedAt = <time of generation at Charging Station> - seqNo = 0 and an eventData element with: - eventId = <id1> - timestamp = <time of event at Charging Station> - trigger = Alerting - actualValue = <current power> (must be > <Configured threshold monitor value>) - cleared is absent or cleared = false - transactionId = <transaction id> (delivery of power is always in transaction) - variableMonitoringId = <monitor id1> - component.name = <Configured threshold monitor component variable> - component.evse.id = <Configured evseId> - variable.name = <Configured threshold monitor component variable> - component = <Configured threshold monitor component variable> - variable.name = <Configured threshold monitor component variable> Other eventData elements can be ignored.

Tool validations
* Step 7: Message: NotifyEventRequest with: - generatedAt = <time of generation at Charging Station> - seqNo = 0 and an eventData element with: - eventId = <id2> - timestamp = <time of event at Charging Station> - trigger = <i>Alerting</i> - actualValue = <current power> (must be =< <Configured threshold monitor value>) - cleared is true - transactionId = <transaction id> (delivery of power is always in transaction) - variableMonitoringId = <monitor id1> - eventNotificationType = <i>CustomMonitor</i> - component.name = <Configured threshold monitor component variable> - component.evse.id = <Configured evseId> - variable.name = <Configured threshold monitor component variable> - component = <Configured threshold monitor component variable> - variable.name = <Configured threshold monitor component variable> and an eventData element with: - eventId = <id3> - timestamp = <time of event at Charging Station> - trigger = <i>Alerting</i> - actualValue = <current power> (must be < <Configured threshold monitor2 value>) - cleared is absent or cleared is false - transactionId = <transaction id> (delivery of power is always in transaction) - variableMonitoringId = <monitor id2> - eventNotificationType = <i>CustomMonitor</i> - component.name = <Configured threshold monitor component variable> - component.evse.id = <Configured evseId> - variable.name = <Configured threshold monitor component variable> - component = <Configured threshold monitor component variable> - variable.name = <Configured threshold monitor component variable> Other eventData elements can be ignored. This can also be sent in two <i>NotifyEventRequests</i>, instead of one.
Post scenario validations: N/A

6.2.90. Page 468 - (2024-06) TC_N_23_CS Offline Notification - OfflineMonitoringEventQueuingSeverity set higher than severityLevel of the monitor [772]

Test case name	Offline Notification - OfflineMonitoringEventQueuingSeverity set higher than severityLevel of the monitor	
Test case Id	TC_N_23_CS	
Use case Id(s)	N07	
Requirement(s)	N07.FR.04	
System under test	Charging Station	
Description	Charging Station does not queue event notifications when offline.	
Purpose	To test that Charging Station does not queue event notifications with a severity higher than OfflineMonitoringEventQueuingSeverity.	
Prerequisite(s)	Charging Station is online at start of test for configuration. CS has implemented device model monitoring and MonitoringCtrlr::Enabled = true.	
Before (Preparations)	Configuration State: SetConfiguration with: - component.name = "MonitoringCtrlr" - variable.name = "OfflineQueuingSeverity" - attributeValue = <Configured Severity>	
	Memory State: Charging Station has custom or predefined monitors on variable AvailabilityState of Configured EVSE and Configured ConnectorId with severity = <Configured severity> + 1	
	Reusable State(s): N/a	
Main (Test scenario)	Charging Station	CSMS
	<u>Manual Action:</u> Connect the EV and EVSE.	
	1. The Charging Station notifies the CSMS about the status change of the connector.	2. The OCTT responds accordingly.
	<u>Note(s):</u> Step 3, 4, 5, 6, 7, and 8 need to be executed when TxStartPoint contains EVConnected OR ParkingBayOccupancy	
	3. The Charging Station sends a TransactionEventRequest	4. The OCTT responds with a TransactionEventResponse
	<u>Manual Action:</u> Take Charging Station offline.	
	<u>Manual Action:</u> Disconnect the EV and EVSE.	
	<u>Manual Action:</u> Connect the EV and EVSE.	
	<u>Note(s):</u> The tool will now wait for <Configured Transaction Duration> seconds	
	<u>Manual Action:</u> Bring Charging Station back online.	
	5. The Charging Station sends a TransactionEventRequest	6. The OCTT responds with a TransactionEventResponse
	7. The Charging Station sends a TransactionEventRequest	8. The OCTT responds with a TransactionEventResponse
	<u>Note(s):</u> The CS shall not send a NotifyEventRequest for AvailabilityState of EVSE and Connector. A StatusNotification may still be received.	

Test case name	Offline Notification - OfflineMonitoringEventQueuingSeverity set higher than severityLevel of the monitor
Tool validations	* Step 1: (Optional:) Message: StatusNotificationRequest - evseld <configured evseld> - connectorId <configured connectorId> - connectorStatus must be <i>Occupied</i> (Required, but can be combined into one NotifyEventRequest:) Message: NotifyEventRequest - eventData[0].trigger must be <i>Delta</i> - eventData[0].actualValue must be <i>Occupied</i> - eventData[0].component.name must be <i>Connector</i> - eventData[0].component.evse.id must be <i>Configured EVSE</i> - eventData[0].component.evse.connectorId must be <i>Configured ConnectorId</i> - eventData[0].variable.name must be <i>AvailabilityState</i> Message: NotifyEventRequest - eventData[0].trigger must be <i>Delta</i> - eventData[0].actualValue must be <i>Occupied</i> - eventData[0].component.name must be <i>EVSE</i> - eventData[0].component.evse.id must be <i>Configured EVSE</i> - eventData[0].variable.name must be <i>AvailabilityState</i>
	* Step 3: Message: TransactionEventRequest - triggerReason must be <i>CablePluggedIn</i> - transactionInfo.chargingState must be <i>EVConnected</i>
	* Step 5: Message: TransactionEventRequest - triggerReason must be <i>EVCommunicationLost</i> - transactionInfo.chargingState must be <i>Idle</i>
	* Step 7: Message: TransactionEventRequest - triggerReason must be <i>CablePluggedIn</i> - transactionInfo.chargingState must be <i>EVConnected</i>
	Post scenario validations: N/A

6.2.91. Page 470 - (2024-09) - TC_N_24_CS - Referring to incorrect use case and requirements [O20-4793]

Test case name	Set Variable Monitoring - Periodic event	
Test case Id	TC_N_24_CS	
Use case Id(s)	N04, N08	
Requirement(s)	N04.FR.01, N04.FR.08, N08.FR.05 and N08.FR.06	
...		
Main (Test scenario)	Charging Station	CSMS
	...	...
Tool validations	...	
	Post scenario validations: N/A	

6.2.92. Page 492 - (2025-02) - TC_N_39_CS - Test case now searches suitable variables to do test with

TC_N_39_CS: Set Variable Monitoring - Component/Variable combination does NOT correspond

Test case name	Set Variable Monitoring - Component/Variable combination does NOT correspond
Test case Id	TC_N_39_CS
...	
Prerequisite(s)	Charging Station supports Monitoring This test case assumes the Charging Station has a non-numeric Component Variable <Configured non-numeric delta component variable> and numeric Component Variable <Configured threshold monitor component variable> .

Before (Preparations)
...
Memory State: Variable monitor is already set with component.name – EVSE, variable.name – AvailabilityState, type – Delta N/a
Reusable State: N/a

Main (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a SetVariableMonitoringResponse	1. The Test System sends a SetVariableMonitoringRequest with setMonitoringData[0].type Delta setMonitoringData[0].value 1 setMonitoringData[0].component = <Configured non-numeric delta component variable> setMonitoringData[0].variable = <Configured non-numeric delta component variable>
2. The Charging Station responds with a SetVariableMonitoringResponse 4. The Charging Station responds with a SetVariableMonitoringResponse	1. The Test System sends a SetVariableMonitoringRequest with setMonitoringData.type UpperThreshold setMonitoringData.variable.name Power setMonitoringData.component.name ChargingStation 3. The Test System sends a SetVariableMonitoringRequest with setMonitoringData[0].id <SetVariableMonitoringResponse.setMonitoringResult[0].id of step 2> setMonitoringData[0].type Delta setMonitoringData[0].value 1 setMonitoringData[0].component = <Configured numeric delta component variable> setMonitoringData[0].variable = <Configured numeric delta component variable>

Main (Test scenario)	
4. The Charging Station responds with a GetMonitoringReportResponse 6. The Charging Station responds with a GetMonitoringReportResponse	3. The Test System sends a GetMonitoringReportRequest with requestId <Generated requestId> 5. The Test System sends a GetMonitoringReportRequest with requestId <Generated requestId>
5. The Charging Station sends a NotifyMonitoringReportRequest 7. The Charging Station sends a NotifyMonitoringReportRequest	6. The Test System responds with a NotifyMonitoringReportResponse 8. The Test System responds with a NotifyMonitoringReportResponse
Note(s): - If tbc is True at Step 3 then step 3 and 4 will be repeated - If tbc is True at Step 7 then step 7 and 8 will be repeated	

Tool validations

* Step 2:

Message **SetVariableMonitoringResponse**

- **setMonitoringResult[0].status** *Rejected*
- **setMonitoringResult[0].type** *UpperThreshold*
- **setMonitoringResult[0].severity** *<Configured severity>*
- **setMonitoringResult[0].component.name** *ChargingStation*
- **setMonitoringResult[0].variable.name** *Power*

* Step 4:

Message **GetMonitoringReportResponse**

- **status** *Accepted*

* Step 5:

Message **NotifyMonitoringReportRequest**

- **monitor.component** *EVSE*
- **monitor.variable** *AvailabilityState*

* Step 2:

Message **SetVariableMonitoringResponse**

- **setMonitoringResult[0].id** *<not omitted>*
- **setMonitoringResult[0].status** *Accepted*
- **setMonitoringResult[0].type** *Delta*
- **setMonitoringResult[0].component** = *<Configured non-numeric delta component variable>*
- **setMonitoringResult[0].variable** = *<Configured non-numeric delta component variable>*

* Step 4:

Message **SetVariableMonitoringResponse**

- **setMonitoringResult[0].status** *Rejected*
- **setMonitoringResult[0].type** *Delta*
- **setMonitoringResult[0].component** = *<Configured numeric delta component variable>*
- **setMonitoringResult[0].variable** = *<Configured numeric delta component variable>*

* Step 6:

Message **GetMonitoringReportResponse**

- **status** *Accepted*

* Step 7:

Message **NotifyMonitoringReportRequest**

Must contain a monitor with

- **monitor[0].component** = *<Configured non-numeric delta component variable>*
- **monitor[0].variable** = *<Configured non-numeric delta component variable>*
- **monitor[0].variableMonitoring[0].id** = *<SetVariableMonitoringResponse.setMonitoringResult[0].id of step 2>*
- **monitor[0].variableMonitoring[0].value** = *1*
- **monitor[0].variableMonitoring[0].type** = *Delta*

Post scenario validations:

- All report parts have been received

6.2.93. Page 472 - (2024-12) - TC_N_26_CS - Made test case more explicit and more time before ending

Note: This erratum was released in tag 2025-01, not the mentioned 2024-12.

TC_N_26_CS: Retrieve Log Information - Diagnostics Log - Upload failed

Main (Test scenario)

Charging Station

CSMS

Main (Test scenario)	
2. The Charging Station responds with a GetLogResponse	1. The Test System sends a GetLogRequest with - logType <i>DiagnosticsLog</i> - retries 3 - retryInterval <Configured retryInterval> - log.remoteLocation <Configured log location with non-existing path>
...	
Note(s): Steps 3 & 4 are optional after the first attempt. The Charging Station will perform step (3,) 5, four times with <Configured retryInterval> seconds in between. - Step 3-4, 5-6 and 3-6 may repeat multiple times depending on Charging Station's implementation. - The Test System waits at least (3 * <Configured retryInterval>), before ending the testcase.	

Tool validations
...
* Step 3:
Must be sent exactly 1 or 4 times
Message LogStatusNotificationRequest
- status <i>Uploading</i>
- requestId Same Id as the GetLogRequest
* Step 5:
Must be sent exactly 1 or 4 times
Message LogStatusNotificationRequest
...
Post scenario validations:
- N/a

6.2.94. Page 470 - (2024-12) - TC_N_24_CS - Test case now searches suitable variable to do test with

Note: This erratum has been superseded by erratum: [Page 470 - \(2025-02\) - TC_N_24_CS - Updating test case for using more specific configuration variables](#)

TC_N_24_CS: Set Variable Monitoring - Periodic event

Test case name	Set Variable Monitoring - Periodic event
Test case Id	TC_N_24_CS
...	
Prerequisite(s)	Charging Station has implemented device model monitoring and MonitoringCtrlr::Enabled = true. This test case assumes the device model exposes at least one component.variable which can be monitored.

Before (Preparations)
...
Reusable State(s):
N/a
State is [csCommunicatedBaseReport]

Main (Test scenario)	
Charging Station	CSMS

Main (Test scenario)	
Search [csCommunicatedBaseReport].baseReportData to get a baseReportData.reportData WHERE reportData.variableCharacteristics.supportsMonitoring is true AND reportData.component.instance is <omitted> AND reportData.instance.instance is <omitted> AS <componentVariable>	
Set the monitor to generate a periodic event notification	
2. Charging Station responds with SetVariableMonitoringResponse	1. Test System sends SetVariableMonitoringRequest with: setMonitoringData[0].value = <Configured Clock Aligned MeterValues Interval> - setMonitoringData[0].value = 2 - setMonitoringData[0].type = Periodic setMonitoringData[0].component.name = "EVSE" setMonitoringData[0].component.evse.id = <Configured evseId> setMonitoringData[0].variable.name = "AvailabilityState" - setMonitoringData[0].component.name = <componentVariable.component.name> - setMonitoringData[0].component.evse.id = <componentVariable.component.evse.id> - setMonitoringData[0].variable.name = <componentVariable.variable.name>
3. Charging Station generates NotifyEventRequest for EVSE #1::AvailabilityState_ every <Configured Clock Aligned MeterValues Interval> seconds. 3. Charging Station sends a NotifyEventRequest	4. Test System responds with a NotifyEventResponse
<u>Note(s)</u> : Step 3 and 4 will repeat every 2 seconds	
Tool validations	
* Step 2: Message: SetVariableMonitoringResponse with: setMonitoringResult[0].status = Accepted setMonitoringResult[0].component.name = "EVSE" setMonitoringResult[0].component.evse.id = <Configured evseId> setMonitoringResult[0].variable.name = "AvailabilityState" setMonitoringResult[0].component.name = <componentVariable.component.name> setMonitoringResult[0].component.evse.id = <componentVariable.component.evse.id> setMonitoringResult[0].variable.name = <componentVariable.variable.name> setMonitoringResult[0].attributeStatusInfo is absent or attributeStatusInfo.reasonCode = "NoError"	
* Step 3: Message: a NotifyEventRequest message every <Configured Clock Aligned MeterValues Interval> seconds with: Message: NotifyEventRequest every 2 seconds with: with an eventData element with: - trigger = Periodic component.name = "EVSE" component.evse.id = 1 variable.name = "AvailabilityState" - component.name = <componentVariable.component.name> - component.evse.id = <componentVariable.component.evse.id> - variable.name = <componentVariable.variable.name>	
Post scenario validations: N/A	

6.2.95. Page 470 - (2025-02) - TC_N_24_CS - Updating test case for using more specific configuration variables

Before (Preparations)
Configuration State: N/a
Memory State: N/a Set MonitoringLevel to 8
Reusable State(s): N/a

Main (Test scenario)	
Charging Station	CSMS
Set the monitor to generate a periodic event notification	
2. Charging Station responds with SetVariableMonitoringResponse	1. Test System sends SetVariableMonitoringRequest with: - setMonitoringData[0].value = <Configured Clock Aligned MeterValues Interval> - setMonitoringData[0].type = Periodic - setMonitoringData[0].component.name = "EVSE" - setMonitoringData[0].component.evse.id = <Configured evseId> - setMonitoringData[0].variable.name = "AvailabilityState" - setMonitoringData[0].severity = 5 - setMonitoringData[0].component = <Configured threshold monitor component variable> - setMonitoringData[0].variable = <Configured threshold monitor component variable>
3. Charging Station generates NotifyEventRequest for EVSE #1:: AvailabilityState_ every <Configured Clock Aligned MeterValues Interval> seconds. 3. Charging Station sends a NotifyEventRequest	4. Test System responds with a NotifyEventResponse
Note(s): Step 3 and 4 will repeat every <Configured Clock Aligned MeterValues Interval> seconds	

Tool validations
* Step 2: Message: SetVariableMonitoringResponse with: setMonitoringResult[0].status = Accepted setMonitoringResult[0].component.name = "EVSE" setMonitoringResult[0].component.evse.id = <Configured evseId> setMonitoringResult[0].variable.name = "AvailabilityState" setMonitoringResult[0].type = Periodic setMonitoringResult[0].severity = 5 setMonitoringResult[0].component = <Configured threshold monitor component variable> setMonitoringResult[0].variable = <Configured threshold monitor component variable> setMonitoringResult[0].attributeStatusInfo is absent or attributeStatusInfo.reasonCode = "NoError"

Tool validations * Step 3: Message: a NotifyEventRequest message every <Configured Clock Aligned MeterValues Interval> seconds with: Message: NotifyEventRequest every <Configured Clock Aligned MeterValues Interval> seconds with: with an eventData element with: - trigger = Periodic - component.name = "EVSE" - component.evse.id = 1 - variable.name = "AvailabilityState" - component = <Configured threshold monitor component variable> - variable = <Configured threshold monitor component variable>
Post scenario validations: N/A

6.2.96. Page 482 - (2025-02) - TC_N_63_CS - Clear Customer Information - add manual action to stop session

Note: This erratum is extended by erratum: [Page 482 - \(2025-04\) - TC_N_63_CS - Added missing reusable state EnergyTransferStarted at before steps](#)

Test case name	Clear Customer Information - Clear and report - customerCertificate
Test case Id	TC_N_63_CS
...	...

Before (Preparations) Configuration State: - AuthCtrlr.Enabled is true - AuthCtrlr.DisableRemoteAuthorization is false - ISO15118Ctrlr.CentralContractValidationAllowed is false - ISO15118Ctrlr.V2GCertificateInstallationEnabled is true - ISO15118Ctrlr.ContractCertificateInstallationEnabled is true - ISO15118Ctrlr.PnCEnabled is true - ISO15118Ctrlr.Secold is NL - ISO15118Ctrlr.CountryName is secold - ISO15118Ctrlr.OrganizationName is OCA
Memory State: RenewV2GChargingStationCertificate (If none are present, when checking with GetInstalledCertificateIds.certificateType = V2GCertificateChain)
Reusable State: Execute Reusable State EVConnectedPreSession Execute Reusable State Authorized15118 (PnC)

Main (Test scenario)	
Charging Station	CSMS
Manual action : EV ends the charging session. Note: The Charging Station receives a SessionStopReq(Terminate) message from the EV to finish the transaction.	
2. The Charging Station responds with a CustomerInformationResponse	1. The Test System sends a CustomerInformationRequest with - report true AND - clear true AND - customerCertificate certificate hash data of contract certificate

Main (Test scenario)	
3. The Charging Station sends a NotifyCustomerInformationRequest	4. The Test System responds with a NotifyCustomerInformationResponse
<u>Note(s):</u> - If tbc is True at Step 3 then step 3 and 4 will be repeated	
6. The Charging Station responds with a CustomerInformationResponse	5. The Test System sends a CustomerInformationRequest with - report true AND - clear false AND - customerCertificate certificate hash data of contract certificate
7. The Charging Station sends a NotifyCustomerInformationRequest	8. The Test System responds with a NotifyCustomerInformationResponse
<u>Note(s):</u> - If tbc is True at Step 7 then step 7 and 8 will be repeated	

Tool validations
...
Post scenario validations: - All report parts have been received

6.2.97. Page 487 - (2025-04) - TC_N_36_CS - LogStatusNotification(AcceptedCanceled) allowed before GetLogResponse

It is allowed that the LogStatusNotification(AcceptedCanceled) for the canceled GetLog is sent before the new GetLogRequest is responded to by a GetLogResponse.

Test case name	Retrieve Log Information - Second Request
Test case Id	TC_N_36_CS
...	...

Before (Preparations)
...

Main (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a GetLogResponse	1. The Test System sends a GetLogRequest with logType <Configured logType> and requestId <#1>
<u>Note(s):</u> - Charging Station is uploading log file	
3. The Charging Station sends a LogStatusNotificationRequest	4. The Test System responds with a LogStatusNotificationResponse .
<u>Note(s):</u> - Charging Station cancels uploading the first log file	
6. The Charging Station responds with a GetLogResponse	5. The Test System sends a GetLogRequest with logType <Configured logType> and requestId <#2>
Step 7 is allowed to occur before step 6 7. The Charging Station sends a LogStatusNotificationRequest	8. The Test System responds with a LogStatusNotificationResponse .
<u>Note(s):</u> - Charging Station is uploading log file	

Main (Test scenario)	
9. The Charging Station sends a LogStatusNotificationRequest	10. The Test System responds with a LogStatusNotificationResponse .
<u>Note(s):</u> - Log file is uploaded	
11. The Charging Station sends a LogStatusNotificationRequest	12. The Test System responds with a LogStatusNotificationResponse .

Tool validations
* Step 2: Message GetLogResponse - status <i>Accepted</i> * Step 3: Message LogStatusNotificationRequest - status <i>Uploading</i> - requestId <#1> Step 7 is allowed to occur before step 6 * Step 6: Message GetLogResponse - status <i>AcceptedCanceled</i> * Step 7: Message LogStatusNotificationRequest - status <i>AcceptedCanceled</i> - requestId <#1> * Step 9: Message LogStatusNotificationRequest - status <i>Uploading</i> - requestId <#2> * Step 11: Message LogStatusNotificationRequest - status <i>Uploaded</i> - requestId <#2> Post scenario validations: - N/a

6.2.98. Page 482 - (2025-04) - TC_N_63_CS - Added missing reusable state **EnergyTransferStarted** at before steps

Note: This erratum extends erratum: [Page 482 - \(2025-02\) - TC_N_63_CS - Clear Customer Information - add manual action to stop session](#)

Test case name	Clear Customer Information - Clear and report - customerCertificate
Test case Id	TC_N_63_CS
...	

Before (Preparations)
Configuration State: ... Memory State: RenewV2GChargingStationCertificate (If none are present, when checking with <code>GetInstalledCertificateIds.certificateType = V2GCertificateChain</code>)

Before (Preparations)
Reusable State: Execute Reusable State <i>EVConnectedPreSession</i> Execute Reusable State <i>Authorized15118</i> (PnC) Execute Reusable State <i>EnergyTransferStarted</i>

6.2.99. Page 493 - (2024-09) - TC_N_41_CS - Set Variable Monitoring - Return to FactoryDefault

Moved preconfigured monitor to Prerequisite.

Test case name	Set Variable Monitoring - Return to FactoryDefault	
Test case Id	TC_N_41_CS	
...		
Prerequisite(s)	Charging Station supports Monitoring and a preconfigured monitor exists with <i>id</i> <Preconfigured monitor id> for component <i>EVSE</i> and variable <i>AvailabilityState</i> and <i>type</i> = <i>Delta</i> and <i>severity</i> = <Preconfigured severity>	
Before (Preparations)	Configuration state: N/a	
	Memory state: a preconfigured monitor exists with <i>id</i> <Preconfigured monitor id> for component <i>EVSE</i> and variable <i>AvailabilityState</i> and <i>type</i> = <i>Delta</i> and <i>severity</i> = <Preconfigured severity>	
	Charging State: N/a	
Main (Test scenario)	Charging Station	CSMS
	...	

6.2.100. Page 482 - (2024-09) - TC_N_63_CS - Clear Customer Information - Clear and report - customerCertificate

Test case design top stop transaction was not correct for an ISO 15118 session.

Test case name	Clear Customer Information - Clear and report - customerCertificate	
Test case Id	TC_N_63_CS	
...		
Before (Preparations)	Configuration State: N/a	
	Memory State: N/a	
	Charging State: Execute Reusable State <i>EVConnectedPreSession</i> Execute Reusable State <i>Authorized15118</i> Execute Reusable State <i>ParkingBayUnoccupied</i>	

Test case name	Clear Customer Information - Clear and report - customerCertificate	
Main (Test scenario)	Charging Station	CSMS
	Note : The Charging Station receives a SessionStopReq(Terminate) message from the EV to finish the transaction.	
	2. The Charging Station responds with a CustomerInformationResponse	1. The OCTT sends a CustomerInformationRequest with - report <i>true</i> AND - clear <i>true</i> AND - customerCertificate <i>customer information used in the transaction</i>
	3. The Charging Station sends a NotifyCustomerInformationRequest	4. The OCTT responds with a NotifyCustomerInformationResponse
	Note(s): - If tbc is <i>True</i> at Step 3 then step 3 and 4 will be repeated	
	6. The Charging Station responds with a CustomerInformationResponse	5. The OCTT sends a CustomerInformationRequest with - report <i>true</i> AND - clear <i>false</i> AND - customerCertificate <i>customer information used in the transaction</i>
	7. The Charging Station sends a NotifyCustomerInformationRequest	8. The OCTT responds with a NotifyCustomerInformationResponse
	Note(s): - If tbc is <i>True</i> at Step 7 then step 7 and 8 will be repeated	
Tool validations	...	

6.2.101. Page 482 - (2025-02) - TC_N_63_CS - Added missing configuration state and authorize explicit using Plug and Charge (PnC)

Before (Preparations)
Configuration State: N/a - AuthCtrlr.Enabled is <i>true</i> - AuthCtrlr.DisableRemoteAuthorization is <i>false</i> - ISO15118Ctrlr.CentralContractValidationAllowed is <i>false</i> - ISO15118Ctrlr.V2GCertificateInstallationEnabled is <i>true</i> - ISO15118Ctrlr.ContractCertificateInstallationEnabled is <i>true</i> - ISO15118Ctrlr.PnCEnabled is <i>true</i> - ISO15118Ctrlr.SecCId is <i>NL</i> - ISO15118Ctrlr.CountryName is <i>secCId</i> - ISO15118Ctrlr.OrganizationName is <i>OCA</i>
Memory State: N/a
Reusable State: Execute Reusable State <i>EVConnectedPreSession</i> Execute Reusable State <i>Authorized15118</i> (PnC)

6.2.102. Page 493 - (2024-09) - TC_N_41_CS - Set Variable Monitoring - Return to FactoryDefault

Moved preconfigured monitor to Prerequisite.

Test case name	Set Variable Monitoring - Return to FactoryDefault	
Test case Id	TC_N_41_CS	
...		
Prerequisite(s)	Charging Station supports Monitoring and a preconfigured monitor exists with id <Preconfigured monitor id> for component EVSE and variable AvailabilityState and type = Delta and severity = <Preconfigured severity>	
Before (Preparations)	Configuration state: N/a	
	Memory state: a preconfigured monitor exists with id <Preconfigured monitor id> for component EVSE and variable AvailabilityState and type = Delta and severity = <Preconfigured severity>	
	Charging State: N/a	
Main (Test scenario)	Charging Station	CSMS
	...	

6.2.103. Page 493 - (2025-02) - TC_N_41_CS - Made less dependent on test case configuration variables, enables predefines monitors

TC_N_41_CS: Set Variable Monitoring - Return to FactoryDefault

Test case name	Set Variable Monitoring - Return to FactoryDefault
Test case Id	TC_N_41_CS
Use case Id(s)	N03
Requirement(s)	N03.FR.04, N04.FR.15
System under test	Charging Station
Description	This test case describes how the CSMS requests the Charging Station to overrule a preconfigured monitor by a custom monitor. When monitoringBase is set to FactoryDefault the preconfigured monitor must return.
Purpose	To verify if the Charging station is able to correctly restore monitors to FactoryDefault.
Prerequisite(s)	Charging Station supports Monitoring and a preconfigured monitor exists with id <Preconfigured monitor id> for component EVSE and variable AvailabilityState and type = Delta and severity = <Preconfigured severity> - Charging Station supports Monitoring - A preconfigured monitor exists with id <Configured preconfigured monitor id> .

Before (Preparations)
Configuration State: N/a
Memory State: N/a MonitoringBase has been set to FactoryDefault
Reusable State: N/a

Main (Test scenario)	
Charging Station	CSMS

Main (Test scenario)	
2. The Charging Station responds with GetMonitoringReportResponse	1. The Test System sends GetMonitoringReportRequest with requestId = <Generated requestId>
3. The Charging Station sends NotifyMonitoringReportRequest	4. The Test System responds with NotifyMonitoringReportResponse
Note(s): - If NotifyMonitoringReportRequest.tbc is True in Step 3 then step 3 and 4 will be repeated	
Search NotifyMonitoringReportRequest.monitoringReportData of step 3 to get a monitoringReportData.monitor WHERE monitor.variableMonitoring.id is <Configured preconfigured monitor id> AS <preconfiguredMonitor>	
6. The Charging Station responds with a SetVariableMonitoringResponse	5. The Test System sends a SetVariableMonitoringRequest with setMonitoringData.id <Preconfigured monitor id> AND setMonitoringData.type Delta setMonitoringData.severity <Preconfigured severity> + 1 setMonitoringData.id <preconfiguredMonitor.variableMonitoring.id> setMonitoringData.transaction <preconfiguredMonitor.variableMonitoring.transaction> setMonitoringData.value <preconfiguredMonitor.variableMonitoring.value> setMonitoringData.type <preconfiguredMonitor.variableMonitoring.type> setMonitoringData.severity IF <preconfiguredMonitor.variableMonitoring.severity> < 9 THEN <preconfiguredMonitor.variableMonitoring.severity> + 1 ELSE 5 END IF setMonitoringData.component <preconfiguredMonitor.component> setMonitoringData.variable <preconfiguredMonitor.variable>
8. The Charging Station responds with a GetMonitoringReportResponse	7. The Test System sends a GetMonitoringReportRequest with - requestId <Generated requestId> - id <Preconfigured monitor id> - componentVariable.component.name EVSE - componentVariable.component.evse.id evseId - componentVariable.variable.name AvailabilityState - monitoringCriteria DeltaMonitoring - id <preconfiguredMonitor.variableMonitoring.id> - componentVariable.component <preconfiguredMonitor.component> - componentVariable.variable <preconfiguredMonitor.variable>
9. The Charging Station sends a NotifyMonitoringReportRequest	10. The Test System responds with a NotifyMonitoringReportResponse .
12. The Charging Station responds with a SetMonitoringBaseResponse with - status Accepted	11. The Test System sends a SetMonitoringBaseRequest with - monitoringBase FactoryDefault

Main (Test scenario)	
14. The Charging Station responds with a GetMonitoringReportResponse	13. The Test System sends a GetMonitoringReportRequest with - requestId <Generated requestId> - id <Preconfigured monitor id> - componentVariable.component.name = EVSE - componentVariable.component.evse.id evseId - componentVariable.variable.name AvailabilityState - monitoringCriteria DeltaMonitoring - id <preconfiguredMonitor.variableMonitoring.id> - componentVariable.component <preconfiguredMonitor.component> - componentVariable.variable <preconfiguredMonitor.variable>
15. The Charging Station sends a NotifyMonitoringReportRequest	16. The Test System responds with a NotifyMonitoringReportResponse .

Tool validations

* Step 2:

Message **GetMonitoringReportResponse**

- **status** *Accepted*

* Step 6:

Message **SetVariableMonitoringResponse**

- **setMonitoringResult[0].status** *Accepted*

- **setMonitoringResult[0].type** *Delta*

~~- **setMonitoringResult[0].component.name** *EVSE*~~

~~- **setMonitoringResult[0].variable.name** *AvailabilityState*~~

- **setMonitoringResult[0].component** *<preconfiguredMonitor.component>*

- **setMonitoringResult[0].variable** *<preconfiguredMonitor.variable>*

* Step 8:

Message **GetMonitoringReportResponse**

- **status** *Accepted*

* Step 9:

Message **NotifyMonitoringReportRequest**

~~- **monitor.component.name** *EVSE*~~

~~- **monitor.variable.name** *AvailabilityState*~~

~~- **monitor.variableMonitoring.id** *<Preconfigured id>*~~

~~- **monitor.variableMonitoring.severity** *<Preconfigured severity> + 1*~~

Should contain monitor:

- **monitor.component** *<preconfiguredMonitor.component>*

- **monitor.variable** *<preconfiguredMonitor.variable>*

- **monitor.variableMonitoring.id** *<preconfiguredMonitor.variableMonitoring.id>*

- **monitor.variableMonitoring.severity** IF *<preconfiguredMonitor.variableMonitoring.severity> < 9 THEN*

<preconfiguredMonitor.variableMonitoring.severity> + 1

ELSE

5

END IF

* Step 15:

Message **NotifyMonitoringReportRequest**

~~- **monitor.component.name** *EVSE*~~

~~- **monitor.variable.name** *AvailabilityState*~~

~~- **monitor.variableMonitoring.severity** *<Preconfigured severity> + 1*~~

~~- **monitor.variableMonitoring.id** *<Preconfigured id>*~~

~~- **monitor.variableMonitoring.severity** *<Preconfigured severity> + 1*~~

Should contain monitor:

- **monitor.component** *<preconfiguredMonitor.component>*

- **monitor.variable** *<preconfiguredMonitor.variable>*

- **monitor.variableMonitoring.id** *<preconfiguredMonitor.variableMonitoring.id>*

- **monitor.variableMonitoring.severity** *<preconfiguredMonitor.variableMonitoring.severity>*

Post scenario validations:

- All report parts have been received

6.2.104. Page 495 - (2024-11) - TC_N_43_CS - Remove incorrect tool validation StatusInfo

Test case name	Set Variable Monitoring - First SetMonitoringData and third SetMonitoringData are valid, but the second contains an out of range value
Test case Id	TC_N_43_CS
...	...

Tool validations
* Step 2: Message: SetVariableMonitoringResponse with (in arbitrary order): setMonitoringResult[1] = { - status = <i>Accepted</i> - type = <i>UpperThreshold</i> - statusInfo is absent or statusInfo.reasonCode = <i>"NoError"</i> } setMonitoringResult[2] = { - status = <i>Rejected</i> - type = <i>Delta</i> - statusInfo is absent or statusInfo.reasonCode = <i>"NoError"</i> (Removed) } setMonitoringResult[3] = { - status = <i>Accepted</i> - type = <i>LowerThreshold</i> - statusInfo is absent or statusInfo.reasonCode = <i>"NoError"</i> }
Post scenario validations:
- N/a

6.2.105. Page 495 - (2025-02) - TC_N_43_CS - Updating test case for using more specific configuration variables

TC_N_43_CS: Set Variable Monitoring - First SetMonitoringData and third SetMonitoringData are valid, but the second contains an out of range value

Test case name	Set Variable Monitoring - First SetMonitoringData and third SetMonitoringData are valid, but the second contains an out of range value
Test case Id	TC_N_43_CS
Use case Id(s)	N04
Requirement(s)	N/a
System under test	Charging Station
Description	This test case describes how the CSMS requests the Charging Station to set monitoring triggers on Variables. Multiple triggers can be set for upper or lower thresholds, delta changes or periodic reporting.
Purpose	To verify if the Charging station is able to correctly respond when one of requested variable monitor data is out of range replace as described at the OCPP specification.
Prerequisite(s)	Charging Station supports Monitoring This test case assumes the Charging Station has a numeric Component Variable <i><Configured numeric delta component variable></i> and numeric Component Variable <i><Configured threshold monitor component variable></i> .

Before (Preparations)
Configuration State: N/a

Before (Preparations)
Memory State: N/a
Reusable State: N/a

Main (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a SetVariableMonitoringResponse	1. The Test System sends a SetVariableMonitoringRequest with - setMonitoringData.component.name = <Configured threshold monitor component variable> - setMonitoringData.variable.name = <Configured threshold monitor component variable> - setMonitoringData[0].value = <Configured threshold monitor value> - setMonitoringData[0].component = <Configured threshold monitor component variable> - setMonitoringData[0].variable = <Configured threshold monitor component variable> - setMonitoringData[0].value = <Configured threshold monitor component variable UpperThreshold trigger value> - setMonitoringData[0].type = UpperThreshold - setMonitoringData[1].component = <Configured numeric delta component variable> - setMonitoringData[1].variable = <Configured numeric delta component variable> - setMonitoringData[1].value = -1.0 - setMonitoringData[1].type = Delta - setMonitoringData[2].value = <Configured threshold monitor2 value> - setMonitoringData[2].component = <Configured threshold monitor component variable> - setMonitoringData[2].variable = <Configured threshold monitor component variable> - setMonitoringData[2].value = <Configured threshold monitor component variable LowerThreshold trigger value> - setMonitoringData[2].type = LowerThreshold

Tool validations

* Step 2:

Message: **SetVariableMonitoringResponse** with (in arbitrary order):

setMonitoringResult[1] = {

- **status** = *Accepted*

- **type** = *UpperThreshold*

- **statusInfo** is absent or **statusInfo.reasonCode** = *"NoError"*

}

setMonitoringResult[2] = {

- **status** = *Rejected*

- **type** = *Delta*

}

setMonitoringResult[3] = {

- **status** = *Accepted*

- **type** = *LowerThreshold*

- **statusInfo** is absent or **statusInfo.reasonCode** = *"NoError"*

}

Post scenario validations:

- N/a

6.2.106. Page 497 - (2025-02) - TC_N_45_CS - Updating test case for using more specific configuration variables

TC_N_45_CS: Alert Event - Delta value exceeded

Test case name	Alert Event - Delta value exceeded
Test case Id	TC_N_45_CS
Use case Id(s)	N07
Requirement(s)	N07.FR.06, N07.FR.07, N07.FR.18, N07.FR.19
System under test	Charging Station
Description	NotifyEventRequest reports every Component/Variable for which a VariableMonitoring setting was triggered. Only the VariableMonitoring settings that are responsible for triggering an event are included.
Purpose	To verify if the Charging station is correctly communicating when a delta value has exceeded as described at the OCPP specification.
Prerequisite(s)	n/a

Before (Preparations)
Configuration State: N/a
Memory State: Variable monitor is configured with: - setMonitoringData.component.name = <Configured threshold monitor component variable> - setMonitoringData.component.evse.id = <Configured EVSEId> - setMonitoringData.value = <Configured threshold monitor value> - setMonitoringData.type = Delta - setMonitoringData.variable.name = <Configured delta monitor component variable> - setMonitoringData.component = <Configured numeric delta component variable> - setMonitoringData.variable = <Configured numeric delta component variable Delta numeric trigger value> - setMonitoringData.value = <Configured numeric delta component variable> - setMonitoringData.type = Delta - setMonitoringData.severity = 5 Set MonitoringLevel to 8 Notes: If componentVariable is set to "Power" or "Current", the value is set to 100.0
Reusable State: N/a

Main (Test scenario)	
Charging Station	CSMS
<u>Manual Action:</u> If componentVariable is set to "Power" or "Current" EnergyTransferStarted will trigger the monitor. If another componentvariable is chosen a manual action is needed to trigger the monitor.	
1. Execute Reusable State <i>EnergyTransferStarted</i> or manually trigger the monitor.	
2. The Charging Station sends a NotifyEventRequest	3. The Test System responds with a NotifyEventResponse .
<u>Note(s):</u> - If tbc is True at Step 2 then step 1 and 3 will be repeated	

Tool validations

* Step 2:

Message **NotifyEventRequest**

- **eventData[0].trigger** *Delta*
- ~~**eventData[0].component.name** <Configured threshold monitor component variable>~~
- ~~**eventData[0].variable.name** <Configured threshold monitor component variable>~~
- ~~**eventData[0].variableMonitoringId** <Configured variableMonitoringId>~~
- **eventData[0].component** <Configured numeric delta component variable>
- **eventData[0].variable** <Configured numeric delta component variable>
- **eventData[0].variableMonitoringId** <SetVariableMonitoringResponse.setMonitoringResult.id in preparation phase>

Post scenario validations:

- N/a

6.2.107. Page 501 - (2025-02) - TC_N_51_CS - Updating test case for using more specific configuration variables

TC_N_51_CS: Set Variable Monitoring - ~~Replace Variable Monitor~~ **Modifying a VariableMonitor and trigger**

Test case name	Set Variable Monitoring - Replace Variable Monitor Set Variable Monitoring - Modifying a VariableMonitor and trigger
Test case Id	TC_N_51_CS
Use case Id(s)	N07
Requirement(s)	N07.FR.11
System under test	Charging Station
Description	NotifyEventRequest reports every Component/Variable for which a VariableMonitoring setting was triggered. Only the VariableMonitoring settings that are responsible for triggering an event are included.
Purpose	To verify if the Charging station is able to correctly check if the current value exceeds the new threshold as described at the OCPP specification.
Prerequisite(s)	Charging Station supports Monitoring

Before (Preparations)	
Configuration State: N/a	
Memory State: Variable monitor is already set with: setMonitoringData.component.name <Configured threshold monitor component variable> AND setMonitoringData.component.evse.id <Configured EVSEId> AND setMonitoringData.value <Configured threshold monitor value> AND setMonitoringData.type UpperThreshold AND setMonitoringData.variable.name <Configured threshold monitor component variable> setMonitoringData.component <Configured threshold monitor component variable> setMonitoringData.variable <Configured threshold monitor component variable> setMonitoringData.value <Configured threshold monitor component variable UpperThreshold non-trigger value> setMonitoringData.type UpperThreshold setMonitoringData.severity 5 Set MonitoringLevel to 8 <u>Notes:</u> If componentVariable is set to "Power" or "Current", the value is set to the configured maxLimit -1	
Reusable State: N/a	

Main (Test scenario)	
Charging Station	CSMS
<u>Notes:</u> If componentVariable is set to "Power" or "Current" EnergyTransferStarted will trigger the monitor. If another componentvariable is chosen a manual action is needed to trigger the monitor.	
1. Execute Reusable State <i>EnergyTransferStarted</i> or manually trigger the monitor.	

Main (Test scenario)	
3. The Charging Station responds with a SetVariableMonitoringResponse	2. The Test System sends a SetVariableMonitoringRequest with setMonitoringData.component.name <Configured threshold monitor component variable> AND setMonitoringData.component.evse.id <Configured EVSEId> AND setMonitoringData.id <Configured variableMonitoringId> AND setMonitoringData.value <Configured threshold monitor value2> AND setMonitoringData.type UpperThreshold setMonitoringData.variable.name <Configured threshold monitor component variable> setMonitoringData.component <Configured threshold monitor component variable> setMonitoringData.variable <Configured threshold monitor component variable> setMonitoringData.id <SetVariableMonitoringResponse.setMonitoringResult.id in preparation phase> setMonitoringData.value <Configured threshold monitor component variable UpperThreshold trigger value> setMonitoringData.type UpperThreshold <u>Notes:</u> If componentVariable is set to "Power" or "Current", the value is set to 0.0
4. The Charging station sends a NotifyEventRequest	5. The Test System responds with a NotifyEventResponse .

Tool validations	
* Step 3: Message SetVariableMonitoringResponse - setMonitoringResult[0].status Accepted - setMonitoringResult[0].type UpperThreshold - setMonitoringResult[0].severity <Configured severity> - setMonitoringResult[0].component.name <Configured threshold monitor component variable> - setMonitoringResult[0].variable.name <Configured threshold monitor component variable> - setMonitoringResult[0].severity 5 - setMonitoringResult[0].component <Configured threshold monitor component variable> - setMonitoringResult[0].variable <Configured threshold monitor component variable> * Step 4: Message NotifyEventRequest - eventData[0].trigger Alerting - eventData[0].actualValue <Configured threshold monitor value> - eventData[0].actualValue > <Configured threshold monitor component variable UpperThreshold non-trigger value>	
Post scenario validations: - All report parts have been received	

6.2.108. Page 503 - (2025-02) - TC_N_52_CS - Updating test case for using more specific configuration variables

TC_N_52_CS: Set Variable Monitoring - Removing a VariableMonitor

Test case name	Set Variable Monitoring - Removing a VariableMonitor
Test case Id	TC_N_52_CS
Use case Id(s)	N07
Requirement(s)	N07.FR.12
System under test	Charging Station
Description	NotifyEventRequest reports every Component/Variable for which a VariableMonitoring setting was triggered. Only the VariableMonitoring settings that are responsible for triggering an event are included.
Purpose	To verify if the Charging station is able to correctly communicate when a threshold has been exceeded and the applicable monitor is removed as described at the OCPP specification.
Prerequisite(s)	Charging Station supports Monitoring

Before (Preparations)
Configuration State: N/a Memory State: Variable monitor is already set with: setMonitoringData.component.name <Configured threshold monitor component variable> AND setMonitoringData.component.evse.id <Configured EVSEId> AND setMonitoringData.value <Configured threshold monitor value> AND setMonitoringData.type UpperThreshold AND setMonitoringData.variable.name <Configured threshold monitor component variable> setMonitoringData.component <Configured threshold monitor component variable> setMonitoringData.variable <Configured threshold monitor component variable> setMonitoringData.value <Configured threshold monitor component variable UpperThreshold trigger value> setMonitoringData.type UpperThreshold setMonitoringData.severity = 5 Set MonitoringLevel to 8 Notes: If componentVariable is set to "Power" or "Current", the value is set to 0.0 Reusable State: Execute Reusable State <i>EnergyTransferStarted</i> or manually trigger the monitor. Notes: If componentVariable is set to "Power" or "Current" EnergyTransferStarted will trigger the monitor. If another componentvariable is chosen a manual action is needed to trigger the monitor.

Main (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a ClearVariableMonitoringResponse	1. The Test System sends a ClearVariableMonitoringRequest with id <Configured variableMonitoringId> id <SetVariableMonitoringResponse.setMonitoringResult.id in preparation phase>
4. The Charging Station responds with a GetMonitoringReportResponse	3. The Test System sends a GetMonitoringReportRequest with componentVariable.component <Configured threshold monitor component variable> componentVariable.variable <Configured threshold monitor component variable> monitoringCriteria ThresholdMonitoring

Main (Test scenario)	
5. Execute Reusable State <i>StopAuthorized</i> or manually trigger the monitor. <i>Notes: If componentVariable is set to "Power" or "Current" EnergyTransferStarted will trigger the monitor. If another componentvariable is chosen a manual action is needed to trigger the monitor.</i>	
6. The Charging Station should not send a request for the cleared monitor	

Tool validations	
* Step 2: Message ClearVariableMonitoringResponse - clearMonitoringResult[0].status <i>Accepted</i> AND - clearMonitoringResult[0].id <i><Configured variableMonitoringId></i> - clearMonitoringResult[0].id <SetVariableMonitoringResponse.setMonitoringResult.id in preparation phase>	
* Step 4: Message GetMonitoringReportResponse - getMonitoringResult[0].status <i>EmptyResultSet</i>	
* Step 6: - No NotifyEventRequest with variableMontioringId <i><Configured variableMonitoringId></i> is send - No NotifyEventRequest with variableMontioringId <SetVariableMonitoringResponse.setMonitoringResult.id in preparation phase> is send	
Post scenario validations: - N/a	

6.2.109. Page 503 - (2025-11) - TC_N_52_CS - Updating test case to check configured monitor works

A prerequisite has been added and a check that the configured monitor works has been added as the first step.

TC_N_52_CS: Set Variable Monitoring - Removing a VariableMonitor

Test case name	Set Variable Monitoring - Removing a VariableMonitor
...	...
Prerequisite(s)	Charging Station supports Monitoring. If <i>Power</i> or <i>Current</i> is used as the monitored variable, then power must flow when a transaction reaches <i>EnergyTransfer</i> , or else the monitor (power or current > 0) is not triggered.

Before (Preparations)

Configuration State:

...

Memory State:

...

Reusable State:

Execute **Reusable State** *EnergyTransferStarted* or manually trigger the monitor.

Notes: If *componentVariable* is set to "Power" or "Current" *EnergyTransferStarted* will trigger the monitor. If another *componentVariable* is chosen a manual action is needed to trigger the monitor.

Main (Test scenario)

Charging Station	CSMS
1. Execute Reusable State <i>EnergyTransferStarted</i> or manually trigger the monitor. <u>Notes:</u> If <i>componentVariable</i> is set to "Power" or "Current" <i>EnergyTransferStarted</i> will trigger the monitor. If another <i>componentVariable</i> is chosen a manual action is needed to trigger the monitor.	
3. The Charging Station responds with a ClearVariableMonitoringResponse	2. The Test System sends a ClearVariableMonitoringRequest with id <monitoringId of monitor set in Memory State>
5. The Charging Station responds with a GetMonitoringReportResponse	4. The Test System sends a GetMonitoringReportRequest with componentVariable.component <Configured threshold monitor component variable> componentVariable.variable <Configured threshold monitor component variable> monitoringCriteria <i>ThresholdMonitoring</i>
6. Execute Reusable State <i>StopAuthorized</i> or manually trigger the monitor. <u>Notes:</u> If <i>componentVariable</i> is set to "Power" or "Current" <i>EnergyTransferStarted</i> will trigger the monitor. If another <i>componentvariable</i> is chosen a manual action is needed to trigger the monitor.	
<u>Note:</u> The Charging Station should not send a request for the cleared monitor	

Tool validations

* Step 1:

Message **NotifyEventRequest**

- **eventData[0].trigger** *Alerting*

- **eventData[0].component** <Configured threshold monitor component variable>

- **eventData[0].variable** <Configured threshold monitor component variable>

- **eventData[0].variableMonitoringId** <monitoringId of monitor set in Memory State> +

* Step 3:

Message **ClearVariableMonitoringResponse**

- **clearMonitoringResult[0].status** *Accepted* AND

- **clearMonitoringResult[0].id** <monitoringId of monitor set in Memory State>

Tool validations

* Step 5:

Message **GetMonitoringReportResponse**

- **getMonitoringResult[0].status** *EmptyResultSet*

Post scenario validations:

- No **NotifyEventRequest** that **variableMontioringId** <monitoringId of monitor set in Memory State> is cleared, is sent.

DRAFT

6.2.110. Page 504 - (2025-02) - TC_N_53_CS - Updating test case for using more specific configuration variables

TC_N_53_CS: Alert Event - Persistent over reboot

Test case name	Alert Event - Persistent over reboot
Test case Id	TC_N_53_CS
Use case Id(s)	N07
Requirement(s)	N07.FR.13
System under test	Charging Station
Description	NotifyEventRequest reports every Component/Variable for which a VariableMonitoring setting was triggered. Only the VariableMonitoring settings that are responsible for triggering an event are included.
Purpose	To verify if the Charging station is able to save the variableMonitor data persistent across reboot as described at the OCPP specification.
Prerequisite(s)	n/a

Before (Preparations)
Configuration State: N/a
Memory State: Variable monitor is already set with: setMonitoringData.component.name <Configured threshold monitor component variable> AND setMonitoringData.component.evse.id <Configured EVSEId> AND setMonitoringData.value <Configured threshold monitor value> AND setMonitoringData.type UpperThreshold AND setMonitoringData.variable.name <Configured threshold monitor component variable> setMonitoringData.component <Configured threshold monitor component variable> setMonitoringData.variable <Configured threshold monitor component variable> setMonitoringData.value <Configured threshold monitor component variable UpperThreshold trigger value> setMonitoringData.type UpperThreshold
Reusable State: Execute Reusable State <i>Booted</i>

Main (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a GetMonitoringReportResponse	1. The Test System sends a GetMonitoringReportRequest with monitoringCriteria <i>ThresholdMonitoring</i>
3. The Charging Station sends a NotifyMonitoringReportRequest	4. The Test System responds with a NotifyMonitoringReportResponse .
Note(s): - If <i>tbc</i> is True at Step 3 then step 3 and 4 will be repeated	

Tool validations
* Step 3: Message NotifyMonitoringReportRequest - requestId <The Id of the request> AND monitor.variableMonitoring.id <Received monitorId from set monitor> - monitor.variableMonitoring.id <SetVariableMonitoringResponse.setMonitoringResult.id in preparation phase> - monitor.variableMonitoring.type <i>UpperThreshold</i>
Post scenario validations: - All reports have been received

6.2.111. Page 505 - (2025-02) - TC_N_56_CS - Made test case configurable using configuration variables

Before (Preparations)
Configuration State: N/a
Memory State: Variable monitor is configured with: component.evse.id <Configured EVSEId> component.name EVSE severity <Configured severity> type Delta value 1.0 variable.name AvailabilityState Variable monitor is configured with: - setMonitoringData[0].value = 1 - setMonitoringData[0].type = Delta - setMonitoringData[0].severity = 5 - setMonitoringData[0].component = <Configured non-numeric delta component variable> - setMonitoringData[0].variable = <Configured non-numeric delta component variable> Set MonitoringLevel to 8 <u>Notes:</u> - Take a non-numeric component variable which can easily modified to trigger the alert.
Reusable State: N/a
Tool validations * Step 1: Message NotifyEventRequest - eventData[0].trigger Delta - eventData[0].component.name EVSE - eventData[0].variable.name AvailabilityState - eventData[0].component <Configured non-numeric delta component variable> - eventData[0].variable <Configured non-numeric delta component variable> - eventData[0].variableMonitoringId monitoringId of monitor set in Memory State Post scenario validations: - N/a

6.2.112. Page 506 - (2025-06) - TC_O_XX_CS - Updated configurations

Some generic changes regarding the configuration of the display message testcases were needed:

- Identifying which display to use has been made configurable.
- The Start / End date time configurations have been changed to offset configurations, so the current time can be used as a base.

Test case Id	TC_O_01_CS, TC_O_06_CS, TC_O_10_CS, TC_O_13_CS, TC_O_14_CS, TC_O_17_CS, TC_O_18_CS, TC_O_19_CS, TC_O_20_CS, TC_O_22_CS, TC_O_28_CS, TC_O_30_CS, TC_O_32_CS, TC_O_36_CS, TC_O_37_CS, TC_O_38_CS, TC_O_39_CS
...	

Main (Test scenario)	
Charging Station	CSMS
n. The Charging Station responds with a SetDisplayMessageResponse	n. The Test System sends a SetDisplayMessageRequest with message.display <Configured display component variable> (Omitted when left empty)

Test case Id	TC_O_02_CS, TC_O_04_CS, TC_O_07_CS, TC_O_08_CS, TC_O_09_CS, TC_O_11_CS, TC_O_12_CS, TC_O_34_CS, TC_O_35_CS
...	

Before (Preparations)	
Configuration State: ...	
Memory State: SetDisplayMessage	
Reusable State: ...	

New Memory state:

Table 7. SetDisplayMessage

State	SetDisplayMessage
System under test	Charging Station
Description	This will set a display message at the Charging Station.

Before (Preparations)	
Configuration State: N/a	
Memory State: N/a	
Reusable State(s): N/a	

Main (Test scenario)	
Charging Station	CSMS

Main (Test scenario)	
2. The Charging Station responds with a SetDisplayMessageResponse	1. The Test System sends a SetDisplayMessageRequest with message.id <Generated displayMessageId> message.priority <Configured priority> message.state <Omitted, unless specifically described at the testcase> message.display <Configured display component variable> (Omitted when left empty) message.message.format <Configured Message Format> message.message.content <Configured Message>

Tool validations
* Step 2: (Message: SetDisplayMessageResponse) status is <i>Accepted</i>
Post scenario validations: N/a

Specific O testcase changes:

Test case name	Set Display Message - Replace DisplayMessage
Test case Id	TC_O_12_CS
...	

Main (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a SetDisplayMessageResponse	1. The Test System sends a SetDisplayMessageRequest with message.id <Generated displayMessageId from before (preperation) steps> message.message.content <Different message to indicate the message was replaced>
Note(s): - The display message is replaced by a new one.	

Test case name	Set Display Message - State Faulted
Test case Id	TC_O_39_CS
...	

Main (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a SetDisplayMessageResponse	1. The Test System sends a SetDisplayMessageRequest with message.id <Generated displayMessageId> message.priority <Configured Priority> message.state Faulted message.message.content <Message indicating the Charging Station is in a Faulted state> message.display <Configured display component variable> (Omitted when left empty)

Test case name	Set Display Message - Display message at StartTime
Test case Id	TC_O_13_CS
...	

Main (Test scenario)	
Charging Station	CSMS

Main (Test scenario)	
2. The Charging Station responds with a SetDisplayMessageResponse	1. The Test System sends a SetDisplayMessageRequest with message.id <Generated displayMessageId> message.priority <Configured Priority> message.startDateTime <Current dateTime + Configured Start Date Time Offset> message.display <Configured display component variable> (Omitted when left empty)
4. The Charging Station responds with a GetDisplayMessagesResponse	3. The Test System sends a GetDisplayMessagesRequest with id <Generated displayMessageId>
5. The Charging Station sends a NotifyDisplayMessagesRequest	6. The Test System responds with a NotifyDisplayMessagesResponse .
<u>Note(s):</u> - If tbc is True at Step 5 then step 5 and 6 will be repeated - Wait till <Configured Start Date Time Offset> seconds have passed - The display message should be displayed after <Configured Start Date Time Offset> seconds.	

Test case name	Set Display Message - Remove message after EndTime
Test case Id	TC_O_14_CS
...	

Main (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a SetDisplayMessageResponse	1. The Test System sends a SetDisplayMessageRequest with message.id <Generated displayMessageId> message.priority <Configured Priority> message.endDateTime <Current dateTime + Configured End Date Time Offset> message.display <Configured display component variable> (Omitted when left empty)
4. The Charging Station responds with a GetDisplayMessagesResponse	3. The Test System sends a GetDisplayMessagesRequest with id <Generated displayMessageId>
5. The Charging Station sends a NotifyDisplayMessagesRequest	6. The Test System responds with a NotifyDisplayMessagesResponse .
<u>Note(s):</u> - If tbc is True at Step 5 then step 5 and 6 will be repeated - Wait till <Configured End Date Time Offset> seconds have passed - The display message is displayed and removed after <Configured End Date Time Offset> seconds.	
8. The Charging Station responds with a GetDisplayMessagesResponse	7. The Test System sends a GetDisplayMessagesRequest with id <Generated displayMessageId> requestId <Generated requestId>

Test case name	Set Display Message - Specific transaction - Remove message after EndTime
Test case Id	TC_O_28_CS
...	

Main (Test scenario)	
Charging Station	CSMS

Main (Test scenario)	
2. The Charging Station responds with a SetDisplayMessageResponse	1. The Test System sends a SetDisplayMessageRequest with message.id <Generated displayMessageId> message.transactionId <Generated transactionId> message.priority <Configured Priority> message.endDateTime <Current dateTime + Configured End Date Time Offset> message.display <Configured display component variable> (Omitted when left empty)
Note(s): - The display message should be displayed. - Waiting <Configured End Date Time Offset> seconds. - The display message is not being displayed anymore after <Configured End Date Time Offset> seconds.	
4. The Charging Station responds with a GetDisplayMessagesResponse	3. The Test System sends a GetDisplayMessagesRequest with id <Generated displayMessageId>

Test case name	Set Display Message - NotSupportedMessageFormat
Test case Id	TC_O_19_CS
...	

Main (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a SetDisplayMessageResponse	1. The Test System sends a SetDisplayMessageRequest with message.id <Generated displayMessageId> message.message.format <Configured Unsupported Message Format> message.display <Configured display component variable> (Omitted when left empty)

Test case name	Set Display Message - Second Alwaysfront priority
Test case Id	TC_O_24_CS
...	

Main (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a SetDisplayMessageResponse	1. The Test System sends a SetDisplayMessageRequest with message.id <Generated displayMessageId> message.priority AlwaysFront
4. The Charging Station responds with a SetDisplayMessageResponse	3. The Test System sends a SetDisplayMessageRequest with message.id <Configured displayMessage2Id> message.priority AlwaysFront
6. The Charging Station responds with a GetDisplayMessagesResponse	5. The Test System sends a GetDisplayMessagesRequest with id <Configured displayMessageId>
6. The Charging Station responds with a GetDisplayMessagesResponse	5. The Test System sends a GetDisplayMessagesRequest with id <Configured displayMessage2Id>
7. The Charging Station sends a NotifyDisplayMessagesRequest	8. The Test System responds with a NotifyDisplayMessagesResponse .
Note(s): - If tbc is True at Step 7 then step 7 and 8 will be repeated - The message from step 1 is NOT displayed anymore and is replaced by the message from step 5.	

Tool validations
* Step 2: Message SetDisplayMessageResponse - status <i>Accepted</i> * Step 4: Message SetDisplayMessageResponse - status <i>Accepted</i> * Step 6: Message GetDisplayMessagesResponse - status <i>Unknown</i> * Step 6: Message GetDisplayMessagesResponse - status <i>Accepted</i> * Step 7: Message NotifyDisplayMessagesRequest - requestId <i><Generated requestId></i>
Post scenario validations: - N/a

6.2.113. Page 520 - (2025-02) - TC_O_15_CS - Test case removed

This test case has been removed.

Test case name	Set Display Message - Language preference of the EV Driver
Test case Id	TC_O_15_CS
...	...

6.2.114. Page 530 - (2025-02) - TC_O_28_CS - Transaction id should be specified for DisplayMessage

Main (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a SetDisplayMessageResponse	1. The Test System sends a SetDisplayMessageRequest with message.id <i><Generated displayMessageId></i> message.transactionId <i><Generated transactionId></i> message.priority <i><Configured Priority></i> message.endDateTime <i><Current dateTime + 60 seconds></i>
[...]	

6.2.115. Page 533 - (2025-02) - TC_O_32_CS - Made notes about display behaviour more explicit

Main (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a SetDisplayMessageResponse	1. The Test System sends a SetDisplayMessageRequest with message.id <i><Generated displayMessageId></i> message.transactionId <i><Received transactionId></i> AND message.priority <i>AlwaysFront</i>
Note(s): - Display message <i><Generated displayMessageId></i> is shown	

Main (Test scenario)	
4. The Charging Station responds with a SetDisplayMessageResponse	3. The Test System sends a SetDisplayMessageRequest with message.id <Configured displayMessageId> message.transactionId <Received transactionId> AND message.priority AlwaysFront
Note(s): - Display message <Generated displayMessage1Id> is not displayed anymore - Display message <Generated displayMessage2Id> is shown	
6. The Charging Station responds with a GetDisplayMessagesResponse	5. The Test System sends a GetDisplayMessagesRequest with id <Generated displayMessageId>
8. The Charging Station responds with a GetDisplayMessagesResponse	7. The Test System sends a GetDisplayMessagesRequest with id <Generated displayMessageId2> requestId <Generated requestId>
9. The Charging Station sends a NotifyDisplayMessagesRequest	10. The Test System responds with a NotifyDisplayMessagesResponse .
11. Execute Reusable State <i>StopAuthorized</i>	
12. Execute Reusable State <i>EVConnectedPostSession</i>	
13. Execute Reusable State <i>EVDisconnected</i>	
14. Execute Reusable State <i>ParkingBayUnoccupied</i>	
Note(s): The display message is not displayed anymore - Display message <Generated displayMessage2Id> is not displayed anymore	
16. The Charging Station responds with a GetDisplayMessagesResponse	15. The Test System sends a GetDisplayMessagesRequest with id <Generated displayMessageId2>

6.2.116. Page 544 - (2025-02) - TC_O_39_CS - Wait for StatusNotificationRequest or NotifyEventRequest

Main (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a SetDisplayMessageResponse	1. The Test System sends a SetDisplayMessageRequest with message.id <Generated displayMessageId> message.priority <Configured Priority> message.state <Configured State> message.message Faulted
Note(s): The display message should NOT be displayed.	
Manual Action: Set the Charging Station to state Faulted.	
3. The Charging Station notifies the CSMS about the status change of the Charging Station.	4. The Test System responds accordingly.
Note(s): The display message should be displayed now. - Step 3/4 are used to detect if the Charging Station status has changed. - The display message should be displayed now.	
Manual Action: Set the Charging Station back to state Available.	
5. The Charging Station notifies the CSMS about the status change of the Charging Station.	6. The Test System responds accordingly.

Main (Test scenario)	
<u>Note(s):</u> The display message should NOT be displayed anymore. - Step 5/6 are used to detect if the Charging Station status has changed. - The display message should NOT be displayed anymore.	
8. The Charging Station responds with a GetDisplayMessagesResponse	7. The Test System sends a GetDisplayMessagesRequest with id <Generated displayMessageId> requestId <Generated requestId>
9. The Charging Station sends a NotifyDisplayMessagesRequest	10. The Test System responds with a NotifyDisplayMessagesResponse .
<u>Note(s):</u> If tbc is True at Step 9 then step 9 and 10 will be repeated	

Tool validations

* Step 2:

Message **SetDisplayMessageResponse**

- **status** *Accepted*

* Step 3:

At least one of the following messages must be sent:

Message: **StatusNotificationRequest**

- **connectorStatus** *Faulted* or *Unavailable*

Message: **NotifyEventRequest**

- **eventData[0].trigger** must be *Delta*

- **eventData[0].actualValue** must be *Faulted* or *Unavailable*

- **eventData[0].component.name** must be *Connector*

- **eventData[0].variable.name** must be *AvailabilityState*

- **evse.id** *<not omitted>*

- **connector.id** *<not omitted>*

Message: **NotifyEventRequest**

- **eventData[0].trigger** must be *Delta*

- **eventData[0].actualValue** must be *Faulted*

- **eventData[0].component.name** must be *ChargingStation*

- **eventData[0].variable.name** must be *AvailabilityState*

- **evse.id** *<omitted>*

- **connector.id** *<omitted>*

* Step 5:

At least one of the following messages must be sent:

Message: **StatusNotificationRequest**

- **connectorStatus** *Available*

Message: **NotifyEventRequest**

- **eventData[0].trigger** must be *Delta*

- **eventData[0].actualValue** must be *Available*

- **eventData[0].component.name** must be *Connector*

- **eventData[0].variable.name** must be *AvailabilityState*

- **evse.id** *<not omitted>*

- **connector.id** *<not omitted>*

Message: **NotifyEventRequest**

- **eventData[0].trigger** must be *Delta*

- **eventData[0].actualValue** must be *Available*

- **eventData[0].component.name** must be *ChargingStation*

- **eventData[0].variable.name** must be *AvailabilityState* + * Step 8:

Message **GetDisplayMessagesResponse**

- **status** *Accepted*

* Step 9:

Message **NotifyDisplayMessagesRequest**

- **requestId** *<Generated requestId>*

- **state** *Faulted*

Post scenario validations:

- N/a

6.2.117. Page 555 - (2024-11) - Remove StatusNotificationRequest from Authorized reusable state Main B steps

Main B (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a RequestStartTransactionResponse	1. The OCTT sends a RequestStartTransactionRequest with idToken.idToken <Configured valid_idtoken_idtoken> idToken.type <Configured valid_idtoken_type> evseld <Configured evseld>
3. The Charging Station sends an AuthorizeRequest <u>Note(s):</u> - This step needs to be executed when AuthCtrlr.AuthorizeRemoteStart is true, unless (AuthEnabled is implemented with mutability ReadOnly AND the value is set to false) OR the idToken is cached. In case the idToken is used for a reservation, sending the AuthorizeRequest message is optional.	4. The OCTT responds with an AuthorizeResponse with idTokenInfo.status Accepted
<Removed>	
5. The Charging Station sends a TransactionEventRequest <u>Note(s):</u> - This step needs to be executed when TxStartPoint contains Authorized OR the transaction already started. So in the case TxStartPoint contains ParkingBayOccupancy or (EVConnected , in the case this testcase was initiated from state EVConnectedPreSession .)	6. The OCTT responds with a TransactionEventResponse <u>Note(s):</u> - The first TransactionEventRequest sent after authorization contains the idToken field. The TransactionEventResponse of this request message contains idTokenInfo with status Accepted

6.2.118. Page 559 - (2025-11) - Reusable state EnergyTransferSuspended - Removed invalid note

Step 1 contains a note that incorrectly describes that the **TransactionEvent** only should be sent in case **TxStopPoint** contains **EnergyTransfer**. This is not correct, the **TransactionEvent** always needs to be transmitted in this case, however the values of the fields it contains might be different.

Removed Note from step 1 and improved tool validation text from step 1:

EnergyTransferSuspended

State	EnergyTransferSuspended
	...

Main (Test scenario)	
Charging Station	CSMS
<u>Notes(s):</u> The tool will wait for <Configured Transaction Duration> seconds	
<u>Manual Action:</u> The EV suspends the energy transfer.	
1. The Charging Station sends a TransactionEventRequest <u>Note(s):</u> - This step needs to be executed unless the transaction was already stopped. So in the case TxStopPoint contains EnergyTransfer	2. The Test System responds with a TransactionEventResponse

Tool validations * Step 1: Message: TransactionEventRequest - triggerReason must be <i>ChargingStateChanged</i> (If chargingState = <i>SuspendedEV</i>) - transactionInfo.chargingState must be <i>EVConnected</i> OR <i>SuspendedEV</i> - transactionInfo.stoppedReason must be <i>StoppedByEV</i> (if eventType = <i>Ended</i>) - eventType must be <i>Ended</i> OR <i>Updated</i> * Step 1: Message: TransactionEventRequest IF TxStopPoint contains <i>EnergyTransfer</i> THEN: - triggerReason must be <i>ChargingStateChanged</i> (If chargingState = <i>SuspendedEV</i>) - transactionInfo.chargingState must be <i>EVConnected</i> OR <i>SuspendedEV</i> - transactionInfo.stoppedReason must be <i>StoppedByEV</i> - eventType must be <i>Ended</i> ELSE: - triggerReason must be <i>ChargingStateChanged</i> - transactionInfo.chargingState must be <i>SuspendedEV</i> - eventType must be <i>Updated</i> END Post scenario validations: State is <i>EnergyTransferSuspended</i>
--

6.2.119. Page 560 - (2025-04) - Reusable states StopAuthorized & Deauthorized

Based on TWG and CWG discussions the transaction validations of the Test system have been made more flexible and validate accordingly based on the different TxStartPoint and TxStopPoint combinations. However, the reusable states described at part 6 did not reflect this yet.

State	StopAuthorized
System under test	Charging Station
Description	This state will prepare the Charging Station, so that it is in a state where the charging session is authorized to stop. This can be done in two ways (Configurable at Test System): A. Using local authorization B. Using a RequestStopTransactionRequest

Main A (Test scenario)	
Charging Station	CSMS
Notes(s): The tool will wait for <Configured Transaction Duration> seconds	
Manual Action: Present the same idToken as used to start the transaction.	
1. The Charging Station sends a TransactionEventRequest	2. The Test System responds with a TransactionEventResponse With idTokenInfo.status is <i>Accepted</i>
Note(s): This step is optional	
3. The Charging Station sends a TransactionEventRequest	4. The Test System responds with a TransactionEventResponse With idTokenInfo.status is <i>Accepted</i>

Tool validations
* Step 1: Message: TransactionEventRequest - triggerReason must be <i>StopAuthorized</i> - idToken omit OR - idToken.idToken <Configured valid_idtoken_idtoken> AND - idToken.type <Configured valid_idtoken_type> * Step 3: Message: TransactionEventRequest - triggerReason must be <i>ChargingStateChanged</i> - transactionInfo.chargingState must be <i>EVConnected</i> If TxStopPoint contains Authorized or PowerPathClosed or EnergyTransfer Then the last of the two TransactionEventRequest messages from step 1 and 3 needs to contain: - eventType must be <i>Ended</i> - transactionInfo.stoppedReason must be <i>Local</i> or omitted Else Then both TransactionEventRequest messages need to contain: - eventType must be <i>Updated</i>

Main B (Test scenario)	
Charging Station	CSMS
2. The Charging Station responds with a RequestStopTransactionResponse	1. The Test System sends a RequestStopTransactionRequest with transactionId <transactionId provided by the Charging Station in <i>TransactionEventRequest</i> >
3. The Charging Station sends a TransactionEventRequest	4. The Test System responds with a TransactionEventResponse
<u>Note(s)</u> : This step is optional	
5. The Charging Station sends a TransactionEventRequest	6. The Test System responds with a TransactionEventResponse With idTokenInfo.status is <i>Accepted</i>

Tool validations
* Step 2: Message: RequestStopTransactionResponse - status must be <i>Accepted</i> * Step 3: Message: TransactionEventRequest - triggerReason must be <i>RemoteStop</i> * Step 5: Message: TransactionEventRequest - triggerReason must be <i>ChargingStateChanged</i> - transactionInfo.chargingState must be <i>EVConnected</i> If TxStopPoint contains Authorized or PowerPathClosed or EnergyTransfer Then the last of the two TransactionEventRequest messages from step 3 and 5 needs to contain: - eventType must be <i>Ended</i> - transactionInfo.stoppedReason must be <i>Remote</i> Else Then both TransactionEventRequest messages need to contain: - eventType must be <i>Updated</i> Post scenario validations: State is <i>StopAuthorized</i>

New reusable state Deauthorized. This is not a change to the testcases It already existed on the background, but is now formally defined at part 6.

6.2.120. Deauthorized

State	Deauthorized
System under test	Charging Station
Description	This reusable state will set the Charging Station to a state in which the transaction will be deauthorized.
Prerequisite	Reusable State <i>Authorized</i> is executed.

Before (Preparations)
Configuration State: N/a
Memory State: N/a
Reusable State(s): Continues executing transaction reusable states in order until the first TransactionEventRequest is received based on the configured TxStartPoint.

Main (Test scenario)	
Charging Station	CSMS
1. The Charging Station sends a TransactionEventRequest	2. The Test System responds with a TransactionEventResponse With idTokenInfo.status is <i>Invalid</i>
3. The Charging Station sends a TransactionEventRequest <u>Note(s):</u> - Step 3 and 4 are relevant when AuthCtrlr.StopTxOnInvalidId is <i>true</i>	4. The Test System responds with a TransactionEventResponse
6. The Charging Station sends a UnlockConnectorResponse	5. The Test System sends a UnlockConnectorRequest <u>Note(s):</u> - Step 5 and 6 are executed when the connector is locked and the transaction gets deauthorized.

Tool validations
* Step 1: Message: TransactionEventRequest - idToken.idToken <Configured valid_idtoken_idtoken> AND - idToken.type <Configured valid_idtoken_type> * Step 3: Message: TransactionEventRequest - triggerReason must be <i>Deauthorized</i> - transactionInfo.chargingState must be <i>EVConnected</i> or <i>SuspendedEVSE</i> If TxStopPoint contains <i>Authorized</i> or <i>PowerPathClosed</i> or <i>EnergyTransfer</i> Then : - eventType must be <i>Ended</i> - stoppedReason must be <i>DeAuthorized</i> Else: - eventType must be <i>Updated</i>

6.2.121. Page 566 - (2025-02) - Reusable state RenegotiateChargingLimits

Main (Test scenario)	
Charging Station	CSMS
[...]	
7. The Charging Station sends a TransactionEventRequest	8. The Test System responds with a TransactionEventResponse
Note: Steps 7 and 8 are optional, but can also repeat until chargingState is Charging.	

Tool validations
[...]
* Step 5: Message: NotifyEVChargingScheduleRequest - evseld <Configured evseld>
* Step 7: Message: TransactionEventRequest - triggerReason must be ChargingStateChanged - transactionInfo.chargingState must be Charging - evseld <Configured evseld>
Post scenario validations: N/a

6.2.122. Page 573/151 - (2025-04) - Removed Main steps B from IdTokenCached reusable state and added IdTokenCached15118

The described steps at step B were incorrect and the originally intended split does not exist in any of the currently existing testcases.

In addition, a separate IdTokenCached15118 reusable state has been added for ISO 15118 sessions. The affected testcases are TC_C_54_CS and TC_C_55_CS.

IdTokenCached

State	IdTokenCached
System under test	Charging Station
Description	An idToken is stored in the Authorization Cache of the Charging Station.

Before (Preparations)
Configuration State: N/a
Memory State: N/a
Reusable State(s): N/a

Main (Test scenario)	
Charging Station	CSMS
1. Execute Reusable State ParkingBayoccupied	
2. Execute Reusable State Authorized	
Note(s): Step 3 and onwards are executed in case the idToken at step 2 was Accepted.	
3. Execute Reusable State EVConnectedPreSession	
4. Execute Reusable State EnergyTransferStarted	
5. Execute Reusable State StopAuthorized	
6. Execute Reusable State EVDisconnected	
7. Execute Reusable State ParkingBayUnoccupied	

Tool validations
N/a

IdTokenCached15118

State	IdTokenCached15118
System under test	Charging Station
Description	A 15118-idToken is stored in the Authorization Cache of the Charging Station.

Before (Preparations)
Configuration State: N/a
Memory State: N/a
Reusable State(s): N/a

Main (Test scenario)	
Charging Station	CSMS
1. Execute Reusable State <i>ParkingBayoccupied</i>	
2. Execute Reusable State <i>EVConnectedPreSession</i>	
3. Execute Reusable State <i>Authorized15118</i>	
4. Execute Reusable State <i>EnergyTransferStarted</i>	
5. Execute Reusable State <i>StopAuthorized</i> (Remote)	
6. Execute Reusable State <i>EVDisconnected</i>	
7. Execute Reusable State <i>ParkingBayUnoccupied</i>	

Tool validations
N/a

Test case name	Authorization using Contract Certificates 15118 - Offline - ContractValidationOffline is true
Test case Id	TC_C_54_CS
...	

Before (Preparations)
Configuration State: ...
Memory State: <i>CertificateInstalled</i> for certificateType V2GRootCertificate <i>CertificateInstalled</i> for certificateType MORootCertificate <i>RenewV2GChargingStationCertificate</i> (If none are present, when checking with GetInstalledCertificateIds.certificateType = V2GCertificateChain) <i>IdTokenCached15118</i> for <i><Configured valid ISO 15118 IdToken></i> (If implemented) <i>IdTokenLocalAuthList</i> for <i><Configured valid ISO 15118 IdToken></i> (If implemented)
Reusable State(s): N/a

Test case name	Authorization using Contract Certificates 15118 - Offline - ContractValidationOffline is false
Test case Id	TC_C_55_CS
...	

Before (Preparations)
Configuration State: ...
Memory State: CertificateInstalled for certificateType V2GRootCertificate CertificateInstalled for certificateType MORootCertificate RenewV2GChargingStationCertificate (If none are present, when checking with GetInstalledCertificateIds.certificateType = V2GCertificateChain) IdTokenCached15118 for <Configured valid ISO 15118 IdToken> (If implemented) IdTokenLocalAuthList for <Configured valid ISO 15118 IdToken> (If implemented)
Reusable State(s): N/a

6.2.123. Page 574 - (2025-04) - IdTokenLocalAuthList memory state - set Enable to true if implemented

State	IdTokenLocalAuthList
System under test	Charging Station
Description	An valid idToken is stored in the Local Authorization List of the Charging Station.

Before (Preparations)
Configuration State: LocalAuthListCtrlr.Enabled is true (If implemented)
Memory State: N/a
Reusable State(s): N/a

6.2.124. Page 575 - (2024-09) - Reusable state RenewChargingStationCertificate expects a reconnection [784]

If a valid certificate is installed, then charging station must use it. This involves reconnecting to set up a new TLS with the new certificate. If the charging station does not do so automatically, then OCTT will force it by sending a Reset command.

State	RenewChargingStationCertificate
System under test	Charging Station
Description	The ChargingStationCertificate is renewed using A02/A03
...	

State	RenewChargingStationCertificate	
Main (Test scenario)	Charging Station	CSMS
	2. The Charging Station responds with a TriggerMessageResponse	1. The OCTT sends a TriggerMessageRequest With requestedMessage <i>SignChargingStationCertificate</i>
	3 The Charging Station sends a SignCertificateRequest	4. The OCTT responds with a SignCertificateResponse With status <i>Accepted</i>
	6. The Charging Station responds with a CertificateSignedResponse	5. The OCTT sends a CertificateSignedRequest With certificateChain <Certificate generated from the received CSR from step 3 and signed by the provided CSMS Root certificate> certificateType <i>ChargingStationCertificate</i>
	If the certificate is valid, then Charging Station should reconnect with the new certificate. OCTT waits some time for a reconnection, and if that does not occur, will send a Reset command to Charging Station to force a reconnection.	
	7. The Charging Station reconnects.	
	8 . If the reconnect was forced by a Reset: The Charging Station sends a BootNotificationRequest	9. OCTT responds with a BootNotificationResponse .
Tool validations	* Step 2: Message: TriggerMessageResponse - status must be <i>Accepted</i> * Step 3: Message: SignCertificateRequest - csr must contain <An CSR that meets the following requirements: <i>When using RSA or DSA the key must be at least 2048 bits long. and when using elliptic curve cryptography the key must be at least 224 bits long. The received CSR must be transmitted as described in RFC 2986 and then encoded in Privacy-Enhanced Mail (PEM) format.></i> * Step 6: Message: CertificateSignedResponse - status must be <i>Accepted</i> * Step 7: Charging Station must reconnect with new certificate.	
	Post scenario validations: N/a	

6.2.125. Page 575 - (2025-02) - Reusable state RenewChargingStationCertificate must not do a Reset [5281]

This memory state was sending a Reset when Charging Station was dropping the connection in time, but according to A02.FR.08 a Reset is not needed. It is enough to just close the websocket connection.

State	RenewChargingStationCertificate
System under test	Charging Station
Description	The ChargingStationCertificate is renewed using A02/A03
...	

State	RenewChargingStationCertificate	
Main (Test scenario)	Charging Station	CSMS
	...	...
	6. The Charging Station responds with a CertificateSignedResponse	5. The OCTT sends a CertificateSignedRequest With certificateChain <Certificate generated from the received CSR from step 3 and signed by the provided CSMS Root certificate> certificateType ChargingStationCertificate
	If the certificate is valid, then Charging Station should reconnect with the new certificate. Test System waits some time for a reconnection, and if that does not occur, will drop the connection to force a reconnection.	
	7. The Charging Station reconnects.	
Tool validations	8. If Charging Station rebooted: The Charging Station sends a BootNotificationRequest	9. Test System responds with a BootNotificationResponse .
	... Post scenario validations: N/a	

6.3. CSMS

6.3.1. Page 593 - (2025-04) - TC_A_11_CSMS - Added post scenario validation for clarification

Note: This erratum extends erratum: [Page 593 - \(2024-09\) - TC_A_11_CSMS - Reconnect using new client certificate](#)

Test case name	Update Charging Station Certificate by request of CSMS - Success - Charging Station Certificate
Test case Id	TC_A_11_CSMS
...	

Tool validations
N/a
Post scenario validations:
The Test System and the CSMS are connected.

6.3.2. Page 593 - (2024-09) - TC_A_11_CSMS - Reconnect using new client certificate

Note: This erratum is extended by erratum: [Page 593 - \(2025-04\) - TC_A_11_CSMS - Added post scenario validation for clarification](#)

The testcase is missing steps to reconnect using the new client certificate.

Test case name	Update Charging Station Certificate by request of CSMS - Success - Charging Station Certificate
Test case Id	TC_A_11_CSMS
...	...

Main (Test scenario)	
Charging Station	CSMS
1. Execute Reusable State RenewChargingStationCertificate	
2. The OCTT disconnects its current connection and reconnects to the CSMS with the new certificate.	3. The CSMS accepts the incoming connection request using the new certificate.

6.3.3. Page 596 - (2024-09) - TC_A_14_CSMS - Update Charging Station Certificate by request of CSMS - Invalid certificate

SecurityEventNotification(InvalidChargingStationCertificate) has been added.

Test case name	Update Charging Station Certificate by request of CSMS - Invalid certificate
Test case Id	TC_A_14_CSMS
...	

Test case name	Update Charging Station Certificate by request of CSMS - Invalid certificate	
Main (Test scenario)	Charging Station	CSMS
	2. The OCTT responds with a TriggerMessageResponse With status Accepted	1. The CSMS sends a TriggerMessageRequest
	3 The OCTT sends a SignCertificateRequest With csr <Configured CSR> certificateType ChargingStationCertificate	4. The CSMS responds with a SignCertificateResponse
	6. The OCTT responds with a CertificateSignedResponse With status Rejected	5. The CSMS sends a CertificateSignedRequest
	7. The OCTT sends a SecurityEventNotificationRequest with type = InvalidChargingStationCertificate	8. The CSMS responds with a SecurityEventNotificationResponse
Tool validations	...	

6.3.4. Page 597 - (2025-04) - TC_A_19_CSMS - Added main steps and clarified tool validations

Clarified the validation steps regarding how the CSMS rejects the connection and added reconnect at the end of the testcase, so it does not end without a connection.

Test case name	Upgrade Charging Station Security Profile - Accepted
Test case Id	TC_A_19_CSMS
...	

Main (Test scenario)	
Charging Station	CSMS
<u>Manual Action:</u> Request the CSMS to set a new <i>NetworkConnectionProfile</i> with a security profile level one higher than currently configured	
2. The Test System responds with a SetNetworkProfileResponse With status Accepted	1. The CSMS sends a SetNetworkProfileRequest
<u>Manual Action:</u> Request the CSMS to change the <i>NetworkConfigurationPriority</i> to one that contains the <i>configurationSlot</i> of the new <i>NetworkConnectionProfile</i> from step 1	
4. The Test System responds with a SetVariablesResponse with status Accepted	3. The CSMS sends a SetVariablesRequest
<u>Manual Action:</u> Request the CSMS to reboot the Charging Station	
6. The Test System responds with a ResetResponse with status Accepted	5. The CSMS sends a ResetRequest
7. The Test System reconnects to the CSMS using the new NetworkProfile, containing the upgraded security profile <Configured securityProfile + 1> .	8. The CSMS accepts the connection attempt.
9. Execute Reusable State Booted	
10. The Test System reconnects to the CSMS using the original NetworkProfile, containing the lower security profile.	11. The CSMS shall not accept the connection attempt.
<u>Note(s) :</u> - This is done to ensure that the CSMS does not accept a connection using the lower security profile anymore.	

Main (Test scenario)	
12. The Test System reconnects to the CSMS using the new NetworkProfile, containing the upgraded security profile <Configured securityProfile + 1>. Note(s) : - This is done to restore the connection before ending the testcase.	13. The CSMS accepts the connection attempt.

Tool validations
* Step 1: Message SetNetworkProfileRequest - connectionData.messageTimeout <Configured messageTimeout> - connectionData.ocppInterface <Configured ocppInterface> - connectionData.ocppTransport JSON - connectionData.ocppVersion OCPP20 - connectionData.securityProfile <Configured securityProfile + 1> * Step 3: Message SetVariablesRequest setVariableData: - variable.name = "NetworkConfigurationPriority" - component.name = "OCPPCommCtrlr" - attributeValue = <contains configurationSlot provided at step 1> * Step 11: When upgrading a Charging Station to a higher security profile, a CSMS has several options regarding which endpoint to use. This affects the way the CSMS is able to detect it needs to reject the incoming connection attempt. In case of having upgraded from security profile 2 to 3, but there is an incoming connection attempt using security profile 2: When the same endpoint is used, then it depends on the CSMS endpoint configuration. - When the CSMS does a full switch and only allows TLS handshakes when a client certificate is provided, then the TLS handshake is rejected. - When the CSMS only requires this Charging Station to use a client certificate, then it accepts the TLS handshake (because it will be unable to detect which Charging Station is connecting) and it rejects the HTTP request to establish the WebSocket connection. When a different port or a whole different endpoint is used for the upgrade, then on the original endpoint the CSMS accepts the TLS handshake and it rejects the HTTP request to establish the WebSocket connection (because this Charging Station is not allowed to connect with security profile 2 anymore). In case of security profile 1, the case is always the same. The CSMS shall always reject the HTTP request to establish the WebSocket connection, because TLS is required for this Charging Station. Post scenario validations: The Test System and the CSMS are connected.

6.3.5. Page 597 - (2024-09) - TC_A_19_CSMS - Added additional information regarding the use of the client certificates

Added additional information regarding the use of the client certificates.

Test case name	Upgrade Charging Station Security Profile - Accepted
Test case Id	TC_A_19_CSMS
...	...

Before (Preparations)
Configuration State: N/a
Memory State: If configured <Security profile> is 2, then RenewChargingStationCertificate The OCTT uses this certificate during the TLS handshake when connecting with security profile 3.
Reusable State(s): N/a

6.3.6. Page 597 - (2024-09) - TC_A_19_CSMS - Removed validation of OcCppCsmsUrl [020-4355]

Validation of OcCppCsmsUrl has been removed, because in some implementations the URL changes with the security profile.

Test case name	Upgrade Charging Station Security Profile - Accepted	
Test case Id	TC_A_19_CSMS	
...		
Main (Test scenario)	Charging Station	CSMS
	...	...
Tool validations	* Step 1: Message SetNetworkProfileRequest - connectionData.messageTimeout <Configured messageTimeout> - connectionData.ocppCsmsUrl <Configured ocppCsmsUrl> - connectionData.ocppInterface <Configured ocppInterface> - connectionData.ocppTransport JSON - connectionData.ocppVersion OCPP20 - connectionData.securityProfile <Configured securityProfile + 1> * Step 3: Message SetVariablesRequest setVariableData: - variable.name = "NetworkConfigurationPriority" - component.name = "OCPPCommCtrlr" - attributeValue = <contains configurationSlot provided at step 1>	
	Post scenario validations: - N/a	

6.3.7. Page 637 - (2024-11) - TC_C_50_CSMS - Changed reference to configured valid idToken to a specific eMAID idToken

For ISO 15118 plug & charge the Charging Station always needs to use an eMAID idToken, which equals the CN of the configured contract certificate.

Test case name	Authorization using Contract Certificates 15118 - Online - Local contract certificate validation - Accepted
Test case Id	TC_C_50_CSMS
Use case Id(s)	C07
Requirement(s)	C07.FR.04
System under test	CSMS
Description	The Charging Station is able to authorize with contract certificates when it supports ISO 15118.
Purpose	To verify if the CSMS is able to validate the certificate hash data and the provided eMAID.

Test case name	Authorization using Contract Certificates 15118 - Online - Local contract certificate validation - Accepted
Prerequisite(s)	- The configured eMAID is known by the CSMS as valid. - The configured contract certificate is valid. - The CN of the configured contract certificate equals the configured eMAID. - iso15118CertificateHashData has a responder URL that points to an OCSP service for OCTT. - CSMS does not have a cached OCSP response for the contract certificate.

Before (Preparations)
Configuration State: N/a
Memory State: N/a
Reusable State(s): State is <i>EVConnectedPreSession</i>

Main (Test scenario)	
Charging Station	CSMS
1. The OCTT sends an AuthorizeRequest With idToken.idToken <i><Configured eMAID></i> idToken.type <i>eMAID</i> iso15118CertificateHashData contains <i><hashes from configured (V2G) certificate chain</i>	...
...	

6.3.8. Page 639 - (2024-09) - TC_C_52_CSMS - TC does not use <Configured contract_certificate>

OCTT already has a keystore that contains the certificate. The pdf should not mention the <Configured contract_certificate> as the testcase does not use it

Test case name	Authorization using Contract Certificates 15118 - Online - Central contract certificate validation - Accepted	
Test case Id	TC_C_52_CSMS	
...		
Prerequisite(s)	- The configured eMAID is known by the CSMS as valid. - The configured contract certificate is signed by the configured V2GRoot or MORoot certificate at the CSMS. - Contract certificate has a responder URL that points to an OCSP service for OCTT. - CSMS does not have a cached OCSP response for the contract certificate.	
...		
Main (Test scenario)	Charging Station	CSMS
	1. The OCTT sends an AuthorizeRequest With idToken.idToken <i><Configured valid_idtoken_idtoken></i> idToken.type <i><Configured valid_idtoken_type></i> iso15118CertificateHashData is absent certificate <i>from keystore</i>	2. The CSMS sends an OCSP request to responder URL of certificate to check validity
...		

6.3.9. Page 639 - (2025-02) - TC_C_52_CSMS - Certificate needs at least one subCA

Test case name	Authorization using Contract Certificates 15118 - Online - Central contract certificate validation - Accepted
Test case Id	TC_C_52_CSMS
...	...

Before (Preparations)
...

Main (Test scenario)	
Charging Station	CSMS
...	...

Tool validations
* Step 2: CSMS sends an OCSP request for certificate * Step 3: Test System checks that received request for certificate is valid AND key type = ECDSA AND certificate chain contains at least one subCA * Step 4: Message: AuthorizeResponse - idTokenInfo.status <i>Accepted</i> - certificateStatus <i>Accepted</i> * Step 6: Message: TransactionEventResponse - idTokenInfo.status <i>Accepted</i> Post scenario validations: N/a

6.3.10. Page 640 - (2025-04) - TC_D_01_CSMS - Missing tool validation that the idTokenInfo must be provided for all list entries

Test case name	Send Local Authorization List - Full
Test case Id	TC_D_01_CSMS
...	...

Tool validations
* Step 1: Message SendLocalListRequest - updateType <i>Full</i> - versionNumber <i><Bigger than 0></i> - localAuthorizationList <i><Not empty></i> - localAuthorizationList[n].idTokenInfo <i><Not empty></i> Post scenario validations: - N/a

6.3.11. Page 712 - (2024-09) - TC_I_01_CSMS - Show EV Driver running total cost

Test case name	Show EV Driver running total cost during charging - costUpdatedRequest
Test case Id	TC_I_01_CSMS
...	...

Test case name	Show EV Driver running total cost during charging - costUpdatedRequest	
Main (Test scenario)	Charging Station	CSMS
	...	
	7. The OCTT sends a TransactionEventRequest With triggerReason is <i>MeterValuePeriodic</i> eventType is <i>Updated</i> timestamp <The intervals between the timestamps of the received Meter Value messages equals the configured sampled Meter Values interval>. sampledValue.context is <i>Sample.Periodic</i> <u>Note(s):</u> - This step will be executed every _<Configured sampled Meter Values interval> - The OCTT will end the testcase after two MeterValues.	8. The CSMS responds with a TransactionEventResponse
	...	
Tool validations	...	
	Post scenario validations: - N/a	

6.3.12. Page 715 - (2025-02) - TC_I_02_CSMS - Added explicit information about CSMS tariff configuration and sending in needed metervalue

Before (Preparations)	
Configuration State: N/a	
Memory State: N/a - CSMS is configured with a tariff which is based on energy consumed.	
Reusable State(s): state is EVConnectedPostSession - N/a	
Main (Test scenario)	
Charging Station	CSMS
1. Execute Reusable State <i>EVConnectedPreSession</i> - The TransactionEventRequest contains the MeterValue field. - sampledValue[0].value 1000 - sampledValue[0].context <i>Transaction.Begin</i>	
2. Execute Reusable State <i>Authorized</i>	
3. Execute Reusable State <i>EnergyTransferStarted</i>	
4. Execute Reusable State <i>StopAuthorized</i>	
5. Execute Reusable State <i>EVConnectedPostSession</i>	

Main (Test scenario)	
6. The Test System notifies the CSMS about the current state of the configured connector. Message: StatusNotificationRequest - connectorStatus <i>Available</i> Message: NotifyEventRequest - trigger <i>Delta</i> - actualValue <i>"Available"</i> - component.name <i>"Connector"</i> - variable.name <i>"AvailabilityState"</i>	7. The CSMS responds accordingly.
8. The Test System sends a TransactionEventRequest with - triggerReason <i>EVCommunicationLost</i> - eventType <i>Ended</i> - transactionInfo.chargingState <i>Idle</i> - transactionInfo.stoppedReason <i>EVDisconnected</i> - meterValue[0].sampledValue[0].value <i>6000</i> - meterValue[0].sampledValue[0].context <i>Transaction.End</i>	9. The CSMS responds with a TransactionEventResponse

6.3.13. Page 726-760 - (2025-04) - TC_K_XX_CSMS - Improved tool validations to be sure valid Charging Profiles are used

Test case name	Set Charging Profile - TxDefaultProfile - Specific EVSE
Test case Id	TC_K_01_CSMS
...	

Tool validations
* Step 1: Message SetChargingProfileRequest evseld <Configured evseld> AND chargingProfile.stackLevel <Configured stackLevel> AND chargingProfile.chargingProfilePurpose <i>TxDefaultProfile</i> AND chargingProfile.chargingProfileKind <i>Absolute</i> AND chargingProfile.validFrom <i>now</i> AND chargingProfile.validTo <i>now + <Configured Charging Schedule Duration></i> AND chargingProfile.chargingSchedule.startSchedule <i>now</i> AND chargingProfile.chargingSchedule.chargingRateUnit <Configured chargingRateUnit> AND chargingProfile.chargingSchedule.duration <Configured duration> AND chargingProfile.chargingSchedule.chargingSchedulePeriod.startPeriod <i>0</i> AND chargingProfile.chargingSchedule.chargingSchedulePeriod.limit <i>6.0 or 6000.0</i> AND chargingProfile.chargingSchedule.chargingSchedulePeriod.numberPhases <Configured numberPhases> where <Configured numberPhases> not 3 OR chargingProfile.chargingSchedule.chargingSchedulePeriod.numberPhases <Configured numberPhases> or <omit> where <Configured numberPhases> 3 Post scenario validations: - N/a

Test case name	Set Charging Profile - TxProfile without ongoing transaction on the specified EVSE
Test case Id	TC_K_02_CSMS
...	

Tool validations
* Step 1: Message SetChargingProfileRequest - evseld <Configured evseld> AND - chargingProfile.chargingProfilePurpose TxProfile AND - chargingProfile.stackLevel <Configured stackLevel> AND - chargingProfile.chargingProfileKind Relative AND - chargingProfile.chargingSchedule.startSchedule must be omitted AND - chargingProfile.chargingSchedule.chargingRateUnit <Configured chargingRateUnit> AND - chargingProfile.chargingSchedule.chargingSchedulePeriod.startPeriod 0 AND - chargingProfile.chargingSchedule.chargingSchedulePeriod.limit 7.0 or 7000.0 AND - chargingProfile.chargingSchedule.chargingSchedulePeriod.numberPhases <Configured numberPhases> where <Configured numberPhases> not 3 OR - chargingProfile.chargingSchedule.chargingSchedulePeriod.numberPhases <Configured numberPhases> or <omit> where <Configured numberPhases> 3
Post scenario validations: - N/a

Test case name	Replace charging profile - With chargingProfileId
Test case Id	TC_K_04_CSMS
...	

Tool validations
* Step 1: Message SetChargingProfileRequest chargingProfile.chargingSchedule.chargingSchedulePeriod.startPeriod 0 AND chargingProfile.chargingSchedule.chargingSchedulePeriod.limit 8.0 (A) or 8000.0 (W) The chargingSchedule contains only one chargingSchedulePeriod * Step 3: Message SetChargingProfileRequest chargingProfile.chargingSchedule.chargingSchedulePeriod.startPeriod 0 AND chargingProfile.chargingSchedule.chargingSchedulePeriod.limit 6.0 (A) or 6000.0 (W) The chargingSchedule contains only one chargingSchedulePeriod * Step 1/3: Message SetChargingProfileRequest chargingProfile.id <Same id for both chargingProfiles> chargingProfile.chargingSchedule.startSchedule must NOT be omitted. chargingProfile.stackLevel <Configured stackLevel> AND chargingProfile.chargingProfilePurpose <Equal value for both profiles> AND (TxDefaultProfile OR ChargingStationMaxProfile) chargingProfile.chargingProfileKind <Equal value for both profiles> AND If chargingProfile.chargingProfilePurpose is TxDefaultProfile then chargingProfile.chargingProfileKind must be Absolute OR Recurring If chargingProfile.chargingProfilePurpose is ChargingStationMaxProfile then chargingProfile.chargingProfileKind must be Absolute If chargingProfile.chargingProfileKind is Recurring then chargingProfile.recurrencyKind must NOT be omitted, else omitted The received Charging Profiles must comply with the requirements defined at part 2 specification.
Post scenario validations: - N/a

Test case name	Clear Charging Profile - With chargingProfileId
Test case Id	TC_K_05_CSMS
...	

Tool validations	
* Step 1: Message ClearChargingProfileRequest chargingProfileId <Generated <i>chargingProfileId</i> > AND chargingProfileCriteria omit	
Post scenario validations: - N/a	

Test case name	Clear Charging Profile - With stackLevel/purpose combination for one profile
Test case Id	TC_K_06_CSMS
...	

Tool validations	
* Step 1: Message ClearChargingProfileRequest chargingProfileCriteria.chargingProfilePurpose <i>TxDefaultProfile</i> AND chargingProfileCriteria.stackLevel <Configured <i>stackLevel</i> > AND chargingProfileCriteria.evseId <Configured <i>evseId</i> > AND chargingProfileId must be omitted.	
Post scenario validations: - N/a	

Test case name	Clear Charging Profile - Without previous charging profile
Test case Id	TC_K_08_CSMS
...	

Tool validations	
* Step 1: Message ClearChargingProfileRequest chargingProfileCriteria.chargingProfilePurpose <i>TxDefaultProfile</i> AND chargingProfileCriteria.stackLevel <Configured <i>stackLevel</i> > AND chargingProfileCriteria.evseId <Configured <i>evseId</i> > AND chargingProfileId must be omitted.	
Post scenario validations: - N/a	

Test case name	Set Charging Profile - Not Supported
Test case Id	TC_K_15_CSMS
...	

Tool validations	
* Step 1: Message SetChargingProfileRequest evseld <Configured evseld> AND chargingProfile.stackLevel <Configured stackLevel> AND chargingProfile.chargingProfilePurpose TxDefaultProfile AND chargingProfile.chargingProfileKind Absolute AND chargingProfile.chargingSchedule.startSchedule <Not omitted> AND chargingProfile.chargingSchedule.chargingRateUnit <Configured ChargingRateUnit> AND chargingProfile.chargingSchedule.chargingSchedulePeriod.startPeriod 0 AND chargingProfile.chargingSchedule.duration <Configured duration> chargingProfile.chargingSchedule.chargingSchedulePeriod.limit 6.0 or 6000.0 AND chargingProfile.chargingSchedule.chargingSchedulePeriod.numberPhases <Configured numberPhases> where <Configured numberPhases> not 3 OR chargingProfile.chargingSchedule.chargingSchedulePeriod.numberPhases <Configured numberPhases> or <omit> where <Configured numberPhases> 3 +	
Post scenario validations: - N/a	

Test case name	Get Charging Profile - Evseld 0
Test case Id	TC_K_29_CSMS
...	

Tool validations	
* Step 1: Message GetChargingProfilesRequest - evseld 0 AND - chargingProfile.chargingProfilePurpose <Configured chargingProfilePurpose> AND <u>Note</u>: chargingProfilePurpose is included, because the chargingProfile field is required and may not be left empty. - chargingProfile.stackLevel must be omitted AND - chargingProfile.chargingLimitSource must be omitted AND - chargingProfile.chargingProfileId must be omitted	
Post scenario validations: - N/a	

Test case name	Get Charging Profile - Evseld > 0
Test case Id	TC_K_30_CSMS
...	

Tool validations	
* Step 1: Message GetChargingProfilesRequest - evseld <Configured evseld> AND - chargingProfile.chargingProfilePurpose <Configured chargingProfilePurpose> AND <u>Note</u>: chargingProfilePurpose is included, because the chargingProfile field is required and may not be left empty. - chargingProfile.stackLevel must be omitted AND - chargingProfile.chargingLimitSource must be omitted AND - chargingProfile.chargingProfileId must be omitted	
Post scenario validations: - N/a	

Test case name	Get Charging Profile - No Evseld
Test case Id	TC_K_31_CSMS
...	

Tool validations	
* Step 1: Message GetChargingProfilesRequest - evseld must be omitted AND - chargingProfile.chargingProfilePurpose <Configured chargingProfilePurpose> AND <i>Note : chargingProfilePurpose is included, because the chargingProfile field is required and may not be left empty.</i> - chargingProfile.stackLevel must be omitted AND - chargingProfile.chargingLimitSource must be omitted AND - chargingProfile.chargingProfileId must be omitted	
Post scenario validations: - N/a	

Test case name	Get Charging Profile - chargingProfileId
Test case Id	TC_K_32_CSMS
...	

Tool validations	
* Step 1: Message GetChargingProfilesRequest - evseld must be omitted AND - chargingProfile.chargingProfilePurpose must be omitted AND - chargingProfile.stackLevel must be omitted AND - chargingProfile.chargingLimitSource must be omitted AND - chargingProfile.chargingProfileId <received chargingProfileId>	
Post scenario validations: - N/a	

Test case name	Get Charging Profile - Evseld > 0 + stackLevel
Test case Id	TC_K_33_CSMS
...	

Tool validations	
* Step 1: Message GetChargingProfilesRequest - evseld <Configured evseld> AND - chargingProfile.stackLevel <Configured stackLevel> AND - chargingProfile.chargingProfilePurpose must be omitted AND - chargingProfile.chargingLimitSource must be omitted AND - chargingProfile.chargingProfileId must be omitted	
Post scenario validations: - N/a	

Test case name	Get Charging Profile - Evseld > 0 + chargingLimitSource
Test case Id	TC_K_34_CSMS
...	

Tool validations	
* Step 1: Message GetChargingProfilesRequest - evseld <Configured evseld> AND - chargingProfile.chargingLimitSource <Configured chargingLimitSource> AND - chargingProfile.stackLevel must be omitted AND - chargingProfile.chargingProfilePurpose must be omitted AND - chargingProfile.chargingProfileId must be omitted	

Tool validations	
Post scenario validations: - N/a	

Test case name	Get Charging Profile - Evseld > 0 + chargingProfilePurpose
Test case Id	TC_K_35_CSMS
...	

Tool validations	
* Step 1: Message GetChargingProfilesRequest - evseld <Configured evseld> AND - chargingProfile.chargingProfilePurpose <Configured chargingProfilePurpose> AND - chargingProfile.stackLevel must be omitted AND - chargingProfile.chargingLimitSource must be omitted AND - chargingProfile.chargingProfileId must be omitted	
Post scenario validations: - N/a	

Test case name	Get Charging Profile - Evseld > 0 + chargingProfilePurpose + stackLevel
Test case Id	TC_K_36_CSMS
...	

Tool validations	
* Step 1: Message GetChargingProfilesRequest - evseld <Configured evseld> AND - chargingProfile.chargingProfilePurpose <Configured chargingProfilePurpose> AND - chargingProfile.chargingLimitSource must be omitted AND - chargingProfile.stackLevel <Configured stackLevel> AND - chargingProfile.chargingProfileId must be omitted	
Post scenario validations: - N/a	

Test case name	Set Charging Profile - TxProfile with ongoing transaction on the specified EVSE
Test case Id	TC_K_60_CSMS
...	

Tool validations	
* Step 1: (Message: SetChargingProfileRequest) chargingProfile.chargingProfilePurpose is TxProfile AND chargingProfile.evseld is <Configured evseld> AND chargingProfile.transactionId <Generated transactionId> AND chargingProfile.chargingProfileKind is Relative OR Absolute If chargingProfileKind is Relative then chargingSchedule.startSchedule must be omitted. If chargingProfileKind is Absolute then chargingSchedule.startSchedule must NOT be omitted. The received Charging Profile must comply with the requirements defined at part 2 specification.	
Post scenario validations: N/a	

Test case name	Remote start transaction with charging profile - Success
Test case Id	TC_K_37_CSMS
...	

Tool validations
* Step 1: Message: RequestStartTransactionRequest idToken.idToken <Configured valid idToken> idToken.type <Configured valid idToken type> evseld <Configured evseld> chargingProfile contains: chargingProfile.chargingProfilePurpose is <i>TxProfile</i> chargingProfile.transactionId is omitted chargingProfile.chargingProfileKind is <i>Relative</i> OR <i>Absolute</i> If chargingProfileKind is <i>Relative</i> then chargingSchedule.startSchedule must be omitted. If chargingProfileKind is <i>Absolute</i> then chargingSchedule.startSchedule must NOT be omitted. The received Charging Profile must comply with the requirements defined at part 2 specification. Post scenario validations: N/a

Test case name	Set Charging Profile - Multiple Profiles
Test case Id	TC_K_70_CSMS
Use case Id(s)	n/a
Requirement(s)	n/a
System under test	CSMS
Description	To enable the CSMS to influence the charging power or current drawn from a specific EVSE or the entire Charging Station over a period of time. The CSMS sends a SetChargingProfileRequest to the Charging Station to influence the power or current drawn by EVs. The CSMS calculates a ChargingSchedule to stay within certain limits, which MAY be imposed by any external system.
Purpose	To verify if the CSMS is able to set multiple Charging Profiles.
Prerequisite(s)	n/a

Tool validations
* Step 1: Message SetChargingProfileRequest chargingProfile.chargingProfilePurpose <i>TxDefaultProfile</i> chargingProfile.chargingProfileKind is <i>Absolute</i> OR <i>Recurring</i> chargingProfile.chargingSchedule.startSchedule must NOT be omitted. If chargingProfile.chargingProfileKind is <i>Recurring</i> then chargingProfile.recurrencyKind must NOT be omitted. * Step 3: Message SetChargingProfileRequest chargingProfile.id <different id from chargingProfile at step 1> chargingProfile.chargingProfilePurpose <i>ChargingStationMaxProfile</i> chargingProfile.chargingProfileKind is <i>Absolute</i> chargingProfile.chargingSchedule.startSchedule must NOT be omitted. The received Charging Profiles must comply with the requirements defined at part 2 specification. Post scenario validations: - N/a

6.3.14. Page 728 - (2024-09) - TC_K_03_CSMS - Not requiring validFrom/To fields in charging profile [020-4592] and chargingProfileKind must be Absolute [020-4591]

Test case name	Set Charging Profile - ChargingStationMaxProfile	
Test case Id	TC_K_03_CSMS	
...		
Main (Test scenario)	Charging Station	CSMS
	...	...
Tool validations	* Step 1: Message SetChargingProfileRequest evseld 0 AND chargingProfile.stackLevel <Configured stackLevel> AND chargingProfile.chargingProfilePurpose ChargingStationMaxProfile_ AND chargingProfile.chargingProfileKind Absolute OR Relative chargingProfile.chargingSchedule.chargingRateUnit <Configured ChargingRateUnit> chargingProfile.chargingSchedule.duration <Configured duration> chargingProfile.chargingSchedule.chargingSchedulePeriod.startPeriod 0 chargingProfile.chargingSchedule.chargingSchedulePeriod.limit 8.0 or 8000.0 chargingProfile.chargingSchedule.chargingSchedulePeriod.numberPhases <Configured numberPhases> where <Configured numberPhases> not 3 OR chargingProfile.chargingSchedule.chargingSchedulePeriod.numberPhases <Configured numberPhases> or <omit> where <Configured numberPhases> 3 chargingProfile.validFrom <Not omitted> chargingProfile.validTo <Not omitted> chargingProfile.chargingSchedule.startSchedule <Not omitted> Post scenario validations: - N/a	

6.3.15. Page 733 - (2024-09) - TC_K_10_CSMS - Not requiring validFrom/To fields in charging profile [020-4592]

Test case name	Set Charging Profile - TxDefaultProfile - All EVSE	
Test case Id	TC_K_10_CSMS	
...		
Main (Test scenario)	Charging Station	CSMS
	...	...

Test case name	Set Charging Profile - TxDefaultProfile - All EVSE
Tool validations	* Step 1: Message SetChargingProfileRequest evseld 0 AND chargingProfile.stackLevel <Configured stackLevel> AND chargingProfile.chargingProfilePurpose TxDefaultProfile AND chargingProfile.chargingProfileKind Absolute AND chargingProfile.validFrom <Not omitted> AND chargingProfile.validTo <Not omitted> AND chargingProfile.chargingSchedule.startSchedule <Not omitted> AND chargingProfile.chargingSchedule.chargingRateUnit <Configured ChargingRateUnit> AND chargingProfile.chargingSchedule.chargingSchedulePeriod.startPeriod 0 AND chargingProfile.chargingSchedule.duration <Configured duration> chargingProfile.chargingSchedule.chargingSchedulePeriod.limit 6.0 or 6000.0 AND chargingProfile.chargingSchedule.chargingSchedulePeriod.numberPhases <Configured numberPhases> where <Configured numberPhases> not 3 OR chargingProfile.chargingSchedule.chargingSchedulePeriod.numberPhases <Configured numberPhases> or <omit> where <Configured numberPhases> 3 Post scenario validations: - N/a

6.3.16. Page 734 - (2024-09) - TC_K_15_CSMS - Not requiring validFrom/To fields in charging profile [020-4592]

Test case name	Set Charging Profile - Not Supported	
Test case Id	TC_K_15_CSMS	
...		
Main (Test scenario)	Charging Station	CSMS
	2. The OCTT responds with RPC Framework: CALLERROR: NotSupported.	1. The CSMS sends a SetChargingProfileRequest with: evseld <Configured evseld> AND chargingProfile.stackLevel <Configured stackLevel> AND chargingProfile.chargingProfilePurpose <i>TxDefaultProfile</i> AND chargingProfile.chargingProfileKind <i>Absolute</i> AND chargingProfile.validFrom <Not omitted> AND chargingProfile.validTo <Not omitted> AND chargingProfile.chargingSchedule.startSchedule <Not omitted> AND chargingProfile.chargingSchedule.chargingRateUnit <Configured ChargingRateUnit> AND chargingProfile.chargingSchedule.chargingSchedulePeriod.startPeriod 0 AND chargingProfile.chargingSchedule.duration <Configured duration> chargingProfile.chargingSchedule.chargingSchedulePeriod.limit 6.0 or 6000.0 AND chargingProfile.chargingSchedule.chargingSchedulePeriod.numberPhases <Configured numberPhases>
Tool validations	...	

6.3.17. Page 752 - (2025-06) - TC_K_53_CSMS - Validate that the CSMS let's the Charging Station charging, according to the negotiated limits

Test case name	Charging with load leveling based on High Level Communication - Success
Test case Id	TC_K_53_CSMS
...	

Main (Test scenario)	
Charging Station	CSMS
1. Execute reusable state <i>ISO15118SmartCharging</i>	
2. The CSMS does NOT send a SetChargingProfileRequest	
Note(s) : - The CSMS must NOT initiate a renegotiate after starting the transaction, without cause. For example; a smart charging algorithm or an external trigger, etc.	

6.3.18. Page 753 - (2024-09) - TC_K_55_CSMS, TC_K_57_CSMS, TC_K_58_CSMS, TC_K_59_CSMS Removed triggerReason = ChargingRateChanged [776]

A trigger reason **ChargingStateChange** must only be sent, when an external actor (not CSMS) changes the charging rate. Therefore, removed the sending of a **triggerReason=ChargingStateChange** by OCTT. This does not affect tool validations, but it was incorrect behavior of OCTT.

The step that sends a **TransactionEventRequest** with **triggerReason=ChargingRateChanged** has been removed from:

- TC_K_55_CSMS
- TC_K_57_CSMS
- TC_K_58_CSMS
- TC_K_59_CSMS

6.3.19. Page 760 - (2024-11) - TC_K_70_CSMS - Updated tool validation chargingProfiles and added preparation step.

We require a CSMS to install multiple **ChargingProfiles** with the same purpose and for the same **connectorId** with a different **stackLevel**. However there are CSMSs that do the stacking themselves and are unable to do this. Therefore it was decided to use different purposes instead.

Test case name	Set Charging Profile - Multiple Profiles	
Test case Id	TC_K_70_CSMS	
...		
Before (Preparations)	...	
	Charging State: State is <i>EnergyTransferStarted</i>	
Main (Test scenario)	Charging Station	CSMS
	2. The OCTT responds with a SetChargingProfileResponse with status Accepted	1. The CSMS sends a SetChargingProfileRequest with chargingProfilePurpose TxDefaultProfile
	4. The OCTT responds with a SetChargingProfileResponse with status Accepted	3. The CSMS sends a SetChargingProfileRequest with chargingProfilePurpose ChargingStationMaxProfile

Test case name	Set Charging Profile - Multiple Profiles
Tool validations	* Step 1: Message SetChargingProfileRequest chargingProfile.chargingProfilePurpose <i>TxDefaultProfile</i> * Step 3: Message SetChargingProfileRequest chargingProfile.id <i><different id for both chargingProfiles></i> chargingProfile.chargingProfilePurpose <i>ChargingStationMaxProfile</i> Post scenario validations: - N/a

6.3.20. Page 805/806 - (2025-06) - TC_N_01_CSMS & TC_N_02_CSMS - omit filter fields that are not tested

These testcases specifically test the use of the different uses of the fields that are used to filter the result. monitoringCriteria in case of TC_N_01_CSMS and componentVariable in case of TC_N_02_CSMS. The fields that are not part of the testcase need to be omitted.

Test case name	Get Monitoring Report - with monitoringCriteria
Test case Id	TC_N_01_CSMS
...	

Tool validations
* Step 1: Message: GetMonitoringReportRequest - monitoringCriteria = <i>DeltaMonitoring</i> - componentVariable is omitted.
* Step 3: Message: GetMonitoringReportRequest - monitoringCriteria = <i>ThresholdMonitoring</i> - componentVariable is omitted.
Post scenario validations: Check that CSMS shows the <i>Threshold</i> monitors.

Test case name	Get Monitoring Report - with component/variable
Test case Id	TC_N_02_CSMS
...	

Tool validations
* Step 1: Message: GetMonitoringReportRequest - componentVariable[0].component.name = <i>"ChargingStation"</i> - componentVariable[0].variable.name = <i>"Power"</i> - monitoringCriteria is omitted.
* Step 3: Message: GetMonitoringReportRequest - componentVariable[1].component.name = <i>"EVSE"</i> - componentVariable[1].component.evse.id = <i>1</i> - componentVariable[1].variable.name = <i>"AvailabilityState"</i> - monitoringCriteria is omitted.
Post scenario validations: Check that CSMS shows the monitor for AvailabilityState for EVSE #1.

6.3.21. Page 806 - (2025-06) - TC_N_02_CSMS - component and variable instance need to be omitted

The testcase specifically requests two component variables that do not have an instance.

Test case name	Get Monitoring Report - with component/variable
Test case Id	TC_N_02_CSMS
...	

Tool validations
* Step 1: Message: GetMonitoringReportRequest - componentVariable[0].component.name = "ChargingStation" - componentVariable[0].component.instance is omitted. - componentVariable[0].variable.name = "Power" - componentVariable[0].variable.instance is omitted. - monitoringCriteria is omitted.
* Step 3: Message: GetMonitoringReportRequest - componentVariable[1].component.name = "EVSE" - componentVariable[1].component.instance is omitted. - componentVariable[1].component.evse.id = 1 - componentVariable[1].variable.name = "AvailabilityState" - componentVariable[1].variable.instance is omitted. - monitoringCriteria is omitted.
Post scenario validations: Check that CSMS shows the monitor for AvailabilityState for EVSE #1.

6.3.22. Page 824 - (2025-04) - TC_N_62_CSMS - Check only single identifier is provided

Added validation that idToken and customerCertificate are omitted.

TC_N_62_CSMS: Clear Customer Information - Clear and report - customerIdentifier

Test case name	Clear Customer Information - Clear and report - customerIdentifier
Test case Id	TC_N_62_CSMS
...	

Before (Preparations)
..

Main (Test scenario)	
Charging Station	CSMS
...	...

Tool validations
* Step 1: Message CustomerInformationRequest - report true - clear true - customerIdentifier "OpenChargeAlliance" - idToken is omitted - customerCertificate is omitted

Tool validations
Post scenario validations: - N/a

6.3.23. Page 830 - (2024-11) - TC_N_46_CSMS - Updated tool validation sendLocalListRequest

Test case name	Clear Customer Information - Update Local Authorization List	
Test case Id	TC_N_46_CSMS	
...		
Before (Preparations)	...	
	Memory State: A local authorization list with <Configured valid_idtoken_idtoken> is configured.	
	...	
Main (Test scenario)	Charging Station	CSMS
	...	...
Tool validations	* Step 1: Message CustomerInformationRequest - report true AND - clear true AND - idToken.idToken <Configured valid_idtoken_idtoken> - idToken.type <Configured valid_idtoken_type> * Step 5: Message SendLocalListRequest - updateType Differential - versionNumber <Bigger than currently configured in OCTT> - localAuthorizationList <Contains only the configured valid_idtoken_idtoken, without idTokenInfo> OR - updateType Full - localAuthorizationList <Does NOT contain configured valid_idtoken_idtoken>	
	Post scenario validations: - N/a	

6.3.24. Page 830 - (2025-02) - TC_N_46_CSMS - Aligning configuration variable usage

Main (Test scenario)	
Charging Station	CSMS
Manual action: Trigger CSMS to CustomerInformationRequest to both report and clear token <Configured valid_idtoken_idtoken> and <Configured valid_idtoken_type>	
2. The Test System responds with a CustomerInformationResponse with status Accepted	1. The CSMS sends a CustomerInformationRequest
3. The Test System sends a NotifyCustomerInformationRequest	4. The CSMS responds with a NotifyCustomerInformationResponse .
Manual action: If not triggered automatically, trigger CSMS to send SendLocalListRequest with version = <configured local list version> + 1 and updateType = Differential and localAuthorizationList = [{ idToken = { <Configured_valid_idtoken>, <Configured valid_idtoken_type> } }]	

Main (Test scenario)	
6 The Test System responds with a SendLocalListResponse with status <i>Accepted</i>	5. The CSMS sends a SendLocalListRequest
Note(s): <i>If the Local Authorization List is too big for one message, step 5 and 6 will be repeated</i>	

Tool validations
* Step 1: Message CustomerInformationRequest - report <i>true</i> AND - clear <i>true</i> AND - idToken.idToken <i><Configured valid_idtoken_idtoken></i> - idToken.type <i><Configured valid_idtoken_type></i> * Step 5: Message SendLocalListRequest - updateType <i>Differential</i> - versionNumber <i><configured local list version> + 1</i> - localAuthorizationList <i><Contains only the configured valid_idtoken_idtoken, without idTokenInfo></i> - localAuthorizationList[0].idToken contains <i><configured_valid_idtoken_idtoken></i> and <i><configured valid_idtoken_type></i> - localAuthorizationList[0].idTokenInfo <i><omitted></i>
Post scenario validations: - All messages have been received

6.3.25. Page 854 - (2025-02) - TC_O_27_CSMS - Fixing validations to be more specific for test case

Tool validations
* Step 1: Message SetDisplayMessageRequest - message.id <Generated Id> - message.startDateTime <Configured startDateTime> - message.transactionId is present - message.state is <omitted> - message.startDateTime is <Configured startDateTime> - message.endDateTime is <omitted> - message.transactionId is <Generated transactionId from Before>
Post scenario validations: - N/a

6.3.26. Page 855 - (2025-02) - TC_O_28_CSMS - Fixing validations to be more specific for test case

A test case needs to start in Preparations phase:

Before (Preparations)
Configuration State: N/a
Memory State: N/a
Reusable State: State is EnergyTransferStarted
Tool validations
* Step 1: Message SetDisplayMessageRequest - message.id <Generated Id> - message.priority <Configured Priority> - message.endDateTime <Configured endDateTime> - message.state <Configured State> - message.transactionId is present - message.state is <omitted> - message.startDateTime is <omitted> - message.endDateTime is <Configured endDateTime> - message.transactionId is <Generated transactionId from Before>
Post scenario validations: - N/a